

structure, when $n = 1$). Within the intestinal cells, folate is converted principally to N^5 -methyl FH_4 , which enters the portal vein and goes to the liver. Smaller amounts of other forms of folate also follow this route.

The liver, which stores half of the body's folate, takes up much of the folate from the portal circulation; uptake may be through active transport or receptor-mediated endocytosis. Within the liver, FH_4 is reconjugated to the polyglutamate form before being used in reactions. A small amount of the folate is partially degraded, and the components enter the urine. A relatively large portion of the folate enters the bile and is subsequently reabsorbed (very similar to the fate of bile salts in the enterohepatic circulation).

N^5 -Methyl- FH_4 , the major form of folate in the blood, is loosely bound to plasma proteins, particularly serum albumin.

C. Oxidation and Reduction of the One-Carbon Groups of Tetrahydrofolate

One-carbon groups transferred by FH_4 are attached either to nitrogen N^5 or N^{10} or they form a bridge between N^5 and N^{10} . The collection of one-carbon groups attached to FH_4 is known as the one-carbon pool. While attached to FH_4 , these one-carbon units can be oxidized and reduced (Fig. 40.3). Thus, reactions requiring a carbon at a particular oxidation state may use carbon from the one-carbon pool that was donated at a different oxidation state.

The individual steps for reduction of the one-carbon group are shown in Fig. 40.4. The most oxidized form is N^{10} -formyl FH_4 . The most reduced form is N^5 -methyl- FH_4 . Once the methyl group is formed, it is *not readily reoxidized back to N^5 , N^{10} methylene FH_4 , and thus N^5 -methyl- FH_4 will tend to accumulate in the cell.*

D. Sources of One-Carbon Groups Carried by FH_4

Carbon sources for the one-carbon pool include serine, glycine, formaldehyde, histidine, and formate (see Fig. 40.4). These donors transfer the carbons to folate at different oxidation states. Serine is the major carbon source of one-carbon groups in the human. Its hydroxymethyl group is transferred to FH_4 in a reversible reaction, catalyzed by the enzyme serine hydroxymethyltransferase. This reaction produces glycine and N^5 , N^{10} -methylene- FH_4 . Because serine can be synthesized from 3-phosphoglycerate, an intermediate of glycolysis, dietary carbohydrate can serve as a source of carbon for the one-carbon pool. The glycine that is produced may be further degraded by donation of a carbon to folate. Additional donors that form N^5 , N^{10} methylene- FH_4 are listed in Table 40.1.

Table 40.1. One-Carbon Pool: Sources and Recipients of Carbon

Source ^a	Form of One-Carbon Donor Produced ^b	Recipient	Final Product
Formate	N^{10} -formyl- FH_4	Purine precursor	Purine (C2 and C8)
Serine	N^5 , N^{10} methylene FH_4	dUMP	dTMP
Glycine		Glycine	Serine
Formaldehyde			
N^5 , N^{10} methylene FH_4	N^5 -methyl FH_4	Vitamin B12	Methylcobalamin
Histidine	N^5 -formimino FH_4 is converted to N^5 , N^{10} methylene FH_4		
Choline	Betaine	Homocysteine	Methionine and Dimethylglycine
Methionine	S-adenosylmethionine (SAM)	Glycine (there are many others; see figure 40.9B)	N-methylglycine (sarcosine)

^aThe major source of carbon is serine.

^bThe carbon unit attached to FH_4 can be oxidized and reduced (see Fig. 40.3). At the methyl level, reoxidation does not occur.

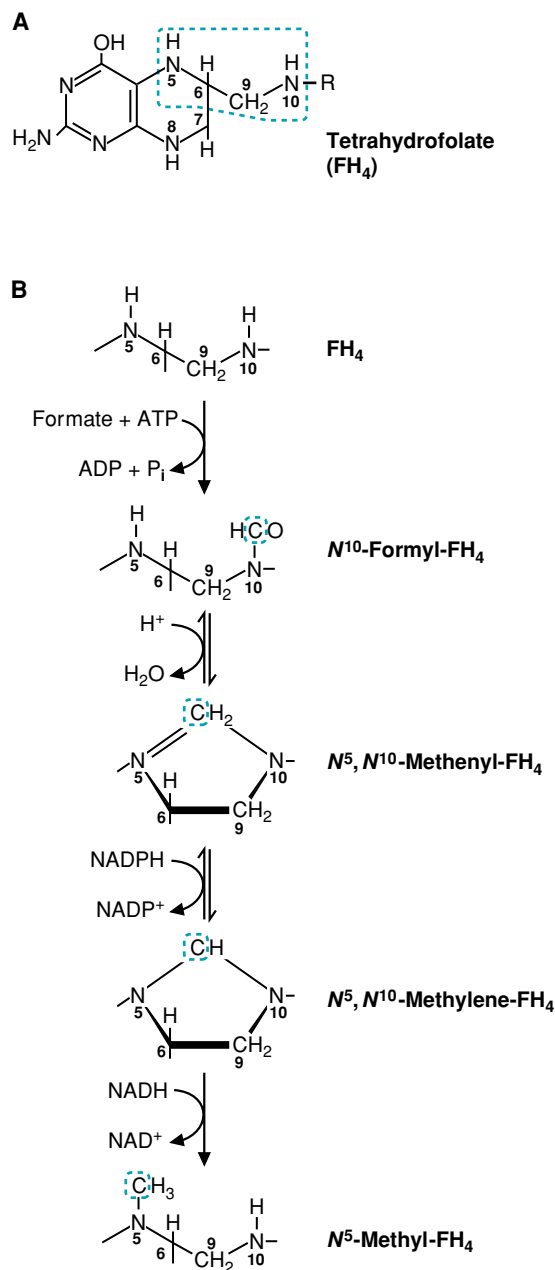


Fig. 40.3. One-carbon units attached to FH₄. A. The active form of FH₄. For definition of R, see Figure 40.2. B. Interconversions of one-carbon units of FH₄. Only the portion of FH₄ from N⁵ to N¹⁰ is shown, which is indicated by the dashed line in Part A. After a formyl group forms a bridge between N⁵ and N¹⁰, two reductions can occur. Note that N⁵-methyl-FH₄ cannot be reoxidized. The most oxidized form of FH₄ is at the top of the figure, whereas the most reduced form is at the bottom.



A deficiency of folate results in the accumulation of FIGLU, which is excreted in the urine. A histidine load test can be used for detecting folate deficiencies. Patients were given a test dose of histidine (a histidine load), and the amount of FIGLU that appeared in the urine was measured.

Histidine and formate provide examples of compounds that donate carbon at different oxidation levels (see Fig. 40.4). Degradation of histidine produces formiminoglutamate (FIGLU), which reacts with FH₄ to donate a carbon and nitrogen (generating N⁵-formimino-FH₄), thereby releasing glutamate. Formate, produced from tryptophan oxidation, can react with FH₄ and generate N¹⁰-formyl-FH₄, the most oxidized folate derivative.

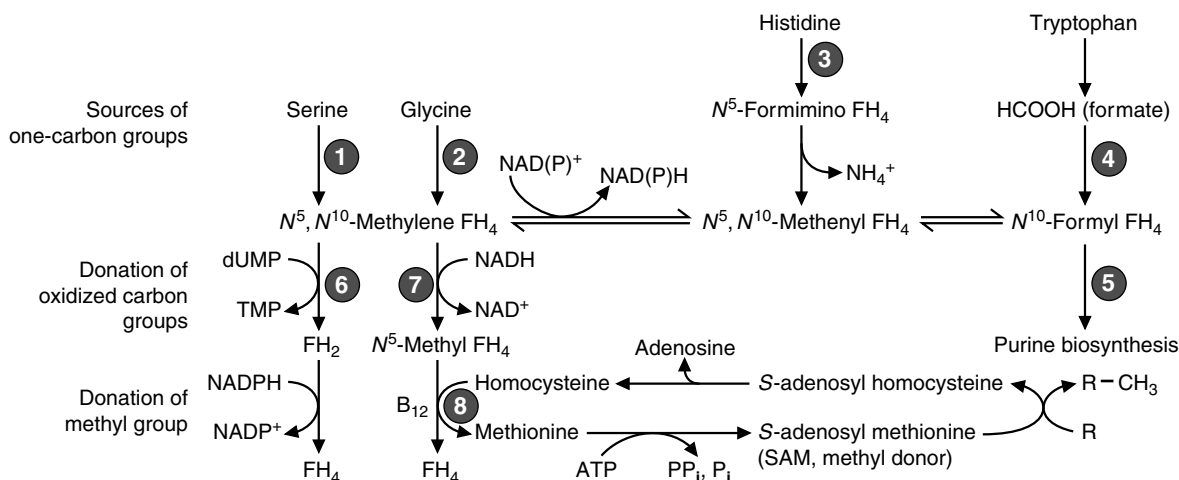


Fig. 40.4. Sources of carbon (reactions 1–4) for the FH_4 pool and the recipients of carbon (reactions 5–8) from the pool. See Figure 40.3 to view the FH_4 derivatives involved in each reaction.

E. Recipients of One-Carbon Groups

The one-carbon groups on FH_4 may be oxidized or reduced (see Fig. 40.3) and then transferred to other compounds (see Fig. 40.4 and Table 40.1). Transfers of this sort are involved in the synthesis of glycine from serine, the synthesis of the base thymine required for DNA synthesis, the purine bases required for both DNA and RNA synthesis, and the transfer of methyl groups to vitamin B12.

Because the conversion of serine to glycine is readily reversible, glycine can be converted to serine by drawing carbon from the one-carbon pool.

The nucleotide deoxythymidine monophosphate (dTMP) is produced from deoxyuridine monophosphate (dUMP) by a reaction in which dUMP is methylated to form dTMP (Fig. 40.5). The source of carbon is N^5, N^{10} -methylene FH_4 . Two hydrogen atoms from FH_4 are used to reduce the donated carbon to the methyl level. Consequently, dihydrofolate (FH_2) is produced. Reduction of FH_2 by NADPH in a



FH_4 is required for the synthesis of deoxythymidine monophosphate and the purine bases used to produce the precursors for DNA replication. Therefore, FH_4 is required for cell division. Blockage of the synthesis of thymine and the purine bases either by a dietary deficiency of folate or by drugs that interfere with folate metabolism results in a decreased rate of cell division and growth.



A better understanding of the structure and function of the purine and pyrimidine bases and of folate metabolism led to the development of compounds having antimetabolic and antifolate action useful for treatment of neoplastic disease. For example, **Colin Tuma** was successfully treated for colon cancer with 5-fluorouracil (5-FU) (see Chapter 12 and Fig. 40.5). 5-FU is a pyrimidine analog, which is converted in cells to the nucleotide fluorodeoxyuridylylate (FdUMP). FdUMP causes a “thymineless death,” especially for tumor cells having a rapid turnover rate. It prevents the growth of cancer cells by blocking the thymidylate synthase reaction, i.e., the conversion of dUMP to dTMP.

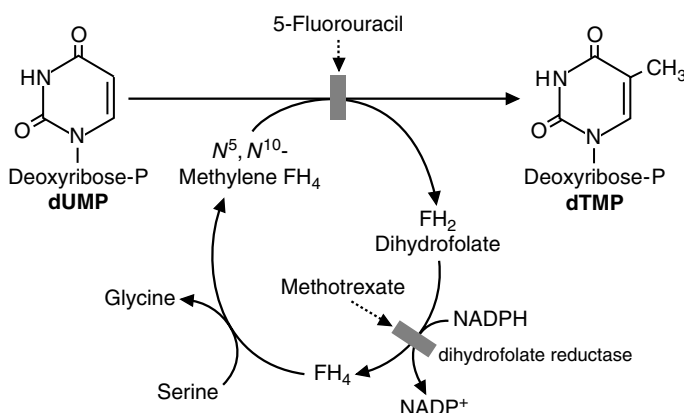


Fig. 40.5. Transfer of a one-carbon unit from N^5, N^{10} methylene FH_4 to dUMP to form dTMP. FH_4 is oxidized to FH_2 (dihydrofolate) in this reaction. FH_2 is reduced to FH_4 by dihydrofolate reductase and FH_4 is converted to N^5, N^{10} methylene FH_4 using serine as a carbon donor. Shaded bars indicate the steps at which the antimetabolites 5-fluorouracil (5-FU) and methotrexate act. 5-FU inhibits thymidylate synthase. Methotrexate inhibits dihydrofolate reductase.



Jean Ann Tonich's megaloblastic anemia was treated, in part, with folate supplements (see Clinical Comments). Within 48 hours of the initiation of folate therapy, megaloblastic or “ineffective” erythropoiesis usually subsides, and effective erythropoiesis begins.

A megaloblastic anemia is caused by a decrease in the synthesis of thymine and the purine bases. These deficiencies lead to an inability of hematopoietic (and other) cells to synthesize DNA and, therefore, to divide. Their persistently thwarted attempts at normal DNA replication, DNA repair, and cell division produce abnormally large cells (called megaloblasts) with abundant cytoplasm capable of RNA and protein synthesis, but with clumping and fragmentation of nuclear chromatin (see Chapter 44). Some of these large cells, although immature, are released early from the marrow in an attempt to compensate for the anemia. Thus, peripheral blood smears will also contain megaloblasts. Many of the large immature cells, however, are destroyed in the marrow and never reach the circulation.



The average daily diet in Western countries contains 5–30 μg vitamin B12, of which 1–5 μg is absorbed into the blood. (The RDA is 2.4 $\mu\text{g}/\text{day}$.) Total body content of this vitamin in an adult is approximately 2–5 mg, of which 1 mg is present in the liver. As a result, a dietary deficiency of B12 is uncommon and is only observed after a number of years on a diet deficient in this vitamin.

In spite of **Jean Ann Tonich's** relatively malnourished state because of chronic alcoholism, her serum cobalamin level was still within the low-to-normal range. If her undernourished state had continued, a cobalamin deficiency would eventually have developed.

reaction catalyzed by dihydrofolate reductase (DHFR) regenerates FH_4 . This is the only reaction involving FH_4 in which the folate group is oxidized as the one-carbon group is donated to the recipient. Recall that DHFR is also required to reduce the oxidized form of the vitamin, which is obtained from the diet (see Fig. 40.2). Thus, DHFR is essential for both regenerating FH_2 in the tissues and from the diet. These reactions contribute to the effect of folate deficiency on DNA synthesis because dTMP is only required for the synthesis of DNA.

During the synthesis of the purine bases, carbons 2 and 8 are obtained from the one-carbon pool (see Chapter 41). N^{10} -Formyl- FH_4 provides both carbons. Folate deficiency would also hinder these reactions, contributing to an inability to replicate DNA because of the lack of precursors.

After the carbon group carried by FH_4 is reduced to the methyl level, it is transferred to vitamin B12. This is the only reaction through which the methyl group can leave FH_4 (recall that the reaction creating N^5 -methyl FH_4 is not reversible).

II. VITAMIN B12

A. Structure and Forms of Vitamin B12

The structure of vitamin B12 (also known as cobalamin) is complex (Fig. 40.6). It contains a corrin ring, which is similar to the porphyrin ring found in heme. The corrin ring differs from heme, however, in that two of the four pyrrole rings are joined directly rather than by a methylene bridge. Its most unusual feature is the presence of cobalt, coordinated with the corrin ring (similar to the iron coordinated with the porphyrin ring). This cobalt can form a bond with a carbon atom. In the body, it reacts with the carbon of a methyl group, forming methylcobalamin, or with the 5'-carbon of 5'-deoxyadenosine, forming 5'-deoxyadenosylcobalamin (note that in this case the deoxy designation refers to the 5' carbon, not the 2' carbon as is the case in the sugar found in DNA). The form of B12 found in vitamin supplements is cyanocobalamin, in which a CN group is linked to the cobalt.

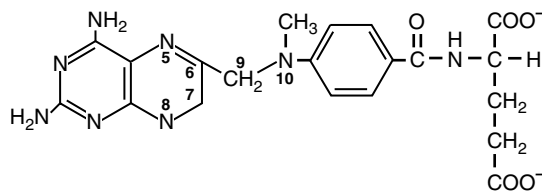
B. Absorption and Transport of Vitamin B12

Although vitamin B12 is produced by bacteria, it cannot be synthesized by higher plants or animals. The major source of vitamin B12 is dietary meat, eggs, dairy products, fish, poultry, and seafood. The animals that serve as the source of these foods obtain B12 mainly from the bacteria in their food supply. The absorption of B12 from the diet is a complex process (Fig. 40.7).

Ingested B12 can exist in two forms, either free or bound to dietary proteins. If free, the B12 binds to proteins known as R-binders (haptocorrins, also known as transcobalamin I), which are secreted by the salivary glands and the gastric mucosa, in either the saliva or the stomach. If the ingested B12 is bound to proteins, it must be released from the proteins by the action of digestive proteases both in the stomach



Individuals with non-Hodgkin's lymphoma receive a number of drugs to treat the tumor, including the use of methotrexate. The structure of methotrexate is shown below.



Methotrexate

What compound does methotrexate resemble?

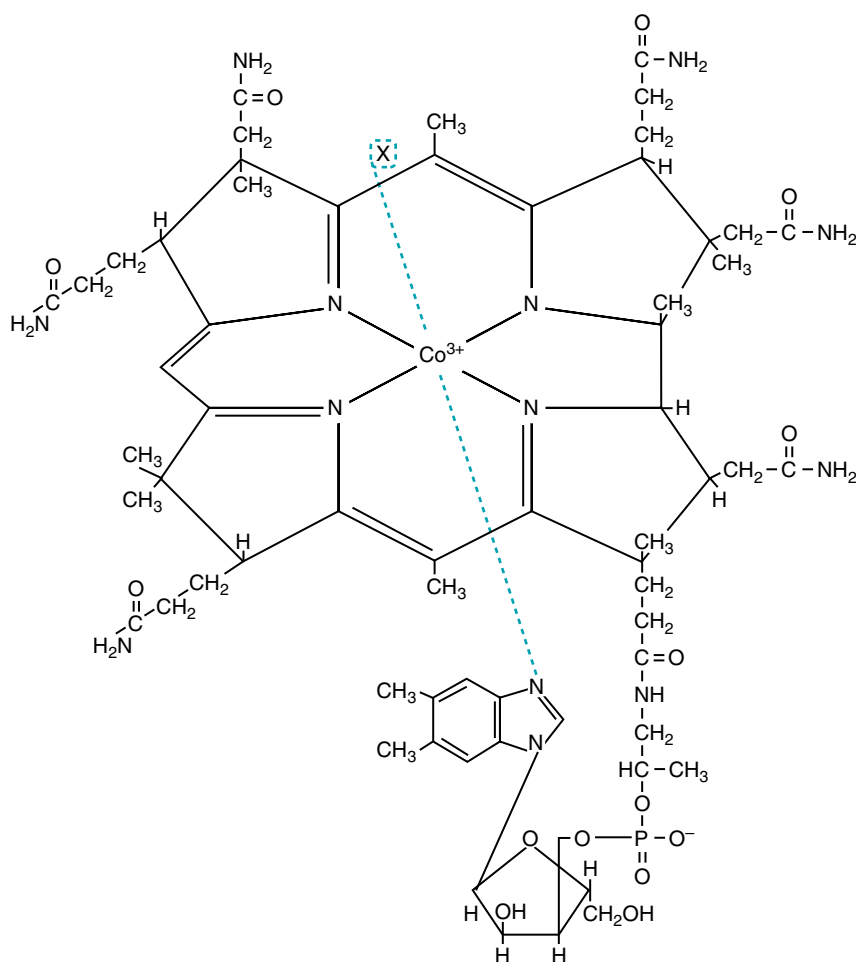


Fig. 40.6. Vitamin B12. X = 5'-deoxyadenosine in deoxyadenosylcobalamin; X = CH₃ in methylcobalamin; X = CN in cyanocobalamin (the commercial form found in vitamin tablets).

and small intestine. Once the B12 is released from its bound protein, it will bind to the haptocorrins. In the small intestine, the pancreatic proteases digest the haptocorrins, and the released B12 then binds to intrinsic factor, a glycoprotein secreted by the parietal cells of the stomach when food enters the stomach. The intrinsic factor–B12 complex attaches to specific receptors in the terminal segment of the small intestine known as the ileum, after which the complex is internalized.

The B12 within the enterocyte complexes with transcobalamin II and then is released into circulation. The transcobalamin II–B12 complex delivers B12 to the tissues, which contain specific receptors for this complex. The liver takes up approximately 50% of the vitamin B12, and the remainder is transported to other tissues. The amount of the vitamin stored in the liver is large enough that 3 to 6 years pass before symptoms of a dietary deficiency occur.

C. Functions of Vitamin B12

Vitamin B12 is involved in two reactions in the body: the transfer of a methyl group from N⁵-methyl FH₄ to homocysteine to form methionine and the rearrangement of the methyl group of L-methylmalonyl CoA to form succinyl CoA (Fig. 40.8).

Tetrahydrofolate receives a one-carbon group from serine or from other sources. This carbon is reduced to the methyl level and transferred to vitamin B12, forming

A: Methotrexate has the same structure as folate except that it has an amino group on C4 and a methyl group on N10. Anticancer drugs such as methotrexate are folate analogs that act by inhibiting dihydrofolate reductase, thereby preventing the conversion of FH₂ to FH₄ (see Fig. 40.5). Thus, the cellular pools of FH₄ are not replenished, and reactions requiring FH₄ cannot proceed.



Pernicious anemia, a deficiency of intrinsic factor, is a relatively common problem caused by malabsorption of dietary cobalamin. It may result from an inherited defect that leads to a decreased ability of gastric parietal cells to synthesize intrinsic factor or from partial resection of the stomach or of the ileum. Production of intrinsic factor often declines with age and may be low in elderly individuals. An alternative circumstance that leads to the development of a B12 deficiency is pancreatic insufficiency or a high intestinal pH, which would result from too little acid being produced by the stomach. Both of these conditions prevent the degradation of the R-binder-B12 complex; as a result, B12 will not be released from the R-binder protein and, therefore, cannot bind to intrinsic factor.

Q: How should vitamin B12 be administered to a patient with pernicious anemia?



Because the problem in pernicious anemia is a lack of intrinsic factor, which results in an inability to absorb vitamin B₁₂ from the gastrointestinal tract, B₁₂ cannot be administered orally to treat this condition. In the past, it was usually given by injection. An effective nasal spray containing B₁₂ has recently been marketed, however, and its use precludes the need for lifelong injections of this vitamin.

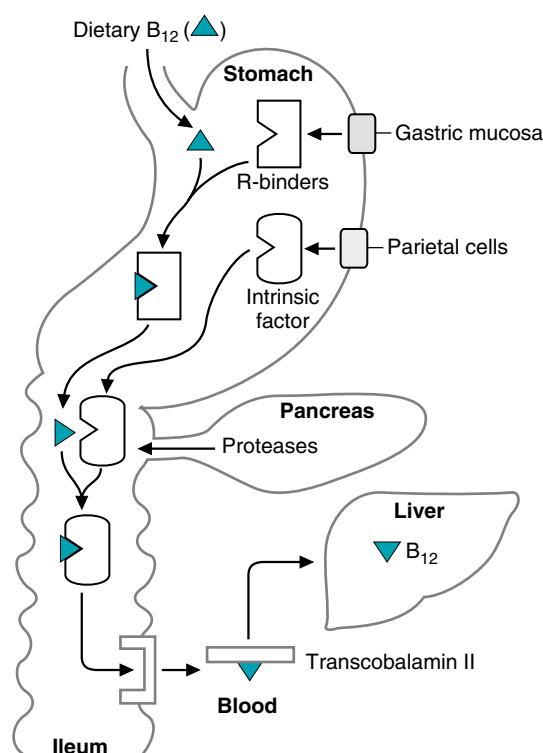


Fig. 40.7. Absorption, transport, and storage of vitamin B₁₂. Dietary B₁₂ binds to R-binders (haptocorrins) in the stomach and travels to the intestine, where the R-binders are destroyed by pancreatic proteases. The freed B₁₂ then binds to intrinsic factor (IF). B₁₂ is absorbed in the ileum and carried by called transcobalamins (TC) to the liver, where B₁₂ is stored.

methyl-B₁₂ (or methylcobalamin). Methylcobalamin transfers the methyl group to homocysteine, which is converted to methionine by the enzyme methionine synthase. Methionine can then be activated to SAM to transfer the methyl group to other compounds (Fig. 40.9).

Vitamin B₁₂ also participates in the conversion of L-methylmalonyl CoA to succinyl CoA. In this case, the active form of the coenzyme is 5'-deoxyadenosylcobalamin. This reaction is part of the metabolic route for the conversion of carbons from valine, isoleucine, threonine, thymine, and the last three carbons of odd-chain fatty acids, all of which form propionyl CoA, to the TCA cycle intermediate succinyl CoA (see Chapter 39).



There are two major clinical manifestations of cobalamin (B₁₂) deficiency. One such presentation is hematopoietic (caused by the adverse effects of a B₁₂ deficiency on folate metabolism), and the other is neurologic (caused by hypomethylation in the nervous system).

The hemopoietic problems associated with a B₁₂ deficiency are identical to those observed in a folate deficiency and, in fact, result from a folate deficiency secondary to (i.e., caused by) the B₁₂ deficiency (i.e., the methyl trap hypothesis). As the FH₄ pool is exhausted, deficiencies of the tetrahydrofolate derivatives needed for purine and dTMP biosynthesis develop, leading to the characteristic megaloblastic anemia.

The classical clinical presentation of the neurologic dysfunction associated with a B₁₂ deficiency includes symmetric numbness and tingling of the hands and feet, diminishing vibratory and position sense, and progression to a spastic gait disturbance. The patient may become somnolent or may become extremely irritable ("megaloblastic madness"). Eventually, blind spots in the central portions of the visual fields develop, accompanied by alterations in gustatory (taste) and olfactory (smell) function. This is believed to be caused by hypomethylation within the nervous system, brought about by an inability to recycle homocysteine to methionine and from there to S-adenosylmethionine. The latter is the required methyl donor in these reactions. The nervous system lacks the betaine pathway of methionine regeneration and is dependent on the B₁₂ system. With a B₁₂ deficiency, this pathway is inoperable in the nervous system.

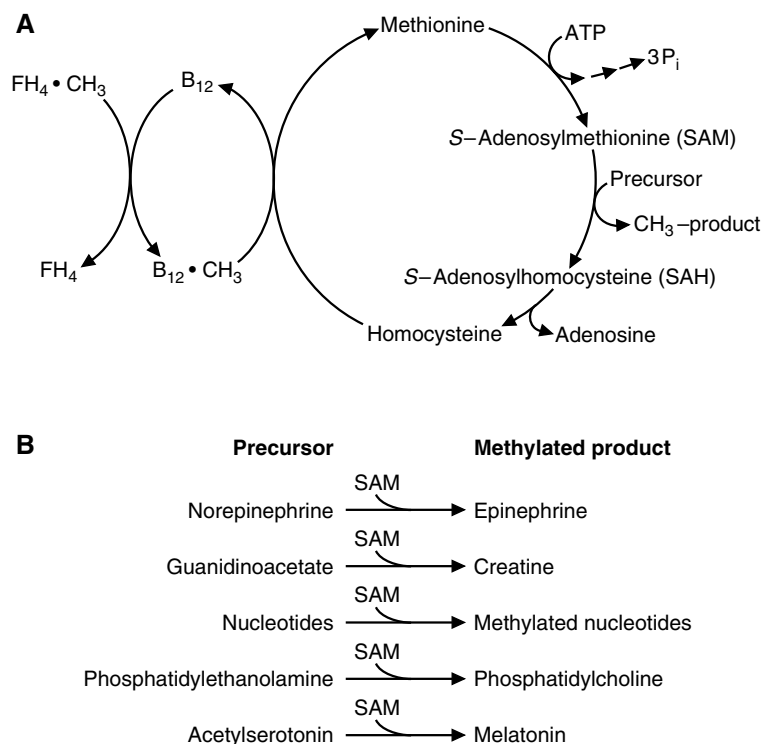


Fig. 40.9. Relationship between FH_4 , B12, and SAM. A. Overall scheme. B. Some specific reactions requiring SAM.

III. S-ADENOSYLMETHIONINE

S-Adenosylmethionine (SAM) participates in the synthesis of many compounds that contain methyl groups. It is used in reactions that add methyl groups to either oxygen or nitrogen atoms in the acceptor (contrast that to folate derivatives, which can add one-carbon groups to sulfur or to carbon). As examples, SAM is required for the conversion of phosphatidylethanolamine to phosphatidylcholine, guanidinoacetate to creatine, norepinephrine to epinephrine, acetylserotonin to melatonin, and nucleotides to methylated nucleotides (see Fig. 40.9B). It is also required for the inactivation of catecholamines and serotonin (see Chapter 48). More than 35 reactions in humans require methyl donation from SAM.

SAM is synthesized from methionine and ATP. As with the activation of vitamin B12, ATP donates the adenosine. With the transfer of its methyl group, SAM forms S-adenosylhomocysteine, which is subsequently hydrolyzed to form homocysteine and adenosine.

Methionine, required for the synthesis of SAM, is obtained from the diet or produced from homocysteine, which accepts a methyl group from vitamin B12

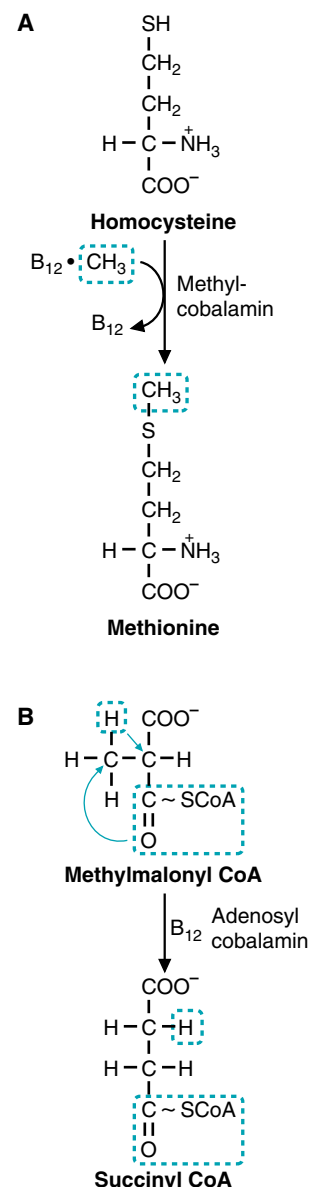


Fig. 40.8. The two reactions involving vitamin B12 in humans.



Many health food stores now sell SAME, a stabilized version of S-adenosylmethionine. SAME has been hypothesized to relieve depression because the synthesis of certain neurotransmitters requires methylation by SAM (see Chapter 47). This has led to the hypothesis that by increasing SAM levels within the nervous system, the biosynthesis of these neurotransmitters will be accelerated. This in turn might alleviate the feelings of depression. There have been reports in the literature indicating that this may occur, but its efficacy as an antidepressant must be confirmed. The major questions that must be addressed include the stability of SAME in the digestive system and the level of uptake of SAME by cells of the nervous system.

(see Fig. 40.9A). Thus, the methyl group of methionine is regenerated. The portion of methionine that is essential in the diet is the homocysteine moiety. If we had an adequate dietary source of homocysteine, methionine would not be required in the diet. However, there is no good dietary source of homocysteine, whereas methionine is plentiful in the diet.

Homocysteine provides the sulfur atom for the synthesis of cysteine (see Chapter 39). In this case, homocysteine reacts with serine to form cystathionine, which is cleaved, yielding cysteine and α -ketobutyrate. The first reaction in this sequence is inhibited by cysteine. Thus, methionine, via homocysteine, is not used for cysteine synthesis unless the levels of cysteine in the body are lower than required for its metabolic functions. An adequate dietary supply of cysteine, therefore, can “spare” (or reduce) the dietary requirement for methionine.

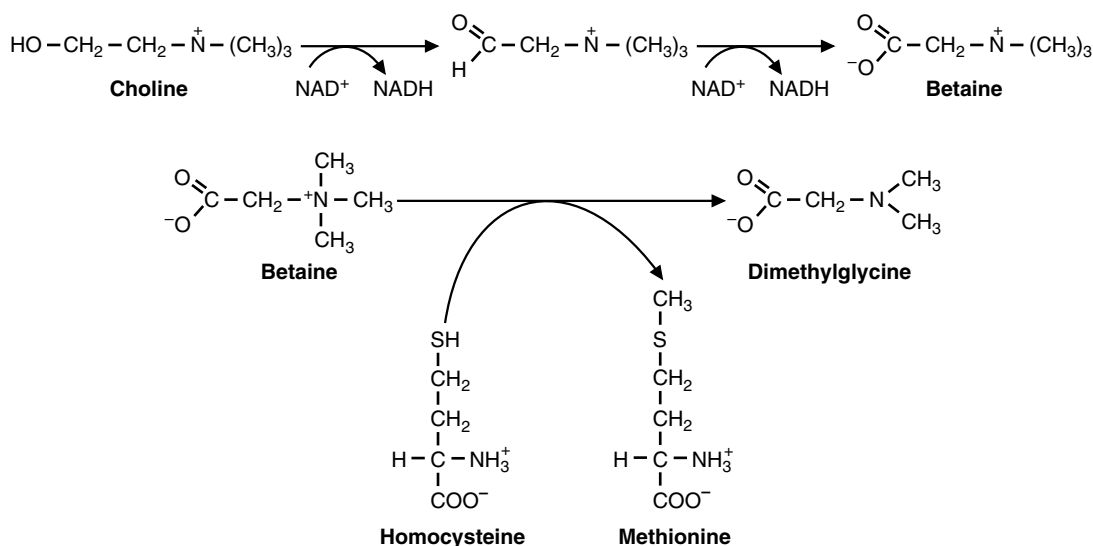
IV. RELATIONSHIPS BETWEEN FOLATE, VITAMIN B12, AND SAM

A. The Methyl-Trap Hypothesis

If one analyzes the flow of carbon in the folate cycle, the equilibrium lies in the direction of the N^5 -methyl FH_4 form. This appears to be the most stable form of carbon attached to the vitamin. However, in only one reaction can the methyl group be removed from N^5 -methyl FH_4 , and that is the methionine synthase reaction, which requires vitamin B12. Thus, if vitamin B12 is deficient, or if the methionine synthase enzyme is defective, N^5 -methyl FH_4 will accumulate. Eventually most folate forms in the body will become “trapped” in the N^5 -methyl form. A functional folate deficiency results because the carbons cannot be removed from the folate. The appearance of a functional folate deficiency caused by a lack of vitamin B12 is known as the “methyl-trap” hypothesis, and its clinical implications are discussed in following sections.



Other compounds involved in one-carbon metabolism are derived from degradation products of choline. Choline, an essential component of certain phospholipids, is oxidized to form betaine aldehyde, which is further oxidized to betaine (trimethylglycine). In the liver, betaine can donate a methyl group to homocysteine to form methionine and dimethyl glycine. This allows the liver to have two routes for homocysteine conversion to methionine. Under conditions in which SAM accumulates, glycine can be methylated to form sarcosine (N-methyl glycine). This route is used when methionine levels are high and excess methionine needs to be metabolized.



B. Hyperhomocysteinemia

Elevated homocysteine levels have been linked to cardiovascular and neurologic disease. Homocysteine levels can accumulate in a number of ways, which are related to both folic acid and vitamin B12 metabolism. Homocysteine is derived from S-adenosyl homocysteine, which arises when SAM donates a methyl group (Fig. 40.10). Because SAM is frequently donating methyl groups, there is a constant production of S-adenosyl homocysteine, which leads to a constant production of homocysteine. Recall from Chapter 39 that homocysteine has two biochemical fates. The homocysteine produced can either be remethylated to methionine or condensed with serine to form cystathionine. There are two routes to methionine production. The major one is methylation by N^5 -methyl FH_4 , requiring vitamin B12. The liver also contains a second pathway in which betaine (a degradation product of choline) can donate a methyl group to homocysteine to form methionine, but this is a minor pathway. The conversion of homocysteine to cystathionine requires pyridoxal phosphate. Thus, if an individual is deficient in vitamin B12, the conversion of homocysteine to methionine by the major route is inhibited. This will direct homocysteine to produce cystathionine, which eventually produces cysteine. As cysteine levels accumulate, the enzyme that makes cystathionine undergoes feedback inhibition, and that pathway is also inhibited (see Fig. 40.10). This, overall, leads to an accumulation of homocysteine, which is released into the blood.

Homocysteine also accumulates in the blood if a mutation is present in the enzyme that converts N^5 , N^{10} methylene FH_4 to N^5 -methyl FH_4 . When this occurs, the levels of N^5 -methyl FH_4 are too low to allow homocysteine to be converted to methionine. The loss of this pathway, coupled with the feedback inhibition by cysteine on cystathionine formation, will also lead to elevated homocysteine levels in the blood.

A third way in which serum homocysteine levels can be elevated is by a mutated cystathionine- β -synthase or a deficiency in vitamin B6, the required cofactor for that enzyme. These defects block the ability of homocysteine to be converted to cystathionine, and the homocysteine that does accumulate cannot all be accommodated by conversion to methionine. Thus, an accumulation of homocysteine results.

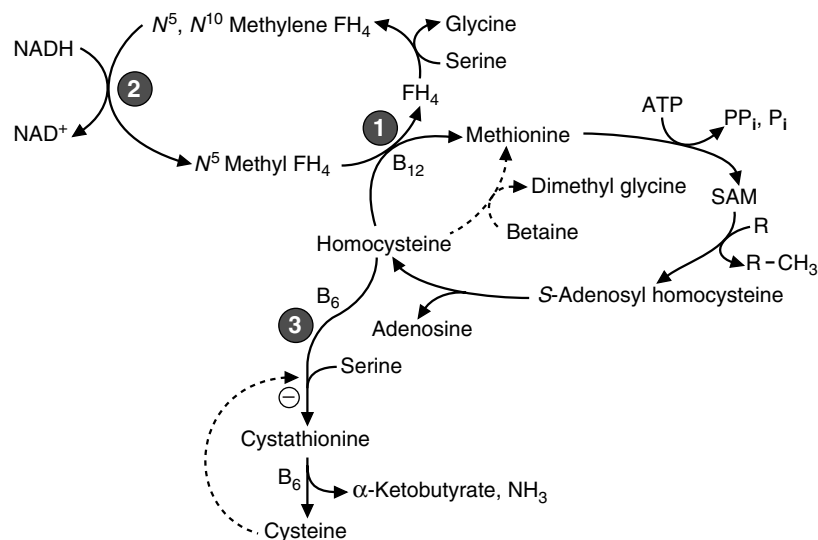


Fig. 40.10. Reaction pathways involving homocysteine. Defects in numbered enzymes (1 = methionine synthase, 2 = N^5 , N^{10} methylene FH_4 reductase, 3 = cystathionine- β -synthase) lead to elevated homocysteine. Recall that as cysteine accumulates, there is feedback inhibition on cystathionine- β -synthase to stop further cysteine production.

C. Neural Tube Defects

Folate deficiency during pregnancy has been associated with an increased risk for neural tube defects in the developing fetus. This risk is significantly reduced if women take folic acid supplements periconceptually. The link between folate deficiency and neural tube defects was first observed in women with hyperhomocysteinemia brought about by a thermolabile variant of N^5, N^{10} methylene tetrahydrofolate reductase. This form of the enzyme, which results from a single nucleotide change (C to T) in position 677 of the gene encoding the protein, is less active at body temperature than at lower temperatures. This results in a reduced level of N^5 -methyl tetrahydrofolate being generated and, therefore, an increase in the levels of homocysteine. Along with the elevated homocysteine, the women were also folate deficient. The folate deficiency and the subsequent inhibition of DNA synthesis leads to neural tube defects. The elevated homocysteine is one indication that such a deficit is present. These findings have led to the recommendation that women considering getting pregnant begin taking folate supplements before conception occurs, and for at least 1 month after conception. The Department of Agriculture has, in fact, mandated that folate be added to flour-containing products in the United States.

CLINICAL COMMENTS



Jean Ann Tonich developed a folate deficiency and is on the verge of developing a cobalamin (vitamin B12) deficiency as a consequence of prolonged moderately severe malnutrition related to chronic alcoholism. Before folate therapy is started, the physician must ascertain that the megaloblastic anemia is not caused by a pure B12 deficiency or a combined deficiency of folate and B12.

If folate is given without cobalamin to a B12-deficient patient, the drug only partially corrects the megaloblastic anemia because it will “bypass” the methyl-folate trap and provide adequate FH_4 coenzyme for the conversion of dUMP to dTMP and for a resurgence of purine synthesis. As a result, normal DNA synthesis, DNA repair, and cell division occur. However, the neurologic syndrome, resulting from hypomethylation in nervous tissue, may progress unless the physician realizes that B12 supplementation is required. In Jean Ann’s case, in which the serum B12 concentration was borderline low and in which the dietary history supported the possibility of a B12 deficiency, a combination of folate and B12 supplements is required to avoid this potential therapeutic trap.



Colin Tuma continued to do well and faithfully returned for his regular colonoscopic examinations.



Bea Twelvwlow was diagnosed with an inability to absorb dietary B12 but not crystalline B12 (the Schilling test results were normal). One of the consequences of aging is a reduced acid production by the gastric mucosa (atrophic gastritis), which limits the ability of pepsin to work on dietary protein. A reduced pepsin efficiency would then reduce the amount of bound B12 released from dietary protein, as a result of which the B12 would be unavailable for absorption. Because Bea absorbs crystalline B12 without a problem, her condition can be easily treated by taking vitamin B12 supplements orally.

BIOCHEMICAL COMMENTS



Folate Deficiencies and DNA Synthesis. Folate deficiencies result in decreased availability of the deoxythymidine and purine nucleotides that serve as precursors for DNA synthesis. The decreased concentrations of these precursors affect not only the DNA synthesis that occurs during replication

before cell division, but also the DNA synthesis that occurs as a step in the processes that repair damaged DNA.

Decreased methylation of deoxyuridine monophosphate (dUMP) to form deoxythymidine monophosphate (dTTP), a reaction that requires N^5, N^{10} -methylene tetrahydrofolate as a coenzyme (see Fig. 40.5), leads to an increase in the intracellular dUTP/dTTP ratio. This change causes a significant increase in the incorporation of uracil into DNA. Although much of this uracil can be removed by DNA repair enzymes, the lack of available dTTP blocks the step of DNA repair catalyzed by DNA polymerase. The result is fragmentation of DNA as well as blockade of normal DNA replication.

These abnormal nuclear processes are responsible for the clumping and polysegmentation seen in the nuclei of neutrophilic leukocytes in the bone marrow and in the peripheral blood of patients with a megaloblastic anemia caused either by a primary folate deficiency or one that is secondary to a B12 deficiency. The abnormalities in DNA synthesis and repair lead to an irreversible loss of the capacity for cell division and eventually to cell death.

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REVIEW QUESTIONS—CHAPTER 40

- Which of the following reactions requires N^5, N^{10} -methylene FH_4 as a carbon donor?
 - Homocysteine to methionine
 - Serine to glycine
 - Betaine to dimethylglycine
 - dUMP to dTMP
 - The de novo biosynthesis of the purine ring
- Propionic acid accumulation from amino acid degradation would result from a deficiency of which of the following vitamins?
 - Vitamin B6
 - Biotin
 - Folic acid
 - Vitamin B12
 - Vitamin B1
 - Vitamin B2
- A dietary vitamin B12 deficiency can result from which of the following?
 - Excessive intrinsic factor production by the gastric parietal cells
 - Eating a diet high in animal protein
 - Pancreatic insufficiency
 - Increased absorption of folic acid
 - Inability to conjugate the vitamin with glutamic acid

4. Which of the following forms of tetrahydrofolate is required for the synthesis of methionine from homocysteine?
- (A) N^5, N^{10} -methylene tetrahydrofolate
 - (B) N^5 -methyl tetrahydrofolate
 - (C) N^5, N^{10} -methenyl tetrahydrofolate
 - (D) N^{10} -formyl tetrahydrofolate
 - (E) N^5 -formimino tetrahydrofolate
5. An alternative method to methylate homocysteine to form methionine is which of the following?
- (A) Using glycine and FH_4 as the methyl donor
 - (B) Using dimethylglycine as the methyl donor
 - (C) Using choline as the methyl donor
 - (D) Using sarcosine as the methyl donor
 - (E) Using betaine as the methyl donor

41 Purine and Pyrimidine Metabolism

Purines and pyrimidines are required for synthesizing **nucleotides** and **nucleic acids**. These molecules can be synthesized either from scratch, **de novo**, or **salvaged** from existing bases. Dietary uptake of purine and pyrimidine bases is low, because most of the ingested nucleic acids are metabolized by the intestinal epithelial cells.

The *de novo* pathway of purine synthesis is complex, consisting of 11 steps, and requiring 6 molecules of ATP for every purine synthesized. The precursors that donate components to produce purine nucleotides include **glycine**, **ribose 5-phosphate**, **glutamine**, **aspartate**, **carbon dioxide**, and **N¹⁰-formyl FH₄** (Fig. 41.1). Purines are synthesized as **ribonucleotides**, with the initial purine synthesized being **inosine monophosphate (IMP)**. Adenosine monophosphate (AMP) and guanosine monophosphate (GMP) are each derived from IMP in two-step reaction pathways.

The purine nucleotide salvage pathway allows free purine bases to be converted into nucleotides, nucleotides into nucleosides, and nucleosides into free bases. Enzymes included in this pathway are **AMP** and **adenosine deaminase**, **adenosine kinase**, **purine nucleoside phosphorylase**, **adenine phosphoribosyltransferase (APRT)**, and **hypoxanthine guanine phosphoribosyltransferase (HGPRT)**. Mutations in a number of these enzymes lead to serious diseases. Deficiencies in purine nucleoside phosphorylase and adenosine deaminase lead to **immunodeficiency disorders**. A deficiency in HGPRT leads to **Lesch-Nyhan syndrome**. The **purine nucleotide cycle**, in which aspartate carbons are converted to fumarate to replenish TCA cycle intermediates in working muscle, and the aspartate nitrogen is released as ammonia, uses components of the purine nucleotide salvage pathway.

Pyrimidine bases are first synthesized as the free base and then converted to a nucleotide. **Aspartate** and **carbamoyl phosphate** form all components of the pyrimidine ring. Ribose 5-phosphate, which is converted to phosphoribosyl pyrophosphate (PRPP), is required to donate the sugar phosphate to form a nucleotide. The first pyrimidine nucleotide produced is **orotate monophosphate (OMP)**. The OMP is converted to **uridine monophosphate (UMP)**, which will become the precursor for both **cytidine triphosphate (CTP)** and **deoxythymidine monophosphate (dTMP)** production.

The formation of deoxyribonucleotides requires **ribonucleotide reductase** activity, which catalyzes the reduction of ribose on nucleotide diphosphate substrates to 2'-deoxyribose. Substrates for the enzyme include adenosine diphosphate (ADP), guanosine diphosphate (GDP), cytidine diphosphate (CDP), and uridine diphosphate (UDP). Regulation of the enzyme is complex. There are two major allosteric sites. One controls the overall activity of the enzyme, whereas the other determines the substrate specificity of the enzyme. All deoxyribonucleotides are synthesized using this one enzyme.

The **regulation** of purine nucleotide biosynthesis occurs at four points in the pathway. The enzymes **PRPP synthetase**, **amidophosphoribosyl transferase**, **IMP**

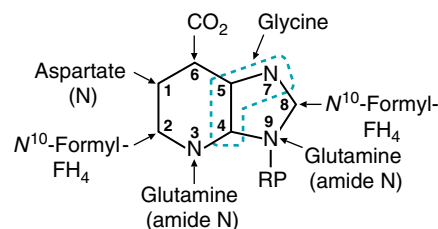


Fig. 41.1. Origin of the atoms of the purine base. FH₄ = tetrahydrofolate.

dehydrogenase, and *adenylosuccinate synthetase* are regulated by allosteric modifiers, as they occur at key branch points through the pathway. Pyrimidine synthesis is regulated at the first committed step, which is the synthesis of cytoplasmic carbamoyl-phosphate, by the enzyme **carbamoyl phosphate synthetase II (CPS-II)**.

Purines, when degraded, cannot generate energy, nor can the purine ring be substantially modified. The end product of purine ring degradation is **uric acid**, which is excreted in the urine. Uric acid has a limited solubility, and if it were to accumulate, uric acid crystals would precipitate in tissues of the body with a reduced temperature (such as the big toe). This condition of acute painful inflammation of specific soft tissues and joints is called **gout**. Pyrimidines, when degraded, however, give rise to water-soluble compounds, such as urea, carbon dioxide, and water and do not lead to a disease state if pyrimidine catabolism is increased.



THE WAITING ROOM



The initial acute inflammatory process that caused **Lotta Topaigne** to experience a painful attack of gouty arthritis responded quickly to colchicine therapy (see Chapter 10). Several weeks after the inflammatory signs and symptoms in her right great toe subsided, Lotta was placed on allopurinol, a drug that reduces uric acid synthesis. Her serum uric acid level gradually fell from a pretreatment level of 9.2 mg/dL into the normal range (2.5–8.0 mg/dL). She remained free of gouty symptoms when she returned to her physician for a follow-up office visit.

I. PURINES AND PYRIMIDINES

As has been seen in previous chapters of this text, nucleotides serve numerous functions in different reaction pathways. For example, nucleotides are the activated precursors of DNA and RNA. Nucleotides form the structural moieties of many coenzymes (examples include NADH, FAD, and coenzyme A). Nucleotides are critical elements in energy metabolism (ATP, GTP). Nucleotide derivatives are frequently activated intermediates in many biosyntheses. For example, UDP-glucose and CDP-diacylglycerol are precursors of glycogen and phosphoglycerides, respectively. S-Adenosylmethionine carries an activated methyl group. In addition, nucleotides act as second messengers in intracellular signaling (e.g., cAMP, cGMP). Finally, nucleotides and nucleosides act as metabolic allosteric regulators. Think about all of the enzymes that have been studied that are regulated by levels of ATP, ADP, and AMP.

Dietary uptake of purine and pyrimidine bases is minimal. The diet contains nucleic acids and the exocrine pancreas secretes deoxyribonuclease and ribonuclease, along with the proteolytic and lipolytic enzymes. This enables digested nucleic acids to be converted to nucleotides. The intestinal epithelial cells contain alkaline phosphatase activity, which will convert nucleotides to nucleosides. Other enzymes within the epithelial cells tend to metabolize the nucleosides to uric acid, or to salvage them for their own needs. Approximately 5% of ingested nucleotides will make it into the circulation, either as the free base or as a nucleoside. Because of the minimal dietary uptake of these important molecules, de novo synthesis of purines and pyrimidines is required.

II. PURINE BIOSYNTHESIS

The purine bases are produced *de novo* by pathways that use amino acids as precursors and produce nucleotides. Most *de novo* synthesis occurs in the liver (Fig. 41.2), and the nitrogenous bases and nucleosides are then transported to other tissues by red blood cells. The brain also synthesizes significant amounts of nucleotides. Because the *de novo* pathway requires six high-energy bonds per purine produced, a salvage pathway, which is used by many cell types, can convert free bases and nucleosides to nucleotides.

A. De Novo Synthesis of the Purine Nucleotides

1. SYNTHESIS OF IMP

As purines are built on a ribose base (see Fig. 41.2), an activated form of ribose is used to initiate the purine biosynthetic pathway. 5-Phosphoribosyl-1-pyrophosphate (PRPP) is the activated source of the ribose moiety. It is synthesized from ATP and ribose 5'-phosphate (Fig. 41.3), which is produced from glucose through the pentose phosphate pathway (see Chapter 29). The enzyme that catalyzes this reaction, PRPP synthetase, is a regulated enzyme (see section II.A.5); however, this step is not the committed step of purine biosynthesis. PRPP has many other uses, which are described as the chapter progresses.

In the first committed step of the purine biosynthetic pathway, PRPP reacts with glutamine to form phosphoribosylamine (Fig. 41.4). This reaction, which produces nitrogen 9 of the purine ring, is catalyzed by glutamine phosphoribosyl amidotransferase, a highly regulated enzyme.

In the next step of the pathway, the entire glycine molecule is added to the growing precursor. Glycine provides carbons 4 and 5 and nitrogen 7 of the purine ring (Fig. 41.5).

Subsequently, carbon 8 is provided by N^{10} -formyl FH_4 , nitrogen 3 by glutamine, carbon 6 by CO_2 , nitrogen 1 by aspartate, and carbon 2 by formyl tetrahydrofolate (see Fig. 41.1). Note that six molecules of ATP are required (starting with ribose 5-phosphate) to synthesize the first purine nucleotide, inosine monophosphate (IMP). This nucleotide contains the base hypoxanthine joined by an *N*-glycosidic bond from nitrogen 9 of the purine ring to carbon 1 of the ribose (Fig. 41.6).

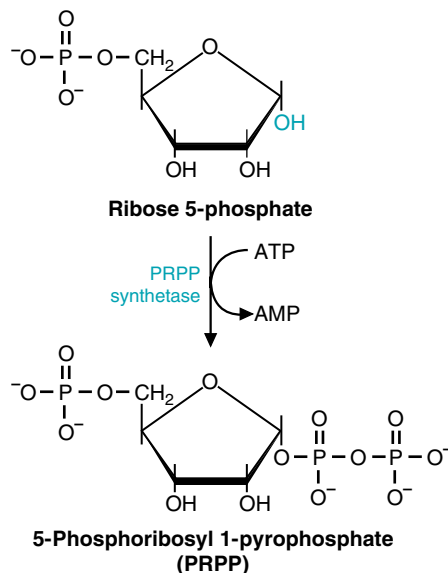


Fig. 41.3. Synthesis of PRPP. Ribose 5-phosphate is produced from glucose by the pentose phosphate pathway.

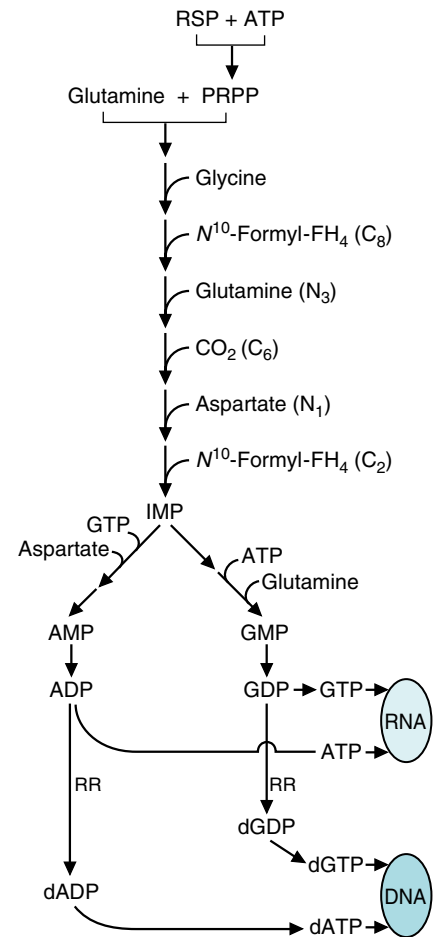


Fig. 41.2. Overview of purine production, starting with glutamine, ribose 5-phosphate, and ATP. The steps that require ATP are also indicated in this figure. RR = ribonucleotide reductase. FH_4 = tetrahydrofolate. PRPP = 5-phosphoribosyl 1-pyrophosphate.



Cellular concentrations of PRPP and glutamine are usually below their K_m for glutamine phosphoribosyl amidotransferase. Thus, any situation which leads to an increase in their concentration can lead to an increase in *de novo* purine biosynthesis.



The base hypoxanthine is found in the anticodon of tRNA molecules (it is formed by the deamination of an adenine base). Hypoxanthine's role in tRNA is to allow wobble base-pairs to form, as the base hypoxanthine can base pair with adenine, cytosine, or uracil. The wobbling allows one tRNA molecule to potentially form base pairs with three different codons.

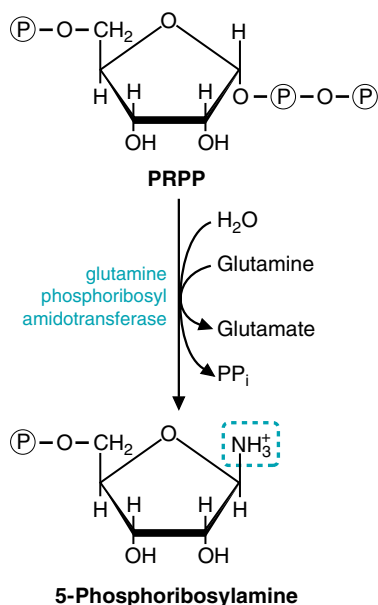


Fig. 41.4. The first step in purine biosynthesis. The purine base is built on the ribose moiety. The availability of the substrate PRPP is a major determinant of the rate of this reaction.



The aspartate to fumarate conversion also occurs in the urea cycle. In both cases, aspartate donates a nitrogen to the product, while the carbons of aspartate are released as fumarate.

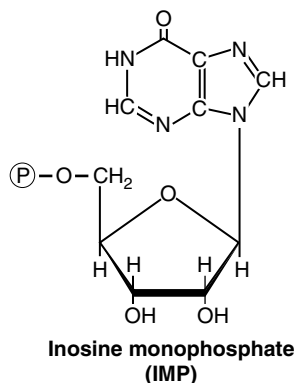


Fig. 41.6. Structure of inosine monophosphate (IMP). The base is hypoxanthine.

2. SYNTHESIS OF AMP

IMP serves as the branchpoint from which both adenine and guanine nucleotides can be produced (see Fig. 41.2). Adenosine monophosphate (AMP) is derived from IMP in two steps (Fig. 41.7). In the first step, aspartate is added to IMP to form adenylosuccinate, a reaction similar to the one catalyzed by argininosuccinate synthetase in the urea cycle. Note how this reaction requires a high-energy bond, donated by GTP. Fumarate is then released from the adenylosuccinate by the enzyme adenylosuccinase to form AMP.

3. SYNTHESIS OF GMP

GMP is also synthesized from IMP in two steps (Fig. 41.8). In the first step, the hypoxanthine base is oxidized by IMP dehydrogenase to produce the base xanthine and the nucleotide xanthosine monophosphate (XMP). Glutamine then donates the amide nitrogen to XMP to form GMP in a reaction catalyzed by GMP synthetase. This second reaction requires energy, in the form of ATP.

4. PHOSPHORYLATION OF AMP AND GMP

AMP and GMP can be phosphorylated to the di- and triphosphate levels. The production of nucleoside diphosphates requires specific nucleoside monophosphate kinases, whereas the production of nucleoside triphosphates requires nucleoside diphosphate kinases, which are active with a wide range of nucleoside diphosphates. The purine nucleoside triphosphates are also used for energy-requiring processes in the cell and also as precursors for RNA synthesis (see Fig. 41.2).

5. REGULATION OF PURINE SYNTHESIS

Regulation of purine synthesis occurs at several sites (Fig. 41.9). Four key enzymes are regulated: PRPP synthetase, amidophosphoribosyl transferase,

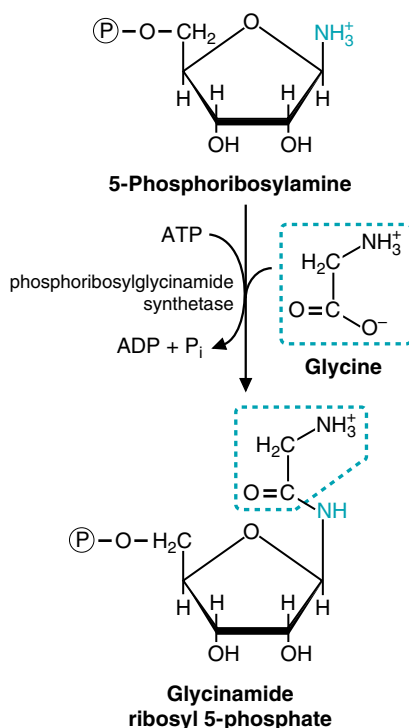


Fig. 41.5. Incorporation of glycine into the purine precursor. The ATP is required for the condensation of the glycine carboxylic acid group with the 1'-amino group of phosphoribosyl 1-amine.

adenylosuccinate synthetase, and IMP dehydrogenase. The first two enzymes regulate IMP synthesis; the last two regulate the production of AMP and GMP, respectively.

A primary site of regulation is the synthesis of PRPP. PRPP synthetase is negatively affected by GDP and, at a distinct allosteric site, by ADP. Thus, the simultaneous binding of an oxypurine (eg., GDP) and an aminopurine (eg., ADP) can occur with the result being a synergistic inhibition of the enzyme. This enzyme is not the committed step of purine biosynthesis; PRPP is also used in pyrimidine synthesis and both the purine and pyrimidine salvage pathways.

The committed step of purine synthesis is the formation of 5-phosphoribosyl 1-amine by glutamine phosphoribosyl amidotransferase. This enzyme is strongly inhibited by GMP and AMP (the end products of the purine biosynthetic pathway). The enzyme is also inhibited by the corresponding nucleoside di- and triphosphates, but under cellular conditions, these compounds probably do not play a central role in regulation. The active enzyme is a monomer of 133,000 daltons but is converted to an inactive dimer (270,000 daltons) by binding of the end products.

The enzymes that convert IMP to XMP and adenylosuccinate are both regulated. GMP inhibits the activity of IMP dehydrogenase, and AMP inhibits adenylosuccinate synthetase. Note that the synthesis of AMP is dependent on GTP (of which GMP is a precursor), whereas the synthesis of GMP is dependent on ATP (which is made from AMP). This serves as a type of positive regulatory mechanism to balance the pools of these precursors: when the levels of ATP are high, GMP will be

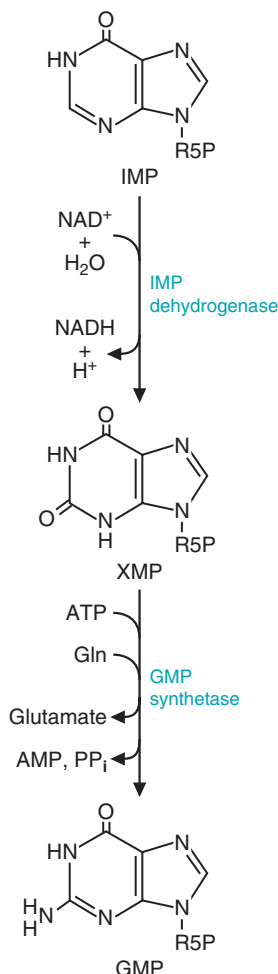


Fig. 41.8. The conversion of IMP to GMP. Note that ATP is required for the synthesis of GMP.

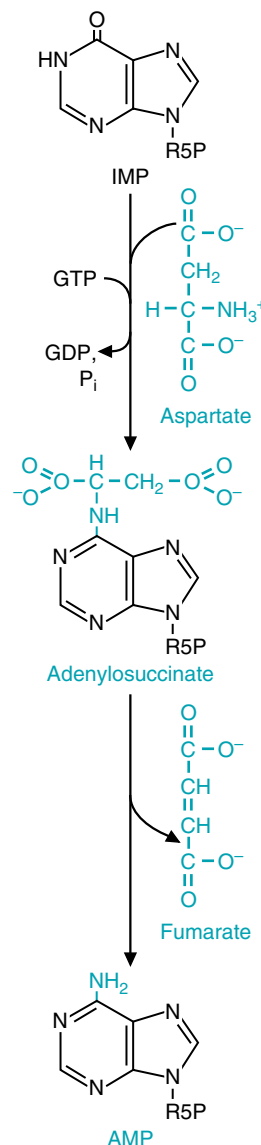


Fig. 41.7. The conversion of IMP to AMP. Note that GTP is required for the synthesis of AMP.

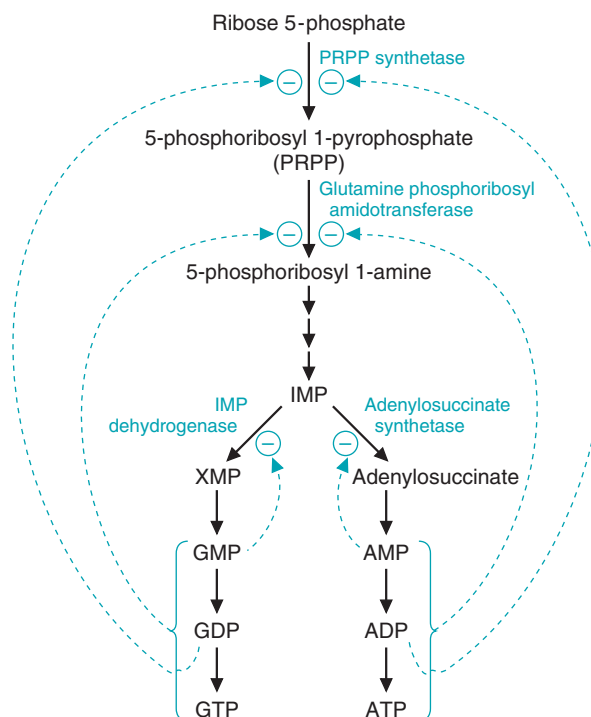


Fig. 41.9. The regulation of purine synthesis. PRPP synthetase has two distinct allosteric sites, one for ADP, the other for GDP. Glutamine phosphoribosyl amidotransferase contains adenine nucleotide and guanine nucleotide binding sites; the monophosphates are the most important, although the di- and tri-phosphates will also bind to and inhibit the enzyme. Adenylosuccinate synthetase is inhibited by AMP; IMP dehydrogenase is inhibited by GMP.

made; when the levels of GTP are high, AMP synthesis will take place. GMP and AMP act as negative effectors at these branch points, a classic example of feedback inhibition.

B. Purine Salvage Pathways

Most of the de novo synthesis of the bases of nucleotides occurs in the liver, and to some extent in the brain, neutrophils, and other cells of the immune system. Within the liver, nucleotides can be converted to nucleosides or free bases, which can be transported to other tissues via the red blood cell in the circulation. In addition, the small amounts of dietary bases or nucleosides that are absorbed also enter cells in this form. Thus, most cells can salvage these bases to generate nucleotides for RNA and DNA synthesis. For certain cell types, such as the lymphocytes, the salvage of bases is the major form of nucleotide generation.

The overall picture of salvage is shown in Figure 41.10. The pathways allow free bases, nucleosides, and nucleotides to be easily interconverted. The major enzymes required are purine nucleoside phosphorylase, phosphoribosyl transferases, and deaminases.

Purine nucleoside phosphorylase catalyzes a phosphorolysis reaction of the N-glycosidic bond that attaches the base to the sugar moiety in the nucleosides guanosine and inosine (Fig. 41.11). Thus, guanosine and inosine are converted to guanine and hypoxanthine, respectively, along with ribose 1-phosphate. The ribose 1-phosphate can be isomerized to ribose 5-phosphate, and the free bases then salvaged or degraded, depending on cellular needs.



A deficiency in purine nucleoside phosphorylase activity leads to an immune disorder in which T-cell immunity is compromised. B-cell immunity, conversely, may be only slightly compromised or even normal. Children lacking this activity have recurrent infections, and more than half display neurologic complications. Symptoms of the disorder first appear at between 6 months and 4 years of age.

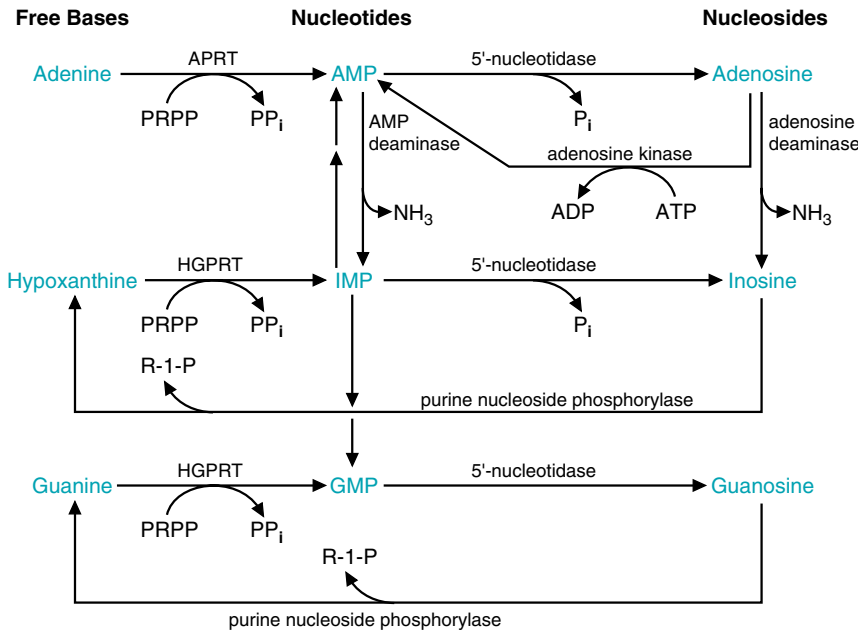


Fig. 41.10. Salvage of bases. The purine bases hypoxanthine and guanine react with PRPP to form the nucleotides inosine and guanosine monophosphate, respectively. The enzyme that catalyzes the reaction is hypoxanthine-guanine phosphoribosyltransferase (HGPRT). Adenine forms AMP in a reaction catalyzed by adenine phosphoribosyltransferase (APRT). Nucleotides are converted to nucleosides by 5'-nucleotidase. Free bases are generated from nucleosides by purine nucleoside phosphorylase. Deamination of the base adenine occurs with AMP and adenosine deaminase. Of the purines, only adenosine can be directly phosphorylated back to a nucleotide, by adenosine kinase.

The phosphoribosyl transferase enzymes catalyze the addition of a ribose 5-phosphate group from PRPP to a free base, generating a nucleotide and pyrophosphate (Fig. 41.12). Two enzymes do this: adenine phosphoribosyl transferase (APRT) and hypoxanthine-guanine phosphoribosyl transferase (HGPRT). The reactions they catalyze are the same, differing only in their substrate specificity.

Adenosine and AMP can be deaminated by adenosine deaminase and AMP deaminase, respectively, to form inosine and IMP (see Fig. 41.10). Adenosine is also the only nucleoside to be directly phosphorylated to a nucleotide by adenosine kinase. Guanosine and inosine must be converted to free bases by purine nucleoside phosphorylase before they can be converted to nucleotides by HGPRT.

A portion of the salvage pathway that is important in muscle is the purine nucleotide cycle (Fig. 41.13). The net effect of these reactions is the deamination of aspartate to fumarate (as AMP is synthesized from IMP and then deaminated back to IMP by AMP deaminase). Under conditions in which the muscle must generate energy, the fumarate derived from the purine nucleotide cycle is used anaplerotically to replenish TCA cycle intermediates and to allow the cycle to operate at a high speed. Deficiencies in enzymes of this cycle lead to muscle fatigue during exercise.



Lesch-Nyhan syndrome is caused by a defective hypoxanthine-guanine phosphoribosyltransferase (HGPRT) (see Fig. 41.12). In this condition, purine bases cannot be salvaged. Instead, they are degraded, forming excessive amounts of uric acid. Individuals with this syndrome suffer from mental retardation. They are also prone to chewing off their fingers and performing other acts of self-mutilation.

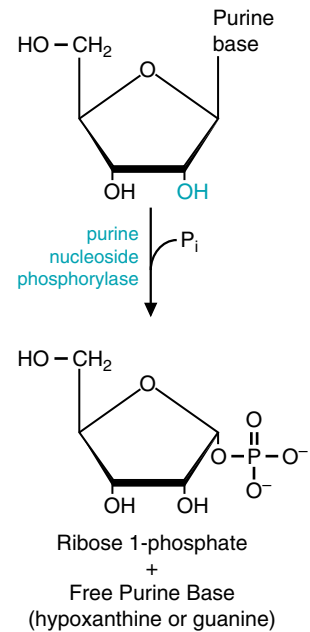


Fig. 41.11. The purine nucleoside phosphorylase reaction, converting guanosine or inosine to ribose 1-phosphate plus the free bases guanine or hypoxanthine.

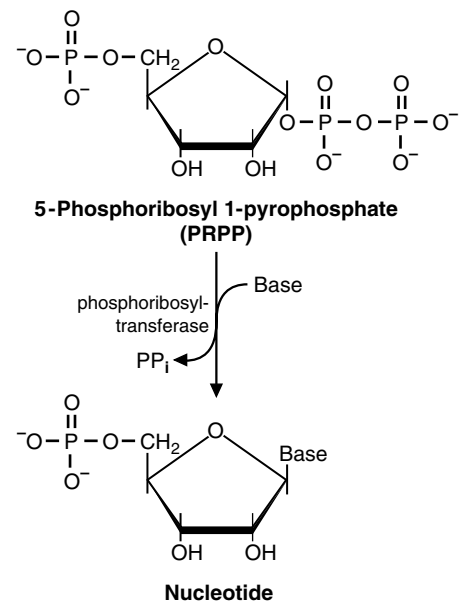


Fig. 41.12. The phosphoribosyltransferase reaction. APRT uses the free base adenine; HGPRT can use either hypoxanthine or guanine as a substrate.

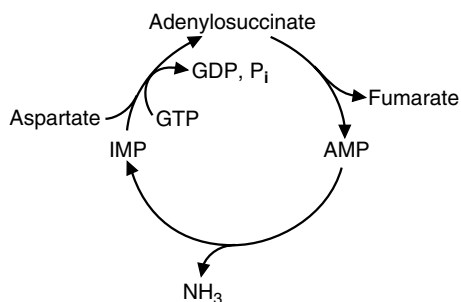


Fig. 41.13. The purine nucleotide cycle. Using a combination of biosynthetic and salvage enzymes, the net effect is the conversion of aspartate to fumarate plus ammonia, with the fumarate playing an anaplerotic role in the muscle.



In bacteria, aspartate transcarbamoylase is the regulated step of pyrimidine production. This is a very complex enzyme and was a model system for understanding how allosteric enzymes were regulated. In humans, however, this enzyme is not regulated.

III. SYNTHESIS OF THE PYRIMIDINE NUCLEOTIDES

A. De Novo Pathways

In the synthesis of the pyrimidine nucleotides, the base is synthesized first, and then it is attached to the ribose 5'-phosphate moiety (Fig. 41.14). The origin of the atoms of the ring (aspartate and carbamoyl-phosphate, which is derived from carbon dioxide and glutamine) is shown in Fig. 41.15. In the initial reaction of the pathway, glutamine combines with bicarbonate and ATP to form carbamoyl phosphate. This reaction is analogous to the first reaction of the urea cycle, except that it uses glutamine as the source of the nitrogen (rather than ammonia) and it occurs in the cytosol (rather than in mitochondria). The reaction is catalyzed by carbamoyl phosphate synthetase II, which is the regulated step of the pathway. The analogous reaction in urea synthesis is catalyzed by carbamoyl phosphate synthetase I, which is activated by *N*-acetylglutamate. The similarities and differences between these two carbamoyl phosphate synthetase enzymes is described in Table 41.1.

In the next step of pyrimidine biosynthesis, the entire aspartate molecule adds to carbamoyl phosphate in a reaction catalyzed by aspartate transcarbamoylase. The molecule subsequently closes to produce a ring (catalyzed by dihydroorotase), which is oxidized to form orotic acid (or its anion, orotate) through the actions of dihydroorotate dehydrogenase. The enzyme orotate phosphoribosyl transferase catalyzes the transfer of ribose 5-phosphate from PRPP to orotate, producing orotidine 5'-phosphate, which is decarboxylated by orotidylic acid dehydrogenase to form

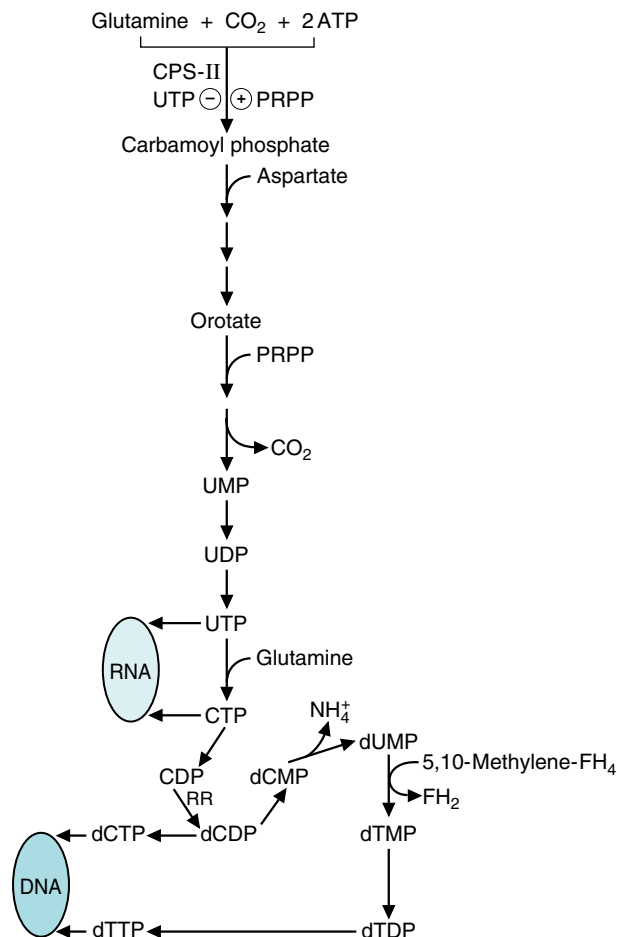


Fig. 41.14. Synthesis of the pyrimidine bases. CPSII = carbamoyl phosphate synthetase II. RR = ribonucleotide reductase; $\oplus$ = stimulated by; $\ominus$ = inhibited by; FH_2 and FH_4 = forms of folate.

Table 41.1. Comparison of Carbamoyl Phosphate Synthetases (CPSI and CPSII)

	CPS-I	CPS-II
Pathway	Urea cycle	Pyrimidine biosynthesis
Source of nitrogen	NH ₄ ⁺	Glutamine
Location	Mitochondria	Cytosol
Activator	N-Acetylglutamate	PRPP
Inhibitor	–	UTP

uridine monophosphate (UMP) (Fig. 41.16). In mammals, the first three enzymes of the pathway (carbamoyl phosphate synthetase II, aspartate transcarbamoylase, and dihydroorotase) are located on the same polypeptide, designated as CAD. The last two enzymes of the pathway are similarly located on a polypeptide known as UMP synthase (the orotate phosphoribosyl transferase and orotidylic acid dehydrogenase activities).

UMP is phosphorylated to UTP. An amino group, derived from the amide of glutamine, is added to carbon 4 to produce CTP by the enzyme CTP synthetase (this reaction cannot occur at the nucleotide monophosphate level). UTP and CTP are precursors for the synthesis of RNA (see Fig. 41.14). The synthesis of thymidine triphosphate (TTP) will be described in section IV.

B. Salvage of Pyrimidine Bases

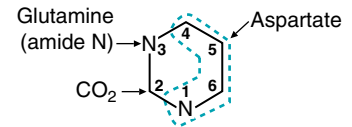
Pyrimidine bases are normally salvaged by a two-step route. First, a relatively non-specific pyrimidine nucleoside phosphorylase converts the pyrimidine bases to their respective nucleosides (Fig. 41.17). Notice that the preferred direction for this reaction is the reverse phosphorylase reaction, in which phosphate is being released and is not being used as a nucleophile to release the pyrimidine base from the nucleoside. The more specific nucleoside kinases then react with the nucleosides, forming nucleotides (Table 41.2). As with purines, further phosphorylation is carried out by increasingly more specific kinases. The nucleoside phosphorylase–nucleoside kinase route for synthesis of pyrimidine nucleoside monophosphates is relatively inefficient for salvage of pyrimidine bases because of the very low concentration of the bases in plasma and tissues.

Pyrimidine phosphorylase can use all of the pyrimidines but has a preference for uracil and is sometimes called uridine phosphorylase. The phosphorylase uses cytosine fairly well but has a very, very low affinity for thymine; therefore, a ribonucleoside containing thymine is almost never made in vivo. A second phosphorylase, thymine phosphorylase, has a much higher affinity for thymine and adds a deoxyribose residue (see Fig. 41.17).

Of the various ribonucleosides and deoxyribonucleoside kinases, one that merits special mention is *thymidine kinase* (TK). This enzyme is allosterically inhibited by dTTP. Activity of thymidine kinase in a given cell is closely related to the proliferative state of that cell. During the cell cycle, the activity of TK rises dramatically as cells enter S phase, and in general rapidly dividing cells have high levels of this enzyme. Radiolabeled thymidine is widely used for isotopic labeling of DNA, for example, in radioautographic investigations or to estimate rates of intracellular DNA synthesis.

Table 41.2. Salvage Reactions for Conversion of Pyrimidine Nucleosides to Nucleotides.

Enzyme	Reaction
Uridine-cytidine kinase	Uridine + ATP → UMP + ADP Cytidine + ATP → CMP + ADP
Deoxythymidine kinase	deoxythymidine + ATP → dTMP + ADP
Deoxycytidine kinase	Deoxycytidine + ATP → dCMP + ADP

**Fig. 41.15.** The origin of the bases in the pyrimidine ring.

In hereditary orotic aciduria, orotic acid is excreted in the urine because the enzymes that convert it to uridine monophosphate, orotate phosphoribosyltransferase and orotidine 5'-phosphate decarboxylase, are defective (see Fig. 41.16). Pyrimidines cannot be synthesized, and, therefore, normal growth does not occur. Oral administration of uridine is used to treat this condition. Uridine, which is converted to UMP, bypasses the metabolic block and provides the body with a source of pyrimidines, as both CTP and dTMP can be produced from UMP.

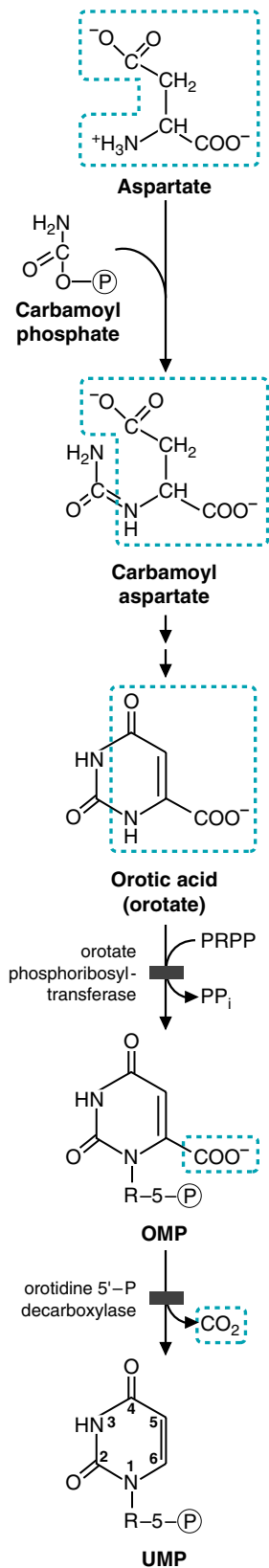


Fig. 41.16. Conversion of carbamoyl phosphate and aspartate to UMP. The defective enzymes in hereditary orotic aciduria are indicated (■).

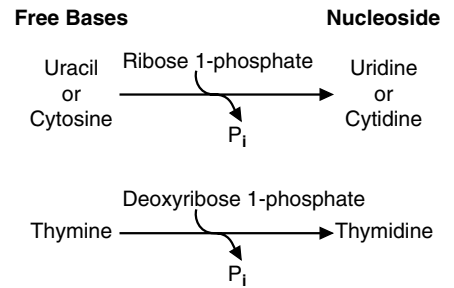


Fig. 41.17. Salvage reactions for pyrimidine nucleoside production. Thymine phosphorylase uses deoxyribose 1-phosphate as a substrate, such that ribothymidine is rarely formed.

C. Regulation of De Novo Pyrimidine Synthesis

The regulated step of pyrimidine synthesis in humans is carbamoyl phosphate synthetase II. The enzyme is inhibited by UTP and activated by PRPP (see Fig. 41.14). Thus, as pyrimidines decrease in concentration (as indicated by UTP levels), CPS-II is activated and pyrimidines are synthesized. The activity is also regulated by the cell cycle. As cells approach S-phase, CPS-II becomes more sensitive to PRPP activation and less sensitive to UTP inhibition. At the end of S-phase, the inhibition by UTP is more pronounced, and the activation by PRPP is reduced. These changes in the allosteric properties of CPS-II are related to its phosphorylation state. Phosphorylation of the enzyme at a specific site by a MAP kinase leads to a more easily activated enzyme. Phosphorylation at a second site by the cAMP-dependent protein kinase leads to a more easily inhibited enzyme.

IV. THE PRODUCTION OF DEOXYRIBONUCLEOTIDES

For DNA synthesis to occur, the ribose moiety must be reduced to deoxyribose (Fig. 41.18). This reduction occurs at the dinucleotide level and is catalyzed by ribonucleotide reductase, which requires the protein thioredoxin. The deoxyribonucleoside diphosphates can be phosphorylated to the triphosphate level and used as precursors for DNA synthesis (see Figs. 41.2 and 41.14).

The regulation of ribonucleotide reductase is quite complex. The enzyme contains two allosteric sites, one controlling the activity of the enzyme and the other controlling the substrate specificity of the enzyme. ATP bound to the activity site activates the enzyme; dATP bound to this site inhibits the enzyme. Substrate specificity is more complex. ATP bound to the substrate site activates the reduction of pyrimidines (CDP and UDP), to form dCDP and dUDP. The dUDP is not used for DNA synthesis; rather, it is used to produce dTMP (see below). Once dTMP is produced, it is phosphorylated to dTTP, which then binds to the substrate site and induces the reduction of GDP. As dGTP accumulates, it replaces dTTP in the substrate site and allows ADP to be reduced to dADP. This leads to the accumulation of dATP, which will inhibit the overall activity of the enzyme. These allosteric changes are summarized in Table 41.3.

dUDP can be dephosphorylated to form dUMP, or, alternatively, dCMP can be deaminated to form dUMP. Methylene tetrahydrofolate transfers a methyl group to dUMP to form dTMP (see Figure 40.5). Phosphorylation reactions produce dTTP, a precursor for DNA synthesis and a regulator of ribonucleotide reductase.

V. DEGRADATION OF PURINE AND PYRIMIDINE BASES

A. Purine Bases

The degradation of the purine nucleotides (AMP and GMP) occurs mainly in the liver (Fig. 41.19). Salvage enzymes are used for most of these reactions. AMP is first deaminated to produce IMP (AMP deaminase). Then IMP and GMP are dephosphorylated (5'-nucleotidase), and the ribose is cleaved from the base by purine nucleoside phosphorylase. Hypoxanthine, the base produced by cleavage of IMP, is converted by xanthine oxidase to xanthine, and guanine is deaminated by

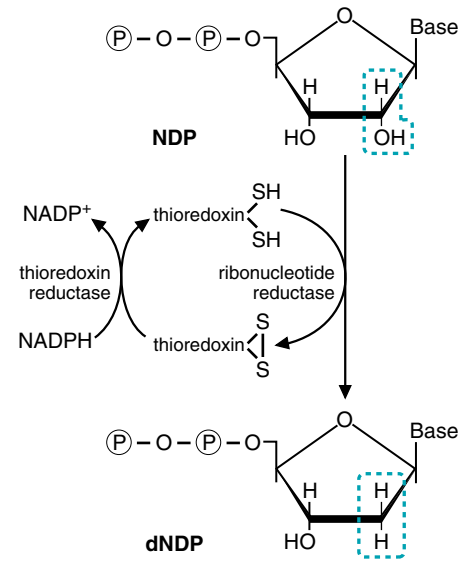


Fig. 41.18. Reduction of ribose to deoxyribose. Reduction occurs at the nucleoside diphosphate level. A ribonucleoside diphosphate (NDP) is converted to a deoxyribonucleoside diphosphate (dNDP). Thioredoxin is oxidized to a disulfide, which must be reduced for the reaction to continue producing dNDP. N = a nitrogenous base.



When ornithine transcarbamoylase is deficient (urea cycle disorder), excess carbamoyl phosphate from the mitochondria leaks into the cytoplasm. The elevated levels of cytoplasmic carbamoyl phosphate lead to pyrimidine production, as the regulated step of the pathway, the reaction catalyzed by carbamoyl synthetase II, is being bypassed. Thus, orotic aciduria results.



Gout is caused by excessive uric acid levels in the blood and tissues. To determine whether a person with gout has developed this problem because of overproduction of purine nucleotides or because of a decreased ability to excrete uric acid, an oral dose of an ^{15}N -labeled amino acid is sometimes used. Which amino acid would be most appropriate to use for this purpose?

Table 41.3. Effectors of Ribonucleotide Reductase Activity

Preferred Substrate	Effector Bound to Overall Activity Site	Effector Bound to Substrate Specificity Site
None	dATP	Any nucleotide
CDP	ATP	ATP or dATP
UDP	ATP	ATP or dATP
ADP	ATP	dGTP
GDP	ATP	dTTP



The entire glycine molecule is incorporated into the precursor of the purine nucleotides. The nitrogen of this glycine also appears in uric acid, the product of purine degradation. ^{15}N -labeled glycine could be used, therefore, to determine whether purines are being over-produced.



Uric acid has a pK_a of 5.4. It is ionized in the body to form urate. Urate is not very soluble in an aqueous environment. The quantity in normal human blood is very close to the solubility constant.



Normally, as cells die, their purine nucleotides are degraded to hypoxanthine and xanthine, which are converted to uric acid by xanthine oxidase (see Fig. 41.15). Allopurinol (a structural analog of hypoxanthine) is a substrate for xanthine oxidase. It is converted to oxypurinol (also called alloxanthine), which remains tightly bound to the enzyme, preventing further catalytic activity (see Fig. 8.19). Thus, allopurinol is a suicide inhibitor. It reduces the production of uric acid and hence its concentration in the blood and tissues (e.g., the synovial lining of the joints in **Lotta Topaigne's** great toe). Xanthine and hypoxanthine accumulate, and urate levels decrease. Overall, the amount of purine being degraded is spread over three products rather than appearing in only one. Therefore, none of the compounds exceeds its solubility constant, precipitation does not occur, and the symptoms of gout gradually subside.

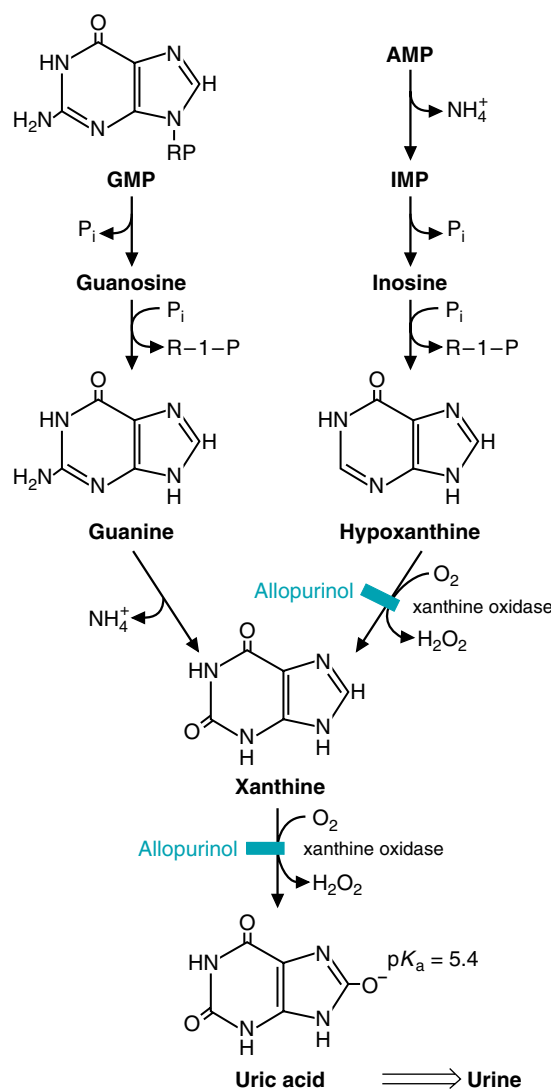


Fig. 41.19. Degradation of the purine bases. The reactions inhibited by allopurinol are indicated. A second form of xanthine oxidase exists that uses NAD^+ instead of O_2 as the electron acceptor.

the enzyme guanase to produce xanthine. The pathways for the degradation of adenine and guanine merge at this point. Xanthine is converted by xanthine oxidase to uric acid, which is excreted in the urine. Xanthine oxidase is a molybdenum-requiring enzyme that uses molecular oxygen and produces hydrogen peroxide (H_2O_2). Another form of xanthine oxidase exists that uses NAD^+ as the electron acceptor (see Chapter 24).

Note how little energy is derived from the degradation of the purine ring. Thus, it is to the cell's advantage to recycle and salvage the ring, because it costs energy to produce and not much is obtained in return.

B. Pyrimidine Bases

The pyrimidine nucleotides are dephosphorylated, and the nucleosides are cleaved to produce ribose 1-phosphate and the free pyrimidine bases cytosine, uracil, and thymine. Cytosine is deaminated, forming uracil, which is converted to CO_2 , NH_4^+ ,

and β -alanine. Thymine is converted to CO_2 , NH_4^+ , and β -aminoisobutyrate (Fig. 41.20). These products of pyrimidine degradation are excreted in the urine or converted to CO_2 , H_2O , and NH_4^+ (which forms urea). They do not cause any problems for the body, in contrast to urate, which is produced from the purines and can precipitate, causing gout. As with the purine degradation pathway, little energy can be generated by pyrimidine degradation.

CLINICAL COMMENTS



Hyperuricemia in **Lotta Topaigne's** case arose as a consequence of overproduction of uric acid. Treatment with allopurinol not only inhibits xanthine oxidase, lowering the formation of uric acid with an increase in the excretion of hypoxanthine and xanthine, but also decreases the overall synthesis of purine nucleotides. Hypoxanthine and xanthine produced by purine degradation are salvaged (i.e., converted to nucleotides) by a process that requires the consumption of PRPP. PRPP is a substrate for the glutamine phosphoribosyl amidotransferase reaction that initiates purine biosynthesis. Because the normal cellular levels of PRPP and glutamine are below the K_m of the enzyme, changes in the level of either substrate can accelerate or reduce the rate of the reaction. Therefore, decreased levels of PRPP cause decreased synthesis of purine nucleotides.

BIOCHEMICAL COMMENTS



A deficiency in adenosine deaminase activity leads to severe combined immunodeficiency disease, or SCID. In the severe form of combined immunodeficiency, both T cells (which provide cell-based immunity, see Chapter 44) and B-cells (which produce antibodies) are deficient, leaving the individual without a functional immune system. Children born with this disorder lack a thymus gland and suffer from many opportunistic infections because of the lack of a functional immune system. Death results if the child is not placed in a sterile environment. Administration of polyethylene glycol–modified adenosine deaminase has been successful in treating the disorder, and the ADA gene was the first to be used in gene therapy in treating the disorder. The question that remains, however, is that even though all cells of the body are lacking ADA activity, why are the immune cells specifically targeted for destruction?

The specific immune disorder is not caused by any defect in purine salvage pathways, as children with Lesch-Nyhan syndrome have a functional immune system, although there are other major problems in those children. This suggests that perhaps the accumulation of precursors to ADA lead to toxic effects. Three hypotheses have been proposed and are outlined below.

In the absence of ADA activity, both adenosine and deoxyadenosine will accumulate. When deoxyadenosine accumulates, adenosine kinase can convert it to dAMP. Other kinases will allow dATP to then accumulate within the lymphocyte. Why specifically the lymphocyte? The other cells of the body are secreting the deoxyadenosine they cannot use, and it is accumulating in the circulation. As the lymphocytes are present in the circulation, they tend to accumulate this compound more so than cells not constantly present within the blood. As dATP accumulates, ribonucleotide reductase becomes inhibited, and the cells can no longer produce deoxyribonucleotides for DNA synthesis. Thus, when cells are supposed to grow and differentiate in response to cytokines, they cannot, and they die.

A second hypothesis suggests that the accumulation of deoxyadenosine in lymphocytes leads to an inhibition of S-adenosylhomocysteine hydrolase, the enzyme that converts S-adenosylhomocysteine to homocysteine and adenosine. This leads

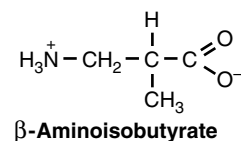
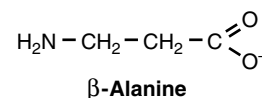


Fig. 41.20. Water-soluble end products of pyrimidine degradation.



Once nucleotide biosynthesis and salvage was understood at the pathway level, it was quickly realized that one way to inhibit cell proliferation would be to block purine or pyrimidine synthesis. Thus, drugs were developed that would interfere with a cell's ability to generate precursors for DNA synthesis, thereby inhibiting cell growth. This is particularly important for cancer cells, which have lost their normal growth regulatory properties. Such drugs have been introduced previously with a number of different patients. **Colin Tuma** was treated with 5-fluorouracil, which inhibits thymidylate synthase (dUMP to TMP synthesis). **Arlyn Foma** was treated with methotrexate for his leukemia; methotrexate inhibits dihydrofolate reductase, thereby blocking the regeneration of tetrahydrofolate and de novo purine synthesis and thymidine synthesis. **Mannie Weitzels** was treated with hydroxyurea to block ribonucleotide reductase activity, with the goal of inhibiting DNA synthesis in the leukemic cells. Development of these drugs would not have been possible without an understanding of the biochemistry of purine and pyrimidine salvage and synthesis. Such drugs also affect rapidly dividing normal cells, which brings about a number of the side effects of chemotherapeutic regimens.

Table 41.4. Gene Disorders in Purine and Pyrimidine Metabolism

Disease	Gene defect	Metabolite that accumulates	Clinical symptoms
Gout	Multiple causes	Uric acid	Painful joints
Severe combined immunodeficiency disease (SCID)	Adenosine deaminase (purine salvage pathway)	Deoxyadenosine and derivatives thereof	Loss of immune system, including no T or B cells
Immunodeficiency disease	Purine nucleoside phosphorylase	Purine nucleosides	Partial loss of immune system; no T cells but B cells are present
Lesch-Nyhan syndrome	Hypoxanthine-guanine phosphoribosyltransferase	Purines, uric acid	Mental retardation, self-mutilation
Hereditary orotic aciduria	UMP synthase	Orotic acid	Growth retardation

to hypo-methylation in the cell and an accumulation of S-adenosylhomocysteine. S-adenosylhomocysteine accumulation has been linked to the triggering of apoptosis.

The third hypothesis suggested is that elevated adenosine levels lead to inappropriate activation of adenosine receptors. Adenosine is also a signaling molecule, and stimulation of the adenosine receptors results in activation of protein kinase A and elevated cAMP levels in thymocytes. Elevated levels of cAMP in these cells triggers both apoptosis and developmental arrest of the cell.

Although it is still not clear which potential mechanism best explains the arrested development of immune cells, it is clear that elevated levels of adenosine and deoxyadenosine are toxic. The biochemical disorders of purine and pyrimidine metabolism discussed in this chapter are summarized in Table 41.4.

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REVIEW QUESTIONS—CHAPTER 41

- Similarities between carbamoyl phosphate synthetase I and carbamoyl phosphate synthetase II include which ONE of the following?
 - Carbon source
 - Intracellular location
 - Nitrogen source
 - Regulation by N-acetyl glutamate
 - Regulation by UMP

2. Gout can result from a reduction in activity of which one of the following enzymes?
 - (A) Glutamine phosphoribosyl amidotransferase
 - (B) Glucose 6-phosphatase
 - (C) Glucose 6-phosphate dehydrogenase
 - (D) PRPP synthetase
 - (E) Purine nucleoside phosphorylase

3. Lesch-Nyhan syndrome is due to an inability to catalyze which of the following reactions?
 - (A) Adenine to AMP
 - (B) Adenosine to AMP
 - (C) Guanine to GMP
 - (D) Guanosine to GMP
 - (E) Thymine to TMP
 - (F) Thymidine to TMP

4. Allopurinol can be used to treat gout because of its ability to inhibit which one of the following reactions?
 - (A) AMP to XMP
 - (B) Xanthine to uric acid
 - (C) Inosine to hypoxanthine
 - (D) IMP to XMP
 - (E) XMP to GMP

5. The regulation of ribonucleotide reductase is quite complex. Assuming that an enzyme deficiency leads to highly elevated levels of dGTP, what effect would you predict on the reduction of ribonucleotides to deoxyribonucleotides under these conditions?
 - (A) Elevated levels of dCDP will be produced.
 - (B) The formation of dADP will be favored.
 - (C) AMP would begin to be reduced.
 - (D) Reduced thioredoxin would become rate-limiting, thereby reducing the activity of ribonucleotide reductase.
 - (E) Deoxy-GTP would bind to the overall activity site and inhibit the functioning of the enzyme.

42 Intertissue Relationships in the Metabolism of Amino Acids

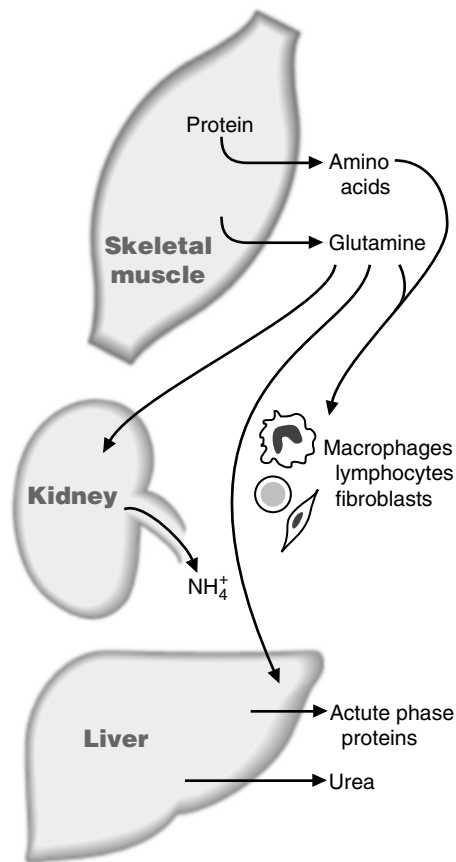


Fig. 42.1. Amino acid flux in sepsis and trauma. In sepsis and traumatic injury, glutamine and other amino acids are released from skeletal muscle for uptake by tissues involved in the immune response and tissue repair, such as macrophages, lymphocytes, fibroblasts, and the liver. Nitrogen excretion as urea and NH_4^+ results in negative nitrogen balance.

The body maintains a relatively large free amino acid pool in the blood, even during fasting. As a result, tissues have continuous access to individual amino acids for the synthesis of proteins and essential amino acid derivatives, such as neurotransmitters. The amino acid pool also provides the liver with **amino acid substrates** for **gluconeogenesis** and provides several other cell types with a source of **fuel**. The **free amino acid pool** is derived from **dietary amino acids** and the **turnover of proteins** in the body. During an **overnight fast** and during **hypercatabolic states**, **degradation** of labile **protein**, particularly that in **skeletal muscle**, is the major source of free amino acids.

The liver is the major site of amino acid metabolism in the body and the major site of **urea synthesis**. The liver is also the major site of amino acid degradation. Hepatocytes partially oxidize most amino acids, converting the carbon skeleton to glucose, ketone bodies, or CO_2 . Because ammonia is toxic, the liver converts most of the nitrogen from amino acid degradation to urea, which is excreted in the urine. The nitrogen derived from amino acid catabolism in other tissues is transported to the liver as **alanine** or **glutamine** and converted to urea.

The **branched-chain amino acids**, or BCAA (valine, isoleucine, and leucine) are oxidized principally in **skeletal muscle** and other tissues and not in the liver. In skeletal muscle, the carbon skeletons and some of the nitrogen are converted to glutamine, which is released into the blood. The remainder of the nitrogen is incorporated into alanine, which is taken up by the liver and converted to urea and glucose.

The formation and release of glutamine from skeletal muscle and other tissues serves several functions. In the kidney, the NH_4^+ carried by glutamine is excreted into the urine. This process removes protons formed during fuel oxidation and helps to maintain the body's pH, especially during metabolic acidosis. **Glutamine** also provides a **fuel** for the **kidney** and **gut**. In rapidly dividing cells (e.g., lymphocytes and macrophages), glutamine is required as a fuel, as a nitrogen donor for biosynthetic reactions, and as a substrate for protein synthesis.

During conditions of **sepsis** (the presence of various pathogenic organisms, or their toxins, in the blood or tissues), **trauma**, **injury**, or **burns**, the body enters a **catabolic state** characterized by a **negative nitrogen balance** (Fig. 42.1). Increased net protein degradation in skeletal muscle increases the availability of glutamine and other amino acids for cell division and protein synthesis in cells involved in the immune response and wound healing. In these conditions, an increased release of glucocorticoids from the adrenal cortex stimulates proteolysis.



THE WAITING ROOM



Katta Bolic, a 62-year-old homeless woman, was found by a neighborhood child who heard Katta's moans coming from an abandoned building. The child's mother called the police, who took Katta to the hospital emergency room. The patient was semicomatose, incontinent of urine, and her clothes were stained with vomitus. She had a fever of 103°F, was trembling uncontrollably, appeared to be severely dehydrated, and had marked muscle wasting. Her heart rate was very rapid, and her blood pressure was low (85/46 mm Hg). Her abdomen was distended and without bowel sounds. She responded to moderate pressure on her abdomen with moaning and grimacing.

Blood was sent for a broad laboratory profile, and cultures of her urine, stool, throat, and blood were taken. Intravenous glucose, saline, and parenteral broad-spectrum antibiotics were begun. X-rays performed after her vital signs were stabilized suggested a bowel perforation. These findings were compatible with a diagnosis of a ruptured viscus (e.g., an infected colonic diverticulum that perforated, allowing colonic bacteria to infect the tissues of the peritoneal cavity, causing peritonitis). Further studies confirmed that a diverticulum had ruptured, and appropriate surgery was performed. All of the arterial blood cultures grew out *Escherichia coli*, indicating that Katta also had a Gram-negative infection of her blood (septicemia) that had been seeded by the proliferating organisms in her peritoneal cavity. Intensive fluid and electrolyte therapy and antibiotic coverage were continued. The medical team (surgeons, internists, and nutritionists) began developing a complex therapeutic plan to reverse Katta's severely catabolic state.

I. MAINTENANCE OF THE FREE AMINO ACID POOL IN BLOOD

The body maintains a relatively large free amino acid pool in the blood, even in the absence of an intake of dietary protein. The large free amino acid pool ensures the continuous availability of individual amino acids to tissues for the synthesis of proteins, neurotransmitters, and other nitrogen-containing compounds (Fig. 42.2). In a normal, well-fed, healthy individual, approximately 300 to 600 g body protein is degraded per day. At the same time, roughly 100 g protein is consumed in the diet per day, which adds additional amino acids. From this pool, tissues use amino acids for the continuous synthesis of new proteins (300–600 g) to replace those degraded. The continuous turnover of proteins in the body makes the complete complement of amino acids available for the synthesis of new and different proteins, such as antibodies. Protein turnover allows shifts in the quantities of different proteins produced in tissues in response to changes in physiologic state and continuously removes modified or damaged proteins. It also provides a complete pool of specific amino acids that can be used as oxidizable substrates; precursors for gluconeogenesis and for heme, creatine phosphate, purine, pyrimidine, and neurotransmitter synthesis; for ammoniogenesis to maintain blood pH levels; and for numerous other functions.

A. Interorgan Flux of Amino Acids in the Postabsorptive State

The fasting state provides an example of the interorgan flux of amino acids necessary to maintain the free amino acid pool in the blood and supply tissues with their required amino acids (Fig. 42.3). During an overnight fast, protein synthesis in the liver and other tissues continues, but at a diminished rate compared with the



The concentration of free amino acids in the blood is not nearly as rigidly controlled as blood glucose levels. The free amino acid pool in the blood is only a small part (0.5%) of the total amino acid pool in whole body protein. Because of the large skeletal muscle mass, approximately 80% of the body's total protein is in skeletal muscle. Consequently, the concentration of individual amino acids in the blood is strongly affected by the rates of protein synthesis and degradation in skeletal muscle, as well as the rate of uptake and utilization of individual amino acids for metabolism in liver and other tissues. For the most part, changes in the rate of protein synthesis and degradation take place over a span of hours.



What changes in hormone levels and fuel metabolism occur during an overnight fast?

A: The hormonal changes that occur during an overnight fast include a decrease of blood insulin levels and an increase of glucagon relative to levels after a high-carbohydrate meal. Glucocorticoid levels also increase in the blood. These hormones coordinate the changes of fat, carbohydrate, and amino acid metabolism. Fatty acids are released from adipose triacylglycerols and are used as the major fuel by heart, skeletal muscle, liver, and other tissues. The liver converts some of the fatty acids to ketone bodies. Liver glycogen stores are diminished and gluconeogenesis becomes the major support of blood glucose levels for glucose-dependent tissues. The major precursors of gluconeogenesis include amino acids released from skeletal muscle, lactate, and glycerol.

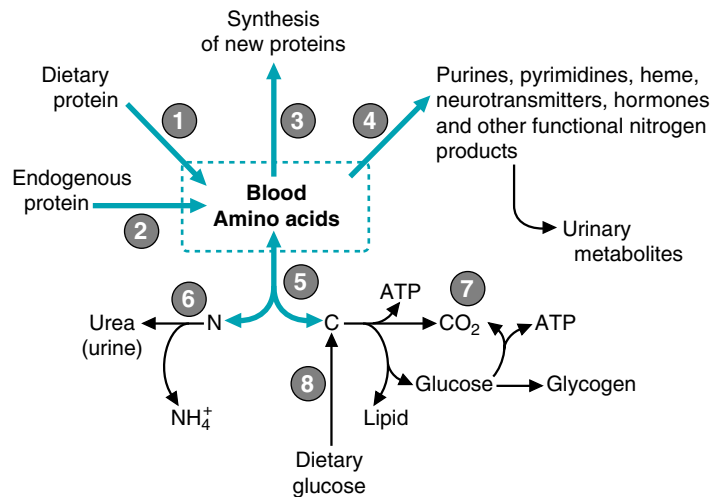


Fig. 42.2. Maintenance of the blood amino acid pool. Dietary protein (1) and degradation of endogenous protein (2) provide a source of essential amino acids (those that cannot be synthesized in the human). 3. The synthesis of new protein is the major use of amino acids from the free amino acid pool. 4. Compounds synthesized from amino acid precursors are essential for physiologic functions. Many of these compounds are degraded to N-containing urinary metabolites and do not return to the free amino acid pool. 5. In tissues, the nitrogen is removed from amino acids by transamination and deamination reactions. 6. The nitrogen from amino acid degradation appears in the urine primarily as urea or NH_4^+ , the ammonium ion. Ammonia excretion is necessary to maintain the pH of the blood. 7. Amino acids are used as fuels either directly or after being converted to glucose by gluconeogenesis. 8. Some amino acids can be synthesized in the human, provided that glucose and a nitrogen source are available.

postprandial state. Net degradation of labile protein occurs in skeletal muscle (which contains the body's largest protein mass) and other tissues.

1. RELEASE OF AMINO ACIDS FROM SKELETAL MUSCLE DURING FASTING

The efflux of amino acids from skeletal muscle supports the essential amino acid pool in the blood (see Fig. 42.3). Skeletal muscle oxidizes the BCAA (valine, leucine, isoleucine) to produce energy and glutamine. The amino groups of the BCAA, and of aspartate and glutamate, are transferred out of skeletal muscle in alanine and glutamine. Alanine and glutamine account for approximately 50% of the total α -amino nitrogen released by skeletal muscle (Fig. 42.4).

The release of amino acids from skeletal muscle is stimulated during an overnight fast by the decrease of insulin and increase of glucocorticoid levels in the blood (see Chapters 31 and 43). Insulin promotes the uptake of amino acids and the general synthesis of proteins. The mechanisms for the stimulation of protein synthesis in human skeletal muscle are not all known, but probably include an activation of the A system for amino acid transport (a modest effect), a general effect on initiation of translation, and an inhibition of lysosomal proteolysis. The fall of blood insulin levels during an overnight fast results in net proteolysis and release of amino acids. As glucocorticoid release from the adrenal cortex increases, an induction of ubiquitin synthesis and an increase of ubiquitin-dependent proteolysis also occur.

2. AMINO ACID METABOLISM IN LIVER DURING FASTING

The major site of alanine uptake is the liver, which disposes of the amino nitrogen by incorporating it into urea (see Fig. 42.3). The liver also extracts free amino acids,

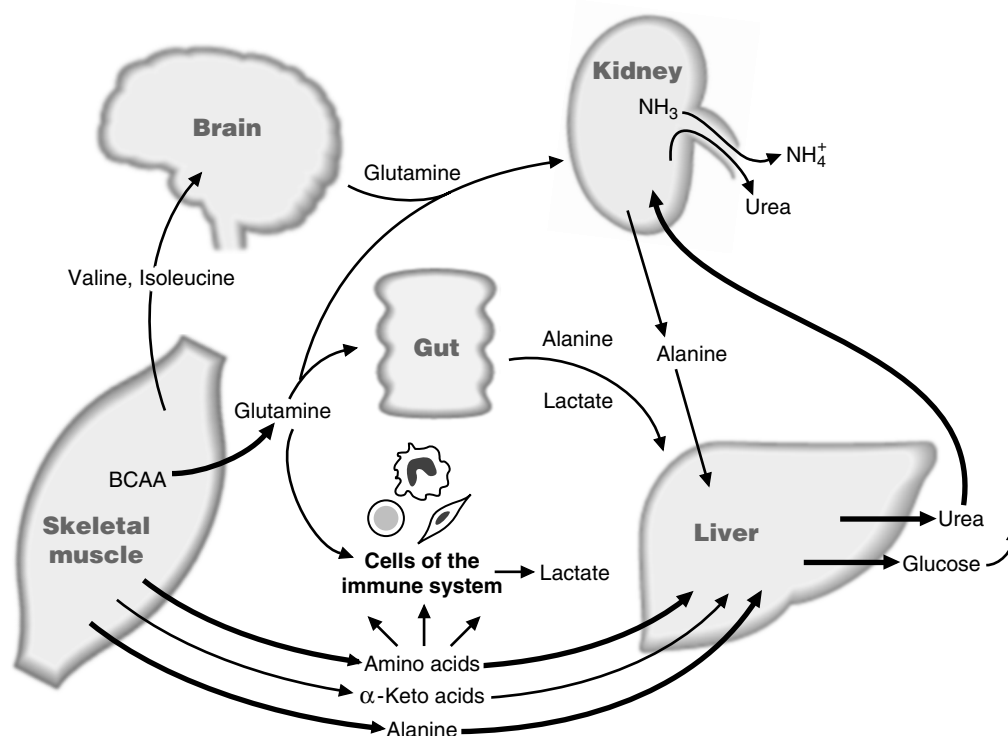


Fig. 42.3. Interorgan amino acid exchange after an overnight fast. After an overnight fast (the postabsorptive state), the utilization of amino acids for protein synthesis, for fuels, and for the synthesis of essential functional compounds continues. The free amino acid pool is supported largely by net degradation of skeletal muscle protein. Glutamine and alanine serve as amino group carriers from skeletal muscle to other tissues. Glutamine brings NH_4^+ to the kidney for the excretion of protons and serves as a fuel for the kidney, gut, and cells of the immune system. Alanine transfers amino groups from skeletal muscle, the kidney, and the gut to the liver, where they are converted to urea for excretion. The brain continues to use amino acids for neurotransmitter synthesis.

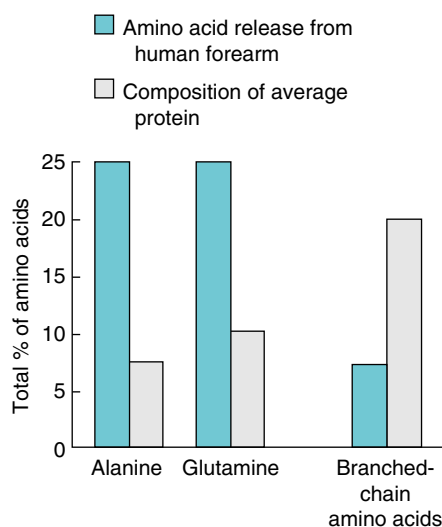


Fig. 42.4. Amino acid release from skeletal muscle. The arteriovenous difference (concentration in arterial blood minus the concentration in venous blood) across the human forearm has been measured for many amino acids. This graph compares the amount of alanine, glutamine, and BCAA released with their composition in the average protein. Alanine and glutamine represent a much higher percentage of total nitrogen released than originally present in the degraded protein, evidence that they are being synthesized in the skeletal muscle. The BCAA (leucine, valine, and isoleucine) are released in much lower amounts than those present in the degraded protein, evidence that they are being catabolized. Aspartate and glutamate also contribute nitrogen to the formation of alanine and glutamine.

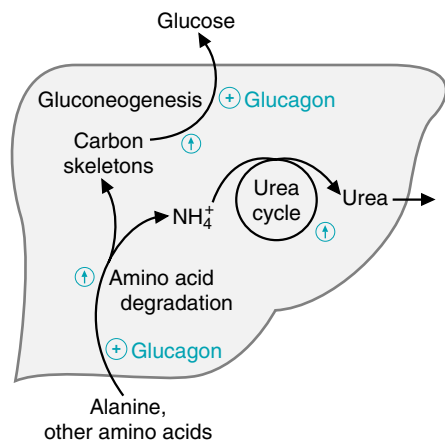


Fig. 42.5. Hormonal regulation of hepatic amino acid metabolism in the postabsorptive state. Circled + = glucagon-mediated activation of enzymes or proteins; circled ↑ = induction of enzyme synthesis mediated by glucagon and glucocorticoids. Induction of urea cycle enzymes occurs both during fasting and after a high-protein meal. Because many individuals in the United States normally have a high-protein diet, the levels of urea cycle enzymes may not fluctuate to any great extent.

Q: The body normally produces approximately 1 mmol of protons per kilogram of body weight per day. Nevertheless, the pH of the blood and extracellular fluid is normally maintained between 7.36 and 7.44. The narrow range is maintained principally by the bicarbonate (HCO_3^-), phosphate (HPO_4^{2-}), and hemoglobin buffering systems, and by the excretion of an amount of acid equal to that produced. The excretion of protons by the kidney regenerates bicarbonate, which can be reclaimed from the glomerular filtrate.

The acids are produced from normal fuel metabolism. The major acid is carbonic acid, which is formed from water and CO_2 produced in the TCA cycle and other oxidative pathways. The oxidation of sulfur-containing amino acids (methionine and cysteine) ultimately produces sulfuric acid (H_2SO_4), which dissociates into $2\text{H}^+ + \text{SO}_4^{2-}$, and the protons and sulfate are excreted. The hydrolysis of phosphate esters produces the equivalent of phosphoric acid. What other acids produced during metabolism appear in the blood?

α -keto acids, and some glutamine from the blood. Alanine and other amino acids are oxidized and their carbon skeletons converted principally to glucose. Glucagon and glucocorticoids stimulate the uptake of amino acids into liver and increase gluconeogenesis and ureagenesis (Fig. 42.5). Alanine transport into the liver, in particular, is enhanced by glucagon. The induction of the synthesis of gluconeogenic enzymes by glucagon and glucocorticoids during the overnight fast correlates with an induction of many of the enzymes of amino acid degradation (e.g., tyrosine aminotransferase) and an induction of urea cycle enzymes (see Chapter 38). Urea synthesis also increases because of the increased supply of NH_4^+ from amino acid degradation in the liver.

3. METABOLISM OF AMINO ACIDS IN OTHER TISSUES DURING FASTING

Glucose, produced by the liver, is used for energy by the brain and other glucose-dependent tissues, such as erythrocytes. The muscle, under conditions of exercise, when the AMP-activated protein kinase is active, also oxidizes some of this glucose to pyruvate, which is used for the carbon skeleton of alanine (the glucose-alanine cycle; see Chapter 38).

Glutamine is generated in skeletal muscle from the oxidation of BCAA, and by the lungs and brain for the removal of NH_4^+ formed from amino acid catabolism or entering from the blood. The kidney, the gut, and cells with rapid turnover rates such as those of the immune system are the major sites of glutamine uptake (see Fig. 42.3). Glutamine serves as a fuel for these tissues, as a nitrogen donor for purine synthesis, and as a substrate for ammoniogenesis in the kidney. Much of the unused nitrogen from glutamine is transferred to pyruvate to form alanine in these tissues. Alanine then carries the unused nitrogen back to the liver.

The brain is glucose dependent, but, like many cells in the body, can use BCAA for energy. The BCAA also provide a source of nitrogen for neurotransmitter synthesis during fasting. Other amino acids released from skeletal muscle protein degradation also serve as precursors of neurotransmitters.

B. Principles Governing Amino Acid Flux between Tissues

The pattern of interorgan flux of amino acids is strongly affected by conditions that change the supply of fuels (for example, the overnight fast, a mixed meal, a high-protein meal) and by conditions that increase the demand for amino acids (metabolic acidosis, surgical stress, traumatic injury, burns, wound healing, and sepsis). The flux of amino acid carbon and nitrogen in these different conditions is dictated by several considerations:

1. Ammonia (NH_3) is toxic. Consequently, it is transported between tissues as alanine or glutamine. Alanine is the principal carrier of amino acid nitrogen from



Katta Bolic was in a severe stage of negative nitrogen balance on admission, which was caused by both her malnourished state and her intra-abdominal infection complicated by sepsis. The physiologic response to her advanced catabolic status includes a degradation of muscle protein with the release of amino acids into the blood. This release is coupled with an increased uptake of amino acids for “acute phase” protein synthesis by the liver (systemic response) and other cells involved in the immune response to general and severe infection.



The differences in amino acid metabolism between tissues are dictated by the types and amounts of different enzyme and transport proteins present in each tissue and the ability of each tissue to respond to different regulatory messages (hormones and neural signals).

other tissues back to the liver, where the nitrogen is converted to urea and subsequently excreted into the urine by the kidneys. The amount of urea synthesized is proportional to the amount of amino acid carbon that is being oxidized as a fuel.

- The pool of glutamine in the blood serves several essential metabolic functions (Table 42.1). It provides ammonia for excretion of protons in the urine as NH_4^+ . It serves as a fuel for the gut, the kidney, and the cells of the immune system. Glutamine is also required by the cells of the immune system and other rapidly dividing cells in which its amide group serves as the source of nitrogen for biosynthetic reactions. In the brain, the formation of glutamine from glutamate and NH_4^+ provides a means of removing ammonia and of transporting glutamate between different cell types within the brain. The utilization of the blood glutamine pool is prioritized. During metabolic acidosis, the kidney becomes the predominant site of glutamine uptake, at the expense of glutamine utilization in other tissues. Conversely, during sepsis, in the absence of acidosis, cells involved in the immune response (macrophages, hepatocytes) become the preferential sites of glutamine uptake.
- The BCAA (valine, leucine, and isoleucine) form a significant portion of the composition of the average protein and can be converted to tricarboxylic acid (TCA) cycle intermediates and used as fuels by almost all tissues. They are also the major precursors of glutamine. Except for the BCAA and alanine, aspartate, and glutamate, the catabolism of amino acids occurs principally in the liver.
- Amino acids are major gluconeogenic substrates, and most of the energy obtained from their oxidation is derived from oxidation of the glucose formed from their carbon skeletons. A much smaller percentage of amino acid carbon is converted to acetyl CoA or to ketone bodies and oxidized. The utilization of amino acids for glucose synthesis for the brain and other glucose-requiring tissues is subject to the hormonal regulatory mechanisms of glucose homeostasis (see Chapters 31 and 36).
- The relative rates of protein synthesis and degradation (protein turnover) determine the size of the free amino acid pools available for the synthesis of new proteins and for other essential functions. For example, the synthesis of new proteins to mount an immune response is supported by the net degradation of other proteins in the body.

Table 42.1. Functions of Glutamine

Protein synthesis
Ammoniogenesis for proton excretion
Nitrogen donor for synthesis of:
Purines
Pyrimidines
NAD^+
Amino sugars
Asparagine
Other compounds
Glutamate donor for synthesis of:
Glutathione
GABA
Ornithine
Arginine
Proline
Other compounds



The ability to convert 4-carbon intermediates of the TCA cycle to pyruvate is required for oxidation of both BCAA and glutamine. This sequence of reactions requires PEP carboxykinase, or decarboxylating malate dehydrogenase (malic enzyme). Most tissues have one, or both, of these enzymes.



Lactic acid is produced from glucose and amino acid metabolism. The ketone bodies (acetoacetate and β -hydroxybutyrate) produced during fatty acid oxidation are also acids. Many α -keto acids, formed from transamination reactions, are also found in the blood.

II. UTILIZATION OF AMINO ACIDS IN INDIVIDUAL TISSUES

Because tissues differ in their physiologic functions, they have different amino acid requirements and contribute differently to whole body nitrogen metabolism. However, all tissues share a common requirement for essential amino acids for protein synthesis, and protein turnover is an ongoing process in all cells.

A. Kidney

One of the primary roles of amino acid nitrogen is to provide ammonia in the kidney for the excretion of protons in the urine. NH_4^+ is released from glutamine by glutaminase and by glutamate dehydrogenase, resulting in the formation of α -ketoglutarate (Fig. 42.6). α -Ketoglutarate is used as a fuel by the kidney and is oxidized to CO_2 , converted to glucose for use in cells in the renal medulla, or converted to alanine to return ammonia to the liver for urea synthesis.

1. USE OF GLUTAMINE NITROGEN TO BUFFER URINE

The rate of glutamine uptake from the blood and its utilization by the kidney depends principally on the amount of acid that must be excreted to maintain a normal pH in the blood. During a metabolic acidosis, the excretion of NH_4^+ by the kidney increases severalfold (Table 42.2). Because glutamine nitrogen provides

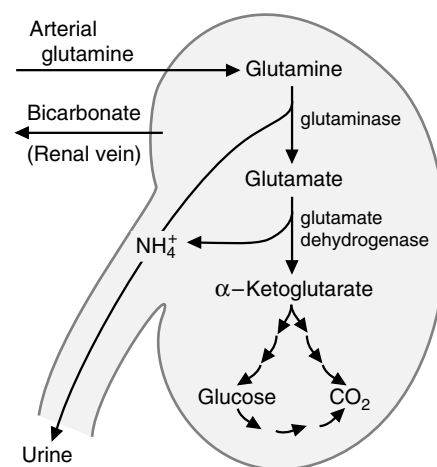


Fig. 42.6. Renal glutamine metabolism. Renal tubule cells preferentially oxidize glutamine. During metabolic acidosis, it is the major fuel for the kidney. Conversion of glutamine to α -ketoglutarate generates NH_4^+ . Ammonium ion excretion helps to buffer systemic acidemia.

Table 42.2. Excretion of Compounds in the Urine

Component	g/24 hr	Nitrogen (mmol)
H ₂ O	1,000	—
SO ₄ ²⁻	2–5	—
PO ₄ ²⁻	2–5	—
K ⁺	1–2	—
Urea	12–20	400–650
Creatinine	1–1.8	25–50
Uric acid	0.2–0.8	4–16
NH ₄ ⁺	0.2–1 (up to 10 in acidosis)	11–55 (up to 550 in acidosis)

approximately two thirds of the NH₄⁺ excreted by the kidney, glutamine uptake by the kidney also increases. Renal glutamine utilization for proton excretion takes precedence over the requirements of other tissues for glutamine.

Ammonia increases proton excretion by providing a buffer for protons that are transported into the renal tubular fluid (which is transformed into urine as it passes through the tubules of the kidney) (Fig. 42.7). Specific transporters in the membranes of the renal tubular cells transport protons from these cells into the tubular lumen in exchange for Na⁺. The protons in the tubular fluid are buffered by

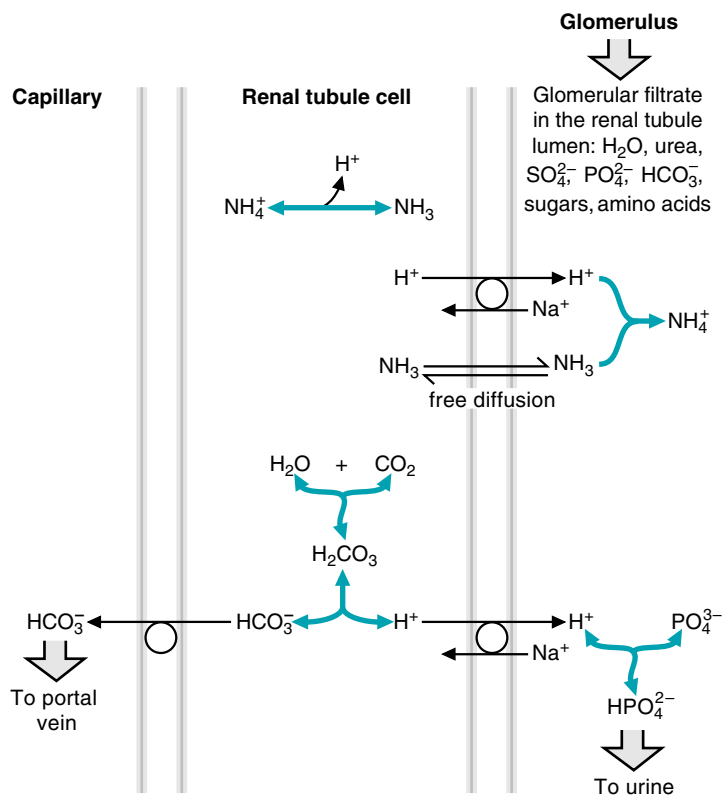


Fig. 42.7. Ammonia excretion by the kidney. Ammonia increases proton excretion by combining with a proton to form ammonium ion in the renal tubular fluid, which is transformed into urine as it passes through the tubules of the kidney. As blood is filtered in the capillary bed of the glomerulus, urea, sugars, amino acids, ions, and H₂O enter the renal tubular fluid (glomerular filtrate). As this fluid passes through a progression of tubules (the proximal convoluted tubule, the loop of Henle, the distal convoluted tubule, and the collecting duct) on its way to becoming urine, various components are reabsorbed or added to the filtrate by the epithelial cells lining the tubules. Specific transporters in the membranes of the renal tubule cells transport protons into the tubule lumen in exchange for Na⁺ so that the glomerular filtrate becomes more acidic as it is transformed into urine. The protons in the tubule fluid are buffered by phosphate, by bicarbonate, and by NH₃. The ammonia, which is uncharged, is able to diffuse through the membrane of the renal tubule cells into the urine. As it combines with a proton in the urine, it forms NH₄⁺, which cannot be transported back into the cells. The removal of protons as NH₄⁺ decreases the requirement for bicarbonate excretion to buffer the urine.

Table 42.3. Major Fuel Sources for the Kidney

% of Total CO ₂ Formed in Different Physiologic States			
Fuel	Normal	Acidosis	Fasted
Lactate	45	20	15
Glucose ^a	25	20	0
Fatty acids	15	20	60
Glutamine	15	40	25

^aGlucose used in the renal medulla is produced in the renal cortex.

phosphate, by bicarbonate, and by ammonia. Ammonia (NH₃), which is uncharged, enters the urine by free diffusion through the cell membrane. As it combines with a proton in the fluid, it forms ammonium ion (NH₄⁺), which cannot be transported back into the cells and is excreted in the urine.

2. GLUTAMINE AS A FUEL FOR THE KIDNEY

Glutamine is used as a fuel by the kidney in the normal fed state and, to a greater extent, during fasting and metabolic acidosis (Table 42.3). The carbon skeleton forms α -ketoglutarate, which is oxidized to CO₂, converted to glucose, or released as the carbon skeleton of serine or alanine (Fig. 42.8). α -Ketoglutarate can be converted to oxaloacetate by TCA cycle reactions, and oxaloacetate is converted to phosphoenolpyruvate (PEP) by PEP carboxykinase. PEP can then be converted to pyruvate and subsequently acetyl CoA, alanine, serine, or glucose. The glucose is used principally by the cells of the renal medulla, which have a relatively high dependence on anaerobic glycolysis because of their lower oxygen supply and mitochondrial capacity. The lactate released from anaerobic glycolysis in these cells is taken up and oxidized in the renal cortical cells, which have a higher mitochondrial capacity and a greater blood supply.

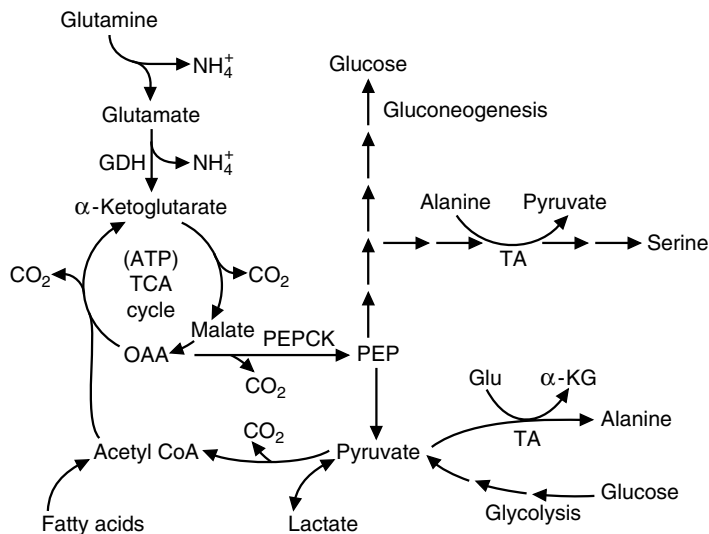


Fig. 42.8. Metabolism of glutamine and other fuels in the kidney. To completely oxidize glutamate carbon to CO₂, it must enter the TCA cycle as acetyl CoA. Carbon entering the TCA cycle as α -Ketoglutarate (α -KG) exits as oxaloacetate and is converted to phosphoenolpyruvate (PEP) by PEP carboxykinase. PEP is converted to pyruvate, which may be oxidized to acetyl CoA. PEP also can be converted to serine, glucose, or alanine. GDH = glutamate dehydrogenase; PEPCK = phosphoenolpyruvate carboxykinase; TA = transaminase; OAA = oxaloacetate.

B. Skeletal Muscle

Skeletal muscle, because of its large mass, is a major site of protein synthesis and degradation in the human. After a high-protein meal, insulin promotes the uptake of certain amino acids and stimulates net protein synthesis. The insulin stimulation of protein synthesis is dependent on an adequate supply of amino acids to undergo protein synthesis. During fasting and other catabolic states, a net degradation of skeletal muscle protein and release of amino acids occur (see Fig. 42.3). The net degradation of protein affects functional proteins, such as myosin, which are sacrificed to meet more urgent demands for amino acids in other tissues. During sepsis, degradation of skeletal muscle protein is stimulated by the glucocorticoid cortisol. The effect of cortisol is exerted through the activation of ubiquitin-dependent proteolysis. During fasting, the decrease of blood insulin levels and the increase of blood cortisol levels increase net protein degradation.

Skeletal muscle is a major site of glutamine synthesis, thereby satisfying the demand for glutamine during the postabsorptive state, during metabolic acidosis, and during septic stress and trauma. The carbon skeleton and nitrogen of glutamine are derived principally from the metabolism of BCAA. Amino acid degradation in skeletal muscle is also accompanied by the formation of alanine, which transfers amino groups from skeletal muscle to the liver in the glucose-alanine cycle.

1. OXIDATION OF BRANCHED-CHAIN AMINO ACIDS IN SKELETAL MUSCLE

The BCAA play a special role in muscle and most other tissues because they are the major amino acids that can be oxidized in tissues other than the liver. However, all tissues can interconvert amino acids and TCA cycle intermediates through transaminase reactions, i.e., alanine $\leftrightarrow$ pyruvate, aspartate $\leftrightarrow$ oxaloacetate, and α -ketoglutarate $\leftrightarrow$ glutamate. The first step of the pathway, transamination of the BCAA to α -keto acids, occurs principally in brain, heart, kidney, and skeletal muscles. These tissues have a high content of BCAA transaminase relative to the low levels in liver. The α -keto acids of the BCAA are then either released into the blood and taken up by liver, or oxidized to CO_2 or glutamine within the muscle or other tissue (Fig. 42.9). They can be oxidized by all tissues that contain mitochondria.

The oxidative pathways of the BCAA convert the carbon skeleton to either succinyl CoA or acetyl CoA (see Chapter 39 and Fig. 42.9). The pathways generate NADH and FAD(2H) for ATP synthesis before the conversion of carbon into intermediates of the TCA cycle, thus providing the muscle with energy without loss of carbon as CO_2 . Leucine is “ketogenic” in that it is converted to acetyl CoA and acetoacetate. Skeletal muscle, adipocytes, and most other tissues are able to use these products and, therefore, directly oxidize leucine to CO_2 . The portion of isoleucine converted to acetyl CoA is also oxidized directly to CO_2 . For the portion of valine and isoleucine that enters the TCA cycle as succinyl CoA to be completely oxidized to CO_2 , it must first be converted to acetyl CoA. To form acetyl CoA, succinyl CoA is oxidized to malate in the TCA cycle, and malate is then converted to pyruvate by malic enzyme (malate + $\text{NADP}^+ \rightarrow$ pyruvate + $\text{NADPH} + \text{H}^+$) (see Fig. 42.9). Pyruvate can then be oxidized to acetyl CoA. Alternatively, pyruvate can form alanine or lactate.

2. CONVERSION OF BRANCHED-CHAIN AMINO ACIDS TO GLUTAMINE

The major route of valine and isoleucine catabolism in skeletal muscle is to enter the TCA cycle as succinyl CoA and exit as α -ketoglutarate to provide the carbon skeleton for glutamine formation (see Fig. 42.9). Some of the glutamine and CO_2 that is formed from net protein degradation in skeletal muscle may also arise from



When the carbon skeleton of alanine is derived from glucose, the efflux of alanine from skeletal muscle and its uptake by liver provide no net transfer of amino acid carbon to the liver for gluconeogenesis. However, some of the alanine carbon is derived from sources other than glucose. Which amino acids can provide carbon for alanine formation? (Hint: See Fig. 42.9.)

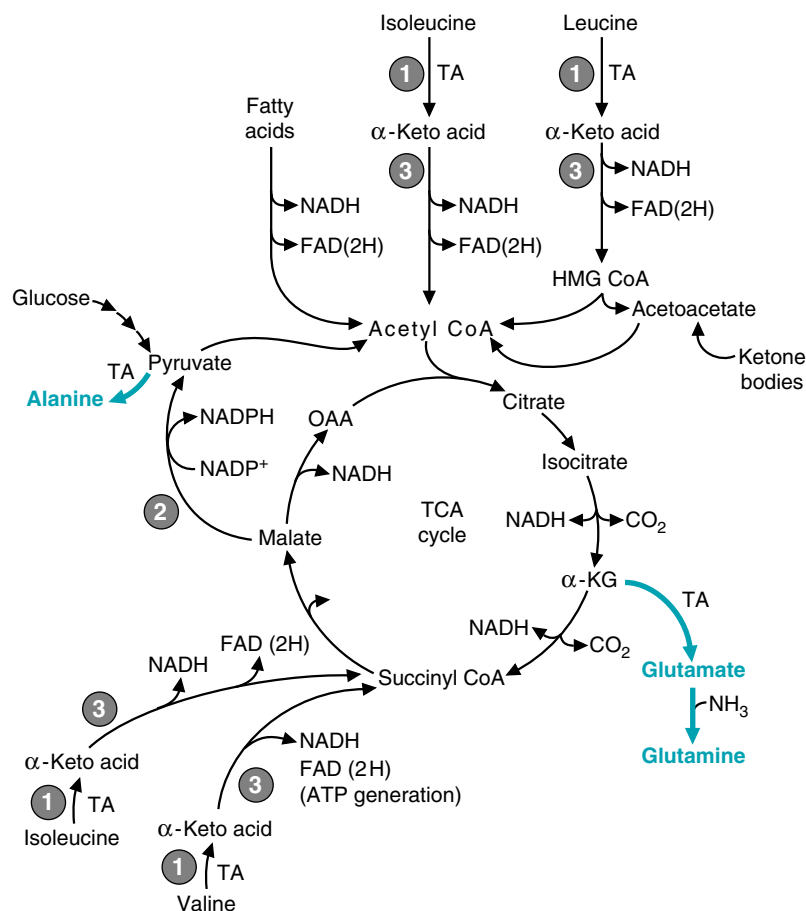


Fig. 42.9. Metabolism of the carbon skeletons of BCAA in skeletal muscle. 1. The first step in the metabolism of BCAA is transamination (TA). 2. Carbon from valine and isoleucine enters the TCA cycle as succinyl CoA and is converted to pyruvate by decarboxylating malate dehydrogenase (malic enzyme). 3. The oxidative pathways generate NADH and FAD(2H) even before the carbon skeleton enters the TCA cycle. The rate-limiting enzyme in the oxidative pathways is the α -keto acid dehydrogenase complex. The carbon skeleton also can be converted to glutamate and alanine, shown in blue.

the carbon skeletons of aspartate and glutamate. These amino acids are transaminated and become part of the pool of 4-carbon intermediates of the TCA cycle.

Glutamine nitrogen is derived principally from the BCAA (Fig. 42.10). The α -amino group arises from transamination reactions that form glutamate from α -ketoglutarate, and the amide nitrogen is formed from the addition of free ammonia to glutamate by glutamine synthetase. Free ammonia in skeletal muscle arises principally from the deamination of glutamate by glutamate dehydrogenase or from the purine nucleotide cycle.

In the purine nucleotide cycle (Fig. 42.11), the deamination of AMP to IMP releases NH_4^+ . AMP is resynthesized with amino groups provided from aspartate. The aspartate amino groups can arise from the BCAA through transamination reactions. The fumarate can be used to replenish TCA cycle intermediates.

3. GLUCOSE-ALANINE CYCLE

The nitrogen arising from the oxidation of BCAA in skeletal muscle can also be transferred back to the liver as alanine in the glucose-alanine cycle (Fig. 42.12, see also Fig. 41.13). The amino group of the BCAA is first transferred to α -ketoglutarate



Some of the alanine released from skeletal muscle is derived directly from protein degradation. The carbon skeletons of valine, isoleucine, aspartate, and glutamate, which are converted to malate and oxaloacetate in the TCA cycle, can be converted to pyruvate and subsequently transaminated to alanine. The extent to which these amino acids contribute carbon to alanine efflux differs between different types of muscles in the human. These amino acids also may contribute to alanine efflux from the gut.



The purine nucleotide cycle is found in skeletal muscle and brain but is absent in liver and many other tissues. One of its functions in skeletal muscle is to respond to the rapid utilization of ATP during exercise. During exercise, the rapid hydrolysis of ATP increases AMP levels, resulting in an activation of AMP deaminase (see Fig. 42.11). As a consequence, the cellular concentration of IMP increases and ammonia is generated. IMP, like AMP, activates muscle glycogen phosphorylase during exercise (see Chapter 22). The ammonia that is generated may help to buffer the increased lactic acid production occurring in skeletal muscles during strenuous exercise.

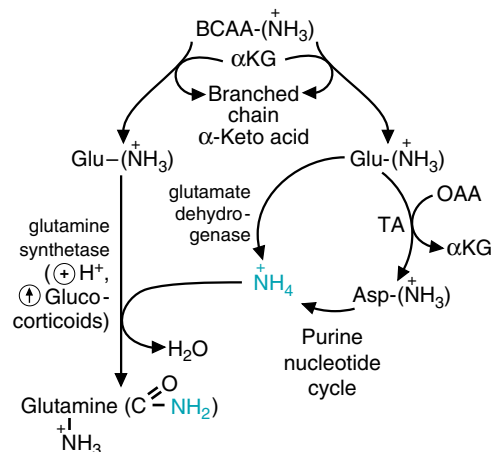


Fig. 42.10. Formation of glutamine from the amino groups of BCAA. The BCAA are first transaminated with α -ketoglutarate to form glutamate and the branched chain α -keto acids. The glutamate nitrogen can then follow either of two paths leading to glutamine formation. TA = transamination; OAA = oxaloacetate; α -KG = α -ketoglutarate.

to form glutamate and then transferred to pyruvate to form alanine by sequential transamination reactions. The pyruvate arises principally from glucose via the glycolytic pathway. The alanine released from skeletal muscle is taken up principally by the liver, where the amino group is incorporated into urea, and the carbon skeleton can be converted back to glucose through gluconeogenesis. Although the amount of alanine formed varies with dietary intake and physiologic state, the transport of nitrogen from skeletal muscle to liver as alanine occurs almost continuously throughout our daily fasting–feeding cycle.

C. Gut

Amino acids are an important fuel for the intestinal mucosal cells after a protein-containing meal and in catabolic states such as fasting or surgical trauma (Fig. 42.13). During fasting, glutamine is one of the major amino acids used by the gut. The principal fates of glutamine carbon in the gut are oxidation to CO₂ and conversion to the carbon

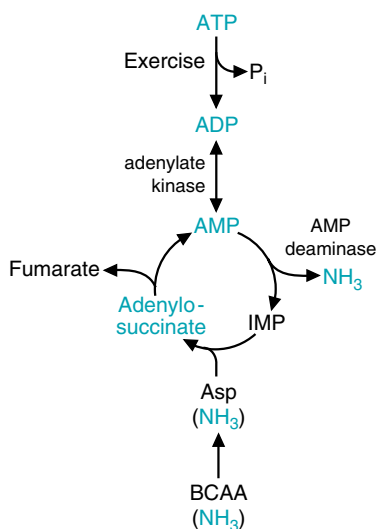


Fig. 42.11. Purine nucleotide cycle. In skeletal muscle, the purine nucleotide cycle can convert the amino groups of the BCAA to NH₃, which is incorporated into glutamine. The compounds containing the amino group released in the purine nucleotide cycle are shown in blue.

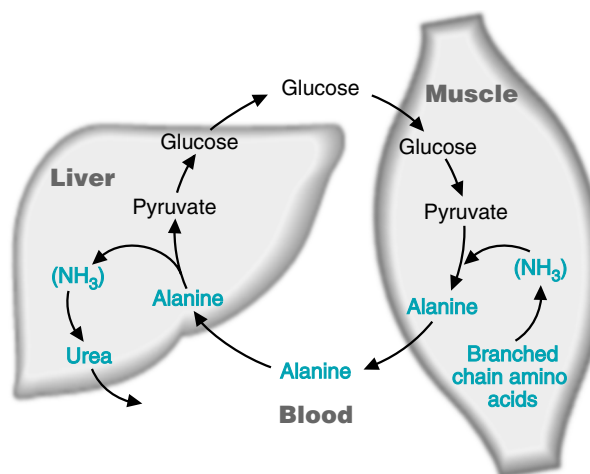


Fig. 42.12. Glucose-alanine cycle. The pathway for transfer of the amino groups from BCAA in skeletal muscle to urea in the liver is shown in blue.

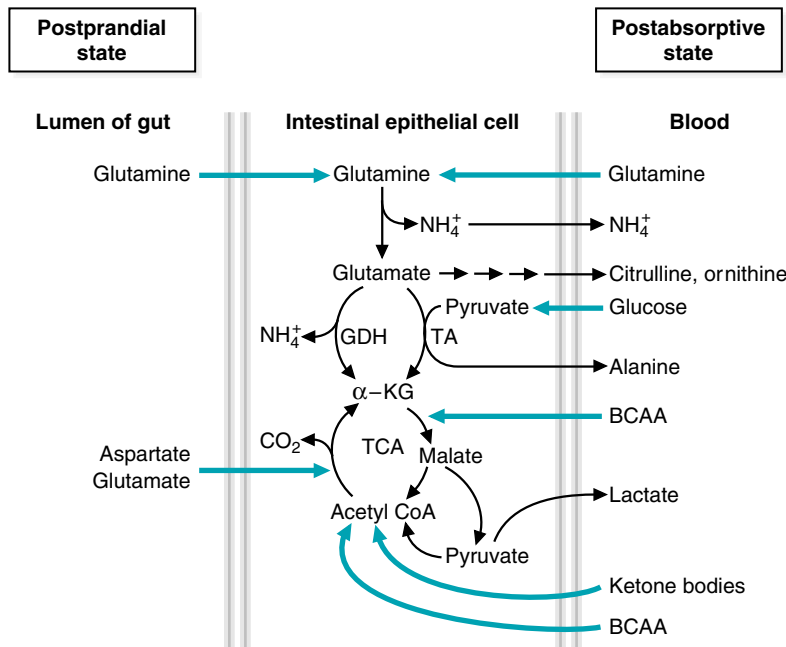


Fig. 42.13. Amino acid metabolism in the gut. The pathways of glutamine metabolism in the gut are the same whether it is supplied by the diet (postprandial state) or from the blood (postabsorptive state). Cells of the gut also metabolize aspartate, glutamate, and BCAA. Glucose is converted principally to the carbon skeleton of alanine. $\alpha\text{-KG}$ = α -ketoglutarate; GDH = glutamate dehydrogenase; TA = transaminase.

skeletons of lactate, citrulline, and ornithine. The gut also oxidizes BCAA. Nitrogen derived from amino acid degradation is converted to citrulline, alanine, NH_4^+ , and other compounds that are released into the blood and taken up by the liver. Although most of the carbon in this alanine is derived from glucose, the oxidation of glucose to CO_2 is not a major fuel pathway for the gut. Fatty acids are also not a significant source of fuel for the intestinal mucosal cells, although they do use ketone bodies.

After a protein meal, dietary glutamine is a major fuel for the gut, and the products of glutamine metabolism are similar to those seen in the postabsorptive state. The gut also uses dietary aspartate and glutamate, which enter the TCA cycle. Colonocytes (the cells of the colon) also use short-chain fatty acids, derived from bacterial action in the lumen.

The importance of the gut in whole body nitrogen metabolism arises from the high rate of division and death of intestinal mucosal cells and the need to continuously provide these cells with amino acids to sustain the high rates of protein synthesis required for cellular division. Not only are these cells important for the uptake of nutrients, but they maintain a barrier against invading bacteria from the gut lumen and are, therefore, part of our passive defense system. As a result of these important functions, the intestinal mucosal cells are supplied with the amino acids required for protein synthesis and fuel oxidation at the expense of the more expendable skeletal muscle protein.

D. Liver

The liver is the major site of amino acid metabolism. It is the major site of amino acid catabolism and converts most of the carbon in amino acids to intermediates of the TCA cycle or the glycolytic pathway (which can be converted to glucose or oxidized to CO_2), or to acetyl CoA and ketone bodies. The liver is also the major site for urea synthesis. It can take up both glutamine and alanine and convert the



The intestine contains the enzymes for the urea cycle, but the V_{max} for argininosuccinate synthetase and argininosuccinate lyase are very low, suggesting that the primary role of the urea cycle enzymes in the gut is to produce citrulline from the carbons of glutamine (glutamine $\rightarrow$ glutamate $\rightarrow$ glutamate semialdehyde $\rightarrow$ ornithine $\rightarrow$ citrulline). The citrulline is released in the circulation for use by the liver.



Glutamine utilization by the gut is diminished by a metabolic acidosis compared with the postabsorptive or postprandial states. During metabolic acidosis, the uptake of glutamine by the kidney is increased, and blood glutamine levels decrease. As a consequence, the gut takes up less glutamine.

nitrogen to urea for disposal (see Chapter 38). Other pathways in the liver provide it with an unusually high amino acid requirement. The liver synthesizes plasma proteins, such as serum albumin, transferrin, and the proteins of the blood coagulation cascade. It is a major site for the synthesis of nonessential amino acids, the conjugation of xenobiotic compounds with glycine, the synthesis of heme and purine nucleotides, and the synthesis of glutathione.

E. Brain and Nervous Tissue

1. AMINO ACID POOL AND NEUROTRANSMITTER SYNTHESIS

A major function of amino acid metabolism in neural tissue is the synthesis of neurotransmitters. More than 40 compounds are believed to function as neurotransmitters, and all of these contain nitrogen derived from precursor amino acids. They include amino acids, which are themselves neurotransmitters (e.g., glutamate, glycine), the catecholamines derived from tyrosine (dopamine and norepinephrine), serotonin (derived from tryptophan), GABA (derived from glutamate), acetylcholine (derived from choline synthesized in the liver and acetyl CoA), and many peptides. In general, neurotransmitters are formed in the presynaptic terminals of axons and stored in vesicles until released by a transient change in electrochemical potential along the axon. Subsequent catabolism of some of the neurotransmitter results in the formation of a urinary excretion product. The rapid metabolism of neurotransmitters requires the continuous availability of a precursor pool of amino acids for de novo neurotransmitter synthesis (see Chapter 47).

2. METABOLISM OF GLUTAMINE IN THE BRAIN

The brain is a net glutamine producer owing principally to the presence of glutamine synthetase in astroglial cells (see Chapter 47). Glutamate and aspartate are synthesized in these cells, using amino groups donated by the BCAA (principally valine) and TCA cycle intermediates formed from glucose and from the carbon skeletons of BCAA (Fig. 42.14). The glutamate is converted to glutamine by glutamine synthetase, which incorporates NH_4^+ released from deamination of amino acids and deamination of AMP in the purine nucleotide cycle in the brain. This glutamine may efflux from the brain, carrying excess NH_4^+ into the blood, or serve as a precursor of glutamate in neuronal cells.

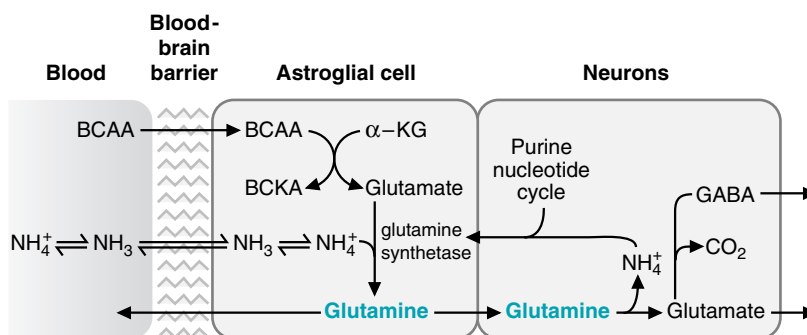


Fig. 42.14. Role of glutamine in the brain. Glutamine serves as a nitrogen transporter in the brain for the synthesis of many different neurotransmitters. Different neurons convert glutamine to γ-aminobutyric acid (GABA) or to glutamate. Glutamine also transports excess NH_4^+ from the brain into the blood. BCKA = branched-chain keto acids; $\alpha\text{-KG}$ = α-ketoglutarate.

Glutamine synthesized in the astroglial cells is a precursor of glutamate (an excitatory neurotransmitter) and GABA (an inhibitory neurotransmitter) in the neuronal cells (see Fig. 42.14). It is converted to glutamate by a neuronal glutaminase isozyme. In GABAergic neurons, glutamate is then decarboxylated to GABA, which is released during excitation of the neuron. GABA is one of the neurotransmitters that is recycled; a transaminase converts it to succinaldehyde, which is then oxidized to succinate. Succinate enters the TCA cycle.

III. CHANGES IN AMINO ACID METABOLISM WITH DIETARY AND PHYSIOLOGIC STATE

The rate and pattern of amino acid utilization by different tissues change with dietary and physiologic state. Two such states, the postprandial period following a high-protein meal and the hypercatabolic state produced by sepsis or surgical trauma, differ from the postabsorptive state with respect to the availability of amino acids and other fuels and the levels of different hormones in the blood. As a result, the pattern of amino acid utilization is somewhat different.

A. A High-Protein Meal

After the ingestion of a high-protein meal, the gut and the liver use most of the absorbed amino acids (Fig. 42.15). Glutamate and aspartate are used as fuels by the gut, and very little enters the portal vein. The gut also may use some BCAA. The liver takes up 60 to 70% of the amino acids present in the portal vein. These amino acids, for the most part, are converted to glucose in the gluconeogenic pathway.

After a pure protein meal, the increased levels of dietary amino acids reaching the pancreas stimulate the release of glucagon above fasting levels, thereby increasing amino acid uptake into liver and stimulating gluconeogenesis. Insulin release is also stimulated, but not nearly to the levels found after a high-carbohydrate meal.

 During hyperammonemia, ammonia (NH_3) can diffuse into the brain from the blood. The ammonia is able to inhibit the neural isozyme of glutaminase, thereby decreasing additional ammonia formation in the brain and inhibiting the formation of glutamate and its subsequent metabolism to GABA. This effect of ammonia might contribute to the lethargy associated with the hyperammonemia found in patients with hepatic disease.

 The levels of transthyretin (binds to vitamin A and thyroid hormones in the blood) and serum albumin in the blood may be used as indicators of the degree of protein malnutrition. In the absence of hepatic disease, decreased levels of these proteins in the blood indicate insufficient availability of amino acids to the liver for synthesis of serum proteins.

Q: In what ways does liver metabolism after a high-protein meal resemble liver metabolism in the fasting state?

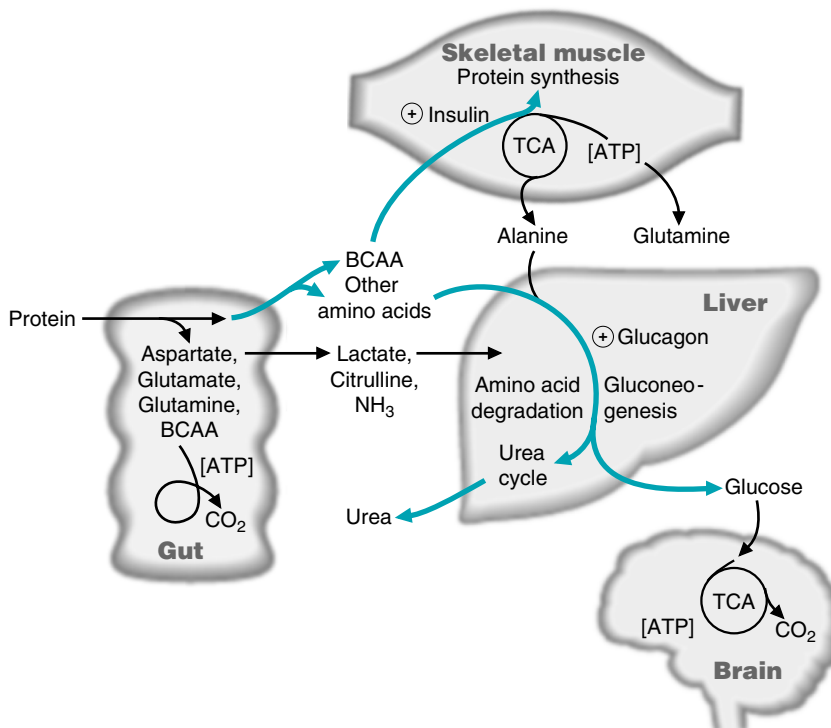


Fig. 42.15. Flux of amino acids after a high-protein meal.

A: Both of these dietary states are characterized by an elevation of glucagon. Glucagon stimulates amino acid transport into the liver, stimulates gluconeogenesis through decreasing levels of fructose 2,6-bisphosphate, and induces the synthesis of enzymes in the urea cycle, the gluconeogenic pathway, and the pathways for degradation of some of the amino acids.

The Atkins high-protein diet is based on the premise that ingesting high-protein, low-carbohydrate meals will keep circulating insulin levels low, such that energy storage is not induced, and glucagon release will point the insulin/glucagon ratio to energy mobilization, particularly fatty acid release from the adipocyte and oxidation by the tissues. The lack of energy storage, coupled with the loss of fat, leads to weight loss.

In general, the insulin released after a high-protein meal is sufficiently high that the uptake of BCAA into skeletal muscle and net protein synthesis is stimulated, but gluconeogenesis in the liver is not inhibited. The higher the carbohydrate content of the meal, the higher the insulin/glucagon ratio and the greater the shift of amino acids away from gluconeogenesis into biosynthetic pathways in the liver such as the synthesis of plasma proteins.

Most of the amino acid nitrogen entering the peripheral circulation after a high-protein meal or a mixed meal is present as the BCAA. Because the liver has low levels of transaminases for these amino acids, it cannot oxidize them to a significant extent, and they enter the systemic circulation. The BCAA are slowly taken up by skeletal muscle and other tissues. These peripheral nonhepatic tissues use the amino acids derived from the diet principally for net protein synthesis.

B. Hypercatabolic States

Surgery, trauma, burns, and septic stress are examples of hypercatabolic states characterized by increased fuel utilization and a negative nitrogen balance (Fig. 42.16). The mobilization of body protein, fat, and carbohydrate stores serves to maintain normal tissue function in the presence of a limited dietary intake, as well as to support the energy and amino acid requirements for the immune response and wound healing. The negative nitrogen balance that occurs in these hypercatabolic states

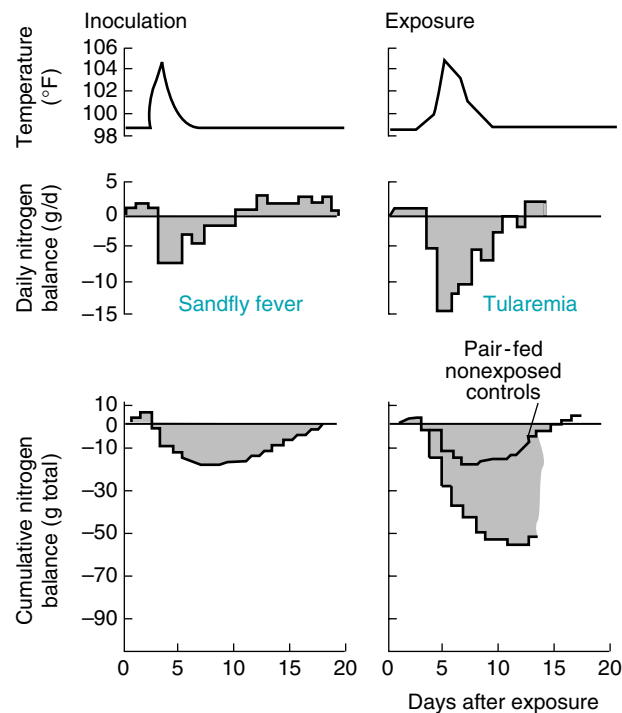


Fig. 42.16. Negative nitrogen balance during infection. The effects of experimentally induced infections on nitrogen balance were determined in human volunteers. After inoculation with sandfly fever, increased amino acid catabolism produced a negative nitrogen balance. A few days after exposure, the daily nitrogen balance became positive until the volunteers returned to their original state. Experiments with patients exposed to tularemia showed that the negative nitrogen balance was much larger than could be expected from a decreased appetite alone. Volunteers who ate the same amount of food as the infected individuals (paired nonexposed controls) had a much smaller cumulative negative nitrogen balance than the infected volunteers. From Beisel WR. *Am J Clin Nutr* 1977;30:1236–1247. © 1977 American Society for Clinical Nutrition.

results from an accelerated protein turnover and an increased rate of net protein degradation, primarily in skeletal muscle.

The catabolic state of sepsis (acute, generalized, febrile infection) is one of enhanced mobilization of fuels and amino acids to provide the energy and precursors required by cells of the immune system, host defense mechanisms, and wound healing. The amino acids must provide the substrates for new protein synthesis and cell division. Glucose synthesis and release are enhanced to provide fuel for these cells, and the patient may become mildly hyperglycemic.

In these hypercatabolic states, skeletal muscle protein synthesis decreases, and protein degradation increases. Oxidation of BCAA is increased and glutamine production enhanced. Amino acid uptake is diminished. Cortisol is the major hormonal mediator of these responses, although certain cytokines may also have direct effects on skeletal muscle metabolism. As occurs during fasting and metabolic acidosis, increased levels of cortisol stimulate ubiquitin-mediated proteolysis, induce the synthesis of glutamine synthetase, and enhance release of amino acids and glutamine from the muscle cells.

The amino acids released from skeletal muscle during periods of hypercatabolic stress are used in a prioritized manner, with the cellular components of the immune system receiving top priority. For example, the uptake of amino acids by the liver for the synthesis of acute phase proteins, which are part of the immune system, is greatly increased. Conversely, during the early phase of the acute response, the synthesis of other plasma proteins (e.g., albumin) is decreased. The increased availability of amino acids and the increased cortisol levels also stimulate gluconeogenesis, thereby providing fuel for the glucose-dependent cells of the immune system (e.g., lymphocytes). An increase of urea synthesis accompanies the acceleration of amino acid degradation.

The increased efflux of glutamine from skeletal muscle during sepsis serves several functions (see Fig. 42.1). It provides the rapidly dividing cells of the immune system with an energy source. Glutamine is available as a nitrogen donor for purine synthesis, for NAD^+ synthesis, and for other biosynthetic functions essential to growth and division of the cells. An increased production of metabolic acids may accompany stress such as sepsis, so there is an increased utilization of glutamine by the kidney.

Under the influence of elevated levels of glucocorticoids, epinephrine, and glucagon, fatty acids are mobilized from adipose tissue to provide alternate fuels for other tissues and spare glucose. Under these conditions, fatty acids are the major energy source for skeletal muscle, and glucose uptake is decreased. These changes may lead to a mild hyperglycemia.



The degree of the body's hypercatabolic response depends on the severity and duration of the trauma or stress. After an uncomplicated surgical procedure in an otherwise healthy patient, the net negative nitrogen balance may be limited to about 1 week. The mild nitrogen losses are usually reversed by dietary protein supplementation as the patient recovers. With more severe traumatic injury or septic stress, the body may catabolize body protein and adipose tissue lipids for a prolonged period, and the negative nitrogen balance may not be corrected for weeks.



Katta Bolic's severe negative nitrogen balance was caused by both her malnourished state and her intra-abdominal infection complicated by sepsis. The systemic and diverse responses the body makes to insults such as an acute febrile illness are termed the "acute phase response." An early event in this response is the stimulation of phagocytic activity (see Fig. 42.17). Stimulated macrophages release cytokines, which are regulatory proteins that stimulate the release of cortisol, insulin, and growth hormone. Cytokines also directly mediate the acute phase response of the liver and skeletal muscle to sepsis.

CLINICAL COMMENTS



The clinician can determine whether a patient such as **Katta Bolic** is mounting an acute phase response to some insult, however subtle, by determining whether several unique acute phase proteins are being secreted by the liver. C-reactive protein, so named because of its ability to interact with the C-polysaccharide of pneumococci, and serum amyloid A protein, a precursor of the amyloid fibril found in secondary amyloidosis, are elevated in patients undergoing the acute phase response and as compared with healthy individuals. Other proteins normally found in the blood of healthy individuals are present in increased concentrations in patients undergoing an acute phase response. These include haptoglobin, certain protease inhibitors, complement components, ceruloplasmin, and fibrinogen. The elevated concentration of these proteins in the blood increases the erythrocyte sedimentation rate (ESR), another laboratory measure of the presence of an acute phase response.

To determine the ESR, the patients' blood is placed vertically in a small-bore glass tube. The speed with which the red blood cells sediment toward the bottom of the tube depends on what percentage of the red blood cells clump together and, thereby, become more dense. The degree of clumping is directly correlated with the presence of one or more of the first-phase proteins listed previously. These proteins interfere with what is known as the zeta-potential of the red blood cells, which normally prevents the red blood cells from clumping. Because many different proteins can individually alter the zeta-potential, the ESR is a nonspecific test for the presence of acute inflammation.

The weight loss often noted in septic patients is primarily caused by a loss of appetite resulting from the effect of certain cytokines on the medullary appetite center. Other causes include increased energy expenditure from fever and enhanced muscle proteolysis.

BIOCHEMICAL COMMENTS



After a catabolic insult such as injury, trauma, infection, or cancer, the interorgan flow of glutamine and fuels is dramatically altered. Teleologically, the changes in metabolism that occur give first priority to cells that are part of the immune system. Evidence suggests that the changes in glutamine and

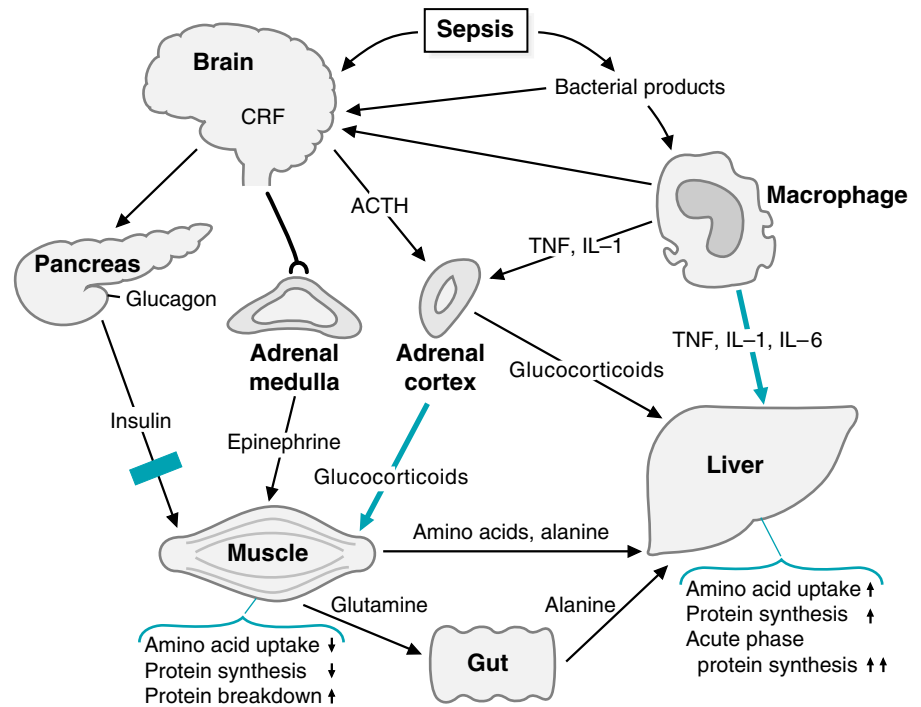


Fig. 42.17. Cytokines and hormones mediate amino acid flux during sepsis. Bacterial products act on macrophages to stimulate the release of cytokines and on the brain to stimulate the sympathoadrenal response. The result is a stimulation of the release of the insulin counter-regulatory hormones, epinephrine, glucagon, and glucocorticoids. The glucocorticoid cortisol may be the principal mediator of net muscle protein degradation during sepsis. Hepatic protein synthesis, particularly that of acute phase proteins, is stimulated both by cortisol and cytokines. Amino acid metabolism in the gut is also probably affected by glucocorticoids and cytokines. Because of the release of the counterregulatory hormones, muscle and other tissues become resistant to insulin action, as indicated by the bar on the figure. Adapted with permission from Fisher J. *Am J Surg* 1991;161:270.

fuel metabolism are mediated by the insulin counterregulatory hormones, such as cortisol and epinephrine, and several different cytokines (see Chapter 11 for a review of cytokines). Cytokines appear to play a more important role than hormones during sepsis, although they exert their effects, in part, through hormones (Fig. 42.17). Although cytokines can be released from a variety of cells, macrophages are the principal source during trauma and sepsis.

Two cytokines that are important in sepsis are interleukin-1 (IL-1) and tumor necrosis factor (TNF). IL-1 and TNF affect amino acid metabolism both through regulation of the release of glucocorticoids and through direct effects on tissues. Although cytokines are generally considered to be paracrine, with their effects being exerted over cells in the immediate vicinity, TNF and IL-1 increase in the blood during sepsis. Other cytokines, such as IL-6, also may be involved.

During sepsis, TNF, IL-1, and possibly other cytokines, bacterial products, or mediators act on the brain to stimulate the release of glucocorticoids from the adrenal cortex (a process mediated by adrenocorticotrophic hormone [ACTH]), epinephrine from the adrenal medulla, and both insulin and glucagon from the pancreas. Although insulin is elevated during sepsis, the tissues exhibit an insulin resistance that is similar to that of the non-insulin-dependent diabetes mellitus patient, possibly resulting from the elevated levels of the insulin counterregulatory hormones (glucocorticoids, epinephrine, and glucagon). Changes in the rate of acute phase protein synthesis are mediated, at least in part, by effects of TNF, IL-1, and IL-6 on the synthesis of groups of proteins in the liver.



Hypercatabolic states may be accompanied by varying degrees of insulin resistance caused, in part, by the release of counterregulatory hormones into the blood. Thus, patients with diabetes mellitus may require higher levels of exogenous insulin during sepsis.

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REVIEW QUESTIONS—CHAPTER 42

- Which of the profiles indicated below would occur within 2 hours after eating a meal very high in protein and low in carbohydrates?

	Blood glucagon levels	Liver gluconeogenesis	BCAA oxidation in muscle
(A)	↓	↓	↑
(B)	↑	↓	↑
(C)	↓	↑	↑
(D)	↑	↑	↑
(E)	↓	↓	↓
(F)	↑	↓	↓
(G)	↓	↑	↓
(H)	↑	↑	↓

2. The gut uses glutamine as an energy source, but can also secrete citrulline, synthesized from the carbons of glutamine. Which of the following compounds is an obligatory intermediate in this conversion (consider only the carbon atoms of glutamine while answering this question)?
 - (A) Aspartate
 - (B) Succinyl CoA
 - (C) Glutamate
 - (D) Serine
 - (E) Fumarate

3. The signal that indicates to muscle that protein degradation needs to be initiated is which of the following?
 - (A) Insulin
 - (B) Glucagon
 - (C) Epinephrine
 - (D) Cortisol
 - (E) Glucose

4. The skeletal muscles convert BCAA carbons to glutamine for export to the rest of the body. An obligatory intermediate, which carries carbons originally from the BCAA, in the conversion of BCAA to glutamine, is which of the following?
 - (A) Urea
 - (B) Pyruvate
 - (C) Lactate
 - (D) Isocitrate
 - (E) Phosphoenolpyruvate

5. An individual in sepsis would display which of the following metabolic patterns?

	Nitrogen balance	Gluconeogenesis	Fatty acid oxidation
(A)	Positive	↑	↑
(B)	Negative	↑	↑
(C)	Positive	↑	↓
(D)	Negative	↑	↓
(E)	Positive	↓	↑
(F)	Negative	↓	↓

Tissue Metabolism

Although many of the pathways described previously are present in all tissues of the body, many tissues also carry out specialized functions and contain unique biochemical pathways. This section describes a number of such tissues and, in some cases, how the tissues interact with the rest of the body to coordinate their functions.

The previous chapters of this text have focused primarily on insulin and glucagon as the major mediators for regulating metabolic pathways; however, a large number of other hormones also regulate the storage and utilization of metabolic fuels (see Chapter 43). These hormones primarily counteract the effects of insulin and are called counter-regulatory hormones. They include growth hormone; thyroid hormone; glucocorticoids, such as cortisol; small peptides, such as the somatostatins; and small molecules, such as the catecholamines. Growth hormone works, in part, by inducing the synthesis of the insulin-like growth factors. These hormones can exert their effects rapidly (through covalent modification of selected enzymes) or long-term (through alterations in the rate of synthesis of selected enzymes). The interplay of these hormones with insulin and glucagon is discussed, as are the synthesis, secretion, and conditions leading to secretion of each hormone.

The proteins and cells in the blood form their own tissue system (see Chapter 44). All blood cells are derived from a common precursor, the stem cell, in the bone marrow. Different cytokine signals trigger differentiation of a particular blood cell lineage. For example, when there is a decreased supply of oxygen to the tissues, the kidney responds by releasing erythropoietin. This hormone specifically stimulates the production of red blood cells.

The red blood cell has limited metabolic functions, owing to its lack of internal organelles. Its main function is to deliver oxygen to the tissues through the binding of oxygen to hemoglobin. When the number of red blood cells is reduced, an anemia is said to have developed. This can be attributable to many causes, including nutritional deficiencies or mutations (hereditary anemias). The morphology of the red blood cell can sometimes aid in distinguishing between the various types of anemia.

Red blood cell metabolism is geared toward preserving the ability of these cells to transport oxygen, as well as to regulate oxygen binding to hemoglobin. Glycolysis provides energy and NADH to protect the oxidation state of the heme-bound iron. The hexose monophosphate shunt pathway generates NADPH to protect red blood cell membranes from oxidation. Heme synthesis, which uses succinyl CoA and glycine for all of the carbon and nitrogen atoms in the structure, occurs in the precursors of red blood cells. Inherited defects in heme synthesis lead to a class of diseases known as the porphyrias. Because the red blood cell normally passes through the very narrow capillaries, its membrane must be easily deformable. This deformability is, in part, attributable to the complex cytoskeletal structure that surrounds the erythrocyte. Mutations in these structural proteins can lead to less deformable cells. This, in turn, can result in a hemolytic anemia.

Among other functions, the hematologic system is responsible for hemostasis as well as for maintaining a constant blood volume (see Chapter 45). A tear in the wall

of a vessel can lead to blood loss, which, when extensive, can be fatal. Repairing vessel damage, whether internal or external, is accomplished by a complicated series of zymogen activations of circulating blood proteins resulting in the formation of a fibrin clot (the coagulation cascade). Platelets play a critical role in hemostasis not only through their release of procoagulants but through their ability to form aggregates within the thrombus (clot) as well. Clots function as a plug, allowing vessel walls to repair and preventing further blood loss. Conversely, inappropriate clot formation in vessels that supply blood to vital organs or tissues can have devastating consequences, such as an acute cerebral or myocardial infarction. Because clotting must be tightly controlled, intricate mechanisms exist that regulate this important hematologic function.

The liver is an altruistic organ that provides multiple services for other tissues (see Chapter 46). It supplies glucose and ketone bodies to the rest of the body when fuel stores are limiting. It disposes of ammonia as urea when amino acid degradation occurs. It is the site of detoxification of xenobiotics, and it synthesizes many of the proteins found in the blood. The liver synthesizes fatty acids and cholesterol and distributes them to other tissues in the form of very-low-density lipoprotein (VLDL). The liver also synthesizes bile acids for fat digestion in the intestine. The liver recycles cholesterol and triglyceride through its uptake of intermediate density lipoprotein (IDL), chylomicron and VLDL remnants, and low-density lipoprotein (LDL) particles. Because of its protective nature and its strategic location between the gut and the systemic circulation, the liver has “first crack” at all compounds that enter the blood through the enterohepatic circulation. Thus, xenobiotic compounds can be detoxified as they enter the liver before they have a chance to reach other tissues.

Muscle cells contain unique pathways that allow them to store energy as creatine phosphate and to closely regulate their use of fatty acids as an energy source (see Chapter 47). The adenosine monophosphate (AMP)-activated protein kinase is an important regulator of muscle energy metabolism. Muscle is comprised of different types of contractile fibers that derive their energy from different sources. For example, the slow-twitch type I fibers use oxidative energy pathways, whereas the type II fast-twitch fibers use the glycolytic pathway for their energy requirements.

The nervous system consists of various cell types that are functionally interconnected so as to allow efficient signal transmission throughout the system (see Chapter 48). The cells of the central nervous system are protected from potentially toxic compounds by the blood-brain barrier, which restricts entry of compounds into the nervous system (ammonia, however, is a notable exception). The brain cells communicate with each other and with other organs, through the synthesis of neurotransmitters and neuropeptides. Many of the neurotransmitters are derived from amino acids, most of which are synthesized within the nerve cell. Because the pathways of amino acid and neurotransmitter biosynthesis require cofactors (such as pyridoxal phosphate, thiamine pyrophosphate, and vitamin B12), deficiencies of these cofactors can lead to neuropathies (dysfunction of specific neurons within the nervous system).

Because of the restrictions imposed by the blood-brain barrier, the brain also must synthesize its own lipids. An adequate supply of lipids is vital to the central nervous system because they are constituents of the myelin sheath that surrounds the neurons and allows them to conduct impulses normally. The neurodegenerative disorders, such as multiple sclerosis, are a consequence of varying degrees of demyelination of the neurons.

Connective tissue, which consists primarily of fibroblasts, produces extracellular matrix materials that surround cells and tissues, determining their appropriate position within the organ (see Chapter 49). These materials include structural proteins (collagen and elastin), adhesive proteins (fibronectin), and glycosaminoglycans (heparan sulfate, chondroitin sulfate). The unique structures of the proteins and carbohydrates found within the extracellular matrix allow tissues and organs to carry out their many functions. A loss of these supportive and barrier functions of connective tissue sometimes leads to significant clinical consequences, such as those that result from the microvascular alterations that lead to blindness or renal failure, or peripheral neuropathies in patients with diabetes mellitus.

43 Actions of Hormones That Regulate Fuel Metabolism

Many hormones affect fuel metabolism, including those that regulate appetite as well as those that influence absorption, transport, and oxidation of foodstuffs. The major hormones that influence nutrient metabolism and their actions on muscle, liver, and adipose tissue are listed in Table 43.1.

Insulin is the major **anabolic hormone**. It promotes the **storage** of **nutrients** as **glycogen** in liver and muscle, and as **triacylglycerols** in adipose tissue. It also stimulates the **synthesis** of **proteins** in tissues such as muscle. At the same time, insulin acts to inhibit fuel mobilization.

Glucagon is the major **counterregulatory** hormone. The term counterregulatory means that its actions are generally opposed to those of insulin (contrainsular). The major action of glucagon is to **mobilize fuel reserves** by stimulating **glycogenolysis** and **gluconeogenesis**. These actions ensure that glucose will be available to glucose-dependent tissues between meals.

Epinephrine, **norepinephrine**, **cortisol**, **somatostatin**, and **growth hormone** also have **contrainsular** activity. **Thyroid hormone** also must be classified as an insulin counterregulatory hormone because it **increases** the rate of **fuel consumption** and also increases the sensitivity of the target cells to other insulin counterregulatory hormones.

Insulin and the counterregulatory hormones exert two types of metabolic regulation (see Chapter 26). The first type of control occurs within minutes to hours of the hormone–receptor interaction and usually results from changes in the catalytic activity or kinetics of key preexisting enzymes, caused by phosphorylation or dephosphorylation of these enzymes. The second type of control involves regulation of the **synthesis of key enzymes** by mechanisms that stimulate or inhibit transcription and translation of mRNA. These processes are slow and require hours to days.



THE WAITING ROOM



Otto Shape, now a third-year medical student, was assigned to do a history and physical examination on a newly admitted 47-year-old patient named **Corti Solemia**. Mr. Solemia had consulted his physician for increasing weakness and fatigue and was found to have a severely elevated serum glucose level. While examining the patient, Otto noted marked redness of the patient's facial skin as well as reddish-purple stripes (striae) in the skin of the patient's lower abdomen and thighs. The patient's body fat was unusually distributed in that it appeared to be excessively

Table 43.1. Major Hormones that Regulate Fuel Metabolism

Hormones	Muscle			Liver						Adipose Tissue	
	Glucose Uptake	Glucose Utilization	Protein Synthesis	Glucose Output	Ketogenesis	Gluconeogenesis	Glycogenolysis	Glycogenesis	Protein Synthesis	Fat Synthesis	Lipolysis
Anabolic hormone											
Insulin	↑↑	↑↑	↑↑	↓↓	↓↓	↓↓	↓↓	↑	↑	↑↑	↓↓
Counterregulatory hormones ^a											
Glucagon	–	–	–	↑↑	↑	↑	↑↑	↓	–	–	↑ (at large doses) ↑↑
Epinephrine and norepinephrine	–	↑	–	↑↑	–	↑	↑↑ (initial)	↑↓	–	–	↑↑
Glucocorticoid	↓	↓	↓	↑	↑	↑ (mainly permissive)	–	↑	↑	–	↑ (permissive)
Growth hormone	↓ (weakly)	↓ (weakly)	↑	↑	↑	↑	–	–	↑	–	↑ (mainly permissive)
Thyroid hormone	–	↑	↑	↑	↑	↑	–	–	–	–	↑ (permissive)
Somatostatin ^b	–	–	–	–	–	–	–	–	–	–	–

^a Hormones with actions that are generally opposed to those of insulin.

^b Somatostatin's effects on metabolism are indirect via suppression of secretion of insulin, glucagon, growth hormone, and thyroid hormone and by effects on gastric acid secretion, gastric emptying time, and pancreatic exocrine secretion (see text).

deposited centrally in his face, neck, upper back, chest and abdomen, while the distal portions of his arms and legs appeared to be almost devoid of fat. The patient's skin appeared thinned, and large bruises were present over many areas of his body, for which Mr. Solemia had no explanation. The neurologic examination showed severe muscle weakness, especially in the proximal arms and legs, where the muscles seemed atrophied.



Sam Atotrope, a 42-year-old jeweler, noted increasingly severe headaches behind his eyes sometimes associated with a “flash of light” in his visual fields. At times his vision seemed blurred, making it difficult to perform some of the intricate work required of a jeweler. He consulted his ophthalmologist, who was impressed with the striking change in Sam's facial features that had occurred since he last saw the patient 5 years earlier. The normal skin creases in Sam's face had grown much deeper, his skin appeared to be thickened, his nose and lips appeared more bulbous, and his jaw seemed more prominent. The doctor also noted that Sam's hands appeared bulky, and his voice had deepened. An eye examination showed that Sam's optic nerves appeared slightly atrophied, and there was a loss of the upper outer quadrants of his visual fields.

I. PHYSIOLOGIC EFFECTS OF INSULIN

The effects of insulin on fuel metabolism and substrate flux were considered in many of the earlier chapters of this book, particularly in Chapter 26. Insulin stimulates the storage of glycogen in liver and muscle and the synthesis of fatty acids and triacylglycerols and their storage in adipose tissue. In addition, insulin stimulates the synthesis in various tissues of more than 50 proteins, some of which contribute to the growth of the organism. In fact, it is difficult to separate the effects of insulin on cell growth from those of a family of proteins known as the somatomedins or the insulin-like growth factors I and II (IGF-I and IGF-II) (see Section III.B.6. of this chapter).

Finally, insulin has paracrine actions within the pancreatic islet cells. When insulin is released from the β cells, it suppresses glucagon release from the α cells.

II. PHYSIOLOGIC EFFECTS OF GLUCAGON

Glucagon is one of several counterregulatory (contrainsular) hormones. It is synthesized as part of a large precursor protein, proglucagon. Proglucagon is produced in the α cells of the islets of Langerhans in the pancreas and in the L cells of the intestine. It contains a number of peptides linked in tandem: glicentin-related peptide, glucagon, glucagon-related peptide 1 (GLP-1), and glucagon-related peptide 2 (GLP-2). Proteolytic cleavage of proglucagon releases various combinations of its constituent peptides. Glucagon is cleaved from proglucagon in the pancreas and constitutes 30 to 40% of the immunoreactive glucagon in the blood. The remaining immunoreactivity is caused by other cleavage products of proglucagon released from the pancreas and the intestine. Pancreatic glucagon has a plasma half-life of 3 to 6 minutes and is removed mainly by the liver and kidney.

Glucagon promotes glycogenolysis, gluconeogenesis, and ketogenesis by stimulating the generation of cyclic adenosine monophosphate (cAMP) in target cells. The liver is the major target organ for glucagon, in part because the concentrations of this hormone bathing the liver cells in the portal blood are higher than in the peripheral circulation. Portal vein levels of glucagon may reach concentrations as high as 500 pg/mL.

Finally, glucagon stimulates insulin release from the β cells of the pancreas. Whether this is a paracrine effect or an endocrine effect has not been established. The pattern of blood flow in the pancreatic islet cells is believed to bathe the β cells

first and then the α -cells. Therefore, the β cells may influence α -cell function by an endocrine mechanism, whereas the influence of β -cell hormone on α -cell function is more likely to be paracrine.

III. PHYSIOLOGIC EFFECTS OF OTHER COUNTERREGULATORY HORMONES

A. Somatostatin

1. BIOCHEMISTRY

Preprosomatostatin, a 116–amino acid peptide, is encoded on the long arm of chromosome 3. Somatostatin (SS-14), a cyclic peptide with a molecular weight of 1,600, is produced from the 14 amino acids at the C-terminus of this precursor molecule. SS-14 was first isolated from the hypothalamus and named for its ability to inhibit the release of growth hormone (GH, somatotropin) from the anterior pituitary. It also inhibits the release of insulin. In addition to the hypothalamus, somatostatin is also secreted from the D cells (δ cells) of the pancreatic islets, many areas of the central nervous system outside of the hypothalamus, and in gastric and duodenal mucosal cells. SS-14 predominates in the central nervous system (CNS) and is the sole form secreted by the δ cells of the pancreas. In the gut, however, prosomatostatin (SS-28), which has 14 additional amino acids extending from the C-terminal portion of the precursor, makes up 70 to 75% of the immunoreactivity (the amount of hormone that reacts with antibodies to SS-14). The prohormone SS-28 is 7 to 10 times more potent in inhibiting the release of GH and insulin than is SS-14.

2. SECRETION OF SOMATOSTATIN

The secretagogues for somatostatin are similar to those that cause secretion of insulin. The metabolites that increase somatostatin release include glucose, arginine, and leucine. The hormones that stimulate somatostatin secretion include glucagon, vasoactive intestinal polypeptide (VIP), and cholecystokinin (CCK). Insulin, however, does not directly influence somatostatin secretion.

3. PHYSIOLOGIC EFFECTS OF SOMATOSTATIN

Five somatostatin receptors have been identified and characterized, all of which are members of the G protein–coupled receptor superfamily. Four of the five receptors do not distinguish between SS-14 and SS-28. Somatostatin binds to its plasma membrane receptors on target cells. These “activated” receptors interact with inhibitory G proteins of adenylate cyclase. As a result, the production of cAMP is inhibited, and protein kinase A is not activated. This inhibitory effect suppresses secretion of GH and thyroid-stimulating hormone (TSH) from the anterior pituitary gland as well as the secretion of insulin and glucagon from the pancreatic islets. If one were to summarize the action of somatostatin in one phrase, it would be “somatostatin inhibits the secretion of many other hormones.” As such, it acts to regulate the effects of those other hormones. In addition to these effects on hormones that regulate fuel metabolism, somatostatin also reduces nutrient absorption from the gut by prolonging gastric emptying time (through a decrease in the secretion of gastrin, which reduces gastric acid secretion), by diminishing pancreatic exocrine secretions (i.e., digestive enzymes, bicarbonate, and water), and by decreasing visceral blood flow. Thus, somatostatin exerts a broad, albeit indirect, influence on nutrient absorption and, therefore, the utilization of fuels.

Somatostatin and its synthetic analogs are used clinically to treat a variety of secretory neoplasms such as GH-secreting tumors of the pituitary. Such tumors can cause gigantism if growth hormone is secreted in excess before the closure of the



Tolbutamide, a sulfonylurea drug that increases insulin secretion, also increases the secretion of pancreatic somatostatin.



In addition to its effects on normal GH secretion, somatostatin also suppresses the pathologic increase in GH that occurs in acromegaly (caused by a GH-secreting pituitary tumor), diabetes mellitus, and carcinoid tumors (tumors that secrete serotonin). Somatostatin also suppresses the basal secretion of TSH, TRH, insulin, and glucagon. The hormone also has a suppressive effect on a wide variety of nonendocrine secretions.

The major limitation in the clinical use of native somatostatin is its short half-life of less than 3 minutes in the circulation. Analogs of native somatostatin, however, have been developed that are resistant to degradation and, therefore, have a longer half-life. One such analog is octreotide, an octapeptide variant of somatostatin with a half-life of approximately 110 minutes.

growth centers of the ends of long bones or acromegaly if excess growth hormone is chronically secreted after the closure of these centers (as in **Sam Atrope**).

B. Growth Hormone

1. BIOCHEMISTRY

Growth hormone is a polypeptide that, as its name implies, stimulates growth. Many of its effects are mediated by insulin-like growth factors (IGFs, also known as somatomedins) that are produced by cells in response to the binding of GH to its cell membrane receptors (see Section 6 below). However, GH also has direct effects on fuel metabolism.

Human growth hormone is a water-soluble 22-kDa polypeptide with a plasma half-life of 20 to 50 minutes. It is composed of a single chain of 191 amino acids having two intramolecular disulfide bonds (Fig. 43.1). The gene for GH is located on chromosome 17. It is secreted by the somatotroph cells (the cells that synthesize and release GH) present in the lateral areas of the anterior pituitary. GH is structurally related to human prolactin and to human chorionic somatomammotropin (hCS) from the placenta, a polypeptide that stimulates growth of the developing fetus. Yet the hCS peptide has only 0.1% of the growth-inducing potency of GH. Growth hormone is the most abundant trophic hormone in the anterior pituitary, being present in concentrations of 5 to 15 mg per gram of tissue. The other anterior pituitary hormones are present in microgram-per-gram of tissue quantities.

The actions of GH can be classified as those that occur as a consequence of the hormone's direct effect on target cells and those that occur indirectly through the ability of GH to generate other factors, particularly IGF-I.

The IGF-I-independent actions of GH are exerted primarily in hepatocytes. GH administration is followed by an early increase in the synthesis of 8 to 10 proteins, among which are IGF-I, α_2 -macroglobulin, and the serine protease inhibitors Spi 2.1 and Spi 2.3. Expression of the gene for ornithine decarboxylase, an enzyme active in polyamine synthesis (and, therefore, in the regulation of cell proliferation), is also significantly increased by GH.

Muscle and adipocyte cell membranes contain GH receptors that mediate direct, rapid metabolic effects on glucose and amino acid transport as well as on lipolysis. These receptors use associated cytoplasmic tyrosine kinases for signal transduction (such as the janus kinases; see Chapter 11, section III.C.). For example, in adipose tissue, GH has acute insulinlike effects followed by increased lipolysis, inhibition of lipoprotein lipase, stimulation of hormone-sensitive lipase, decreased glucose transport, and decreased lipogenesis. In muscle, GH causes increased amino acid transport, increased nitrogen retention, increased fat-free (lean) tissue, and increased energy expenditure. GH also has growth-promoting effects. GH receptors



An MRI scan of **Sam Atrope's** brain showed a macroadenoma (a tumor greater than 10 mm in diameter) in the pituitary gland, with superior extension that compressed the optic nerve as it crossed above the sella turcica, causing his visual problems. The skeletal and visceral changes noted by the ophthalmologist are characteristic of acromegalic patients with chronically elevated serum levels of GH and IGF-I.

Therapeutic alternatives for acromegaly caused by a GH-secreting tumor of the anterior pituitary gland include lifelong medical therapy with the somatostatin analog octreotide or the GH receptor antagonist pegvisomant. Other therapeutic options include stereotactic radiation therapy or surgical resection of the neoplasm. If the excessive secretion of GH is controlled successfully, some of the visceral or soft tissue changes of acromegaly may slowly subside to varying degrees. The skeletal changes, however, cannot be reversed.



The ophthalmologist ordered a morning fasting serum GH level on **Sam Atrope**, which was elevated at 56 ng/mL (normal = 0–5 ng/mL)



Sam Atrope was given an oral dose of 100 g glucose syrup. This dose would suppress serum GH levels to less than 2 ng/mL in normal subjects, but not in patients with acromegaly who have an autonomously secreting pituitary tumor making GH. Because Sam's serum GH level was 43 ng/mL after the oral glucose load, a diagnosis of acromegaly was made. The patient was referred to an endocrinologist for further evaluation.

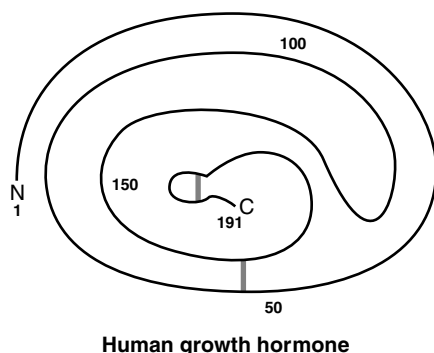


Fig. 43.1. Structure of human growth hormone. From Murray, RK, et al, Harper's Biochemistry, 23rd Ed. Stamford, CT: Appleton & Lange, 1993:502.



GH not only stimulates IGF-I gene expression in the liver but in a number of extrahepatic tissues as well. In acromegals, rising levels of IGF-I cause a gradual generalized increase in skeletal, muscular, and visceral growth. As a consequence, a diffuse increase occurs in the bulk of all tissues (enlargement = “megaly”) especially in the “acral” (most peripheral) tissues of the body, such as the face, the hands, and the feet, hence the term “acromegaly.”

Sam Atotrope's coarse facial features and bulky hands are typical of patients with acromegaly.

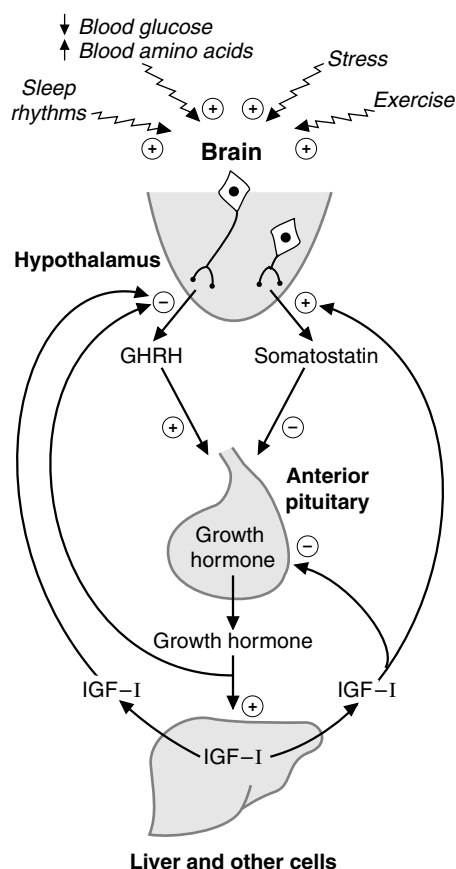


Fig. 43.2. Control of growth hormone secretion. Various factors stimulate the release of GHRH from the hypothalamus. The hypothalamus also releases somatostatin in response to other stimuli. GHRH stimulates and somatostatin inhibits the release of growth hormone from the anterior pituitary. Growth hormone causes the release of IGF-I from liver and other tissues. IGF-I inhibits GHRH release and stimulates somatostatin release.

are present on a variety of tissues in which GH increases IGF-I gene expression. The subsequent rise in IGF-I levels contributes to cell multiplication and differentiation by autocrine or paracrine mechanisms. These, in turn, lead to skeletal, muscular, and visceral growth. These actions are accompanied by a direct anabolic influence of GH on protein metabolism with a diversion of amino acids from oxidation to protein synthesis and a shift to a positive nitrogen balance.

2. CONTROL OF SECRETION OF GROWTH HORMONE

Although the regulation of GH secretion is complex, the major influence is hormonal (Fig. 43.2). The pulsatile pattern of GH secretion reflects the interplay of two hypothalamic regulation peptides. Release is stimulated by growth hormone–releasing hormone (GHRH, also called somatotocin). The structure of GHRH was identified in 1982 (Fig. 43.3). It exists as both a 40– and a 44–amino acid peptide encoded on chromosome 20 and produced exclusively in cells of the arcuate nucleus. Its C-terminal leucine residue is amidated. Full biologic activity of this releasing hormone resides in the first 29 amino acids of the N-terminal portion of the molecule. GHRH interacts with specific receptors on the plasma membranes of the somatotrophs. The intracellular signaling mechanisms that result in GH synthesis and release appear to be multiple, as cAMP and calcium-calmodulin both stimulate GH release.

Conversely, GH secretion is suppressed by growth hormone release-inhibiting hormone (GHRH, also called somatostatin, which has already been discussed). In addition, IGF-I, produced primarily in the liver in response to the action of GH on hepatocytes, feeds back negatively on the somatotrophs to limit GH secretion. Other physiologic factors (e.g., exercise and sleep) and many pathologic factors control its release (Table 43.2).

In addition, GH release is modulated by plasma levels of all of the metabolic fuels, including proteins, fats, and carbohydrates. A rising level of glucose in the blood normally suppresses GH release, whereas hypoglycemia increases GH

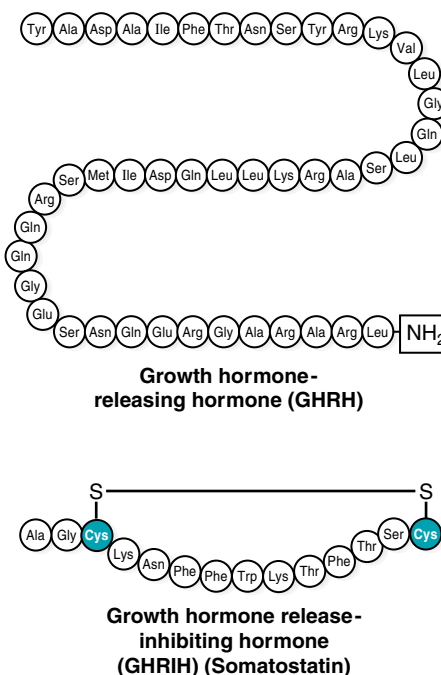


Fig. 43.3. Structures of growth hormone–releasing hormone (GHRH) and growth hormone release-inhibiting hormone (GHRH, also called somatostatin). GHRH has an amide at the C-terminal.

Table 43.2. Some Factors Affecting Growth Hormone Secretion

	Stimulate	Suppress
Physiologic	Low blood glucose after meals High blood amino acids after meals Exercise Sleep Stress	High blood glucose after meals High blood fatty acids
Pharmacologic	GHRH Estrogens α -Adrenergic agonists β -Adrenergic antagonists Dopamine agonists Serotonin precursors K^+ infusion	Somatostatin Progesterone α -Adrenergic antagonists β -Adrenergic agonists Dopamine antagonists
Pathologic	Starvation Anorexia nervosa Ectopic GHRH production Acromegaly Chronic renal failure Hypoglycemia	Growth hormone and IGF-I Obesity Hypothyroidism Hyperthyroidism

secretion in normal subjects. Amino acids, such as arginine, stimulate release of GH when their concentrations rise in the blood. Rising levels of fatty acids may blunt the GH response to arginine or a rapidly dropping blood glucose level. However, prolonged fasting, in which fatty acids are mobilized in an effort to spare protein, is associated with a rise in GH secretion. Some of the physiologic, pharmacologic, and pathologic influences on GH secretion are given in Table 43.2.

3. EFFECTS OF GROWTH HORMONE ON ENERGY METABOLISM

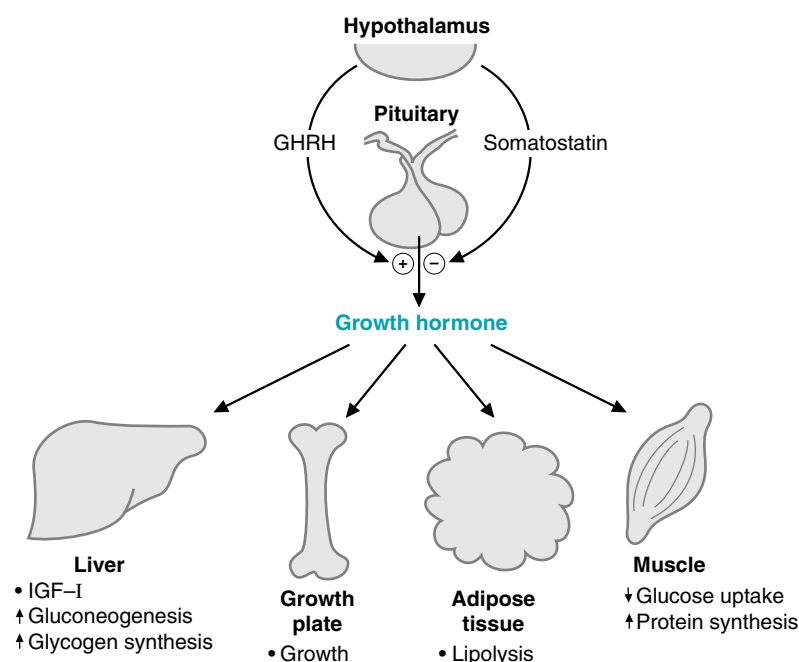
GH affects the uptake and oxidation of fuels in adipose tissue, muscle, and liver and indirectly influences energy metabolism through its actions on the islet cells of the pancreas. In summary, GH increases the availability of fatty acids, which are oxidized for energy. This and other effects of GH spare glucose and protein; that is, GH indirectly decreases the oxidation of glucose and amino acids (Fig. 43.4).



These modulators of GH secretion provide the basis for clinical suppression and stimulation tests in patients suspected of having excessive or deficient GH secretion.



While **Sam Atotrope** was trying to decide which of the major alternatives for the treatment of his growth hormone (GH)-secreting pituitary tumor to choose, he noted progressive fatigue and the onset of increasing urinary frequency associated with a marked increase in thirst. In addition, he had lost 4 lb over the course of the last 6 weeks in spite of a good appetite. His physician suspected that Mr. Atotrope had developed diabetes mellitus, perhaps related to the chronic hypersecretion of GH. This suspicion was confirmed when Sam's serum glucose level, drawn before breakfast, was reported to be 236 mg/dL.

**Fig. 43.4.** Anabolic effects of growth hormone on various tissues.



As a result of the metabolic effects of GH, the clinical course of acromegaly may be complicated by impaired glucose tolerance or even overt diabetes mellitus, as occurred with **Sam Atotrope**.

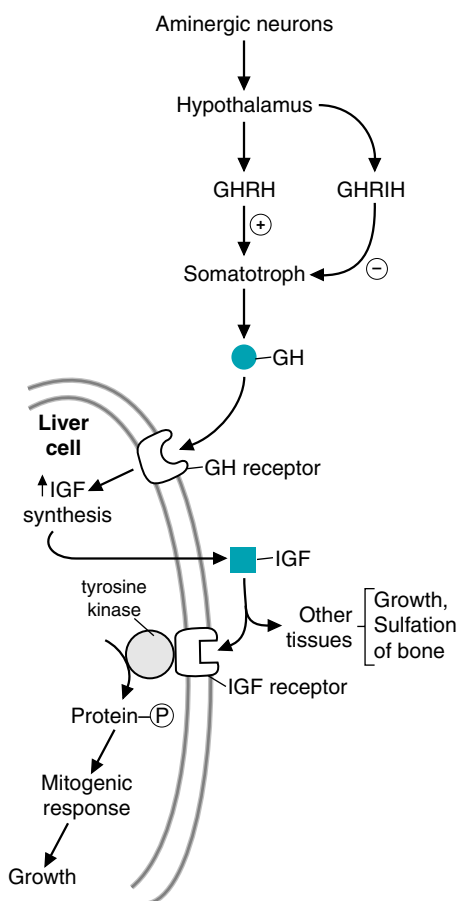


Fig 43.5. Production and action of IGFs. The hypothalamus produces growth hormone-releasing hormone (GHRH), which stimulates somatotrophs in the anterior pituitary to release growth hormone (GH). Growth hormone release-inhibiting hormone (GHRIH) inhibits GH release. GH binds to cell surface receptors and stimulates IGF production and release by liver and other tissues. IGF binds to cell surface receptors and stimulates the phosphorylation of proteins that cause mitosis and growth.



High levels of circulating IGF-1 has been linked to the development of breast, prostate, colon, and lung cancer. Additionally, experimental modulation of IGF-1 receptor activity can alter the growth of different types of tumor cells. Current research is aimed at targeting the interaction of IGF-1 and its receptor to reduce tumor cell proliferation.

4. EFFECTS OF GROWTH HORMONE ON ADIPOSE TISSUE

Growth hormone increases the sensitivity of the adipocyte to the lipolytic action of the catecholamines and decreases its sensitivity to the lipogenic action of insulin. These actions lead to the release of free fatty acids and glycerol into the blood to be metabolized by the liver. GH also decreases esterification of fatty acids, thereby reducing triacylglycerol synthesis within the fat cell. Recent evidence suggests that GH may impair glucose uptake by both fat and muscle cells by a postreceptor inhibition of insulin action.

5. EFFECTS OF GROWTH HORMONE ON MUSCLE

The lipolytic effects of GH increase free fatty acid levels in the blood bathing muscle. These fatty acids are preferentially used as fuel, indirectly suppressing glucose uptake by muscle cells. Through the effects on glucose uptake, the rate of glycolysis is proportionately reduced.

GH increases the transport of amino acids into muscle cells, providing substrate for protein synthesis. Through a separate mechanism, GH increases the synthesis of DNA and RNA. The positive effect on nitrogen balance is reinforced by the protein-sparing effect of GH-induced lipolysis that makes fatty acids available to muscle as an alternative fuel source.

6. EFFECTS OF GROWTH HORMONE ON THE LIVER

When plasma insulin levels are low, as in the fasting state, GH enhances fatty acid oxidation to acetyl CoA. This effect in concert with the increased flow of fatty acids from adipose tissue enhances ketogenesis. The increased amount of glycerol reaching the liver as a consequence of enhanced lipolysis acts as a substrate for gluconeogenesis.

Hepatic glycogen synthesis is also stimulated by GH in part because of the increased gluconeogenesis in the liver. Finally, glucose metabolism is suppressed by GH at several steps in the glycolytic pathway.

A major effect of GH on liver is to stimulate production and release of IGFs. The IGFs are also known as somatomedins. The two somatomedins in humans share structural homologies with proinsulin, and both have substantial insulin-like growth activity; hence the designations, insulin-like growth factor I (human IGF-I, or somatomedin-C) and insulin-like growth factor II (human IGF-II, or somatomedin A). IGF-I is a single-chain basic peptide having 70 amino acids, and IGF-II is slightly acidic with 67 amino acids. These two peptides are identical to insulin in half of their residues. In addition, they contain a structural domain that is homologous to the C-peptide of proinsulin.

A broad spectrum of normal cells respond to high doses of insulin by increasing thymidine uptake and initiating cell propagation. In most instances, IGF-I causes the same response as insulin in these cells but at significantly smaller, more physiologic concentrations. Thus, the IGFs are more potent than insulin in their growth-promoting actions.

Evidence suggests that the IGFs exert their effects through either an endocrine or a paracrine/autocrine mechanism. IGF-I appears to stimulate cell propagation and growth by binding to specific IGF-I receptors on the plasma membrane of target cells, rather than binding to GH receptors (Fig. 43.5).

Like insulin, the intracellular portion of the plasma membrane receptor for IGF-I (but not IGF-II) has intrinsic tyrosine kinase activity. The fact that the receptors for insulin and a number of other growth factors have intrinsic tyrosine kinase activity indicates that tyrosine phosphorylation initiates the process of cellular replication and growth. Subsequently, a chain of kinases is activated, which include a number of proto-oncogene products (see Chapters 11 and 18).

Most cells of the body have mRNA for IGF, but the liver has the greatest concentration of these messengers, followed by kidney and heart. The synthesis of IGF-I is regulated, for the most part, by GH, whereas hepatic production of IGF-II is independent of GH levels in the blood.

C. Catecholamines (Epinephrine, Norepinephrine, Dopamine)

The catecholamines belong to a family of bioamines and are secretory products of the sympathoadrenal system, which are required for the body to adapt to a great variety of acute and chronic stresses. Epinephrine (80–85% of stored catecholamines) is synthesized primarily in the cells of the adrenal medulla, whereas norepinephrine (15–20% of stored catecholamines) is synthesized and stored not only in the adrenal medulla but also in various areas of the central nervous system (CNS) and in the nerve endings of the adrenergic nervous system. Dopamine, another catecholamine, acts primarily as a neurotransmitter and has little effect on fuel metabolism.

The first total chemical synthesis of epinephrine was accomplished by F. Stolz et al in 1904. In 1950, Earl Sutherland was the first to demonstrate that epinephrine (and glucagon) induces glycogenolysis. This marked the beginning of our understanding of the molecular mechanisms through which hormones act.

1. SYNTHESIS OF THE CATECHOLAMINES

Tyrosine is the precursor of the catecholamines. The pathway for the biosynthesis of these molecules is described in Chapter 48.

2. SECRETION OF THE CATECHOLAMINES

Secretion of epinephrine and norepinephrine from the adrenal medulla is stimulated by a variety of stresses, including pain, hemorrhage, exercise, hypoglycemia, and hypoxia. Release is mediated by stress-induced transmission of nerve impulses emanating from adrenergic nuclei in the hypothalamus. These impulses stimulate the release of the neurotransmitter acetylcholine from preganglionic neurons that innervate the adrenomedullary cells. Acetylcholine depolarizes the plasma membranes of these cells, allowing the rapid entry of extracellular calcium (Ca^{2+}) into the cytosol. Ca^{2+} stimulates the synthesis and release of epinephrine and norepinephrine from the chromaffin granules into the extracellular space by exocytosis.

3. PHYSIOLOGIC EFFECTS OF EPINEPHRINE AND NOREPINEPHRINE

The catecholamines act through two major types of receptors on the plasma membrane of target cells, the α -adrenergic and the β -adrenergic receptors (see Chapter 26, section IV.C).

The actions of epinephrine and norepinephrine in the liver, the adipocyte, the skeletal muscle cell, and the α and β cells of the pancreas directly influence fuel metabolism (Fig. 43.6). These catecholamines are counterregulatory hormones that have metabolic effects directed toward mobilization of fuels from their storage sites for oxidation by cells to meet the increased energy requirements of acute and chronic stress. They simultaneously suppress insulin secretion, which ensures that fuel fluxes will continue in the direction of fuel utilization rather than storage as long as the stressful stimulus persists.

In addition, norepinephrine works as a neurotransmitter and affects the sympathetic nervous system in the heart, lungs, blood vessels, bladder, gut, and other



In patients suspected of having a neoplasm of the adrenal medulla that is secreting excessive quantities of epinephrine or norepinephrine (a pheochromocytoma), either the catecholamines themselves (epinephrine, norepinephrine, and dopamine) or their metabolites (the metanephrines and vanillylmandelic acid, VMA) may be measured in a 24-hour urine collection, or the level of catecholamines in the blood may be measured. A patient who has consistently elevated levels in the blood or urine should be considered to have a pheochromocytoma, particularly if the patient has signs and symptoms of catecholamine excess, such as excessive sweating, palpitations, tremulousness, and hypertension.

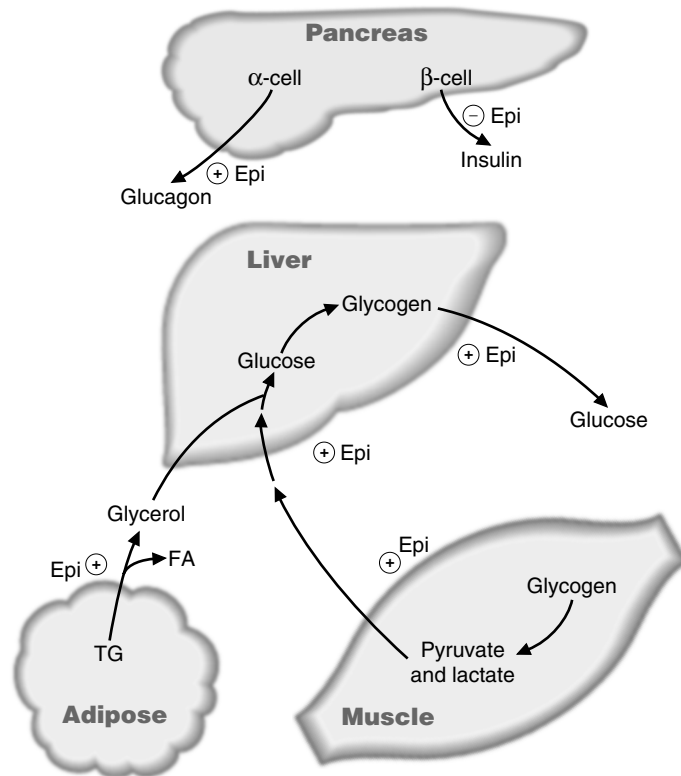


Fig. 43.6. Effects of epinephrine on fuel metabolism and pancreatic endocrine function. Epinephrine (Epi) stimulates glycogen breakdown in muscle and liver, gluconeogenesis in liver, and lipolysis in adipose tissue. Epinephrine further reinforces these effects because it increases the secretion of glucagon, a hormone that shares many of the same effects as epinephrine. Epi also inhibits insulin release but stimulates glucagon release from the pancreas.



A relatively rare form of secondary hypertension (high blood pressure) is caused by a catecholamine-secreting neoplasm of the adrenal medulla, known as a pheochromocytoma. Patients with the tumor periodically secrete large amounts of epinephrine and norepinephrine into the bloodstream. Symptoms related to this secretion include a sudden and often severe increase in blood pressure, heart palpitations, a throbbing headache, and inappropriate and diffuse sweating. In addition, chronic hypersecretion of these catecholamines may lead to impaired glucose tolerance or even overt diabetes mellitus. Describe the actions of these hormones that lead to the significant rise in glucose levels.



Glucocorticoids, such as cortisol, were named for their ability to raise blood glucose levels. These steroids are among the “counterregulatory” hormones that protect the body from insulin-induced hypoglycemia.

organs. These effects of catecholamines on the heart and blood vessels serve to increase cardiac output and systemic blood pressure, hemodynamic changes that facilitate the delivery of blood borne fuels to metabolically active tissues.

Epinephrine has a short half-life in the blood and to be effective pharmacologically must be administered parenterally. It may be used clinically to support the beating of the heart, to dilate inflamed bronchial muscles, and even to decrease bleeding from organs during surgery.

4. METABOLISM AND INACTIVATION OF CATECHOLAMINES

Catecholamines have a relatively low affinity for both α - and β -receptors. After binding, the catecholamine disassociates from its receptor quickly, causing the duration of the biologic response to be brief. The free hormone is degraded in several ways, as outlined in Chapter 48.

D. Glucocorticoids

1. BIOCHEMISTRY

Cortisol (hydrocortisone) is the major physiologic glucocorticoid (GC) in humans, although corticosterone also has some glucocorticoid activity. The biosynthesis of steroid hormones and their basic mechanism of action has been described in Chapters 34 and 17.

2. SECRETION OF GLUCOCORTICOIDS

The synthesis and secretion of cortisol is controlled by a cascade of neural and endocrine signals linked in tandem in the cerebrocortical-hypothalamic-pituitary-adrenocortical axis. Cerebrocortical signals to the midbrain are initiated in the cerebral cortex by “stressful” signals such as pain, hypoglycemia, hemorrhage, and exercise (Fig. 43.7). These nonspecific “stresses” elicit the production of monoamines in the cell bodies of neurons of the midbrain. Those that stimulate the synthesis and release of corticotropin-releasing hormone (CRH) are acetylcholine and serotonin. These neurotransmitters then induce the production of CRH by neurons originating in the paraventricular nucleus. These neurons discharge CRH into the hypothalamo-hypophyseal portal blood. CRH is delivered through these portal vessels to specific receptors on the cell membrane of the adrenocorticotrophic hormone (ACTH)-secreting cells of the anterior pituitary gland (corticotrophs). This hormone–receptor interaction causes ACTH to be released into the general circulation to eventually interact with specific receptors for ACTH on the plasma membranes of cells in the zona fasciculata and zona reticulosa of the adrenal cortex. The major trophic influence of ACTH on cortisol synthesis is at the level of the conversion of cholesterol to pregnenolone, from which the adrenal steroid hormones are derived (see Chapter 34 for the biosynthesis of the steroid hormones).

Cortisol is secreted from the adrenal cortex in response to ACTH. The concentration of free (unbound) cortisol that bathes the CRH-producing cells of the hypothalamus and the ACTH-producing cells of the anterior pituitary acts as a negative feedback signal that has a regulatory influence on the release of CRH and ACTH (see Fig. 43.7). High cortisol levels in the blood suppress CRH and ACTH secretion, and low cortisol levels stimulate secretion. In severe stress, however, the negative feedback signal on ACTH secretion exerted by high cortisol levels in the blood is overridden by the stress-induced activity of the higher portions of the axis (see Fig. 43.7).

The effects of glucocorticoids on fuel metabolism in liver, skeletal muscle, and adipose tissue are outlined in Table 43.1 and in Figure 43.8. Their effects on other tissues are diverse and, in many instances, essential for life. Some of the nonmetabolic actions of GCs are listed in Table 43.3.

3. EFFECTS OF GLUCOCORTICOIDS

Glucocorticoids have diverse actions that affect most tissues of the body. At first glance, some of these effects may appear to be contradictory (such as inhibition of



When Otto was writing his list of differential diagnoses to explain the clinical presentation of **Corti Solemia**, he suddenly thought of a relatively rare endocrine disorder that could explain all of the presenting signs and symptoms. He made a provisional diagnosis of excessive secretion of cortisol secondary to an excess secretion of ACTH (Cushing’s “disease”) or by a primary increase of cortisol production by an adrenocortical tumor (Cushing’s syndrome).

Otto suggested that resting, fasting plasma cortisol and ACTH levels be measured at 8:00 the next morning. These studies showed that Mr. Solemia’s morning plasma ACTH and cortisol levels were both significantly above the reference range. Therefore, Otto concluded that Mr. Solemia probably had a tumor that was producing ACTH autonomously (i.e., not subject to normal feedback inhibition by cortisol). The high plasma levels of ACTH were stimulating the adrenal cortex to produce excessive amounts of cortisol. Additional laboratory and imaging studies indicated that the hypercortisolemia was caused by a benign ACTH-secreting adenoma of the anterior pituitary gland (Cushing’s “disease”).

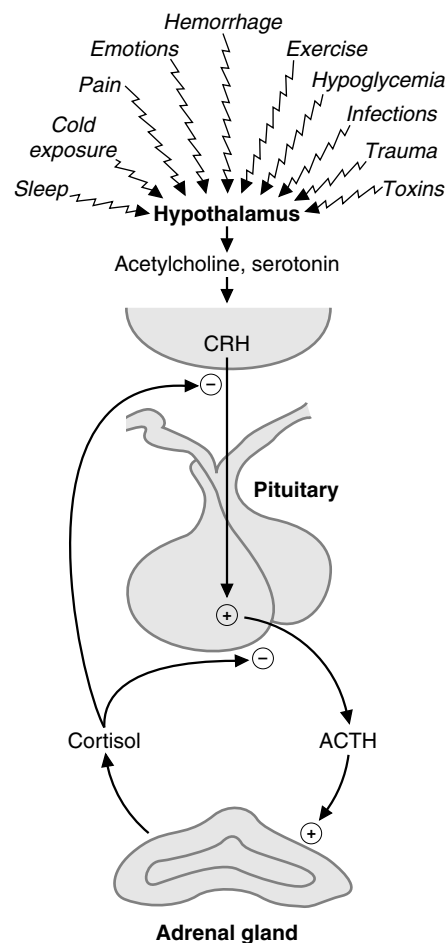


Fig. 43.7. Regulation of cortisol secretion. Various factors act on the hypothalamus to stimulate the release of corticotrophin-releasing hormone (CRH). CRH stimulates the release of ACTH from the anterior pituitary, which stimulates the release of cortisol from the adrenal cortex. Cortisol inhibits the release of CRH and ACTH.



The catecholamines are counterregulatory hormones that mobilize fuels from their storage sites for oxidation in target cells to meet the increased energy requirements that occur during acute or chronic stress, or, in this case, the release of catecholamines by a tumor in the adrenal medulla. These actions provide the liver, for example, with increased levels of substrate needed for gluconeogenesis. Although in normal individuals most of the glucose generated through this mechanism is oxidized, blood glucose levels rise in the process. In addition, the catecholamines suppress insulin secretion to ensure that fuels will continue to flow in the direction of utilization rather than storage under these circumstances. Hence, blood glucose levels may rise in patients who have a pheochromocytoma.

Table 43.3. Some Nonmetabolic Physiologic Actions of Glucocorticoids

On electrolyte and water balance:
Increase sodium and water retention (1/3,000 the potency of aldosterone)
Increase renal glomerular filtration rate to maintain water excretion rate
Suppress arginine vasopressin (ADH) release from posterior pituitary (?)
On cardiovascular system:
Indirect effect of glucocorticoid actions on sodium and water metabolism (above)
Maintain volume of microcirculation to tissues (cardiac output)
Maintain normal vasomotor response to vasoconstricting agents
On skeletal muscle:
Maintain muscle function by providing normal microcirculation to muscle
Influence muscle mass by enhancing protein catabolism and suppressing protein synthesis
On central nervous system:
Indirect
Maintain normal cerebral microcirculation
Direct
Influence mood, behavior
Influence sensitivity of special senses to stimuli
Suppress CRH, ACTH, and ADH secretion
On formed elements in blood:
Increase red blood cell mass and granulocyte proliferation
Decrease lymphocyte, monocyte, and basophil proliferation
Anti-inflammatory actions:
Inhibit early inflammatory process (i.e., edema, fibrin deposition, capillary dilation, leukocyte migration, and phagocytic action)
Inhibit late inflammatory process (proliferation of capillaries and fibroblasts, deposition of collagen, and, later, scar formation)
Immune suppressant actions (of questionable significance at physiologic levels):
Prevent manifestations of humoral and cellular immunity
Interfere with production of cytokines needed for immune competence via cell-to-cell communication

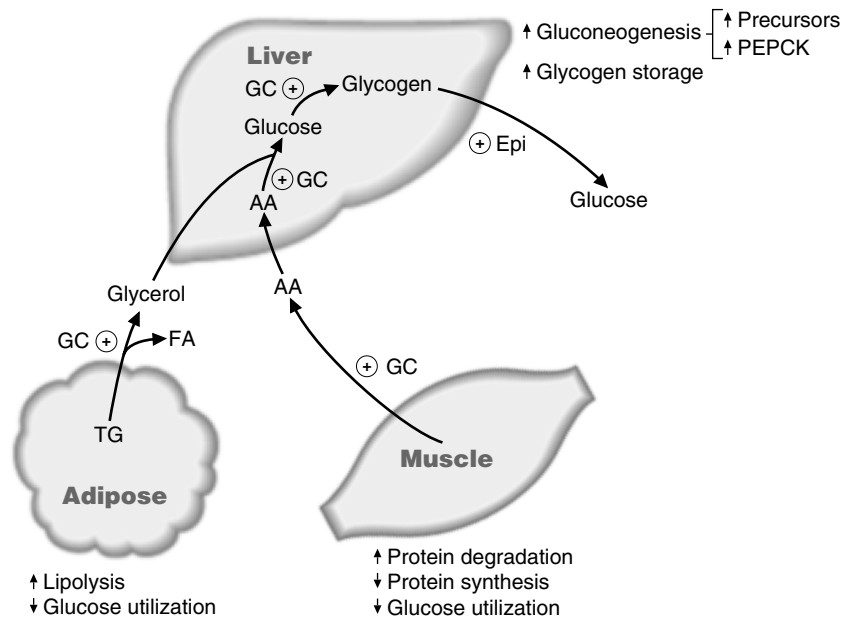


Fig. 43.8. Effects of glucocorticoids (GC) on fuel metabolism. Glucocorticoids stimulate lipolysis in adipose tissue and the release of amino acids from muscle protein. In liver, glucocorticoids stimulate gluconeogenesis and the synthesis of glycogen. The breakdown of liver glycogen is stimulated by epinephrine.

glucose uptake by certain tissues), but taken together they promote survival in times of stress.

In many tissues, GCs inhibit DNA, RNA, and protein synthesis and stimulate the degradation of these macromolecules. In response to chronic stress, GCs act to make fuels available, so that when the acute alarm sounds and epinephrine is released, the organism can fight or flee. When GCs are elevated, glucose uptake by the cells of many tissues is inhibited, lipolysis occurs in peripheral adipose tissue, and proteolysis occurs in skin, lymphoid cells, and muscle. The fatty acids that are released are oxidized by the liver for energy, and the glycerol and amino acids serve in the liver as substrates for the production of glucose, which is converted to glycogen and stored. The alarm signal of epinephrine stimulates liver glycogen breakdown, making glucose available as fuel to combat the acute stress.

The mechanism by which GCs exert these effects involves binding of the steroid to intracellular receptors, interaction of the steroid-receptor complex with GC response elements on DNA, transcription of genes, and synthesis of specific proteins (see Chapter 16, section III.C.2.). In some cases, the specific proteins responsible for the GC effect are known (e.g., the induction of phosphoenolpyruvate carboxykinase that stimulates gluconeogenesis). In other cases, the proteins responsible for the GC effect have not yet been identified.

E. Thyroid Hormone

1. BIOCHEMISTRY

The secretory products of the thyroid acinar cells are tetraiodothyronine (thyroxine, T_4) and triiodothyronine (T_3). Their structures are shown in Figure 43.9. The basic steps in the synthesis of T_3 and T_4 in these cells involve the transport or trapping of iodide from the blood into the thyroid acinar cell against an electrochemical gradient; the oxidation of iodide to form an iodinating species; the iodination of tyrosyl residues on the protein, thyroglobulin, to form iodotyrosines; and the coupling of residues of monoiodo- and diiodotyrosine in thyroglobulin to form residues of T_3 and T_4 (Fig. 43.10). Proteolytic cleavage of thyroglobulin releases free T_3 and T_4 . The steps in thyroid hormone synthesis are stimulated by thyroid-stimulating hormone (TSH), a glycoprotein produced by the anterior pituitary.

Iodide transport from the blood into the thyroid acinar cell is accomplished through an energy-requiring, iodide-trapping mechanism that is poorly defined but may involve the Na^+, K^+ -ATPase coupled to a cotransporter for Na^+ and iodide in the plasma membrane of the acinar cell.



The “central” deposition of fat in patients, such as **Corti Solemia**, with Cushing’s “disease” or syndrome is not readily explained because GCs actually cause lipolysis in adipose tissue. The increased appetite caused by an excess of GC and the lipogenic effects of the hyperinsulinemia that accompanies the GC-induced chronic increase in blood glucose levels have been suggested as possible causes. Why the fat is deposited centrally under these circumstances, however, is not understood. This central deposition leads to the development of a large fat pad at the center of the upper back (“buffalo hump”), to accumulation of fat in the cheeks and jowls (“moon facies”) and neck area, as well as a marked increase in abdominal fat. Simultaneously, there is a loss of adipose and muscle tissue below the elbows and knees, exaggerating the appearance of “central obesity” in Cushing’s “disease” or syndrome.



Approximately 35% of T_4 is deiodinated at the 5' position to form T_3 , and 43% is deiodinated at the 3 position to form the inactive “reverse” T_3 . Further deiodination or oxidative deamination leads to formation of compounds that have no biologic activity.



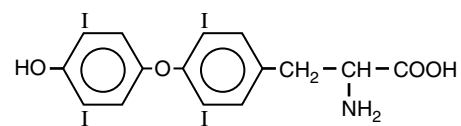
Otto was now able to explain the mechanism for most of **Corti Solemia’s** signs and symptoms. For example, Otto knew the metabolic explanation for the patient’s hyperglycemia. Some of Mr. Solemia’s muscle wasting and weakness were caused by the catabolic effect of hypercortisolemia on protein stores, such as those in skeletal muscle, to provide amino acids as precursors for gluconeogenesis. This catabolic action also resulted in the degradation of elastin, a major supportive protein of the skin, as well as an increased fragility of the walls of the capillaries of the cutaneous tissues. These changes resulted in the easy bruisability and the torn subcutaneous tissues of the lower abdomen, which resulted in red striae or stripes. The plethora (redness) of Mr. Solemia’s facial skin was also caused in part by the thinning of the skin as well as by a cortisol-induced increase in the bone marrow production of red blood cells, which enhanced the “redness” of the subcutaneous tissues.



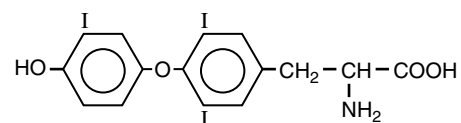
If **Corti Solemia’s** problem had been caused by a neoplasm of the adrenal cortex, what would his levels of blood ACTH and cortisol have been?



Normally, the thyroid gland secretes 80–100 μg T_4 and approximately 5 μg T_3 per day. The additional 22–25 μg T_3 “produced” daily is the result of the deiodination of the 5'-carbon of T_4 in peripheral tissues. T_3 is believed to be the predominant biologically active form of thyroid hormone in the body.



3,5,3',5'-Tetraiodothyronine (T_4)



3,5,3'-Triiodothyronine (T_3)

Fig. 43.9. Thyroid hormones, T_3 and T_4 .

A: If **Corti Solemia's** problem had resulted from primary hypersecretion of cortisol by a neoplasm of the adrenal cortex, his blood cortisol levels would have been elevated. The cortisol would have acted on the CRH-producing cells of the hypothalamus and the ACTH-secreting cells of the anterior pituitary by a negative feedback mechanism to decrease ACTH levels in the blood.

Because his cortisol and ACTH levels were both high, Mr. Solemia's tumor was most likely in the pituitary gland or possibly in neoplastic extrapituitary tissue that was secreting ACTH "ectopically." (Ectopic means that the tumor or neoplasm is producing and secreting a substance that is not ordinarily made or secreted by the tissue from which the tumor developed.) Mr. Solemia's tumor was in the anterior pituitary, not in an extrapituitary ACTH-producing site.

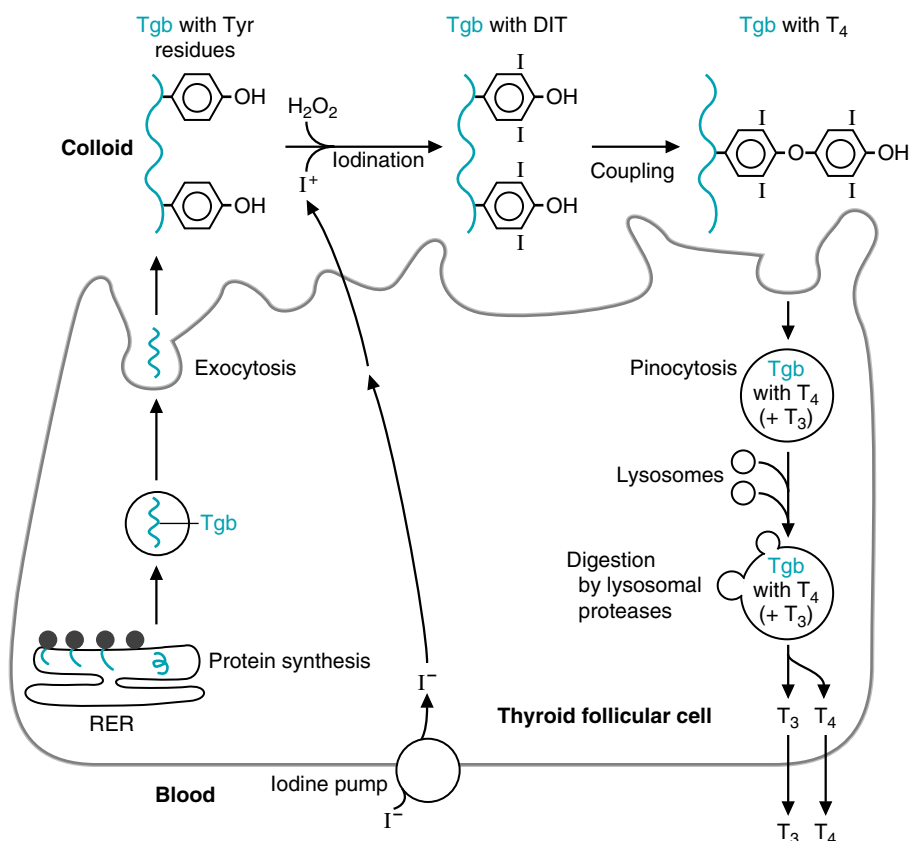


Fig. 43.10. Synthesis of the thyroid hormones (T_3 and T_4). The protein thyroglobulin (Tgb) is synthesized in thyroid follicular cells and secreted into the colloid. Iodination and coupling of tyrosine residues in Tgb produce T_3 and T_4 residues, which are released from Tgb by pinocytosis (endocytosis) and lysosomal action. The coupling of a monoiodotyrosine with a diiodotyrosine (DIT) to form triiodothyronine (T_3) is not depicted here.

The iodide concentrating or trapping process present in the plasma membrane of thyroid acinar cells creates iodide levels within the thyroid cell that are several hundredfold greater than those in the blood, depending on the current size of the total body iodide pool and the present need for new hormone synthesis.

In areas of the world in which the soil is deficient in iodide, hypothyroidism is prevalent. The thyroid gland enlarges (forms a goiter) in an attempt to produce more thyroid hormone. In the United States, table salt (NaCl) enriched with iodide (iodized salt) is used to prevent hypothyroidism caused by iodine deficiency.

The thyroid gland is unique in that it has the capacity to store large amounts of hormone as amino acid residues in thyroglobulin within its colloid space. This storage accounts for the low overall turnover rate of T_3 and T_4 in the body.

The rate of iodide transport is influenced by the absolute concentration of iodide within the thyroid cell. An internal autoregulatory mechanism decreases transport of iodide into the cell when the intracellular iodide concentration exceeds a certain threshold and increases transport when intracellular iodide falls below this threshold level.

The oxidation of intracellular iodide is catalyzed by thyroid peroxidase (located at the apical border of the thyroid acinar cell) in what may be a two-electron oxidation step forming I^+ (iodinium ion). Iodinium ion may react with a tyrosine residue in the protein thyroglobulin to form a tyrosine quinoid and then a 3'-monoiodotyrosine (MIT) residue. It has been suggested that a second iodide is added to the ring by similar mechanisms to form a 3,5-diiodotyrosine (DIT) residue. Because iodide is added to these organic compounds, iodination is also referred to as the "organification of iodide."

The biosynthesis of thyroid hormone proceeds with the coupling of an MIT and a DIT residue to form a triiodothyronine (T_3) residue or of two DIT residues to form a tetraiodothyronine (T_4) residue. T_3 and T_4 are stored in the thyroid follicle as amino acid residues in thyroglobulin. Under most circumstances, the T_4/T_3 ratio in thyroglobulin is approximately 13:1.

The plasma half-life of T_4 is approximately 7 days, and that of T_3 is 1 to 1.5 days. These relatively long plasma half-lives result from binding of T_3 and T_4 to several transport proteins in the blood. Of these transport proteins, thyroid-binding globulin (TBG) has the highest affinity for these hormones and carries approximately 70% of

bound T_3 and T_4 . Only 0.03% of total T_4 and 0.3% of total T_3 in the blood are in the unbound state. This free fraction of hormone has biologic activity because it is the only form that is capable of diffusing across target cell membranes to interact with intracellular receptors. The transport proteins, therefore, serve as a large reservoir of hormone that can release additional free hormone as the metabolic need arises.

The thyroid hormones are degraded in liver, kidney, muscle, and other tissues by deiodination, which produces compounds with no biologic activity.

2. SECRETION OF THYROID HORMONE

The release of T_3 and T_4 from thyroglobulin is controlled by thyroid-stimulating hormone (TSH) from the anterior pituitary. TSH stimulates the endocytosis of thyroglobulin to form endocytic vesicles within the thyroid acinar cells (see Fig. 43.10). Lysosomes fuse with these vesicles, and lysosomal proteases hydrolyze thyroglobulin, releasing free T_4 and T_3 into the blood in a 10:1 ratio. In various tissues, T_4 is deiodinated, forming T_3 , which is the active form of the hormone.

TSH is synthesized in the thyrotropic cells of the anterior pituitary. Its secretion is primarily regulated by a balance between the stimulatory action of hypothalamic thyroid-releasing hormone (TRH) and the inhibitory (negative feedback) influence of thyroid hormone (primarily T_3) at levels above a critical threshold in the blood bathing the pituitary thyrotrophs. TSH secretion occurs in a circadian pattern, a surge beginning late in the afternoon and peaking before the onset of sleep. In addition, TSH is secreted in a pulsatile fashion with intervals of 2 to 6 hours between peaks.

TSH stimulates all phases of thyroid hormone synthesis by the thyroid gland, including iodide trapping from the plasma, organification of iodide, coupling of monoiodotyrosine and diiodotyrosine, endocytosis of thyroglobulin, and proteolysis of thyroglobulin to release triiodothyronine (T_3) and tetraiodothyronine (T_4) (see Fig. 43.10). In addition, the vascularity of the thyroid gland increases as TSH stimulates hypertrophy and hyperplasia of the thyroid acinar cells.

The predominant mechanism of action of TSH is mediated by binding of TSH to its specific receptor on the plasma membrane of the thyroid acinar cell, leading to an increase in the concentration of cytosolic cAMP. Recent evidence indicates, however, that TSH also increases the cellular levels of inositol trisphosphate and diacylglycerol, causing a rise in cytosolic Ca^{2+} within the thyroid cell.

The large protein thyroglobulin, which contains T_3 and T_4 in peptide linkage, is stored extracellularly in the colloid that fills the central space of each thyroid follicle. Each of the biochemical reactions that leads to the release and eventual secretion of T_3 and T_4 , such as those that lead to their formation in thyroglobulins, is TSH-dependent. Rising levels of serum TSH stimulate the endocytosis of stored thyroglobulin into the thyroid acinar cell. Lysosomal enzymes then cleave T_3 and T_4 from thyroglobulin. T_3 and T_4 are secreted into the bloodstream in response to rising levels of TSH.

As the free T_3 level in the blood bathing the thyrotrophs of the anterior pituitary gland rises, the feedback loop is closed. Secretion of TSH is inhibited until the free T_3 levels in the systemic circulation fall just below a critical level, which once again signals the release of TSH. This feedback mechanism ensures an uninterrupted supply of biologically active free T_3 in the blood (Fig. 43.11). High levels of T_3 also inhibit the release of TRH from the hypothalamus.

3. PHYSIOLOGIC EFFECTS OF THYROID HORMONE

Only those physiologic actions of thyroid hormone that influence fuel metabolism are considered here. It is important to stress the term *physiologic*, because the effects of supraphysiologic concentrations of thyroid hormone on fuel metabolism may not be simple extensions of their physiologic effects. In general, the following

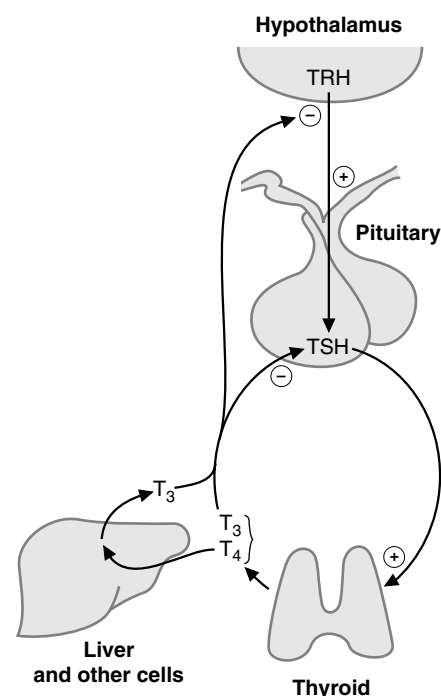


Fig. 43.11. Feedback regulation of thyroid hormone levels. TRH from the hypothalamus stimulates the release of TSH from the anterior pituitary, which stimulates the release of T_3 and T_4 from the thyroid. T_4 is converted to T_3 in the liver and other cells. T_3 and T_4 inhibit the release of TSH from the anterior pituitary and of TRH from the hypothalamus.



A patient presents with the following clinical and laboratory profile: the serum free and total T_3 and T_4 and the serum TSH levels are elevated, but the patient has symptoms of mild hypothyroidism, including a diffuse, palpable goiter. What single abnormality in the pituitary-thyroid-thyroid hormone target cell axis would explain all of these findings?



A generalized (i.e., involving all of the target cells for thyroid hormone in the body), but incomplete resistance of cells to the actions of thyroid hormone could explain the profile of the patient. In Refetoff's disorder, a mutation in the portion of the gene that encodes the ligand binding domain of the β -subunit of the thyroid hormone receptor protein causes a relative resistance to the suppressive action of thyroid hormone on the secretion of TSH by the thyrotrophs of the anterior pituitary gland. Therefore, the gland releases more TSH than normal into the blood. The elevated level of TSH causes an enlargement of the thyroid gland (goiter) as well as an increase in the secretion of thyroid hormone into the blood. As a result, the serum levels of both T_3 and T_4 rise in the blood. The increase in the secretion of thyroid hormone may or may not be adequate to fully compensate for the relative resistance of the peripheral tissues to thyroid hormone. If the compensatory increase in the secretion of thyroid hormone is inadequate, the patient may develop the signs and symptoms of hypothyroidism.



When present in excess, T_3 has severe catabolic effects that increase the flow of amino acids from muscle into the blood and eventually to the liver.

comments apply to the effects of thyroid hormone on energy metabolism in individuals with normal thyroid hormone levels in the blood.

i. Effects of Thyroid Hormone on The Liver

Several of the actions of thyroid hormone affect carbohydrate and lipid metabolism in the liver. Thyroid hormone increases glycolysis and cholesterol synthesis and increases the conversion of cholesterol to bile salts. Through its action of increasing the sensitivity of the hepatocyte to the gluconeogenic and glycogenolytic actions of epinephrine, T_3 indirectly increases hepatic glucose production (permissive or facilitatory action). Because of its ability to sensitize the adipocyte to the lipolytic action of epinephrine, T_3 increases the flow of fatty acids to the liver and thereby indirectly increases hepatic triacylglycerol synthesis. The concurrent increase in the flow of glycerol to the liver (as a result of increased lipolysis) further enhances hepatic gluconeogenesis.

ii. Effects of Thyroid Hormone on The Adipocyte

T_3 has an amplifying or facilitatory effect on the lipolytic action of epinephrine on the fat cell. Yet thyroid hormone has a bipolar effect on lipid storage, because it increases the availability of glucose to the fat cell, which serves as a precursor for fatty acid and glycerol 3-phosphate synthesis. The major determinant of the rate of lipogenesis, however, is not T_3 , but rather the amount of glucose and insulin available to the adipocyte for triacylglycerol synthesis.

iii. Effects of Thyroid Hormone on Muscle

In physiologic concentrations, T_3 increases glucose uptake by muscle cells. It also stimulates protein synthesis, and, therefore, growth of muscle, through its stimulatory actions on gene expression.

In physiologic concentrations, thyroid hormone sensitizes the muscle cell to the glycogenolytic actions of epinephrine. Glycolysis in muscle is increased by this action of T_3 .

iv. Effects of Thyroid Hormone on The Pancreas

Thyroid hormone increases the sensitivity of the β cells of the pancreas to those stimuli that normally promote insulin release and is required for optimal insulin secretion.

4. CALORIGENIC EFFECTS OF THYROID HORMONE

The oxidation of fuels converts approximately 25% of the potential energy present in the foods ingested by humans to ATP. This relative inefficiency of the human "engine" leads to the production of heat as a consequence of fuel utilization. This inefficiency, in part, allows homeothermic animals to maintain a constant body temperature in spite of rapidly changing environmental conditions. The acute response to cold exposure is shivering, which is probably secondary to increased activity of the sympathetic nervous system in response to this "stressful" stimulus.

Thyroid hormone participates in this acute response by sensitizing the sympathetic nervous system to the stimulatory effect of cold exposure. The ability of T_3 to increase heat production is related to its effects on the pathways of fuel oxidation, which both generate ATP and release energy as heat. The effects of T_3 on the sympathetic nervous system increase the release of norepinephrine. Norepinephrine stimulates the uncoupling protein thermogenin in brown adipose tissue (BAT), resulting in increased heat production from the uncoupling of oxidative phosphory-

lation (see Chapter 21). Very little residual brown fat persists in normal adult human beings, however.

Norepinephrine also increases the permeability of BAT and skeletal muscle to sodium. Because an increase of intracellular Na^+ is potentially toxic to cells, Na^+, K^+ -ATPase is stimulated to transport Na^+ out of the cell in exchange for K^+ . The increased hydrolysis of ATP by Na^+, K^+ -ATPase stimulates the oxidation of fuels and the regeneration of more ATP and heat from oxidative phosphorylation. Over a longer time course, thyroid hormone also increases the level of Na^+, K^+ -ATPase and many of the enzymes of fuel oxidation. Because even at normal room temperature ATP utilization by Na^+, K^+ -ATPase accounts for 20% or more of our basal metabolic rate (BMR), changes in its activity can cause relatively large increases in heat production.

Thyroid hormone also may increase heat production by stimulating ATP utilization in futile cycles (in which reversible ATP-consuming conversions of substrate to product and back to substrate use fuels and, therefore, produce heat).

F. Gastrointestinal-Derived Hormones Affecting Fuel Metabolism

In addition to insulin and the counterregulatory hormones discussed, a variety of peptides synthesized in the endocrine cells of the pancreatic islets, or the cells of the enteric nervous system, or the endocrine cells of the stomach, small bowel and large bowel, as well as certain cells of the central and peripheral nervous system, influence fuel metabolism directly. Some of these peptides and their tissue of origin, their actions on fuel metabolism, and the factors that stimulate (or suppress) their secretion are listed in Table 43.4. In addition to these peptides, others such as gastrin, motilin, pancreatic polypeptide (PP), peptide YY (PYY), and secretin may also influence fuel metabolism but by indirect effects on the synthesis or secretion of insulin or the counterregulatory hormones (Table 43.5). For example, gastrin induces gastric acid secretion, which ultimately affects nutrient absorption and metabolism. Motilin, secreted by enteroendocrine M cells of the proximal small bowel, stimulates gastric and pancreatic enzyme secretion, which, in turn, influences nutrient digestion. Pancreatic polypeptide (PP) from the pancreatic islets reduces gastric emptying and slows upper intestinal motility. Peptide YY (PYY) from the alpha cells in the mature pancreatic islets inhibits gastric acid secretion. Finally, secretin, produced by the enteroendocrine S cells in the proximal small bowel, regulates pancreatic enzyme secretion and inhibits gastrin release and gastric acid secretion. Although not directly influencing fuel metabolism, these “gut” hormones have a significant impact on how ingested nutrients are digested and prepared for absorption. If digestion or absorption of fuels is altered through a disturbance in the delicate interplay of all of the peptides, fuel metabolism will be altered as well.

Several of these gastrointestinal peptides such as GLP-1 and GIP do not act as direct insulin secretagogues when blood glucose levels are normal but do so after a meal large enough to cause an increase in the blood glucose concentration.

The release of these peptides may explain why the modest postprandial increase in serum glucose seen in normal subjects has a relatively robust stimulatory effect on insulin release, whereas a similar glucose concentration *in vitro* elicits a significantly smaller increase in insulin secretion. Likewise, this effect (certain factors potentiating insulin release), known as the “incretin effect,” could account for the greater beta cell response seen after an oral glucose load as opposed to that seen after the administration of glucose intravenously.



In hypothyroid patients, insulin release may be suboptimal, although glucose intolerance on this basis alone is uncommon.

In hyperthyroidism, the degradation and the clearance of insulin are increased. These effects, plus the increased demand for insulin caused by the changes in glucose metabolism, may lead to varying degrees of glucose intolerance in these patients (a condition called metathyroid diabetes mellitus). The patient with uncomplicated hyperthyroidism, however, rarely develops significant diabetes mellitus.

Table 43.4. Gastrointestinal-Derived Hormones Directly Affecting Fuel Metabolism

Hormone	Primary Cell/ Tissue of Origin	Actions	Secretory Stimuli (and Inhibitors)
Amylin	Pancreatic beta cell, endocrine cells of stomach and small intestine	1. Inhibits arginine-stimulated and postprandial glucagon secretion 2. Inhibits insulin secretion	Co-secreted with insulin in response to oral nutrients
Calcitonin gene-related peptide (CGRP)	Enteric neurons and enteroendocrine cells of the rectum	Inhibits insulin secretion	Oral glucose intake and gastric acid secretion
Galanin	Nervous system, pituitary, neurons of gut, pancreas, thyroid, and adrenal gland	Inhibits secretion of insulin, somatostatin, enteroglucagon, pancreatic polypeptide, and others	Intestinal distension
Gastric inhibitory polypeptide/ glucose-dependent insulinotropic polypeptide (GIP)	Neuroendocrine K cells of duodenum and proximal jejunum	1. Increases insulin release via an "incretin" effect 2. Regulates glucose and lipid metabolism	Oral nutrient ingestion, especially long-chain fatty acids
Gastrin-releasing peptide (GRP)	Enteric nervous system and pancreas	Stimulates release of cholecystokinin; GIP, gastrin, glucagon, GLP-1, GLP-2, and somatostatin	
Ghrelin	Central nervous system, stomach, small intestine, and colon	Stimulates growth hormone release	Fasting
Glucagon	Pancreatic alpha cell, central nervous system	Primary counter-regulatory hormone that restores glucose levels in hypoglycemic state (increases glycogenolysis and gluconeogenesis as well as protein-lipid flux in liver and muscle)	Neural and humoral factors released in response to hypoglycemia
Glucagon-like peptide-1 (GLP-1)	Enteroendocrine L cells in ileum, colon, and central nervous system	1. Enhances glucose disposal after meals by inhibiting glucagon secretion and stimulating insulin secretion 2. Acts through second messengers in beta cells to increase sensitivity of these cells to glucose (an incretin)	1. Oral nutrient ingestion 2. Vagus nerve 3. GRP and GIP 4. Somatostatin inhibits secretion
Glucagon-like peptide-2 (GLP-2)	The same as for GLP-1	Stimulates intestinal hexose transport	Same as GLP-1
Neuropeptide Y	Central and peripheral nervous system, pancreatic islet cells	Inhibits glucose-stimulated insulin secretion	Oral nutrient ingestion and activation of sympathetic nervous system
Neurotensin (NT)	Small intestinal N cells (especially ileum), enteric nervous system, adrenal gland, pancreas	In brain, modulates dopamine neurotransmission and anterior pituitary secretions	1. Luminal lipid nutrients 2. GRP 3. Somatostatin inhibits secretion
Pituitary adenylate cyclase activating peptide (PACAP)	Brain, lung, and enteric nervous system	Stimulates insulin and catecholamine release	Activation of central nervous system
Somatostatin	Central nervous system, pancreatic delta cells, and enteroendocrine delta cells	1. Inhibits secretion of insulin, glucagon and PP (islets), and gastrin, secretin, GLP-1, and GLP-2 (in gut) 2. Reduces carbohydrate absorption from gut lumen	1. Luminal nutrients 2. GLP-1 3. GIP 4. PACAP 5. VIP 6. Beta-adrenergic stimulation
Vasoactive intestinal peptide (VIP)	Widely expressed in the central and peripheral nervous systems	May regulate release of insulin and pancreatic glucagon	1. Mechanical stimulation of gut 2. Activation of central and peripheral nervous systems

Table 43.5. Gastrointestinal-Derived Hormones Indirectly Affecting Fuel Metabolism

Hormone	Primary Cell/ Tissue of Origin	Actions	Secretory Stimuli (and Inhibitors)
Cholecystokinin (CCK)	Enteroendocrine I cells, enteric nerves, others	1. Inhibits proximal gastric motility 2. Increases antral and pyloric contractions 3. Regulates nutrient-stimulated enzyme secretion and gallbladder contraction 4. Increases postprandial satiety	1. Oral nutrient ingestion 2. GGRP and bombesin from gut
Gastrin	Enteroendocrine G cells of the stomach, duodenal bulb, and other cells	Induces gastric acid secretion	1. Luminal contents, especially aromatic amino acids, calcium, coffee, and ethanol 2. Vagus nerve stimulation; activation of beta-adrenergic and GABA neurons 3. Somatostatin inhibits secretion
Motilin	Enteroendocrine M cells in upper small bowel and other cells	1. Induces phase III contractions in stomach 2. Stimulates gastric secretion and pancreatic enzyme secretion 3. Induces gallbladder contraction	4. Duodenal alkalization 5. Gastric distension 6. Secretion suppressed by nutrients in duodenum
Pancreatic polypeptide (PP)	Endocrine cells in periphery of islets in the head of the pancreas	1. Reduces CCK-mediated gastric acid secretion 2. Increases intestinal transit time (slows motility)	Stimulated by intraluminal nutrients, hypoglycemia, and vagal nerve stimulation
Peptide YY (PYY)	Enteroendocrine cells, developing pancreas; alpha cells in mature islets	1. Inhibits both gastric acid secretion and gastric motility 2. Slows intestinal motility 3. Inhibits pancreatic exocrine secretion	1. Oral nutrient ingestion 2. Bile acids and fatty acids 3. Amino acids in colon
Secretin	Enteroendocrine S cells in upper small bowel	1. Stimulates pancreatic and biliary bicarbonate and water secretion 2. Regulates pancreatic enzyme secretion 3. Inhibits postprandial gastric emptying, gastrin release, and gastric acid secretion	1. Gastric acid, bile salts, fatty acids, peptides, and ethanol 2. Somatostatin inhibits secretion
Tachykinins	Neurons localized in the submucous and myenteric plexuses; enterochromaffin cells in gut epithelium	1. Regulates vasomotor and gastrointestinal smooth muscle contraction 2. Mucus secretion and water absorption	Direct and indirect activation of neurons in submucosa and myenteric plexuses in gut epithelium
Thyrotropin-releasing hormone (TRH)	Enteric nervous system, colon, G cells of stomach, and pancreatic beta cell	1. Suppresses hormone-stimulated gastric acid secretion 2. Inhibits cholesterol synthesis within the intestinal mucosa	In the stomach, histamine and serotonin stimulate secretion

G. Neural Factors Controlling Insulin and Counter-regulatory Hormone Secretion

Although beyond the scope of this text, the gastrointestinal neuroendocrine system is briefly described with regard to its effects on fuel metabolism. The pancreatic islet cells are innervated by both the adrenergic and the cholinergic limbs of the autonomic nervous system. Although stimulation of both the sympathetic and the parasympathetic systems increases glucagon secretion, insulin secretion is increased by vagus nerve fibers and suppressed by sympathetic fibers via the alpha-adrenoreceptors. Evidence also suggests that the sympathetic nervous system indirectly regulates pancreatic beta cell function through stimulation or suppression of the secretion of somatostatin, beta₂-adrenergic receptor number, and the neuropeptides, neuropeptide Y and galanin.

A tightly controlled interaction between the hormonal and neural factors that control nutrient metabolism is necessary to maintain normal fuel and, hence, energy homeostasis.



To establish the diagnosis of a secretory tumor of an endocrine gland, one must first demonstrate that basal serum levels of the hormone in question are regularly elevated. More importantly, one must show that the hypersecretion of the hormone (and, hence, its elevated level in the peripheral blood) cannot be adequately inhibited by “maneuvers” that are known to suppress secretion from a normally functioning gland (i.e., one must show that the hypersecretion is “autonomous”).

To ensure that both the basal and the postsuppression levels of the specific hormone to be tested will reflect the true secretory rate of the suspected endocrine tumor, all of the known factors that can stimulate the synthesis of the hormone must be eliminated. For GH, for example, the secretagogues (stimulants to secretion) include nutritional factors; the patient’s level of activity, consciousness, and stress; and certain drugs. GH secretion is stimulated by a high-protein meal or by a low level of fatty acids or of glucose in the blood. Vigorous exercise, stage III–IV sleep, psychological and physical stress, and levodopa, clonidine, and estrogens also increase GH release.

The suppression test used to demonstrate the autonomous hypersecretion of GH involves giving the patient an oral glucose load and, subsequently, measuring GH levels. A sudden rise in blood glucose suppresses serum GH to 2 ng/mL or less in normal subjects, but not in patients with active acromegaly.

If one attempts to demonstrate autonomous hypersecretion of GH in a patient suspected of having acromegaly, therefore, before drawing the blood for both the basal (pre-glucose load) serum GH level and the post-glucose load serum GH level, one must be certain that the patient has not eaten for 6–8 hours, has not done vigorous exercise for at least 4 hours, remains fully awake during the entire testing period (in a non-stressed state to the extent possible), and has not taken any drugs known to increase GH secretion for at least 1 week.

Under these carefully controlled circumstances, if both the basal and postsuppression serum levels of the suspect hormone are elevated, one can conclude that autonomous hypersecretion is probably present. At this point, localization procedures (such as an MRI of the pituitary gland in an acromegalic suspect) are performed to further confirm the diagnosis.

CLINICAL COMMENTS



One of the functions of cortisol is to prepare the body to deal with periods of stress. In response to cortisol, the body re-sorts its fuel stores so that they can rapidly be made available for the “fight or flight” response to the alarm signal sounded by epinephrine. Cortisol causes gluconeogenic substrates to move from peripheral tissues to the liver, where they are converted to glucose and stored as glycogen. The release of epinephrine stimulates the breakdown of glycogen, increasing the supply of glucose to the blood. Thus fuel becomes available for muscle to fight or flee.

Cushing’s “disease,” the cause of **Corti Solemia’s** current problems, results from prolonged hypersecretion of ACTH from a benign pituitary tumor. ACTH stimulates the adrenal cortex to produce cortisol, and blood levels of this steroid hormone rise.

Other nonpituitary causes of Cushing’s syndrome, however, include a primary tumor of the adrenal cortex secreting excessive amounts of cortisol directly into the bloodstream. This disorder also can result from the release of ACTH from secretory nonendocrine nonpituitary neoplasms (“ectopic” ACTH syndrome). Cushing’s syndrome is often caused by excessive doses of synthetic GCs used to treat a variety of disorders because of their potent anti-inflammatory effects (iatrogenic Cushing’s syndrome).

The diabetogenic potential of chronically elevated GH levels in the blood is manifest by the significant incidence of diabetes mellitus (25%) and impaired glucose tolerance (33%) in patients with acromegaly, such as **Sam Atotrope**. Yet, under normal circumstances, physiologic concentrations of GH (as well as cortisol and thyroid hormone) have a facilitatory or permissive effect on the quantity of insulin released in response to hyperglycemia and other insulin secretagogues. This “proinsular” effect is probably intended to act as a “brake” to dampen any potentially excessive “contrain-sular” effects that increments in GH and the other counterregulatory hormones exert.

BIOCHEMICAL COMMENTS



Most hormones are present in body fluids in picomolar to nanomolar amounts, requiring highly sensitive assays to determine their concentration in the blood or urine. Radioimmunoassays (RIAs), developed in the 1960s, use an antibody, generated in animals, against a specific antigen (the hormone to be measured). Determining the concentration of the hormone in the sample involves incubating the plasma or urine sample with the antibody and then quantifying the level of antigen–antibody complex formed during the incubation by one of several techniques.

The classic RIA uses very high-affinity antibodies, which have been fixed (immobilized) on the inner surface of a test tube, a Teflon bead, or a magnetized particle. A standard curve is prepared, using a set amount of the antibody and various known concentrations of the unlabeled hormone to be measured. In addition to a known concentration of the unlabeled hormone, each tube contains the same small, carefully measured amount of radiolabeled hormone. The labeled hormone and the unlabeled hormone compete for binding to the antibody. The higher the amount of unlabeled hormone in the sample, the less radiolabeled hormone is bound. A standard curve is plotted (Fig. 43.12). The unknown sample from the patient’s blood or urine, containing the unlabeled hormone to be measured, is incubated with the immobilized antibody in the presence of the same small, carefully measured amount of radiolabeled hormone. The amount of radiolabeled hormone bound to the antibody is determined, and the standard curve is used to quantitate the amount of unlabeled hormone in the patient sample.

The same principle is used in immunoradiometric assays (IRMAs), but with this technique the antibody, rather than the antigen to be measured, is radiolabeled.

The sensitivity of RIAs can be enhanced using a “sandwich technique.” This method uses two different monoclonal antibodies (antibodies generated by a single clone of plasma cells rather than multiple clones), each of which recognizes a different specific portion of the hormone’s structure. The first antibody, attached to a solid support matrix such as a plastic culture dish, binds the hormone to be assayed. After exposure of the patient sample to this first antibody, the excess plasma is washed away, and the second antibody (which is radiolabeled) is then incubated with the first antibody–hormone complex. The amount of binding of the second (labeled) antibody to the first complex is proportional to the concentration of the hormone in the sample.

The sandwich technique can be improved even further if the second antibody is attached to an enzyme, such as alkaline phosphatase. The enzyme rapidly converts an added colorless substrate into a colored product, or a nonfluorescent substrate into a highly fluorescent product. These changes can be quantitated if the degree of change in color or fluorescence is proportional to the amount of hormone present in the patient sample. Less than a nanogram (10^{-9} g) of a protein can be measured by such an enzyme-linked immunosorbent assay (ELISA).

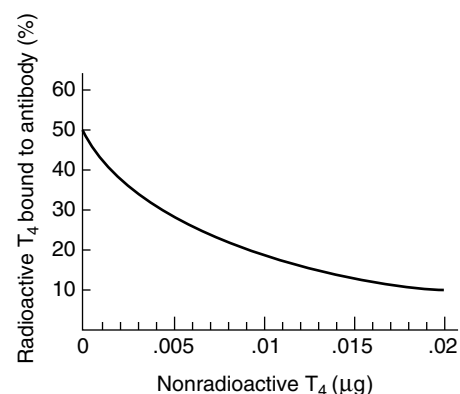


Fig. 43.12. Standard curve for a radioimmunoassay. A constant amount of radioactive T₄ is added to a series of tubes, each of which contains a different amount of nonradioactive T₄. The amount of radioactive hormone that binds to an antibody specific for the hormone is measured and plotted against the nonradioactive hormone concentration. When more nonradioactive T₄ is present in the tube, less radioactive T₄ binds to the antibody.

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REVIEW QUESTIONS—CHAPTER 43

- As a third-year medical student, you examine your first patient. You find that he is 52 years old, has a round face, acne, and a large hump of fat on the back of his neck. He complains that he is too weak to mow his lawn. His fasting blood glucose level is 170 mg/dL (reference range = 80–100 mg/dL). Plasma cortisol levels are 62 μ g/mL (reference range = 3–31 μ g/mL). Plasma ACTH levels are 0 pg/mL (reference range = 0–100 pg/mL).

Based on the information given above, if the patient’s problem is attributable to a single cause, the most likely diagnosis is which of the following?

- Non–insulin-dependent diabetes mellitus
- Insulin-dependent diabetes mellitus
- A secretory tumor of the anterior pituitary
- A secretory tumor of the posterior pituitary
- A secretory tumor of the adrenal cortex

2. A woman was scheduled for a growth hormone suppression test. If each of the following events occurred the morning of the test, which one of the events would be most likely to cause a decrease in growth hormone levels?
- (A) She ate four large doughnuts for breakfast.
 - (B) She was on estrogen replacement therapy and took her tablets after breakfast.
 - (C) While unlocking her car, she was chased by the neighbor's vicious dog.
 - (D) She fell asleep at the start of the test and slept soundly until it was completed 1.5 hours later.
 - (E) She forgot to eat her breakfast before the test.
3. A dietary deficiency of iodine would lead to which of the following?
- (A) A direct effect on the synthesis of thyroglobulin on ribosomes
 - (B) An increased secretion of thyroid stimulatory hormone (TSH)
 - (C) Decreased production of thyrotropin releasing hormone (TRH)
 - (D) Increased heat production
 - (E) Weight loss
4. A woman whose thyroid gland was surgically removed was treated with 0.10 mg thyroxine daily (tablet form). After 3 months of treatment, serial serum TSH levels ranged between 10 and 15 mIU/mL (reference range = 0.3 – 5.0 mIU/mL). She complained of fatigue, weight gain, and hoarseness. Her dose of thyroid hormone should be adjusted in which direction?
- (A) Increased
 - (B) Decreased
 - (C) Remain the same

The next question is based on the following scenario:

A patient complains of nervousness, palpitations, sweating, and weight loss without loss of appetite, and has a goiter. Suspecting a defect in thyroid function, the physician orders a total serum T_4 . The test is performed by radioimmunoassay. The standard curve for the assay, which measures T_4 in 0.1 mL serum, is shown in Fig. 43.12. Normal levels of T_4 = 4–10 $\mu\text{g/dL}$. In an assay of 0.1 mL of the patient's serum, 15% of the radioactive T_4 was bound by the antibody.

5. According to the radioimmunoassay, the approximate blood level of T_4 is which of the following?
- (A) 0.015 $\mu\text{g/dL}$
 - (B) 0.15 $\mu\text{g/dL}$
 - (C) 15 $\mu\text{g/dL}$
 - (D) 20 $\mu\text{g/dL}$
 - (E) 30 $\mu\text{g/dL}$

44 The Biochemistry of the Erythrocyte and other Blood Cells

The cells of the blood are classified as **erythrocytes**, **leukocytes**, or **thrombocytes**. The erythrocytes (**red cells**) carry oxygen to the tissues and are the most numerous cells in the blood. The leukocytes (**white cells**) are involved in defense against infection, and the thrombocytes (**platelets**) function in blood clotting. All of the cells in the blood can be generated from hematopoietic stem cells in the bone marrow on demand. For example, in response to infection, leukocytes secrete **cytokines** called **interleukins** that stimulate the production of additional leukocytes to fight the infection. Decreased supply of oxygen to the tissues signals the kidney to release **erythropoietin**, a hormone that stimulates the production of red cells.

The red cell has limited metabolic function, owing to its lack of internal organelles. **Glycolysis** is the main energy-generating pathway, with lactate production regenerating NAD^+ for glycolysis to continue. The NADH produced in glycolysis is also used to reduce the ferric form of hemoglobin, **methemoglobin**, to the normal ferrous state. Glycolysis also leads to a side pathway in which **2,3-bisphosphoglycerate** is produced, which is a major allosteric effector for oxygen binding to hemoglobin. The **hexose monophosphate shunt** pathway generates **NADPH** to protect red cell membrane lipids and proteins from oxidation, through regeneration of reduced glutathione. **Heme synthesis** occurs in the precursors of red cells and is a complex pathway that originates from succinyl-CoA and glycine. Mutations in any of the steps of heme synthesis lead to a group of diseases known collectively as **porphyrias**.

The red cell membrane must be highly deformable to allow it to travel throughout the capillary system in the body. This is because of a complex **cytoskeletal** structure that consists of the major proteins **spectrin**, **ankyrin**, and **band 3 protein**. Mutations in these proteins lead to improper formation of the membrane cytoskeleton, ultimately resulting in malformed red cells, **spherocytes**, in the circulation. Spherocytes have a shortened life span, leading to loss of blood cells.

When the body does not have sufficient red cells, the patient is said to be **anemic**. Anemia can result from many causes. **Nutritional deficiencies** of **iron**, **folate**, or **vitamin B12** prevent the formation of adequate numbers of red cells. **Mutations** in the genes that encode red cell metabolic **enzymes**, **membrane structural proteins**, and **globins** cause **hereditary anemias**. The appearance of red cells on a blood smear frequently provides clues to the cause of an anemia. Because the mutations that give rise to hereditary anemias also provide some protection against malaria, hereditary anemias are some of the most common genetic diseases known.

The human alters globin gene expression during development, a process known as **hemoglobin switching**. The switch between expression of one gene to another is regulated by transcription factor binding to the promoter regions of these genes. Current research is attempting to reactivate fetal hemoglobin genes to combat sickle-cell disease and thalassemia.



THE WAITING ROOM



Anne Niemick, who has β^+ thalassemia, complains of pain in her lower spine (see Chapters 14 and 15). A quantitative computed tomogram (CT) of the vertebral bodies of the lumbar spine shows evidence of an area of early spinal cord compression in the upper lumbar region. She is suffering from severe anemia, resulting in stimulation of production of red blood cell precursors (the erythroid mass) from the stem cells in her bone marrow. This expansion of marrow volume causes compression of tissues in this area, which, in turn, causes pain. Local irradiation is considered, as is a program of regular blood transfusions to maintain the oxygen-carrying capacity of circulating red blood cells. The results of special studies related to the genetic defect underlying her thalassemia are pending, although preliminary studies have shown that she has elevated levels of fetal hemoglobin, which, in part, moderates the manifestations of her disease. **Anne Niemick's** parents have returned to the clinic to discuss the results of these tests.



Spiro Site is a 21-year-old college student who complains of feeling tired all the time. Two years previously he had had gallstones removed, which consisted mostly of bilirubin. His spleen is palpable, and jaundice is evidenced by yellowing of the whites of his eyes. His hemoglobin was low (8 g/dL; reference value 13.5–17.5 gm/dL). A blood smear showed dark, rounded, abnormally small red cells called spherocytes as well as an increase in the number of circulating immature red blood cells known as reticulocytes.

I. CELLS OF THE BLOOD

The blood, together with the bone marrow, makes up the organ system that makes a significant contribution to achieving **homeostasis**, the maintenance of the normal composition of the body's internal environment. Blood can be considered a liquid tissue consisting of water, proteins, and specialized cells. The most abundant cell in the blood is the **erythrocyte** or red blood cell, which transports oxygen to the tissues and contributes to buffering of the blood through the binding of protons by hemoglobin (see section IV of this chapter, and the material in Chapter 4, section IV.D.2., and Chapter 7, section VII). Red blood cells lose all internal organelles during the process of differentiation. The white blood cells (**leukocytes**) are nucleated cells present in blood that function in the defense against infection. The platelets (**thrombocytes**), which contain cytoplasmic organelles but no nucleus, are involved in the control of bleeding by contributing to normal thrombus (clot) formation within the lumen of the blood vessel. The average concentration of these cells in the blood of normal individuals is presented in Table 44.1.

Table 44.1. Normal Values of Blood Cell Concentrations in Adults

Cell Type	Mean (cells/mm ³)
Erythrocytes	5.2×10^6 (men); 4.6×10^6 women
Neutrophils	4,300
Lymphocytes	2,700
Monocytes	500
Eosinophils	230
Basophils	40

A. Classification and Functions of Leukocytes and Thrombocytes

The leukocytes can be classified either as polymorphonuclear leukocytes (granulocytes) or mononuclear leukocytes, depending on the morphology of the nucleus in these cells. The mononuclear leukocyte has a rounded nucleus, whereas the polymorphonuclear leukocytes have a multilobed nucleus.

1. THE GRANULOCYTES

The granulocytes, so named because of the presence of secretory granules visible on staining, are the **neutrophils**, **eosinophils**, and **basophils**. When these cells are activated in response to chemical stimuli, the vesicle membranes fuse with the cell plasma membrane, resulting in the release of the granule contents (**degranulation**). The granules contain many cell-signaling molecules that mediate inflammatory processes. The granulocytes, in addition to displaying segmented nuclei (are polymorphonuclear), can be distinguished from each other by their staining properties (caused by different granular contents) in standard hematologic blood smears; neutrophils stain pink, eosinophils stain red, and basophils stain blue.

Neutrophils are phagocytic cells that rapidly migrate to areas of infection or tissue damage. As part of the response to acute infection, neutrophils engulf foreign bodies, and destroy them, in part, by initiating the respiratory burst (see Chapter 24). The respiratory burst creates oxygen radicals that rapidly destroy the foreign material found at the site of infection.

A primary function of eosinophils is to destroy parasites such as worms. The eosinophilic granules are lysosomes containing hydrolytic enzymes and cationic proteins, which are toxic to parasitic worms. Eosinophils have also been implicated in asthma and allergic responses, although their exact role in the development of these disorders is still unknown, and this is an active area of research.

Basophils, the least abundant of the leukocytes, participate in hypersensitivity reactions, such as allergic responses. Histamine, produced by the decarboxylation of histidine, is stored in the secretory granules of basophils. Release of histamine during basophil activation stimulates smooth muscle cell contraction and increases vascular permeability. The granules also contain enzymes such as proteases, β -glucuronidase, and lysophospholipase. These enzymes degrade microbial structures and assist in the remodeling of damaged tissue.

2. MONONUCLEAR LEUKOCYTES

The mononuclear leukocytes consist of various classes of lymphocytes and the monocytes. Lymphocytes are small, round cells originally identified in lymph fluid. These cells have a high ratio of nuclear volume to cytoplasmic volume and are the primary antigen (foreign body)-recognizing cells. There are three major types of lymphocytes: T cells, B cells, and NK cells. The precursors of T cells (thymus-derived lymphocytes) are produced in the bone marrow and then migrate to the thymus, where they mature before being released to the circulation. Several subclasses of T cells exist. These subclasses are identified by different surface membrane proteins, the presence of which correlate with the function of the subclass. Lymphocytes that mature in the bone marrow are the B cells, which secrete antibodies in response to antigen binding. The third class of lymphocytes are the natural killer cells (NK cells), which target virally infected and malignant cells for destruction.

Circulatory monocytes are the precursors of tissue macrophages. Macrophages (large eater) are phagocytic cells that enter inflammatory sites and consume microorganisms and necrotic host cell debris left behind by granulocyte attack of the foreign material. Macrophages in the spleen play an important role in maintaining

Table 44.2. Normal Hemoglobin Levels in Blood (g/dL)

Adult	
Males	13.5–17.5
Females	11.5–15.5
Children	
Newborns	15.0–21.0
3–12 mo.	9.5–12.5
1 yr to puberty	11.0–13.5



Other measurements used to classify the type of anemia present include the mean corpuscular volume (MCV) and the mean corpuscular hemoglobin concentration (MCHC). The MCV is the average volume of the red blood cell, expressed in femto (10^{-15}) liters. Normal MCV range from 80 to 100 fL. The MCHC is the average concentration of hemoglobin in each individual erythrocyte, expressed in g/L. The normal range is 32 to 37; a value of less than 32 would indicate hypochromic cells. Thus, microcytic, hypochromic red blood cells have an MCV of less than 80 and an MCHC of less than 32. Macrocytic, normochromic cells have an MCV of greater than 100, with an MCHC between 32 and 37.



The trace amounts of 2,3 BPG found in cells other than erythrocytes is required for the phosphoglycerate mutase reaction of glycolysis, in which 3-phosphoglycerate is isomerized to 2-phosphoglycerate. As the 2,3 BPG is regenerated during each reaction cycle, it is only required in catalytic amounts.

the oxygen-delivering capabilities of the blood by removing damaged red blood cells that have a reduced oxygen-carrying capacity.

3. THE THROMBOCYTES

Platelets are heavily granulated disc-like cells that aid in intravascular clotting. Like the erythrocyte, platelets lack a nucleus. Their function is discussed in the following chapter. Platelets arise by budding of the cytoplasm of megakaryocytes, multi-nucleated cells that reside in the bone marrow.

B. Anemia

The major function of erythrocytes is to deliver oxygen to the tissues. To do this, a sufficient concentration of hemoglobin in the red blood cells is necessary for efficient oxygen delivery to occur. When the hemoglobin concentration falls below normal values (Table 44.2), the patient is classified as anemic. Anemias can be categorized based on red cell size and hemoglobin concentration. Red cells can be of normal size (normocytic), small (microcytic), or large (macrocytic). Cells containing a normal hemoglobin concentration are termed normochromic; those with decreased concentration are hypochromic. This classification system provides important diagnostic tools (Table 44.3) that enable one to properly classify, diagnose, and treat the anemia.

II. ERYTHROCYTE METABOLISM

A. The Mature Erythrocyte

To best understand how the erythrocyte can carry out its major function, a discussion of erythrocyte metabolism is required. Mature erythrocytes contain no intracellular organelles, so the metabolic enzymes of the red blood cell are limited to those found in the cytoplasm. In addition to hemoglobin, the cytosol of the red blood cell contains enzymes necessary for the prevention and repair of damage done by reactive oxygen species (see Chapter 24) and the generation of energy (Fig. 44.1). Erythrocytes can only generate adenosine triphosphate (ATP) by glycolysis (see Chapter 22). The ATP is used for ion transport across the cell membrane (primarily Na^+ , K^+ , and Ca^{+2}), the phosphorylation of membrane proteins, and the priming reactions of glycolysis. Erythrocyte glycolysis also uses the Rapaport-Luebering shunt to generate 2,3-bisphosphoglycerate (2,3-BPG). Red cells contain 4 to 5 mM 2,3-BPG, compared with trace amounts in other cells. As discussed in more detail in Section IV, 2,3-BPG is a modulator of oxygen binding to hemoglobin that stabilizes the deoxy form of hemoglobin, thereby facilitating the release of oxygen to the tissues.

To bind oxygen, the iron of hemoglobin must be in the ferrous (+2) state. Reactive oxygen species can oxidize the iron to the ferric (+3) state, producing

Table 44.3. Classification of the Anemias on the Basis of Red Cell Morphology

Red Cell Morphology	Functional Deficit	Possible Causes
Microcytic, hypochromic	Impaired hemoglobin synthesis	Iron deficiency, thalassemia mutation, lead poisoning
Macrocytic, normochromic	Impaired DNA synthesis	B12 or folic acid deficiency, erythroleukemia
Normocytic, normochromic	Red cell loss	Acute bleeding, sickle cell disease, red cell metabolic defects, red cell membrane defects

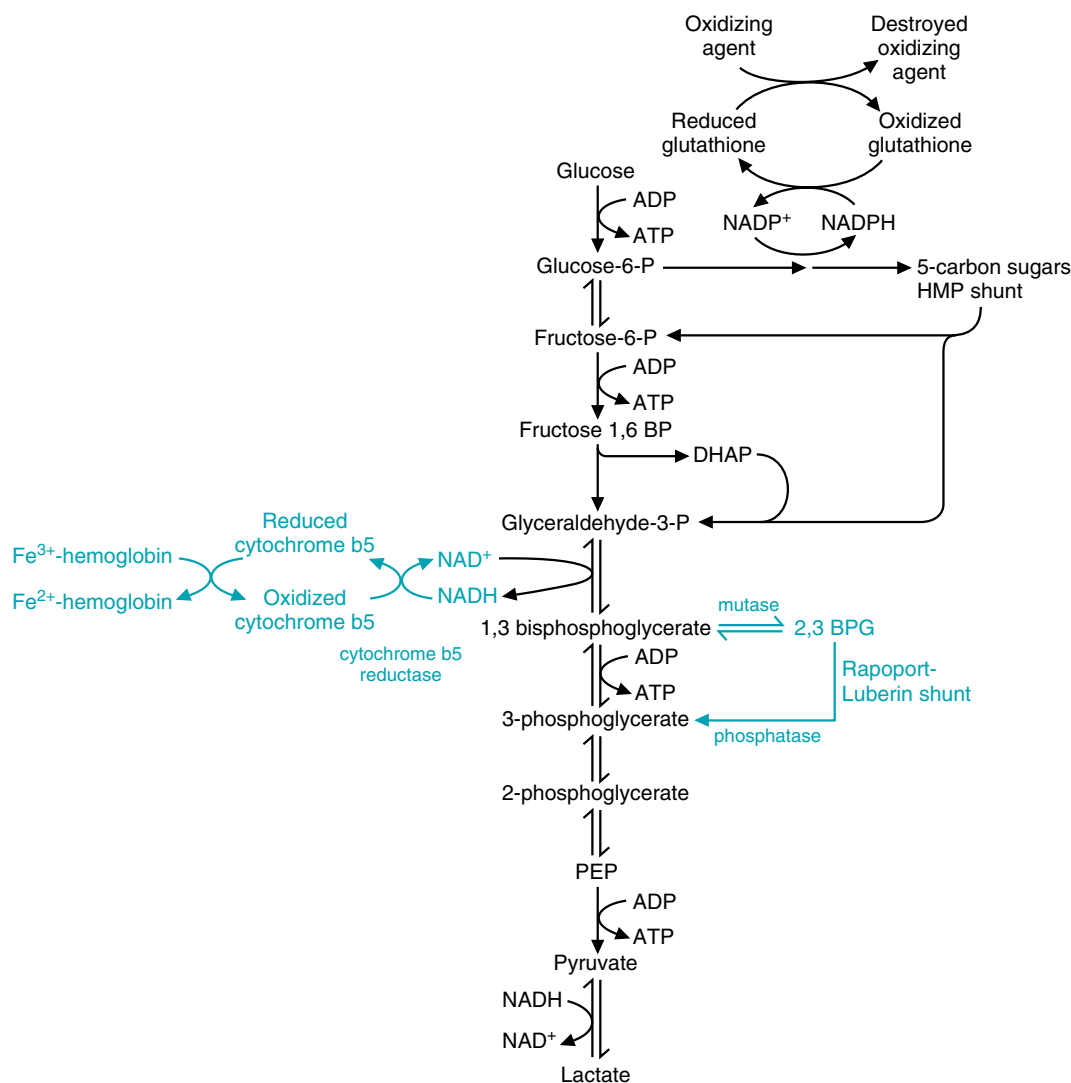


Fig. 44.1. Overview of erythrocyte metabolism. Glycolysis is the major pathway, with branches for the hexose monophosphate shunt (for protection against oxidizing agents) and the Rapoport-Luebering shunt (which generates 2,3 bisphosphoglycerate, which moderates oxygen binding to hemoglobin). The NADH generated from glycolysis can be used to reduce methemoglobin (Fe^{+3}) to normal hemoglobin (Fe^{+2}), or to convert pyruvate to lactate, such that NAD^{+} can be regenerated and used for glycolysis. Pathways unique to the erythrocyte are indicated in blue.

methemoglobin. Some of the NADH produced by glycolysis is used to regenerate hemoglobin from methemoglobin by the NADH-cytochrome b_5 methemoglobin reductase system. Cytochrome b_5 reduces the Fe^{+3} of methemoglobin. The oxidized cytochrome b_5 is then reduced by a flavin-containing enzyme, cytochrome b_5 reductase (also called methemoglobin reductase), using NADH as the reducing agent.



An inherited deficiency in pyruvate kinase leads to hemolytic anemia (an anemia caused by the destruction of red blood cells; hemoglobin values typically drop to 4 to 10 g/dL in this condition). Because the amount of ATP formed from glycolysis is decreased by 50%, red blood cell ion transporters cannot function effectively. The red blood cells tend to gain Ca^{2+} and lose K^{+} and water. The water loss increases the intracellular hemoglobin concentration. With the increase in intracellular hemoglobin concentration, the internal viscosity of the cell is increased to the point that the cell becomes rigid and, therefore, more susceptible to damage by shear forces in the circulation. Once damaged, the red blood cells are removed from circulation, leading to the anemia. However, the effects of the anemia are frequently moderated by the twofold to threefold elevation in 2,3-BPG concentration that results from the blockage of the conversion of phosphoenol pyruvate to pyruvate. Because 2,3-BPG binding to hemoglobin decreases the affinity of hemoglobin of oxygen, the red blood cells that remain in circulation are highly efficient in releasing their bound oxygen to the tissues.



Congenital methemoglobinemia, the presence of excess methemoglobin, is found in people with an enzymatic deficiency in cytochrome *b*₅ reductase or in people who have inherited hemoglobin M. In hemoglobin M, a single amino acid substitution in the heme-binding pocket stabilizes the ferric (Fe³⁺) oxygen. Individuals with congenital methemoglobinemia appear cyanotic but have few clinical problems. Methemoglobinemia can be acquired by ingestion of certain oxidants such as nitrites, quinones, aniline, and sulfonamides. Acquired methemoglobinemia can be treated by the administration of reducing agents, such as ascorbic acid or methylene blue.



G6PD deficiency is the most common enzyme deficiency known in humans, probably, in part, because individuals with G6PD deficiency are resistant to malaria. The resistance to malaria counterbalances the deleterious effects of the deficiency. G6PD-deficient red cells have a shorter life span and are more likely to lyse under conditions of oxidative stress. When soldiers during the Korean War were given the antimalarial drug primaquine prophylactically, approximately 10% of the soldiers of African ancestry developed a spontaneous anemia. Because the gene for G6PD is found on the X chromosome, these men had only one copy of a variant G6PD gene.

All known G6PD variant genes contain small in-frame deletions or missense mutations. The corresponding proteins, therefore, have decreased stability or lowered activity, leading to a reduced half-life or lifespan for the red cell. No mutations have been found that result in complete absence of G6PD. Based on studies with knockout mice, those mutations would be expected to result in embryonic lethality.



Heme, which is red, is responsible for the color of red blood cells and of muscles that contain a large number of mitochondria.

Chlorophyll, the major porphyrin in plants, is similar to heme, except that it is coordinated with magnesium rather than iron, and it contains different substituents on the rings, including a long-chain alcohol (phytol). As a result of these structural differences, chlorophyll is green.

Approximately 5 to 10% of the glucose metabolized by red blood cells is used to generate NADPH by way of the hexose monophosphate shunt. The NADPH is used to maintain glutathione in the reduced state. The glutathione cycle is the red blood cell's chief defense against damage to proteins and lipids by reactive oxygen species (see Chapter 24).

The enzyme that catalyzes the first step of the hexose monophosphate shunt is glucose-6-phosphate dehydrogenase (G6PD). The lifetime of the red blood cell correlates with G6PD activity. Lacking ribosomes, the red blood cell cannot synthesize new G6PD protein. Consequently, as the G6PD activity decreases, oxidative damage accumulates, leading to lysis of the erythrocyte. When red blood cell lysis (hemolysis) substantially exceeds the normal rate of red blood cell production, the number of erythrocytes in the blood drops below normal values, leading to a **hemolytic anemia**.

B. The Erythrocyte Precursor Cells and Heme Synthesis

1. HEME STRUCTURE

Heme consists of a porphyrin ring coordinated with an atom of iron (Fig. 44.2). Four pyrrole rings are joined by methionyl bridges (—CH—) to form the porphyrin ring (see Fig. 7.12). Eight side chains serve as substituents on the porphyrin ring, two on each pyrrole. These side chains may be acetyl (A), propionyl (P), methyl (M), or vinyl (V) groups. In heme, the order of these groups is M V M V M P P M. This order, in which the position of the methyl group is reversed on the fourth ring, is characteristic of the porphyrins of the type III series, the most abundant in nature.

Heme is the most common porphyrin found in the body. It is complexed with proteins to form hemoglobin, myoglobin, and the cytochromes (see Chapters 7 and 21), including cytochrome P450 (see Chapter 24).

2. SYNTHESIS OF HEME

Heme is synthesized from glycine and succinyl CoA (Fig. 44.3), which condense in the initial reaction to form δ -aminolevulinic acid (δ -ALA) (Fig. 44.4). The enzyme that catalyzes this reaction, δ -ALA synthase, requires the participation of pyridoxal phosphate, as the reaction is an amino acid decarboxylation reaction (glycine is decarboxylated; see Chapter 39).

The next reaction of heme synthesis is catalyzed by δ -ALA dehydratase, in which two molecules of δ -ALA condense to form the pyrrole, porphobilinogen (Fig. 44.5). Four of these pyrrole rings condense to form a linear chain and then a series of porphyrinogens. The side chains of these porphyrinogens initially contain

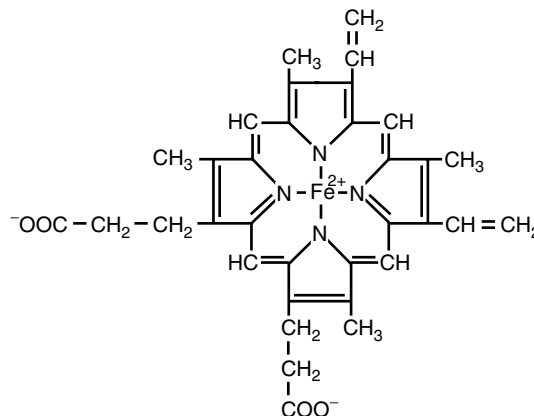


Fig. 44.2. Structure of heme. The side chains can be abbreviated as MVMVMPPM. M = methyl (CH₃); V = vinyl (—CH=CH₂); P = propionyl (—CH₂—CH₂—COO[−]).

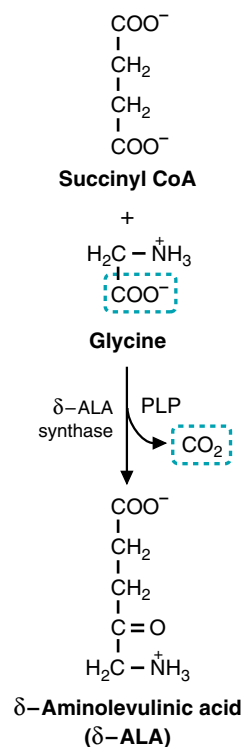
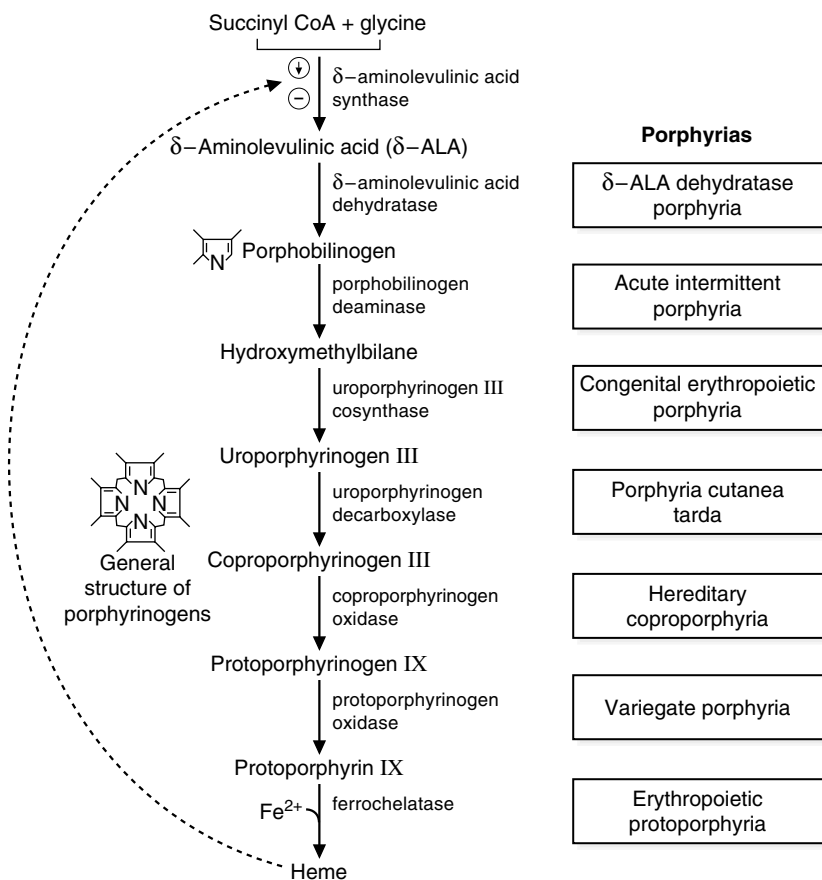


Fig. 44.4. Synthesis of δ -aminolevulinic acid (δ -ALA). PLP = pyridoxal phosphate.

Fig. 44.3. Synthesis of heme. To produce one molecule of heme, 8 molecules each of glycine and succinyl CoA are required. A series of porphyrinogens are generated in sequence. Finally, iron is added to produce heme. Heme regulates its own production by repressing the synthesis of δ -aminolevulinic acid (δ -ALA) synthase (circled $\downarrow$) and by directly inhibiting the activity of this enzyme (circled $-$). Deficiencies of enzymes in the pathway result in a series of diseases known as porphyrias (listed on the right, beside the deficient enzyme).

acetyl (A) and propionyl (P) groups. The acetyl groups are decarboxylated to form methyl groups. Then the first two propionyl side chains are decarboxylated and oxidized to vinyl groups, forming a protoporphyrinogen. The methylene bridges are subsequently oxidized to form protoporphyrin IX (see Fig. 44.3).

In the final step of the pathway, iron (as Fe^{2+}) is incorporated into protoporphyrin IX in a reaction catalyzed by ferrochelatase (also known as heme synthase).

3. SOURCE OF IRON

Iron, which is obtained from the diet, has a Recommended Dietary Allowance (RDA) of 10 mg for men and postmenopausal women, and 15 mg for premenopausal women. The average American diet contains 10 to 50 mg of iron. However, only 10 to 15% is normally absorbed, and iron deficiencies are fairly common.



Pyridoxine (vitamin B6) deficiencies are often associated with a microcytic, hypochromic anemia. Why would a B6 deficiency result in small (microcytic), pale (hypochromic) red blood cells?



δ -ALA dehydratase, which contains zinc, and ferrochelatase are inactivated by lead. Thus, in lead poisoning, δ -ALA and protoporphyrin IX accumulate, and the production of heme is decreased. Anemia results from a lack of hemoglobin, and energy production decreases because of the lack of cytochromes for the electron transport chain.



Porphyrias are a group of rare inherited disorders resulting from deficiencies of enzymes in the pathway for heme biosynthesis (see Fig. 44.3). Intermediates of the pathway accumulate and may have toxic effects on the nervous system that cause neuropsychiatric symptoms. When porphyrinogens accumulate, they may be converted by light to porphyrins, which react with molecular oxygen to form oxygen radicals. These radicals may cause severe damage to the skin. Thus, individuals with excessive production of porphyrins are photosensitive. The scarring and increased growth of facial hair seen in some porphyrias may have contributed to the development of the werewolf legends.

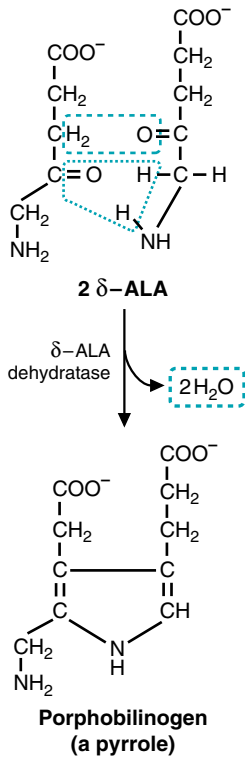


Fig. 44.5. Two molecules of δ -ALA condense to form porphobilinogen.



In a B6 deficiency, the rate of heme production is slow because the first reaction in heme synthesis requires pyridoxal phosphate (see Fig. 44.4). Thus, less heme is synthesized, causing red blood cells to be small and pale. Iron stores are usually elevated.



The iron lost by adult males (approximately 1 mg/day) by desquamation of the skin and in bile, feces, urine, and sweat is replaced by iron absorbed from the diet. Men are not as likely to suffer from iron deficiencies as premenopausal adult women, who also lose iron during menstruation and who must supply iron to meet the needs of the growing fetus during a pregnancy. If a man eating a Western diet has iron-deficiency anemia, his physician should suspect bleeding from the gastrointestinal tract due to ulcers or colon cancer.



Although spinach has been touted as a wonderful source of iron (mostly by the cartoon character Popeye), this iron is not readily absorbed because spinach has a high content of phytate (inositol with a phosphate group attached to each of its 6 hydroxyl groups).

The iron in meats is in the form of heme, which is readily absorbed. The non-heme iron in plants is not as readily absorbed, in part because plants often contain oxalates, phytates, tannins, and other phenolic compounds that chelate or form insoluble precipitates with iron, preventing its absorption. Conversely, vitamin C (ascorbic acid) increases the uptake of non-heme iron from the digestive tract. The uptake of iron is also increased in times of need by mechanisms that are not yet understood. Iron is absorbed in the ferrous (Fe^{2+}) state (Fig. 44.6), but is oxidized to the ferric state by a ferroxidase known as ceruloplasmin (a copper-containing enzyme) for transport through the body.

Because free iron is toxic, it is usually found in the body bound to proteins (see Fig. 44.6). Iron is carried in the blood (as Fe^{3+}) by the protein apotransferrin, with which it forms a complex known as transferrin. Transferrin is usually only one-third saturated with iron. The total iron-binding capacity of blood, mainly due to its content of transferrin, is approximately 300 $\mu\text{g/dL}$.

Storage of iron occurs in most cells but especially those of the liver, spleen, and bone marrow. In these cells, the storage protein, apoferritin, forms a complex with iron (Fe^{3+}) known as ferritin. Normally, little ferritin is present in the blood. This amount increases, however, as iron stores increase. Therefore, the amount of ferritin in the blood is the most sensitive indicator of the amount of iron in the body's stores.

Iron can be drawn from ferritin stores, transported in the blood as transferrin, and taken up via receptor-mediated endocytosis by cells that require iron (e.g., by reticulocytes that are synthesizing hemoglobin). When excess iron is absorbed from the diet, it is stored as hemosiderin, a form of ferritin complexed with additional iron that cannot be readily mobilized.

4. REGULATION OF HEME SYNTHESIS

Heme regulates its own synthesis by mechanisms that affect the first enzyme in the pathway, δ -ALA synthase (see Fig. 44.3). Heme represses the synthesis of this enzyme, and also directly inhibits the activity of the enzyme (an allosteric modifier). Thus, heme is synthesized when heme levels fall. As heme levels rise, the rate of heme synthesis decreases.

Heme also regulates the synthesis of hemoglobin by stimulating synthesis of the protein globin. Heme maintains the ribosomal initiation complex for globin synthesis in an active state (see Chapter 15).

5. DEGRADATION OF HEME

Heme is degraded to form bilirubin, which is conjugated with glucuronic acid and excreted in the bile (Fig. 44.7). Although heme from cytochromes and myoglobin also undergoes conversion to bilirubin, the major source of this bile pigment is hemoglobin. After red blood cells reach the end of their lifespan (approximately 120 days), they are phagocytosed by cells of the reticuloendothelial system. Globin is cleaved to its constituent amino acids, and iron is returned to the body's iron stores. Heme is oxidized and cleaved to produce carbon monoxide and biliverdin (Fig. 44.8). Biliverdin is reduced to bilirubin, which is transported to the liver complexed with serum albumin.

In the liver, bilirubin is converted to a more water-soluble compound by reacting with UDP-glucuronate to form bilirubin monoglucuronide, which is converted to the diglucuronide (see Fig. 30.5). This conjugated form of bilirubin is excreted into the bile.



Drugs, such as phenobarbital, induce enzymes of the drug metabolizing systems of the endoplasmic reticulum that contain cytochrome P450. Because heme is used for synthesis of cytochrome P450, free heme levels will fall and δ -ALA synthase will be induced to increase the rate of heme synthesis.

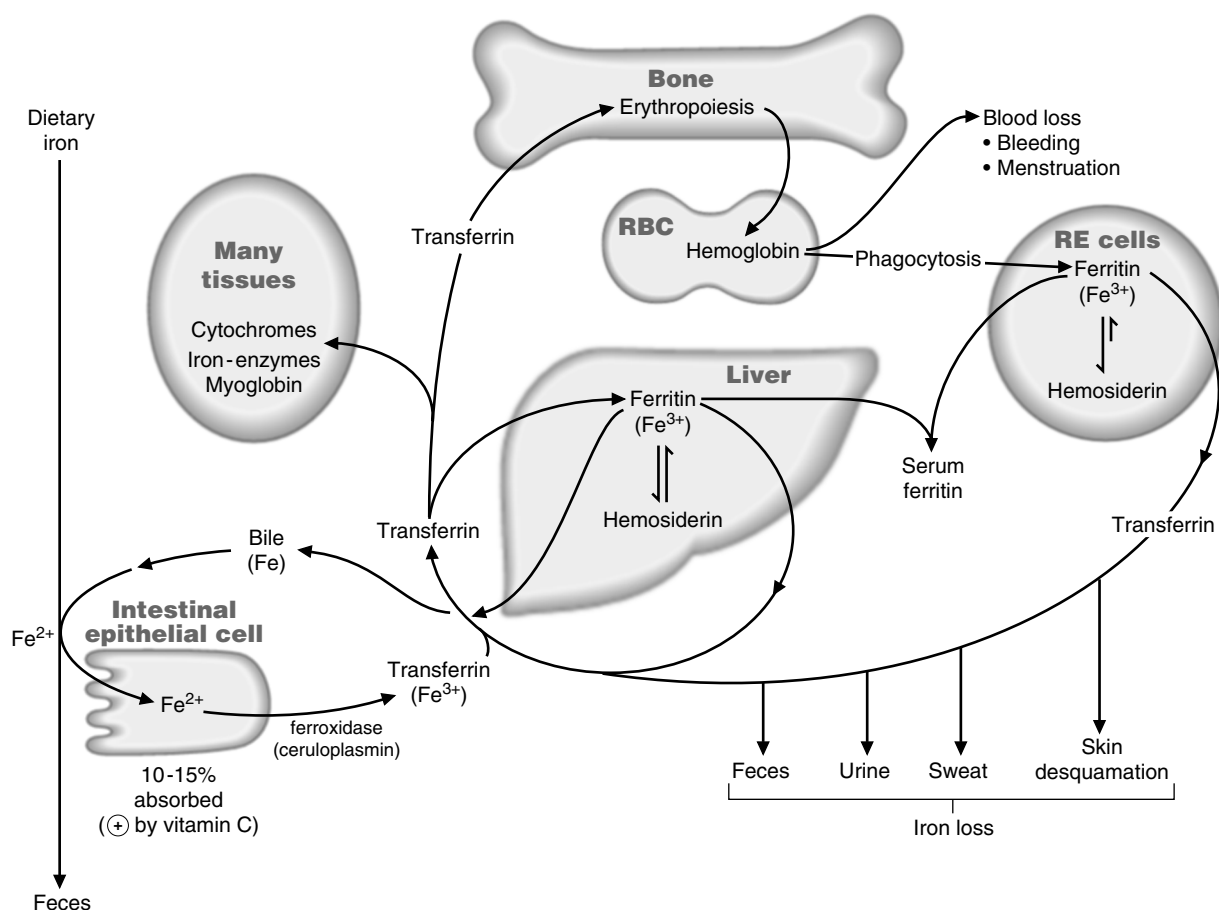


Fig. 44.6. Iron metabolism. Iron is absorbed from the diet, transported in the blood in transferrin, stored in ferritin, and used for the synthesis of cytochromes, iron-containing enzymes, hemoglobin, and myoglobin. It is lost from the body with bleeding and sloughed-off cells, sweat, urine, and feces. Hemosiderin is the protein in which excess iron is stored. Small amounts of ferritin enter the blood and can be used to measure the adequacy of iron stores. RE = reticuloendothelial.

In the intestine, bacteria deconjugate bilirubin diglucuronide and convert the bilirubin to urobilinogens (see Fig. 44.7). Some urobilinogen is absorbed into the blood and excreted in the urine. However, most of the urobilinogen is oxidized to urobilins, such as stercobilin, and excreted in the feces. These pigments give feces their brown color.



Q: In an iron deficiency, what characteristics would blood exhibit?

III. THE RED BLOOD CELL MEMBRANE

Under the microscope, the red blood cell appears to be a red disc with a pale central area (biconcave disc) (Fig. 44.9). The biconcave disc shape (as opposed to a spherical shape) serves to facilitate gas exchange across the cell membrane. The membrane proteins that maintain the shape of the red blood cell also allow the red blood cell to traverse the capillaries with very small luminal diameters to deliver oxygen to the tissues. The interior diameters of many capillaries are smaller than the approximately $7.5\text{-}\mu\text{m}$ diameter of the red cell. Furthermore, in passing through the kidney, red blood cells traverse hypertonic areas that are up to six times the normal isotonicity, and back again, causing the red cell to shrink and expand during its travels. The spleen is the organ responsible for determining the viability of the red blood cells. Erythrocytes pass through the spleen 120 times per day. The elliptical passageways through the spleen are approximately $3\text{ }\mu\text{m}$ in diameter, and

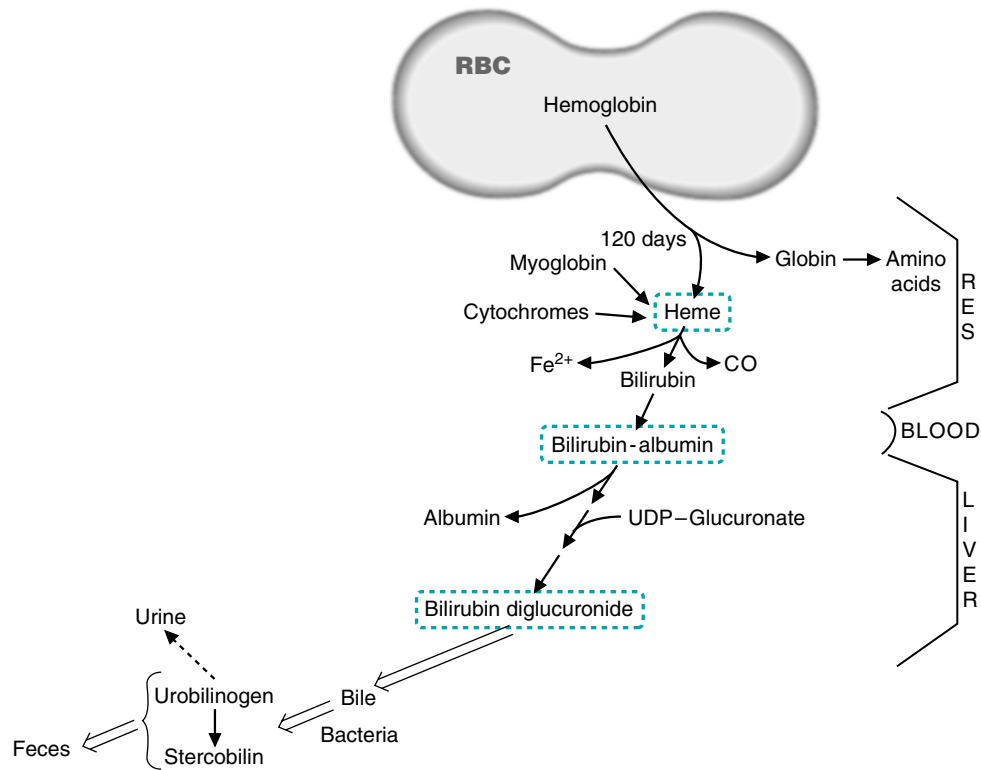


Fig. 44.7. Overview of heme degradation. Heme is degraded to bilirubin, carried in the blood by albumin, conjugated to form the diglucuronide in the liver, and excreted in the bile. The iron is returned to the body's iron stores. RES = reticuloendothelial system; RBC = red blood cells.



Iron deficiency would result in a microcytic, hypochromic anemia. Red blood cells would be small and pale. In contrast to a vitamin B6 deficiency, which also results in a microcytic, hypochromic anemia, iron stores are low in an iron-deficiency anemia.



The unusual names for some erythrocyte membrane proteins, such as band 4.1, arose through analysis of red blood cell membranes by polyacrylamide gel electrophoresis. The stained bands observed in the gel were numbered according to molecular weight (band 1, band 2, and so on), and as functions were assigned to the proteins, more common names were assigned to the proteins (for example, spectrin is actually band 1).

normal red cells traverse them in approximately 30 seconds. Thus, to survive in the circulation, the red cell must be highly deformable. Damaged red cells that are no longer deformable become trapped in the passages in the spleen, where they are destroyed by macrophages. The reason for the erythrocyte's deformability lies in its shape and in the organization of the proteins that make up the red blood cell membrane.

The surface area of the red cell is approximately $140 \mu\text{m}^2$, which is greater than the surface of a sphere needed to enclose the contents of the red cell ($98 \mu\text{m}^2$). The presence of this extra membrane and the cytoskeleton that supports it allows the red cell to be stretched and deformed by mechanical stresses as the cell passes through narrow vascular beds. On the cytoplasmic side of the membrane, proteins form a two-dimensional lattice that gives the red cell its flexibility (Fig. 44.10). The major proteins are spectrin, actin, band 4.1, band 4.2, and ankyrin. Spectrin, the major protein, is a heterodimer composed of α and β subunits wound around each other. The dimers self-associate at the heads. At the opposite end of the spectrin dimers, actin and band 4.1 bind near to each other. Multiple spectrins can bind to each actin filament, resulting in a branched membrane cytoskeleton.

The spectrin cytoskeleton is connected to the membrane lipid bilayer by ankyrin, which interacts with β -spectrin and the integral membrane protein, band 3. Band 4.2 helps to stabilize this connection. Band 4.1 anchors the spectrin skeleton with the membrane by binding the integral membrane protein glycophorin C and the actin complex, which has bound multiple spectrin dimers.

When the red blood cell is subjected to mechanical stress, the spectrin network rearranges. Some spectrin molecules become uncoiled and extended; others

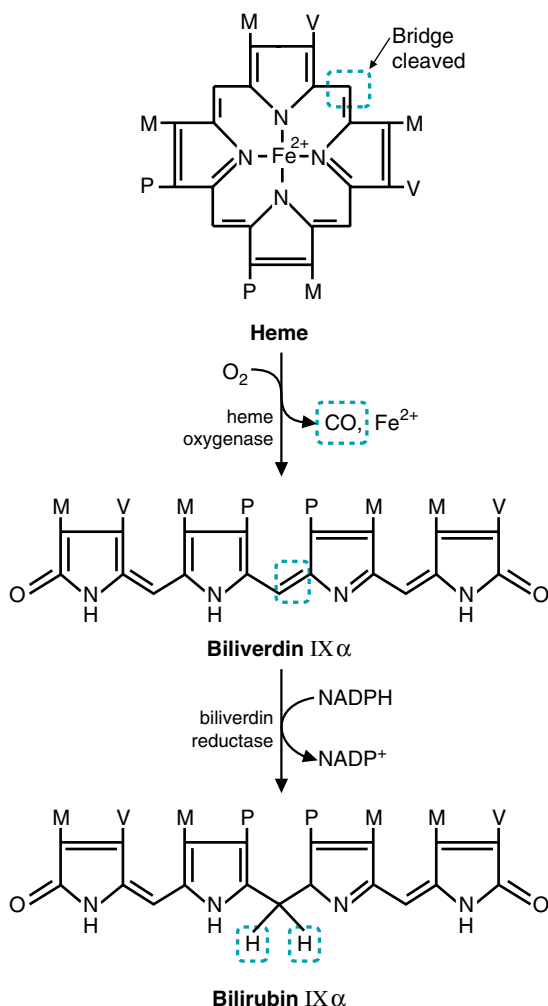


Fig. 44.8. Conversion of heme to bilirubin. A methylene bridge in heme is cleaved, releasing carbon monoxide (CO) and iron. Then, the center methylene bridge is reduced.

become compressed, thereby changing the shape of the cell, but not its surface area.

The mature erythrocyte cannot synthesize new membrane proteins or lipids. However, membrane lipids can be freely exchanged with circulating lipoprotein lipids. The glutathione system protects the proteins and lipids from oxidative damage.

IV. AGENTS THAT AFFECT OXYGEN BINDING

The major agents that affect oxygen binding to hemoglobin are shown in Figure 44.11.

A. 2,3-Bisphosphoglycerate

2,3-Bisphosphoglycerate (2,3-BPG) is formed in red blood cells from the glycolytic intermediate 1,3-bisphosphoglycerate, as indicated in Figure 44.1. 2,3-BPG binds to hemoglobin in the central cavity formed by the four subunits, increasing the energy required for the conformational changes that facilitate the binding of oxygen. Thus, 2,3-BPG lowers the affinity of hemoglobin for oxygen. Therefore, oxygen is less readily bound (i.e., more readily released in tissues) when hemoglobin contains 2,3-BPG.

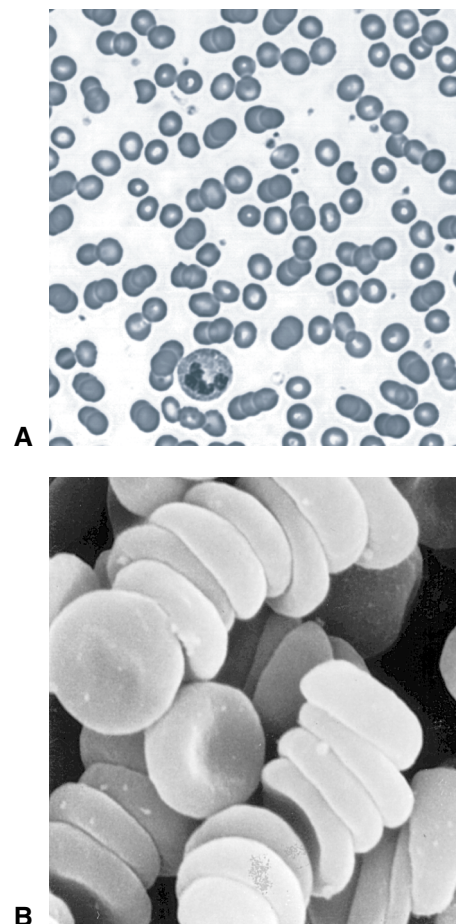


Fig. 44.9. The shape of the red blood cell. A. Wright-stained cells, displaying the pale staining in the center. Three leukocytes also are present in the preparation. The magnification is 350 $\times$. B. Scanning electron micrograph, showing the biconcave disc structure of the cells. The stacks of erythrocytes in this preparation (collected from a blood tube) is not unusual. The magnification is 28,000 $\times$. These photographs were obtained, with permission, from Ross et al, *Histology, A Text and Atlas with Cell and Molecular Biology*, 4th Ed. Philadelphia: Lippincott, 2003:216–217.



Defects in erythrocyte cytoskeletal proteins lead to hemolytic anemia.

Shear stresses in the circulation result in the loss of pieces of the red cell membrane. As the membrane is lost, the red blood cell becomes more spherical and loses its deformability. As these cells become more spherical, they are more likely to lyse in response to mechanical stresses in the circulation, or to be trapped and destroyed in the spleen.



- ① Hydrogen ions
- ② 2,3-Bisphosphoglycerate
- ③ Covalent binding of CO_2

Fig. 44.11. Agents that affect oxygen binding by hemoglobin. Binding of hydrogen ions, 2,3 bisphosphoglycerate, and carbon dioxide to hemoglobin decrease its affinity for oxygen.

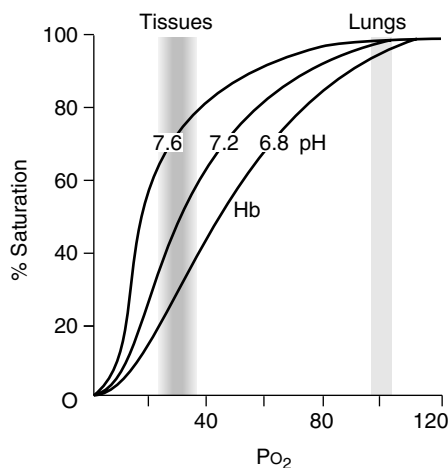


Fig. 44.12. Effect of pH on oxygen saturation curves. As the pH decreases, the affinity of hemoglobin for oxygen decreases, producing the Bohr effect.

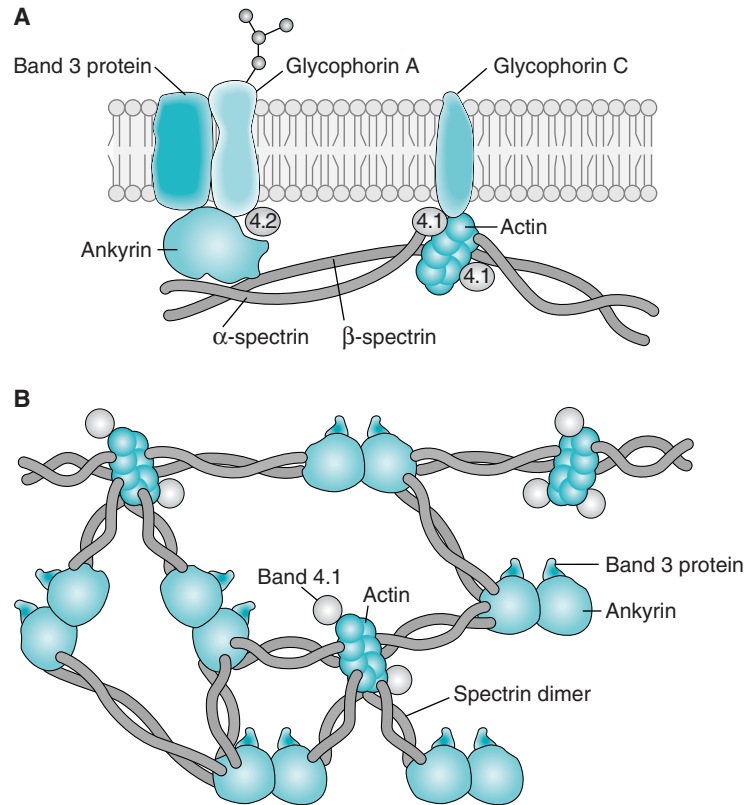


Fig. 44.10. A generalized view of the erythrocyte cytoskeleton. A. The major protein, spectrin, is linked to the plasma membrane either through interactions with ankyrin and band 3, or with actin, band 4.1, and glycophorin. Other proteins in this complex, but not shown, are tropomyosin and adducin. B. A view from inside the cell, looking up at the cytoskeleton. This view displays the cross-linking of the spectrin dimers to actin and band 3 anchor sites.

B. Proton Binding (Bohr effect)

The binding of protons by hemoglobin lowers its affinity for oxygen (Fig. 44.12), contributing to a phenomenon known as the Bohr effect (Fig. 44.13). The pH of the blood decreases as it enters the tissues (and the proton concentration rises) because the CO_2 produced by metabolism is converted to carbonic acid by the reaction catalyzed by carbonic anhydrase in red blood cells. Dissociation of carbonic acid produces protons that react with several amino acid residues in hemoglobin, causing conformational changes that promote the release of oxygen.

In the lungs, this process is reversed. Oxygen binds to hemoglobin, causing a release of protons, which combine with bicarbonate to form carbonic acid. This decrease of protons causes the pH of the blood to rise. Carbonic anhydrase cleaves the carbonic acid to H_2O and CO_2 , and the CO_2 is exhaled. Thus, in tissues in which the pH of the blood is low because of the CO_2 produced by metabolism, oxygen is released from hemoglobin. In the lungs, where the pH of the blood is higher because CO_2 is being exhaled, oxygen binds to hemoglobin.

C. Carbon Dioxide

Although most of the CO_2 produced by metabolism in the tissues is carried to the lungs as bicarbonate, some of the CO_2 is covalently bound to hemoglobin (Fig. 44.14). In the tissues, CO_2 forms carbamate adducts with the N-terminal amino groups of deoxyhemoglobin and stabilizes the deoxy conformation. In the

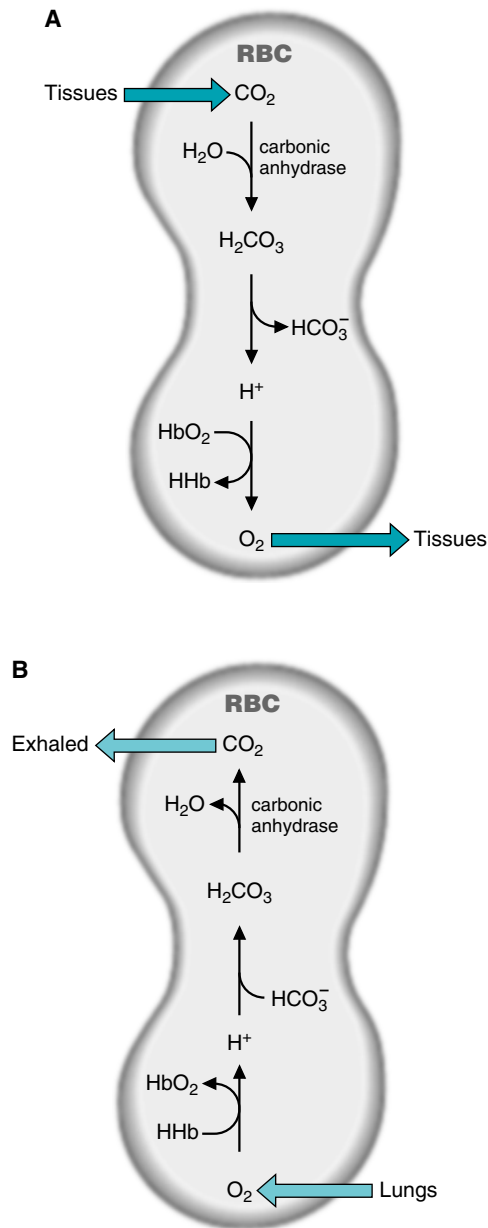


Fig. 44.13. Effect of H^+ on oxygen binding by hemoglobin (Hb). A. In the tissues, CO_2 is released. In the red blood cell, this CO_2 forms carbonic acid, which releases protons. The protons bind to Hb, causing it to release oxygen to the tissues. B. In the lungs, the reactions are reversed. O_2 binds to protonated Hb, causing the release of protons. They bind to bicarbonate (HCO_3^-), forming carbonic acid, which is cleaved to water and CO_2 , which is exhaled.

lungs, where the pO_2 is high, oxygen binds to hemoglobin and this bound CO_2 is released.

V. HEMATOPOIESIS

The various types of cells (lineages) that make up the blood are constantly being produced in the bone marrow. All cell lineages are descended from hematopoietic stem cells, cells that are renewable throughout the life of the host. The population of hematopoietic stem cells is quite small. Estimates vary between 1 to 10 per 10^5 bone marrow cells. In the presence of the appropriate signals, hematopoietic stem

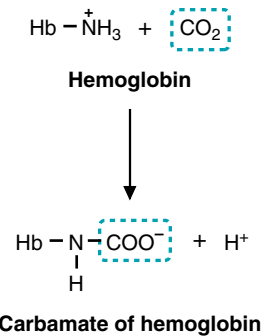


Fig. 44.14. Binding of CO_2 to hemoglobin. CO_2 forms carbamates with the N-terminal amino groups of Hb chains. Approximately 15% of the CO_2 in blood is carried to the lungs bound to Hb. The reaction releases protons, which contribute to the Bohr effect. The overall effect is the stabilization of the deoxy form of hemoglobin.



Populations of hematopoietic cells enriched with stem cells can be isolated by fluorescence activated cell sorting, based on the expression of specific cell surface markers. Increasing the population of stem cells in cells used for a bone marrow transplantation increases the chances of success of the transplantation.

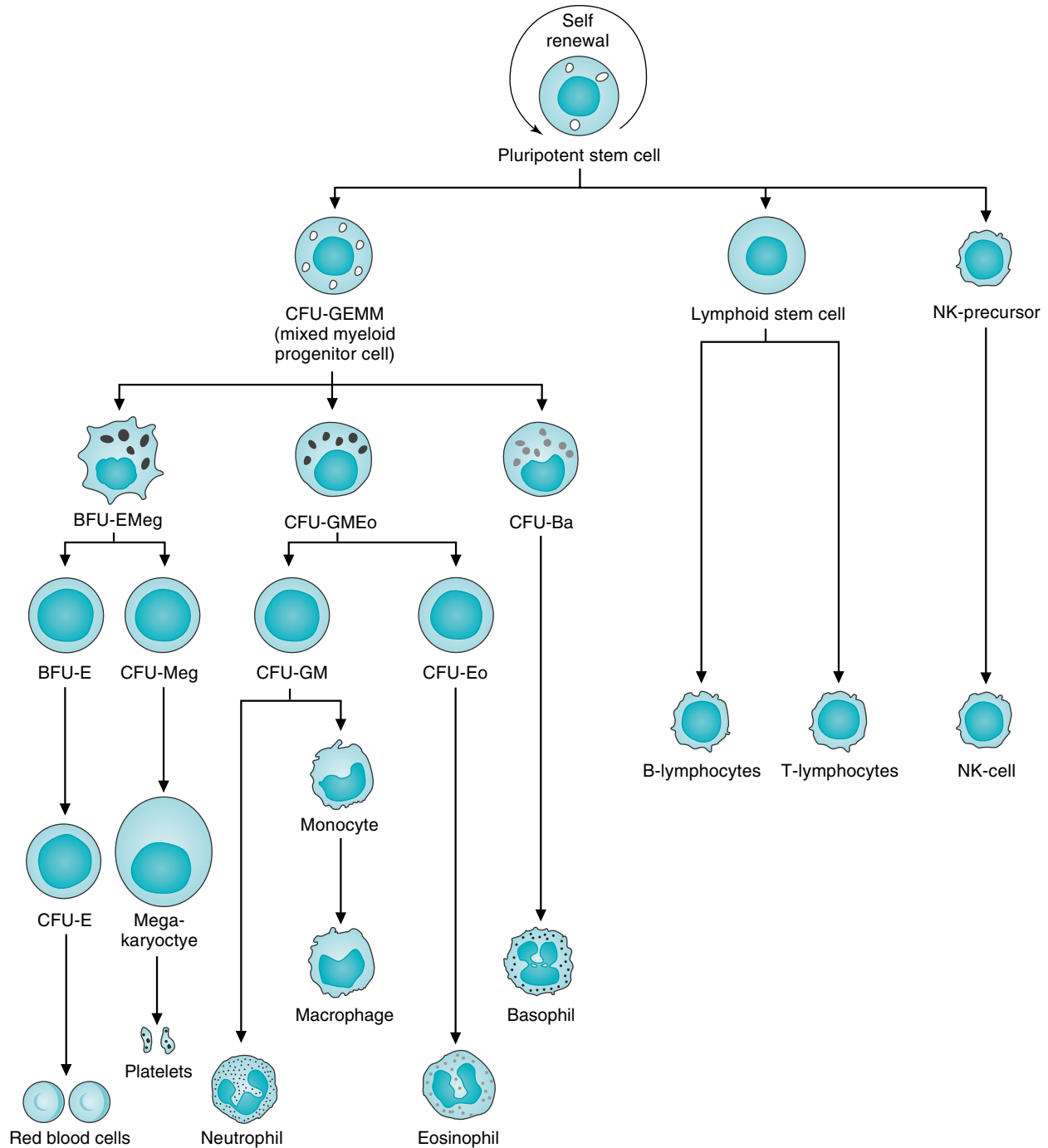


Fig. 44.15. The hematopoietic tree. All blood cells arise from the self-renewing pluripotent stem cell. Different cytokines are required at each step for these events to occur. CFU = colony-forming unit; BFU = burst-forming unit.



Leukemias, malignancies of the blood, arise when a differentiating hematopoietic cell does not complete its developmental program but remains in an immature, proliferative state. Leukemias have been found in every hematopoietic lineage.

cells proliferate, differentiate, and mature into any of the types of cells that make up the blood (Figure 44.15).

Hematopoietic differentiation is hierarchical. The number of fates a developing blood cell may adopt becomes progressively restricted. Hematopoietic progenitors are designated colony-forming unit–lineage, or colony-forming unit–erythroid (CFU-E). Progenitors that form very large colonies are termed burst-forming units.

A. Cytokines and Hematopoiesis

Developing progenitor cells in the marrow grow in proximity with marrow stromal cells. These include fibroblasts, endothelial cells, adipocytes, and macrophages. The stromal cells form an extracellular matrix and secrete growth factors that regulate hematopoietic development.

The hematopoietic growth factors have multiple effects. An individual growth factor may stimulate proliferation, differentiation, and maturation of the progenitor cells and also may prevent apoptosis. These factors also may activate various functions within the mature cell. Some hematopoietic growth factors act on multiple lineages, whereas others have more limited targets.

Most hematopoietic growth factors are recognized by receptors belonging to the cytokine receptor superfamily. Binding of ligand to receptor results in receptor aggregation, which induces phosphorylation of Janus kinases (JAKs). The JAKs are a family of cytoplasmic tyrosine kinases that are active when phosphorylated (see Chapter 11, section III.C., and Fig. 11.15). The activated JAKs then phosphorylate the cytokine receptor. Phosphorylation of the receptor creates docking regions where additional signal transduction molecules bind, including members of the signal transducer and activator of transcription (STAT) family of transcription factors. The JAKs phosphorylate the STATs, which dimerize and translocate to the nucleus, where they activate target genes. Additional signal transduction proteins bind to the phosphorylated cytokine receptor, leading to activation of the Ras/Raf/MAP kinase pathways. Other pathways are also activated, some of which lead to an inhibition of apoptosis (see Chapter 18).

The response to cytokine binding is usually transient because the cell contains multiple negative regulators of cytokine signaling. The family of silencer of cytokine signaling (SOCS) proteins are induced by cytokine binding. One member of the family binds to the phosphorylated receptor and prevents the docking of signal transduction proteins. Other SOCS proteins bind to JAKs and inhibit them. Whether SOCS inhibition of JAKs is a consequence of steric inhibition or whether SOCS recruit phosphatases that then dephosphorylate the JAKs (Figure 44.16) is uncertain.

SHP-1 is a tyrosine phosphatase found primarily in hematopoietic cells that is necessary for proper development of myeloid and lymphoid lineages. Its function is to dephosphorylate JAK2, thereby inactivating it.

STATs are also inactivated. The protein inhibitors of activated STAT (PIAS) family of proteins bind to phosphorylated STATs and prevent their dimerization or promote the dissociation of STAT dimers. STATs also may be inactivated by dephosphorylation, although the specific phosphatases have not yet been identified, or by targeting activated STATs for proteolytic degradation.

B. Erythropoiesis

The production of red cells is regulated by the demands of oxygen delivery to the tissues. In response to reduced tissue oxygenation, the kidney releases the hormone erythropoietin, which stimulates the multiplication and maturation of erythroid progenitors. The progression along the erythroid pathway begins with the stem cell and passes through the mixed myeloid progenitor cell, (CFU-GEMM, colony-forming unit–granulocyte, erythroid, monocyte, megakaryocyte), burst-forming unit–erythroid (BFU-E), colony-forming unit–erythroid (CFU-E), and to the first recognizable red cell precursor, the normoblast. Each normoblast undergoes four more cycles of cell division. During these four cycles, the nucleus becomes smaller and more condensed. After the last division, the nucleus is extruded. The red cell at this state is called a reticulocyte. Reticulocytes still retain ribosomes and mRNA and are capable of synthesizing hemoglobin. They are released from the bone marrow and circulate for 1 to 2 days. Reticulocytes mature in the spleen, where the ribosomes and mRNA are lost (Fig. 44.17).



Bone marrow cells can be cultured in semisolid media with the addition of the appropriate growth factors. After 14 to 18 days in culture, colonies of blood cells can be seen. The type (lineage) of these cells can be determined based on morphological or staining properties. Most colonies will be of single lineage, indicating descent from a hematopoietic progenitor that was committed to a lineage. Occasionally a multilineage colony will be obtained, indicating that it was derived from a more primitive hematopoietic progenitor.



In X-linked severe combined immunodeficiency disease (SCID), the most common form of SCID, circulating T lymphocytes are not formed, and B lymphocytes are not active. The affected gene encodes the gamma chain of the interleukin 2 receptor. Mutant receptors are unable to activate JAK3, and the cells are unresponsive to the cytokines that stimulate growth and differentiation. Recall also that adenosine deaminase deficiency (see Chapter 41), which is not X-linked, also leads to a form of SCID, but for different reasons.



Families have been identified whose members have a mutant erythropoietin (epo) receptor that is unable to bind SHP-1. Erythropoietin is the hematopoietic cytokine that stimulates production of red blood cells. Individuals with the mutant epo receptor have a higher than normal percentage of red blood cells in the circulation, because the mutant epo receptor cannot be deactivated by SHP-1. Erythropoietin causes sustained activation of JAK2 and STAT 5 in this case.



Perturbed JAK/STAT signaling is associated with development of lymphoid and myeloid leukemias, severe congenital neutropenia, a condition in which levels of circulating neutrophils are severely reduced, and Fanconi anemia, which is characterized by bone marrow failure and increased susceptibility to malignancy.

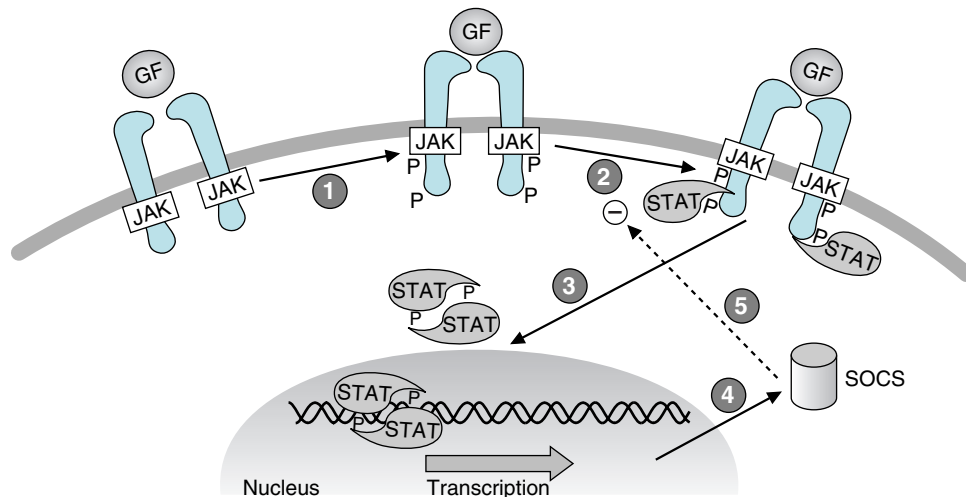


Fig. 44.16. Cytokine signaling through the JAK/STAT pathway. 1. Cytokine binding to receptors initiates dimerization and activation of the JAK kinase, which phosphorylates the receptor on tyrosine residues. 2. STAT proteins bind to the activated receptors and are themselves phosphorylated. 3. Phosphorylated STAT proteins dimerize, travel to the nucleus, and initiate gene transcription. 4. One of the proteins whose synthesis is stimulated by STATs is SOCS (suppressor of cytokine signaling), which inhibits further activation of STAT proteins (circle 5) by a variety of mechanisms.

C. Nutritional Anemias

Each person produces approximately 10^{12} red blood cells per day. Because so many cells must be produced, nutritional deficiencies in iron, vitamin B12, and folate prevent adequate red blood cell formation. The physical appearance of the cells in the case of a nutritional anemia frequently provides a clue as to the nature of the deficiency.

In the case of iron deficiency, the cells are smaller and paler than normal. The lack of iron results in decreased heme synthesis, which in turn affects globin synthesis. Maturing red cells following their normal developmental program divide until their hemoglobin has reached the appropriate concentration. Iron- (and hemoglobin-) deficient developing red blood cells continue dividing past their normal stopping point, resulting in small (microcytic) red cells. The cells are also pale because of the lack of hemoglobin, as compared with normal cells (thus, a pale, microcytic anemia results).

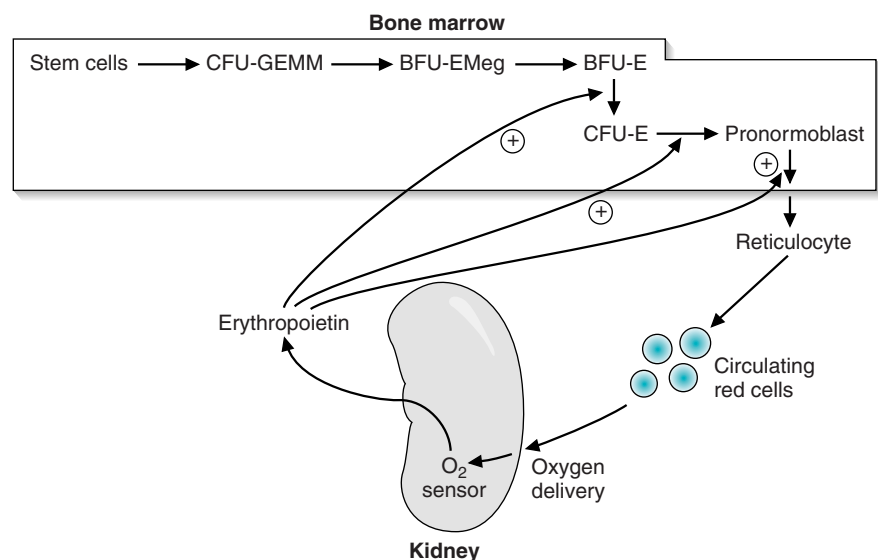


Fig. 44.17. Erythropoietin stimulation of erythrocyte maturation. The abbreviations are described in the text.

Deficiencies of folate or vitamin B12 can cause megaloblastic anemia, in which the cells are larger than normal. Folate and B12 are required for DNA synthesis (see Chapters 40 and 41). When these vitamins are deficient, DNA replication and nuclear division do not keep pace with the maturation of the cytoplasm. Consequently, the nucleus is extruded before the requisite number of cell divisions has taken place, and the cell volume is greater than it should be, and fewer blood cells are produced.

VI. HEMOGLOBINOPATHIES, HEREDITARY PERSISTENCE OF FETAL HEMOGLOBIN, AND HEMOGLOBIN SWITCHING

A. Hemoglobinopathies: Disorders in the Structure or Amount of the Globin Chains

More than 700 different mutant hemoglobins have been discovered. Most arise from a single base substitution, resulting in a single amino acid replacement. Many have been discovered during population screenings and are not clinically significant. However, in patients with hemoglobin S (HbS, sickle cell anemia), the most common hemoglobin mutation, the amino acid substitution has a devastating effect in the homozygote (see **Will Sichel** in Chapter 6). Another common hemoglobin variant, HbC, results from a glu to lys replacement in the same position as the HbS mutation. This mutation has two effects. It promotes water loss from the cell by activating the K^+ transporter by an unknown mechanism, resulting in a higher than normal concentration of hemoglobin within the cell. The amino acid replacement also substantially lowers the hemoglobin solubility in the homozygote, resulting in a tendency of the mutant hemoglobin to precipitate within the red cell, although, unlike sickle cells, the cell does not become deformed. Homozygotes for the HbC mutation have a mild hemolytic anemia. Heterozygous individuals are clinically unaffected.

B. Thalassemias

For optimum function, the hemoglobin α and β -globin chains must have the proper structure and be synthesized in a 1:1 ratio. A large excess of one subunit over the other results in the class of diseases called thalassemias. These anemias are clinically very heterogeneous, as they can arise by multiple mechanisms. Like sickle cell anemia, the thalassemia mutations provide resistance to malaria in the heterozygous state.

Hemoglobin single amino acid replacement mutations that give rise to a globin subunit of decreased stability is one mechanism by which thalassemia arises. More common, however, are mutations that result in decreased synthesis of one subunit. Alpha thalassemias usually arise from complete gene deletions. Two copies of the α -globin gene are found on each chromosome 16, for a total of 4 α -globin genes per precursor cell. If one copy of the gene is deleted, the size and hemoglobin concentration of the individual red blood cells is minimally reduced. If two copies are deleted, the red blood cells are of decreased size (microcytic) and reduced hemoglobin concentration (hypochromic). However, the individual is usually not anemic. The loss of three α -globin genes causes a moderately severe microcytic hypochromic anemia (hemoglobin 7–10 g/dL) with splenomegaly (enlarged spleen). The absence of four α -globin genes (hydrops fetalis) is usually fatal in utero.



There are two ways in which an individual could have two α -globin genes deleted. In one case, one copy of chromosome 16 could have both α -globin genes deleted, whereas the other copy had two functional α genes. In the second case, both chromosomes could have lost one of their two copies of the α -globin gene. The former possibility is more common among Asians; the latter among Africans.



A registry of hemoglobin mutations is found at the International Hemoglobin Information Center <http://e20.manu.edu.mk/rcgeb/ihic/>.



A complication of sickle cell disease is an increased formation of gallstones. A sickle cell crisis accompanied by the intravascular destruction of red blood cells (hemolysis) experienced by patients with sickle cell disease, such as **Will Sichel**, increases the amount of unconjugated bilirubin that is transported to the liver. If the concentration of this unconjugated bilirubin exceeds the capacity of the hepatocytes to conjugate it to the more soluble diglucuronide through interaction with hepatic UDP-glucuronate, both the total and the unconjugated bilirubin levels would rise in the blood. More unconjugated bilirubin would be secreted by the liver into the bile. The increase in unconjugated bilirubin (which is not very water-soluble) results in its precipitation within the gallbladder lumen, leading to the formation of pigmented (calcium bilirubinate) gallstones.



HbC is found in high frequency in West Africa, in regions with a high frequency of HbS. Consequently, compound heterozygotes for HbS and HbC are not uncommon both in some African regions and among African-Americans. HbS/HbC individuals have significantly more hematopathology than individuals with sickle cell trait (HbA/HbS). Polymerization of deoxygenated HbS is dependent on the HbS concentration within the cell. The presence of HbC in the compound heterozygote increases the HbS concentration by stimulating K^+ and water efflux from the cell. Because the HbC globin is produced more slowly than HbA or HbS, the proportion of HbS tends to be higher in HbS/HbC cells than in the cells of individuals with sickle cell trait (HbS/HbA). The way in which multiple mutations ameliorate or exacerbate hematologic diseases has provided insights into the molecular mechanisms of hemoglobin function and developmental regulation.



The difference in amino acid composition between the β -chains of HbA and γ -chains of HbF results in structural changes that cause HbF to have a lower affinity for 2,3-BPG than adult hemoglobin (HbA) and, thus, a greater affinity for oxygen. Therefore, the oxygen released from the mother's hemoglobin (HbA) is readily bound by HbF in the fetus. Thus, the transfer of oxygen from the mother to the fetus is facilitated by the structural difference between the hemoglobin molecule of the mother and that of the fetus.



Individuals with sickle cell anemia or beta thalassemia (usually) have intact γ -globin loci. If a way could be found to reactivate the γ -globin loci (the drug hydroxyurea is a potential candidate for this), it would be an attractive therapeutic option for the treatment of these diseases.



An additional source of variation in the levels of fetal hemoglobin is the FCP (**F**-cell producing) locus on the short arm of the X chromosome in a region thought not to be susceptible to X inactivation. Both normal individuals and individuals with hemoglobinopathies vary in the amount of hemoglobin F they produce. In studies of normal individuals, a high level of hemoglobin F appears to be inherited as an X-linked dominant trait. The FCP locus is responsible for a substantial amount of the variation in Hemoglobin F seen among sickle cell patients. The protein encoded at the FCP locus has not been identified; current speculations are that it is a transcription factor involved in the regulation of the globin locus.

As discussed in Chapter 14, beta thalassemia is a very heterogeneous genetic disease. Insufficient β -globin synthesis can result from deletions, promoter mutations, and splice junction mutations. Heterozygotes for β^+ (some globin chain synthesis) or β null (β^0 , no globin chain synthesis) are generally asymptomatic, though they typically have microcytic, hypochromic red blood cells and may have a mild anemia. β^+/β^+ homozygotes have an anemia of variable severity, β^+/β^0 compound heterozygotes tend to be more severely affected, and β^0/β^0 homozygotes have severe disease. In general, diseases of β chain deficiency are more severe than diseases of α chain deficiency. Excess β chains form a homotetramer, hemoglobin H (HbH), which is useless for delivering oxygen to the tissues because of its high oxygen affinity. As red blood cells age, HbH will precipitate in the cells, forming inclusion bodies. Red blood cells with inclusion bodies have a shortened life span, because they are more likely to be trapped and destroyed in the spleen. Excess α chains are unable to form a stable tetramer. However, excess α chains precipitate in erythrocytes at every developmental stage. The α chain precipitation in erythroid precursors results in their widespread destruction, a process called ineffective erythropoiesis. The precipitated α chains also damage red blood cell membranes through the heme-facilitated lipid oxidation by reactive oxygen species. Both lipids and proteins, particularly band 4.1, are damaged.

C. Hereditary Persistence of Fetal Hemoglobin

Fetal hemoglobin (HbF), the predominant hemoglobin of the fetal period, consists of two alpha chains and two gamma chains, whereas adult Hb consists of two alpha and two beta chains. The process that regulates the conversion of HbF to HbA is called hemoglobin switching. Hb switching is not 100%; most individuals continue to produce a small amount of HbF throughout life. However, some people, who are clinically normal, produce abnormally high levels (up to 100%) of fetal hemoglobin (Hemoglobin F) in place of HbA. Patients with hemoglobinopathies such as β -thalassemia or sickle cell anemia frequently have less severe illnesses if their levels of fetal hemoglobin are elevated. One goal of much research on hemoglobin switching is to discover a way to reactivate transcription of the γ -globin genes to compensate for defective β -globin synthesis. Individuals who express fetal hemoglobin past birth have hereditary persistence of fetal hemoglobin (HPFH).

1. NON-DELETION FORMS OF HPFH

The non-deletion forms of HPFH are those that derive from point mutations in the A γ and G γ promoters. When these mutations are found with sickle cell or beta thalassemia mutations, they have an ameliorating effect on the disease, because of the increased production of gamma chains.

2. DELETION FORMS OF HPFH

In deletion HPFH, both the entire delta and beta genes have been deleted from one copy of chromosome 11 and only HbF can be produced. In some individuals the fetal globins remain activated after birth, and enough HbF is produced that the individuals are clinically normal. Other individuals with similar deletions that remove the entire delta and beta genes do not produce enough fetal hemoglobin to compensate for the deletion and are considered to have $\delta^0\beta^0$ thalassemia. The difference between these two outcomes is believed to be the site at which the deletions end within the β -globin gene cluster. In deletion HPFH, powerful enhancer sequences 3' of the β -globin gene are resituated because of the deletion such that they activate the gamma promoters. In individuals with $\delta^0\beta^0$ thalassemia, the enhancer sequences have not been relocated such that they can interact with the gamma promoters.

D. Hemoglobin Switching: A Developmental Process Controlled by Transcription Factors

In humans, embryonic megaloblasts (the embryonic red blood cell is large and is termed a “blast” because it retains its nucleus) are first produced in the yolk sac approximately 15 days after fertilization. After 6 weeks, the site of erythropoiesis shifts to the liver. The liver and to a lesser extent the spleen are the major sites of fetal erythropoiesis. In the last few weeks before birth, the bone marrow begins producing red blood cells. By 8 to 10 weeks after birth, the bone marrow is the sole site of erythrocyte production. The composition of the hemoglobin also changes with development, because both the α -globin locus and the β -globin locus have multiple genes that are differentially expressed during development (Figure 44.18).

E. Structure and Transcriptional Regulation of the Alpha and Beta Globin Gene Loci

The α -globin locus on chromosome 16 contains the embryonic ζ (zeta) gene and two copies of the alpha gene, α_2 and α_1 . The β -globin locus on chromosome 11 contains the embryonic ϵ gene, two copies of the fetal β -globin gene G_γ and A_γ

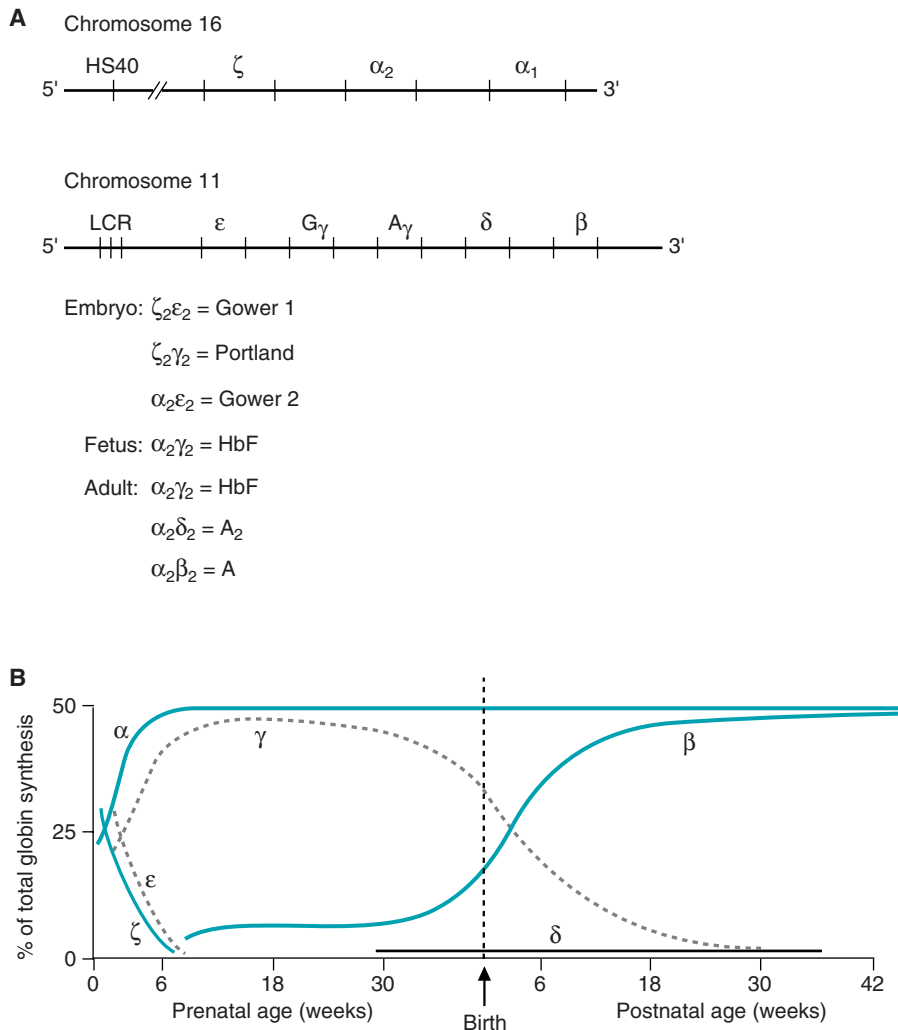


Fig 44.18. Globin gene clusters and expression during development. A. The globin gene clusters with the α genes on chromosome 16 and the β genes on chromosome 11. LCR = locus control region. B. The switching of globin chain synthesis during development.

(which differ by one amino acid), and two adult genes, δ and β . The order of the genes along the chromosome parallels the order of expression of the genes during development (see Fig. 44.18). The embryonic hemoglobins are $\zeta_2\epsilon_2$ (Gower 1), $\zeta_2\gamma_2$ (Portland), and $\alpha_2\gamma_2$ (Gower 2). Fetal hemoglobin is predominantly $\alpha_2\text{G}_{\gamma_2}$. The major adult species is $\alpha_2\beta_2$ (hemoglobin A); the minor adult species is $\alpha_2\delta_2$ (hemoglobin A₂). The fetal hemoglobin found in adult cells is $\alpha_2\text{A}_{\gamma_2}$. The timing of hemoglobin switching is controlled by a developmental clock not significantly altered by environmental conditions and is related to changes in expression of specific transcription factors. Premature newborns convert from HbF to HbA on schedule with their gestational ages.

CLINICAL COMMENTS



Spiro Site's red blood cells are deficient in spectrin. This deficiency impairs the ability of his erythrocytes to maintain the redundant surface area necessary to maintain deformability. Mechanical stresses in the circulation cause progressive loss of pieces of membrane. As membrane components are lost, **Spiro Site's** red blood cells become spherical and unable to deform. His spleen is enlarged because of the large number of red blood cells that have become trapped within it. His erythrocytes are lysed by mechanical stresses in the circulation and by macrophages in the spleen. Consequently, this hemolytic process results in an anemia. His gallstones were the result of the large amounts of bilirubin that were produced and stored in the gallbladder as a result of the hemolysis. The abnormally rounded red cells seen on a blood smear are characteristic of hereditary spherocytosis.

Mutations in the genes for ankyrin, β -spectrin, or band 3 account for three quarters of the cases of hereditary spherocytosis, whereas mutations in the genes for α -spectrin or band 4.2 account for the remainder. The result of defective synthesis of any of the membrane cytoskeletal proteins results in improper formation of the membrane cytoskeleton. Excess membrane proteins are catabolized, resulting in a net deficiency of spectrin. **Spiro Site** underwent a splenectomy. Because the spleen was the major site of destruction of his red blood cells, his anemia significantly improved after surgery. He was discharged with the recommendation to take a folate supplement daily. It was explained to Mr. Site that because the spleen plays a major role in protection against certain bacterial agents, he would require immunizations against pneumococcus, meningococcus, and *Haemophilus influenzae* type b.



Anne Niemick was found to be a compound heterozygote for mutations in the β -globin gene. On one gene, a mutation in position 6 of intron 1 converted a T to a C. The presence of this mutation, for unknown reasons, raises HbF production. The other β -globin gene had a mutation in position 110 of exon 1 (a C to T mutation). Both β -globin chains have reduced activity, but combined with the increased expression of HbF, results in a β^+ thalassemia.

BIOCHEMICAL COMMENTS



How is hemoglobin switching controlled? Although there are still many unanswered questions, some of the molecular mechanisms have been identified. The α -globin locus covers ~100 kb. The major regulatory element, HS 40, is a nuclease-sensitive region of DNA that lies 5' of the ζ gene (see Fig. 44.18). HS 40 acts as an erythroid-specific enhancer that interacts with the upstream regulatory regions of the ζ and α genes, and stimulates their transcription. The region immediately 5' of the ζ gene contains the regulatory sequences responsible for silencing ζ gene transcription. However, the exact sequences and transcription factors responsible for this silencing have not yet been identified. Even after silencing, low levels of

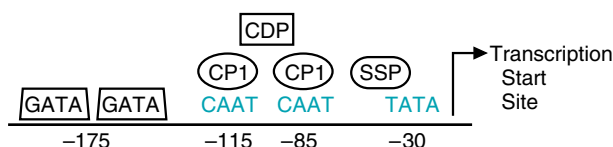


Fig. 44.19. The γ -globin gene promoter indicating some of the transcription factor binding sites associated with hereditary persistence of fetal hemoglobin.

ζ gene transcripts are still produced after the embryonic period; however, they are not translated. This is because both the ζ globin and α -globin transcripts have regions that bind to a messenger ribonucleoprotein (mRNP) stability-determining complex. Binding to this complex prevents the mRNA from being degraded. The α -globin messenger RNA has a much higher affinity for the mRNP than the ζ -globin message, which leads to the ζ -globin message being rapidly degraded.

The β -globin locus covers ~100 kb. From 5 to 25 kb upstream of the ϵ gene is the locus control region (LCR), containing five DNase hypersensitive sites. The LCR is necessary for the function of the β -globin locus. It maintains the chromatin of the entire locus in an active configuration and acts as an enhancer and entry point for the factors that transcribe the genes of the β -globin locus. One model of the control of hemoglobin switching postulates that proteins bound at the promoters of the ϵ -, γ -, and β -globin genes compete to interact with the enhancers of the LCR.

Each gene in the β -globin locus has individual regulatory elements—a promoter, silencers, or enhancers that control its developmental regulation. The promoters controlling the γ and β -globin genes have been intensively studied because of their clinical relevance.

The ϵ -globin gene, like the ζ globin gene, has silencers in the 5' regulatory region. Binding of proteins to these regions turns off the ϵ gene.

The proximal region of the γ -globin gene promoter has multiple transcription factor binding sites (Fig. 44.19). Many HPFH mutations map to these transcription factor-binding sites, either by destroying a site or by creating a new one, but the exact mechanisms are still not understood. Two sites that appear to be significant in the control of hemoglobin switching are the stage selector protein binding (SSP) site and the CAAT box region. When the SSP complex is bound to the promoter, the γ -globin gene has a competitive advantage over the β -globin promoter for interaction with the LCR. A second transcription factor, Sp1, also binds at the SSP-binding site, where it may act as a repressor, and competition between these two protein complexes for the SSP-binding site helps to determine the activity of the γ -globin gene. A similar mechanism appears to be operating at the CAAT box. CP1, thought to be a transcription activator, binds at the CAAT box. CAAT displacement protein (CDP) is a repressor that binds at the CAAT site and displaces CP1. Part of the mechanism of hemoglobin switching appears to be the binding of repressors at the ϵ -globin and γ -globin upstream regulatory regions.

The β -globin gene also has binding sites for multiple transcription factors in its regulatory regions. Mutations that affect binding of transcription factors can produce thalassemia by reducing the activity of the β -globin promoter. There is also an enhancer 3' of the poly A signal that seems to be required for stage-specific activation of the β -globin promoter.



Transgenic mice are an invaluable tool for studying the roles of transcription factors in developmental processes in general and hemoglobin switching in particular. Transgenic mice were created with mutations in the silencer regions of the ϵ -globin genes. These mice continued to express the ϵ -globin gene into adulthood.

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REVIEW QUESTIONS—CHAPTER 44

1. A compensatory mechanism to allow adequate oxygen delivery to the tissues at high altitudes, where oxygen concentrations are low, would be which of the following?
 - (A) An increase in 2,3-bisphosphoglycerate synthesis by the red cell
 - (B) A decrease in 2,3-bisphosphoglycerate synthesis by the red cell
 - (C) An increase in hemoglobin synthesis by the red cell
 - (D) A decrease in hemoglobin synthesis by the red cell
 - (E) Decreasing the blood pH
2. A 2-year-old boy of normal weight and height is brought to a clinic because of excessive fatigue. Blood work indicates an anemia, with microcytic hypochromic red cells. The boy lives in a 100-year-old apartment building and has been caught ingesting paint chips. His parents indicate that the child eats a healthy diet and takes a Flintstones vitamin supplement every day. His anemia is most likely attributable to a deficiency in which of the following?
 - (A) Iron
 - (B) B12
 - (C) Folate
 - (D) Heme
 - (E) B6
3. Drugs are being developed that will induce the transcription of globin genes, which are normally silent in patients affected with sickle cell disease. A good target gene for such therapy in this disease would be which of the following?
 - (A) The $\alpha 1$ gene
 - (B) The $\alpha 2$ gene
 - (C) The γ gene
 - (D) The β gene
 - (E) The ζ gene
4. A mature blood cell that lacks a nucleus is which of the following?
 - (A) Lymphocyte
 - (B) Basophil
 - (C) Eosinophil
 - (D) Platelet
 - (E) Neutrophil
5. A family has two children, one with a mild case of thalassemia, and a second with a severe case of thalassemia, requiring frequent blood transfusions as part of the treatment plan. One parent is of Mediterranean descent, the other is of Asian descent. Neither parent exhibits clinical signs of thalassemia. Both children express 20% of the expected level of β -globin; the more severely affected child expresses normal levels of α -globin, whereas the less severely affected child only expresses 50% of the normal levels of α -globin. Why is the child who has a deficiency in α -globin expression less severely affected?
 - (A) Thalassemia is caused by a mutation in the α gene, and the more severely affected child expresses more of it.
 - (B) The less severely affected child must be synthesizing the ζ gene to make up for the deficiency in a α chain synthesis.
 - (C) The more severely affected child also has HPFH.
 - (D) The more severely affected child produces more inactive globin tetramers than the less severely affected child.
 - (E) Thalassemia is caused by an iron deficiency, and when the child is synthesizing normal levels of α -globin there is insufficient iron to populate all of the heme molecules synthesized.

45 Blood Plasma Proteins, Coagulation and Fibrinolysis

*The blood is the body's main transport system. Although the transport and delivery of oxygen to the cells of the tissues is carried out by specialized cells, other vital components such as nutrients, metabolites, electrolytes, and hormones, are all carried in the noncellular fraction of the blood, the **plasma**. Some components, such as glucose, are dissolved in the plasma; others, for example, lipids and steroid hormones, are bound to carrier proteins for transport. The **osmotic pressure** of the plasma proteins regulates the distribution of water between the blood and the tissues. Plasma proteins in conjunction with platelets maintain the integrity of the circulatory system through the process of clotting.*

*Blood normally circulates through endothelium-lined blood vessels. When a blood vessel is severed, a **blood clot** (called a **thrombus**, which is formed by the process of **thrombosis**) forms as part of **hemostasis**, the physiologic response that stops bleeding. Blood clots also form to repair damage to the endothelial lining, in which components of the **subendothelial layer** become accessible to plasma proteins.*

*In hemostasis and thrombosis, a primary hemostatic plug, consisting of **aggregated platelets** and a **fibrin clot**, forms at the site of injury. Platelets are attached to the subendothelial layer of the vessel principally through the protein **von Willebrand factor**, and are activated to bind fibrinogen. **Fibrinogen** binds the platelets to each other to form a platelet aggregate. The platelet aggregate is rapidly covered with layers of **fibrin**, which are formed from fibrinogen by the proteolytic enzyme **thrombin**.*

*Injury to the endothelium and exposure of **tissue factor** on the **subendothelial layer** to plasma proteins also activate the blood coagulation cascade, which ultimately activates thrombin and **Factor XIII**. Factor XIII **cross-links** strands of polymerized fibrin monomers to form a stable clot (the **secondary hemostatic plug**). The blood coagulation cascade consists of a series of enzymes (such as **Factor X**), which are inactive until proteolytically cleaved by the preceding enzyme in the cascade. Other proteins (**Factor V** and **Factor VIII**) serve as binding proteins, which assemble factor complexes at the site of injury. Ca^{2+} and γ -**carboxyglutamate** residues in the proteins (formed by a vitamin K-dependent process in the liver) attach the factor complexes to **phospholipids** exposed on platelet membranes. Consequently, thrombus formation is rapidly accelerated and localized to the site of injury.*

***Regulatory mechanisms** within the blood coagulation cascade and **antifibrinolytic mechanisms** prevent random coagulation within blood vessels that might obstruct blood flow. Impairments in these mechanisms lead to thrombosis.*



THE WAITING ROOM



Sloe Klotter, a 6-month-old male infant, was brought to his pediatrician's office with a painful, expanding mass in his right upper thigh that was first noted just hours after he fell down three uncarpeted steps in his home. The child appeared to be in severe distress.

An x-ray showed no fractures, but a soft tissue swelling, consistent with a hematoma (bleeding into the tissues), was noted. Sloe's mother related that soon after he began to crawl, his knees occasionally became swollen and seemed painful.

The pediatrician suspected a disorder of coagulation. A screening coagulation profile suggested a possible deficiency of Factor VIII, a protein involved in the formation of blood clots. Sloe's plasma Factor VIII level was found to be only 3% of the average level found in normal subjects. A diagnosis of hemophilia A was made.

I. PLASMA PROTEINS MAINTAIN PROPER DISTRIBUTION OF WATER BETWEEN BLOOD AND TISSUES

When the cells are removed from the blood, the remaining plasma is composed of water, nutrients, metabolites, hormones, electrolytes, and proteins. Plasma has essentially the same electrolyte composition as other extracellular fluids and constitutes approximately 25% of the body's total extracellular fluid. The plasma proteins serve a number of functions, which include maintaining the proper distribution of water between the blood and the tissues, transporting nutrients, metabolites, and hormones throughout the body, defending against infection, and maintaining the integrity of the circulation through clotting. Many diseases alter the amounts of plasma proteins produced and, hence, their concentration in the blood. These changes can be determined by electrophoresis of plasma proteins over the course of a disease.



The hydrostatic pressure in an arteriole is the force that "pushes" fluid out of the capillary and into the interstitial spaces. The plasma protein osmotic pressure, plus the tissue pressure, is the force that "pulls" water from interstitial spaces into the venular side of the capillary. Thus, if the hydrostatic pressure is greater than the osmotic pressure, fluid will leave the circulation; if it is less, fluid will enter the circulation.



In cases of severe protein malnutrition (kwashiorkor), the concentration of the plasma proteins decreases, as a result of which the osmotic pressure of the blood decreases. As a result, fluid is not drawn back to the blood and instead accumulates in the interstitial space (edema). The distended bellies of famine victims are the result of fluid accumulation in the extravascular tissues because of the severely decreased concentration of plasma proteins, particularly albumin. Albumin synthesis decreases fairly early under conditions of protein malnutrition.

A. Body Fluid Maintenance between Tissues and Blood

As the arterial blood enters the capillaries, fluid moves from the intravascular space into the interstitial space (that surrounding the capillaries) because of what are known as Starling's forces. The hydrostatic pressure in the arteriolar end of the capillaries (~37 mm Hg) exceeds the sum of the tissue pressure (~1 mm Hg) and the osmotic pressure of the plasma proteins (~25 mm Hg). Thus, water tends to leave the capillaries and enter extravascular spaces. At the venous end of the capillaries, the hydrostatic pressure falls to approximately 17 mm Hg while the osmotic pressure and the tissue pressure remain constant, resulting in movement of fluid back from the extravascular (interstitial) spaces and into the blood. Thus, most of the force bringing water back from the tissues is the osmotic pressure mediated by the presence of proteins in the plasma.

B. The Major Serum Protein, Albumin

As indicated in Table 45.1, the liver produces a number of transport or binding proteins and secretes them into the blood. The major protein synthesized is albumin, which constitutes approximately 60% of the total plasma protein, but because of its relatively small size (69 kDa) is thought to contribute 70 to 80% of the total osmotic pressure of the plasma. Albumin, like most plasma proteins, is a glycoprotein and is a carrier of free fatty acids, calcium, zinc, steroid hormones, copper, and bilirubin.

Table 45.1. Specific Plasma Binding Proteins Synthesized in the Liver

Ceruloplasmin	Binds copper; appears to be more important as a copper storage pool than as a transport protein; integrates iron and copper homeostasis
Corticosteroid-binding globulin	Binds cortisol
Haptoglobin	Binds extracorporeal heme
Lipoproteins	Transport cholesterol and fatty acids
Retinol-binding protein	Binds vitamin A
Sex hormone-binding globulin	Binds estradiol and testosterone
Transferrin	Transports iron
Transthyretin	Binds thyroxine (T_4); also forms a complex with retinol-binding protein

Many drugs also bind to albumin, which may have important pharmacologic implications. For example, when a drug binds to albumin, such binding will likely lower the effective concentration of that drug and may lengthen its lifetime in the circulation. Drug dosimetry may need to be recalculated if a patient's plasma protein concentration is abnormal.

II. THE PLASMA CONTAINS PROTEINS THAT AID IN IMMUNE DEFENSE

Two different sets of proteins aid in the immune response, the immunoglobulins and the complement proteins. The immunoglobulins are secreted by a subset of differentiated B lymphocytes termed plasma cells and bind antigens at binding sites formed by the hypervariable regions of the proteins (see Chapter 7). Once the antibody–antigen complex is formed, it must be cleared from the circulation. The complement system participates in this function. The complement system consisting of approximately 20 proteins becomes activated in either of two ways. The first is interaction with antigen–antibody complexes, and the second, specific for bacterial infections, is through interaction of bacterial cell polysaccharides with complement protein C3b. Activation of the complement system by either trigger results in a proteolytic activation cascade of the proteins of the complement system, resulting in the release of biologically active peptides or polypeptide fragments. These peptides mediate the inflammatory response, attract phagocytic cells to the area, initiate degranulation of granulocytes, and promote clearance of antigen–antibody complexes.

Protease inhibitors in plasma serve to carefully control the inflammatory response. Activated neutrophils release lysosomal proteases from their granules that can attack elastin, the various types of collagen, and other extracellular matrix proteins. The plasma proteins α 1-antitrypsin (α 1-antitrypsin) and α 2-macroglobulin limit proteolytic damage by forming noncovalent complexes with the proteases, thereby inactivating them. However, the product of neutrophil myeloperoxidase, HOCl, inactivates the protease inhibitors, thereby insuring that the proteases are active at the site of infection.

III. PLASMA PROTEINS MAINTAIN THE INTEGRITY OF THE CIRCULATORY SYSTEM

Blood is lost from the circulation when the endothelial lining of the blood vessels is damaged and the vessel wall is partly or wholly severed. When this occurs, the subendothelial cell layer is exposed, consisting of the basement membrane of the endothelial cells and smooth muscle cells. In response to the damage, a barrier (the hemostatic plug, a fibrin clot), initiated by platelet binding to the damaged area, is formed at the site of injury. Regulatory mechanisms limit clot formation to the site of injury and control its size and stability. As the vessel heals, the clot is degraded (fibrinolysis). Plasma proteins are required for these processes to occur.



In spite of the importance of albumin in the maintenance of osmotic pressure in the blood, individuals lacking albumin (analbuminemia) have only moderate edema. Apparently, the concentration of other plasma proteins is increased to compensate for the lack of albumin. The frequency of albuminemia is less than one per million individuals.



α 1-Antitrypsin (AAP) is the main serine protease inhibitor of human plasma. Individuals with a point mutation in exon 5 of the AAP gene, which results in a single amino acid substitution in the protein, have diminished secretion of AAP from the liver. These individuals are at increased risk for developing emphysema. When neutrophils degranulate in the lungs as part of the body's defense against microorganisms, insufficient levels of AAP are present to neutralize the elastase and other proteases released. The excess proteolytic activity damages lung tissue, leading to loss of alveolar elasticity and emphysema.

Methionine 358 of AAP is necessary for AAP binding to the proteases. Oxidation of this methionine destroys AAP's protease-binding capacity. Cigarette smoke oxidizes Met-358, thereby increasing the risk for emphysema.



Platelets are derived from megakaryocytes in the bone marrow. Megakaryocytes differentiate from the hematopoietic stem cell. As the megakaryocyte matures, it undergoes synchronous nuclear replication without cellular division, to form a cell with a multilobed nucleus and enlarged cytoplasm. At approximately the 8-nucleus stage, the cytoplasm becomes granular, and the platelets are budded off the cytoplasm. A single megakaryocyte gives rise to approximately 4,000 platelets.



Idiopathic thrombocytopenic purpura (ITP) is an autoimmune disease in which antibodies to platelet glycoproteins are produced. Antibody binding to platelets results in their clearance by the spleen. An early symptom of the disorder is the appearance of small red spots on the skin (petechial hemorrhages) caused by blood leakage from capillaries. Minor damage to vascular endothelial cells is constantly being caused by mechanical forces related to blood flow. In patients with ITP, few platelets are available to repair the damage.



Von Willebrand factor is a large multimeric glycoprotein with a subunit molecular weight of 220,000 daltons. Its size in the circulation ranges between 500 and 20,000 kDa, and its role in circulation is to stabilize Factor VIII and protect it from degradation. The high-molecular-weight forms are concentrated in the endothelium of blood vessels and are released in response to stress hormones and endothelial damage. High-molecular-weight vWF released by the endothelium is cleaved by a specific metalloprotease in the serum, reducing the size of the circulating vWF. Large vWF multimers are more effective at forming complexes with platelets than are small vWF multimers.

A. Formation of the Hemostatic Plug

1. THE PLATELET

Platelets are non-nucleated cells present in the blood whose major role is to form mechanical plugs at the site of vessel injury and to secrete regulators of the clotting process and vascular repair. In the absence of platelets, leakage of blood from vents in small vessels is common.

In the nonactivated platelet (a platelet not involved in forming a clot), the plasma membrane invaginates extensively into the interior of the cell, forming an open membrane (canalicular) system. Because the plasma membrane contains receptors and phospholipids that accelerate the clotting process, the canalicular structure substantially increases the membrane surface area potentially available for clotting reactions. The platelet interior contains microfilaments and an extensive actin/myosin system. Platelet activation in response to endothelial injury causes Ca^{2+} -dependent changes in the contractile elements, which, in turn, substantially change the architecture of the plasma membrane. Long pseudopodia are generated, increasing the surface area of the membrane as clot formation is initiated.

Platelets contain three types of granules. The first are electron-dense granules, which contain calcium, adenosine diphosphate (ADP), adenosine triphosphate (ATP), and serotonin. The second type of granule is the α granule, which contains a heparin antagonist (heparin interferes with blood clotting; see biochemical comments), platelet-derived growth factor, β -thromboglobulin, fibrinogen, von Willebrand factor (vWF), and other clotting factors. The third type of granule is the lysosomal granule, which contains hydrolytic enzymes. During activation, the contents of these granules, which modulate platelet aggregation and clotting, are secreted.

2. PLATELET ACTIVATION

Three fundamental mechanisms are involved in platelet function during blood coagulation: adhesion, aggregation, and secretion. Adhesion sets off a series of reactions termed platelet activation, which leads to platelet aggregation and secretion of platelet granule contents.

The adhesion step refers primarily to the platelet–subendothelial interaction that occurs when platelets initially adhere to the sites of blood vessel injury (Fig. 45.1). Blood vessel injury exposes collagen, subendothelial matrix-bound vWF, and other matrix components. vWF is a protein synthesized in endothelial cells and megakaryocytes, and is located in the subendothelial matrix, in specific platelet granules, and in the circulation bound to Factor VIII. The platelet cell membrane contains glycoproteins (GPs) that bind to collagen and to vWF, causing the platelet

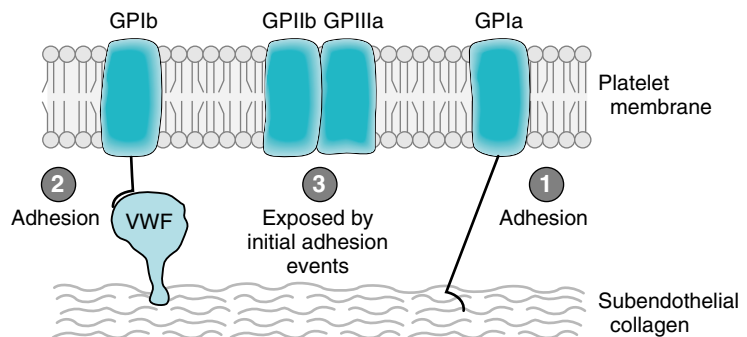


Fig. 45.1. Adhesion of platelets to the subendothelial cell layer. 1. GPIa initially binds to the exposed collagen, which results in changes in the three-dimensional configuration of the complex, allowing GPIb to bind to vWF (2). 3. This second binding event exposes the GPIIb/GPIIIa complex, which also can bind to vWF and fibrinogen.

to adhere to the subendothelium. Binding to collagen by GPIa (integrin $\alpha 2\beta 1$) causes the platelet to change its shape from a flat disc to a spherical cell. The latter extrudes long pseudopods, which promote platelet/platelet interactions. Binding of subendothelial vWF by GPIb causes changes in the platelet membrane that expose GPIIb/IIIa (integrin $\alpha \text{IIb}\beta 3$) binding sites to fibrinogen and vWF.

The initial adherence of platelets sets off a series of reactions (platelet activation) that results in more platelets being recruited and aggregated at the site of injury. After initial adherence, some of the platelets release the contents of their dense granules and their α granules, with ADP release being of particular importance because ADP is a potent platelet activator. ADP released from the platelets and from damaged red blood cells binds to a receptor on the platelet membrane, which leads to the further unmasking of GPIIb/IIIa binding sites. Aggregation of platelets cannot take place without ADP stimulation, because ADP induces swelling of the activated platelets, promoting platelet/platelet contact and adherence.

Fibrinogen is a protein that circulates in the blood and is also found in platelet granules. It consists of two triple helices held together with disulfide bonds. (Fig.45.2) Binding of fibrinogen to activated platelets is necessary for aggregation,



vWF deficiency is the most common cause of inherited bleeding disorders. Both platelet adherence and the clotting cascade are affected because levels of Factor VIII are low. In the absence of vWF, Factor VIII is rapidly cleared from the system. The vWF gene is large, covering approximately 180 kb, and contains 52 exons. Multiple mutations are known, with varying clinical presentations.



Defects in GPIb cause a bleeding disorder known as Bernard-Soulier syndrome. Platelet aggregation is affected, because of the inability of GPIb to adhere to subendothelial vWF.

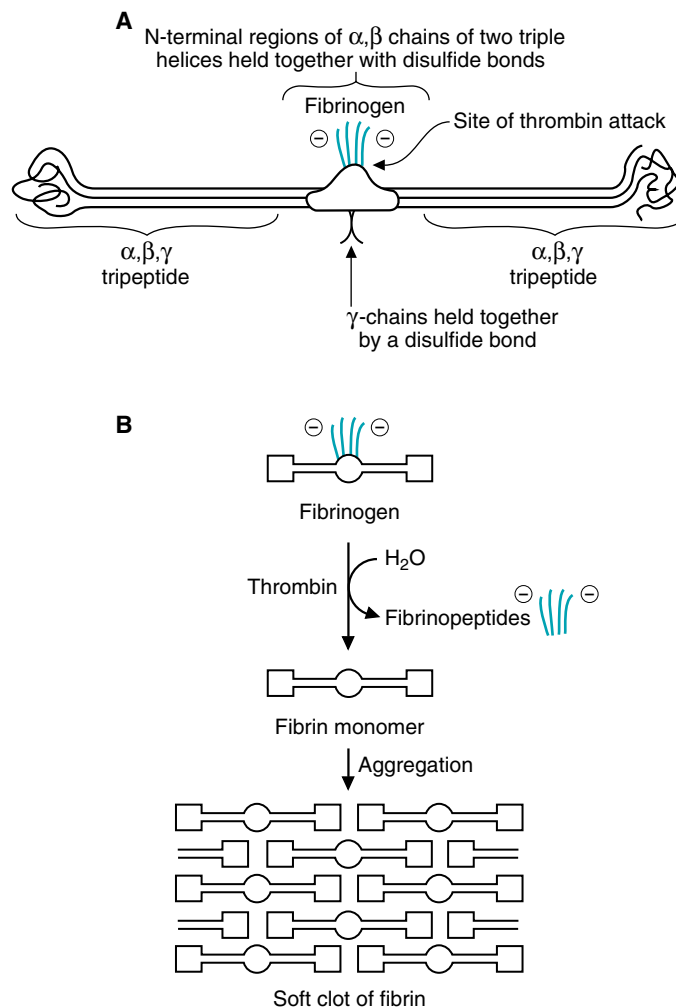


Fig. 45.2. Cleavage of fibrinogen results in clot formation. A. Fibrinogen, the precursor protein of fibrin, is formed from two triple helices joined together at their N-terminal ends. The α , β peptides are held together by disulfide bonds, and the γ -peptides are joined to each other by disulfide bonds. The terminal α , β peptide regions, shown in blue, contain negatively charged glutamate and aspartate residues that repel each other and prevent aggregation. B. Thrombin, a serine protease, cleaves the terminal portions of fibrinogen containing negative charges. The fibrin monomers can then aggregate and form a “soft” clot. The soft clot is then subsequently cross-linked by another enzyme.



Thrombotic thrombocytopenic purpura (TTP) is a disease characterized by the formation in the circulation of microclots (microthrombi) consisting of aggregated platelets. The microthrombi collect in the microvasculature and damage red blood cells, resulting in hemolytic anemia. They also damage vascular endothelium, exposing collagen and releasing high-molecular-weight vWF, promoting more platelet aggregation. The subsequent depletion of platelets renders the patient susceptible to internal hemorrhage. Mortality in untreated TTP can approach 90%.

Familial TTP is associated with mutations in the vWF-specific metalloprotease, although not all individuals with defective protease develop TTP. Sporadic cases of TTP are associated with the development of an antibody to the metalloprotease.



In response to collagen and thrombin, platelets release vasoconstrictors. Serotonin is released from the dense granules of the platelets, and the synthesis of thromboxane A₂ is stimulated. This will reduce blood flow to the damaged area. Platelet-derived growth factor, which stimulates proliferation of vascular cells, is also released into the environment surrounding the damage.



The utilization of an active site serine to cleave a peptide bond is common to a variety of enzymes referred to as serine proteases. Serine proteases are essential for activating the formation of a blood clot from fibrin. Fibrin and many of the other proteins involved in blood coagulation are present in the blood as inactive precursors or zymogens, which must be activated by proteolytic cleavage. Thrombin, the serine protease that converts fibrinogen to fibrin, has the same aspartate-histidine-serine catalytic triad found in chymotrypsin and trypsin.

Thrombin is activated by proteolytic cleavage of its precursor protein, prothrombin. The sequence of proteolytic cleavages leading to thrombin activation requires Factor VIII, the blood-clotting protein deficient in **Sloe Klotter**.

providing, in part, the mechanism by which platelets adhere to one another. Cleavage of fibrinogen by thrombin (a protease that is activated by the coagulation cascade) produces fibrin monomers that polymerize and, together with platelets, form a “soft clot”. Thrombin itself is a potent activator of platelets, through binding to a specific receptor on the platelet surface.

B. The Blood Coagulation Cascade

Thrombus (clot) formation is enhanced by thrombin activation, which is mediated by the complex interaction that constitutes the blood coagulation cascade. This cascade (Fig. 45.3) consists primarily of proteins that serve as enzymes or cofactors, which function to accelerate thrombin formation and localize it at the site of injury. These proteins are listed in Table 45.2. All of these proteins are present in the plasma as pro-proteins (zymogens). These precursor proteins are activated by cleavage of the polypeptide chain at one or more sites. The key to successful and appropriate thrombus formation is the regulation of the proteases that activate these zymogens.

The proenzymes (Factors VII, XI, IX, X, XII, and prothrombin) are serine proteases that, when activated by cleavage, cleave the next proenzyme in the cascade. Because of the sequential activation, a great acceleration and amplification of the response is achieved. That cleavage and activation have occurred is indicated by the addition of an “a” to the name of the proenzyme (e.g., Factor IX is cleaved to form the active Factor IXa).

The cofactor proteins (tissue factor, Factors V and VIII) serve as binding sites for other factors. Tissue factor is not related structurally to the other blood coagulation cofactors and is an integral membrane protein that does not require cleavage for active function. Factors V and VIII serve as procofactors, which, when activated by cleavage, function as binding sites for other factors.

Two additional proteins that are considered part of the blood coagulation cascade, protein S and protein C, are regulatory proteins. Only protein C is regulated by proteolytic cleavage, and when activated, is itself a serine protease.

C. The Process of Blood Coagulation

Activation of the blood coagulation cascade is triggered by the reaction of plasma proteins with the subendothelium at the same time that platelets are adhering to the subendothelial layer. Historically, two different pathways were discovered, one dependent on external stimuli (such as blunt trauma, which initiates the extrinsic pathway) and one using internal stimuli (the intrinsic pathway). As our understanding of blood clotting has expanded, it has become obvious that these distinctions are no longer correct, because there is overlap between the pathways, but the terms have persisted in the description of the pathways.

In the case of external trauma, damaged tissues present tissue factor to the blood at the injured site, thereby triggering the extrinsic phase of blood coagulation. Circulating Factor VII binds to tissue factor, which autocatalyzes its own activation to Factor VIIa. Factor VIIa then activates Factor X (to Xa) in the extrinsic pathway and Factor IX (to IXa) in the intrinsic pathway. Factor IXa, as part of the intrinsic pathway, also activates Factor X. Therefore, activation of both the extrinsic and intrinsic pathways result in the conversion of Factor X to Factor Xa. All of these conversions require access to membranes and calcium; the platelet membrane, which had adhered to the damaged site, is used as a scaffold for the activation reactions to occur. The γ -carboxylated clotting proteins are chelated to membrane surfaces via electrostatic interactions with calcium and negatively charged phospholipids of the platelet membrane. The protein cofactors VIIIa and Va serve as sites for assembling enzyme-cofactor complexes on the platelet surface, thereby accelerating and localizing the reaction. The result is thrombin formation, which augments its own formation by converting Factors V, VIII, and XI into activated cofactors and

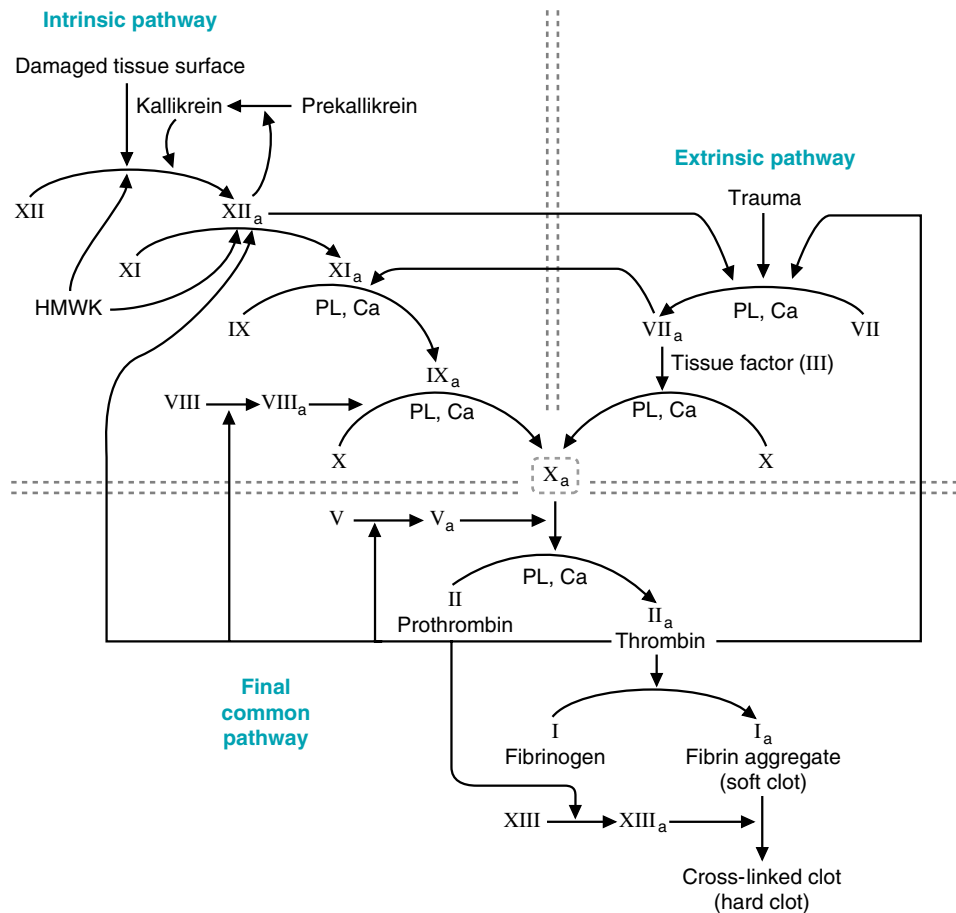


Fig. 45.3. Blood coagulation cascade. Activation of clot formation occurs through two separate but interlocking pathways, termed the intrinsic and extrinsic pathways. The intrinsic pathway is activated when plasma proteins react with the exposed subendothelium of the damaged blood vessel. Platelets and vWF bind to the exposed subendothelium, and the platelets, in turn, bind fibrinogen. The extrinsic pathway is activated by tissue factor. The reactions designated by "PL, Ca" are occurring through cofactors bound to phospholipids (PL) on the cell surface in a Ca^{2+} -coordination complex. Factors XII_a, XI_a, IX_a, VII_a, X_a, and thrombin are serine proteases. Note the positive feedback regulation of thrombin on the activation of proteases earlier in the cascade sequence. HMWK = high-molecular-weight kininogen.

Table 45.2. Proteins of Blood Coagulation

Coagulation Factors		
Factor	Descriptive Name	Function/Active Form
I	Fibrinogen	Fibrin
II	Prothrombin	Serine protease
III	Tissue factor	Receptor and cofactor
IV	Ca^{+2}	Cofactor
V	Proaccelerin, labile factor	Cofactor
VII	Proconvertin	Serine protease
VIII	Antihemophilia factor A	Cofactor
IX	Antihemophilia factor B, Christmas factor	Serine protease
X	Stuart-Prower factor	Serine protease
XI	Plasma thromboplastin antecedent	Serine protease
XII	Hageman (contact) factor	Serine protease
XIII	Fibrin stabilizing factor	Ca^{2+} -dependent transglutaminase
	Prekallikrein	Serine protease
	High-molecular-weight kininogen	Cofactor
Regulatory Proteins		
	Thrombomodulin	Endothelial cell receptor, binds thrombin
	Protein C	Activated by thrombomodulin-bound thrombin; is a serine protease
	Protein S	cofactor; binds activated protein C



The initial activation of prothrombin to thrombin is slow, because the activator cofactors, Factors VIIIa and Va, are only present in small amounts. However, once a small amount of thrombin is activated, it will accelerate its own production by cleaving Factors V and VIII to their active forms.

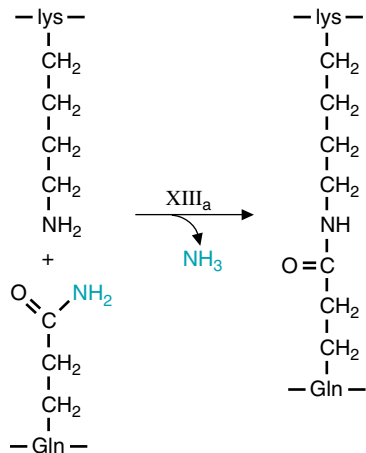


Fig. 45.4. The transamidation reaction catalyzed by Factor XIIIa, transglutaminase. This reaction cross-links fibrin monomers, allowing “hard” clots to form.

stimulating platelet degranulation. Note that these factors are in the intrinsic pathway. The intrinsic pathway is thought to sustain the coagulation response initiated by the extrinsic pathway. The major substrate of thrombin is fibrinogen, which is hydrolyzed to form fibrin monomers that undergo spontaneous polymerization to form the fibrin clot. This is considered a “soft” clot because the fibrin monomers are not cross-linked. Cross-linking requires Factor XIIIa, which is activated by thrombin cleavage of Factor XIII.

1. CROSS-LINKING OF FIBRIN

Factor XIIIa catalyzes a transamidation reaction between Gln and Lys side chains on adjacent fibrin monomers. The covalent cross-linking takes place in three dimensions, creating a strong network of fibers resistant to mechanical and proteolytic damage. This network of fibrin fibers traps the aggregated platelets and other cells, forming the clot that plugs the vent in the vascular wall. (Fig. 45.4) Factor XIIIa is the only enzyme in the blood coagulation cascade that is not a serine protease.

2. KALLIKREIN AND HIGH-MOLECULAR-WEIGHT KININOGEN (HMWK)

The classical intrinsic pathway begins with the assembly of prekallikrein, high-molecular-weight kininogen (HMWK), Factor XII, and Factor XI on a negatively charged surface, presumably an endothelial cell in vivo (see Fig. 45.3). High-molecular-weight kininogen is a glycoprotein that binds prekallikrein and aids in its assembly on the endothelial cell. Prekallikrein is the zymogen form of a serine protease. Factor XII autoactivates, forming Factor XIIa, which converts prekallikrein to kallikrein. Kallikrein then enhances the activation of Factor XII, which leads to the activation of Factors XI and VII.

How important these steps are in the initiation of the coagulation cascade is unknown. Individuals lacking HMWK, prekallikrein, or Factor XII do not suffer from bleeding disorders. Under usual conditions, activation of Factor VII with subsequent activation of Factors IX and X is thought to be sufficient to activate the coagulation pathway.

3. FACTOR COMPLEXES

In several of the essential steps in the blood coagulation cascade, the activated protease is bound in a complex attached to the surface of the platelets that have aggregated at the site of injury. Factors VII, IX, X, and prothrombin contain a domain in which one or more glutamate residues are carboxylated to γ -carboxyglutamate in a reaction requiring vitamin K (Fig 45.5). Prothrombin and Factor X both contain 10 or more γ -carboxyglutamate residues that bind Ca^{2+} . Ca^{2+} forms a coordination complex with the negatively charged platelet membrane phospholipids and the γ -carboxyglutamates, thereby localizing the complex assembly and thrombin formation to the platelet surface.

Cofactor Va contains a binding site for both Factor Xa and prothrombin, the zymogen substrate of Factor Xa. On binding to the Factor Va–platelet complex, prothrombin undergoes a conformational change, rendering it more susceptible to enzymatic cleavage. Binding of Factor Xa to the Factor Va–prothrombin–platelet complex allows the prothrombin-to-thrombin conversion. Complex assembly accelerates the rate of this conversion 10,000- to 15,000-fold as compared with non-complex formation.

Factor VIIIa forms a similar type of complex on the surface of activated platelets, but binds Factor IXa and its zymogen substrate, Factor X. Tissue factor works slightly differently because it is an integral membrane protein. However, once exposed by injury, it also binds Factor VIIa and initiates complex formation.

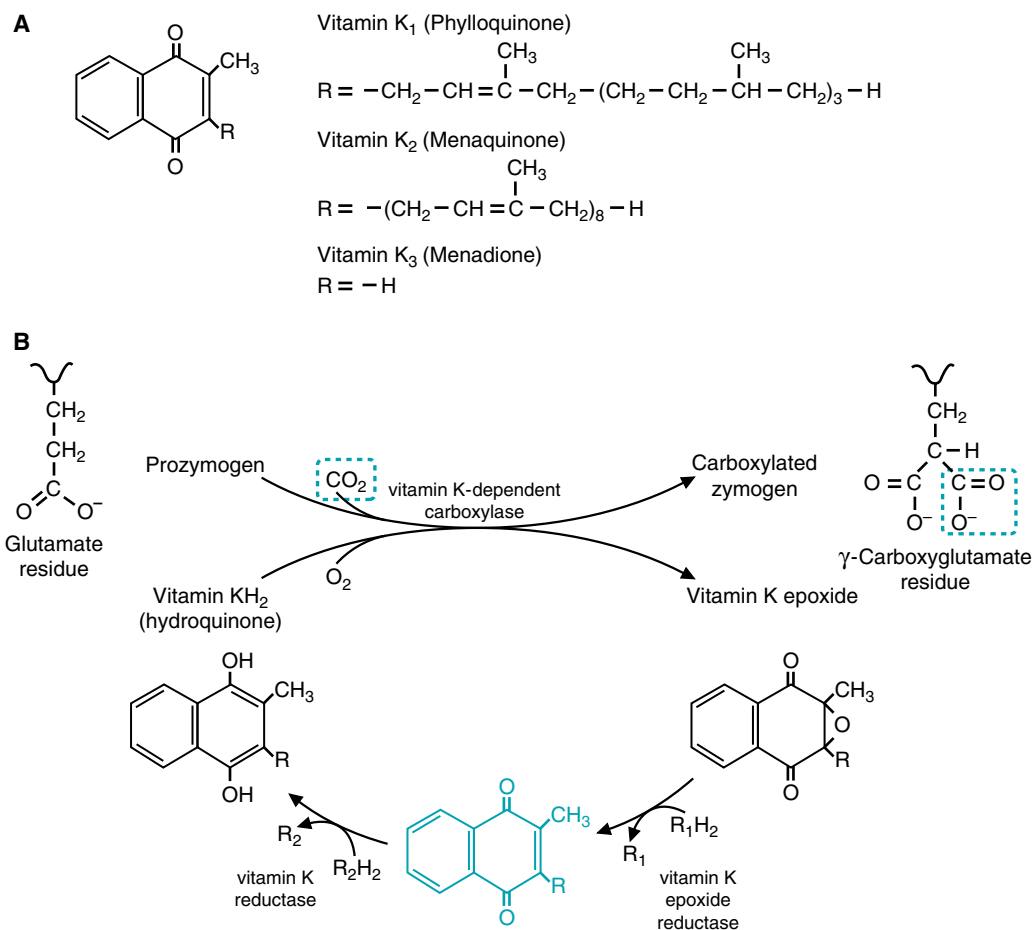


Fig. 45.5. A. Structures of vitamin K derivatives. Phylloquinone is found in green leaves, and intestinal bacteria synthesize menaquinone. Humans will convert menadione to a vitamin K active form. B. Vitamin K-dependent formation of γ -carboxyglutamate residues. Thrombin, Factor VII, Factor IX, and Factor X are bound to their phospholipid activation sites on cell membranes by Ca^{2+} . The vitamin K-dependent carboxylase, which adds the extra carboxyl group, uses a reduced form of vitamin K (KH₂) as the electron donor and converts vitamin K to an epoxide. Vitamin K epoxide is reduced, in two steps, back to its active form by the enzymes vitamin K epoxide reductase and vitamin K reductase.

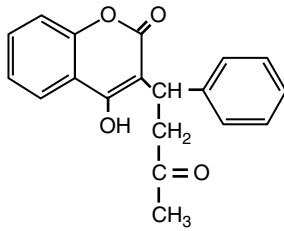
Complex assembly has two physiologically important consequences. First, it enhances the rate of thrombin formation by as much as several hundred thousand-fold, enabling the clot to form rapidly enough to preserve hemostasis. Secondly, such explosive thrombin formation is localized to the site of vascular injury at which the negatively charged phospholipids are exposed. From our knowledge of the location of such phospholipids in cellular and subcellular organelle membranes, these surface-binding sites are only exposed at an injury site in which cell rupture exposes the internal membrane surface (recall that certain phospholipids only face the cytoplasm; if these lipids are now exposed to the environment, substantial cell damage must have occurred).

4. VITAMIN K REQUIREMENT FOR BLOOD COAGULATION

The formation of the γ -carboxyglutamate residues on blood coagulation factors takes place in the hepatocyte before release of the protein. Within the hepatocyte, vitamin K (which is present in the quinone form) is reduced to form vitamin KH₂ by a microsomal quinone reductase (see Fig.45.5). Vitamin KH₂ is a cofactor for carboxylases that add a carboxyl group to the appropriate glutamate residues in the proenzyme to form the carboxylated proenzyme (e.g., prothrombin). In the same



Warfarin (Coumadin®) is a slow- and long-acting blood anticoagulant with a structure resembling that of vitamin K. The structural similarity allows the compound to compete with vitamin K and prevent γ -carboxylation of glutamate residues in Factors II, VII, IX, X, and proteins C and S. The noncarboxylated blood clotting protein precursors increase in both the blood and plasma, but they are unable to promote blood coagulation because they cannot bind calcium and thus cannot bind to their phospholipid membrane sites of activation.



Warfarin

Warfarin is a commonly used rat poison and thus is occasionally encountered in emergency departments in cases of accidental poisoning. It is effective as a rat poison because it takes many hours to develop pathologic symptoms, which allows one poisoned trap to kill more than one rat.



Deficiency in the amount or functionality of protein C or protein S increases the risk for venous thromboembolism. Homozygous individuals for these mutations do not survive the neonatal period unless given replacement therapy.



In European populations, a point mutation in the Factor V gene (Factor V Leiden) causes the replacement of an arg with a gln in the preferred site for cleavage by activated protein C, rendering Factor Va Leiden resistant to APC. Heterozygous individuals have a sixfold to eightfold increased risk of deep vein thromboses, and homozygous individuals have a 30- to 140-fold increased risk. The Factor V Leiden mutation does not appear to be associated with increased risk for arterial thrombosis, such as myocardial infarction, except in young women who smoke.

Genetic studies suggest that the Factor V Leiden mutation arose after the separation of the European, Asian, and African populations. The frequency of this variant indicates that it conferred some selective advantage at one time. In the developed world, inherited APC resistance is the most prevalent risk factor for familial thrombotic disease.

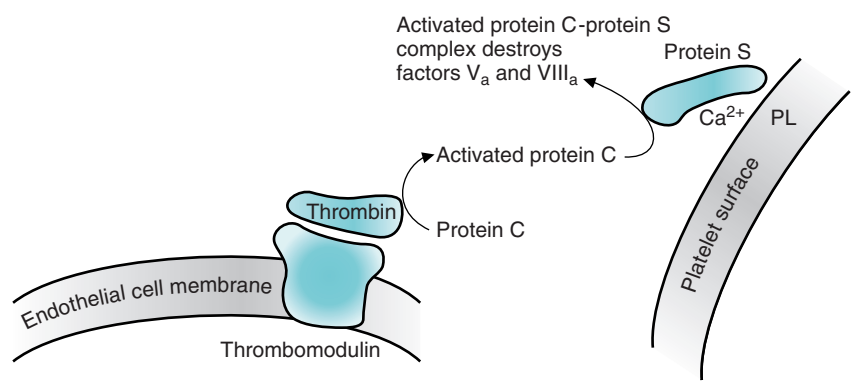


Fig. 45.6. Anti-thrombotic effects of thrombin. Thrombin, bound to thrombomodulin on the endothelial cell surface, activates protein C. Activated protein C, in complex with protein S, binds to the platelet membrane, and the activated complex destroys Factors Va and VIIIa, thereby inhibiting the coagulation cascade.

reaction, vitamin K is converted to vitamin K epoxide. To recover active vitamin KH_2 , vitamin K is first reduced to the quinone form by vitamin K epoxide reductase, and then to the active hydroquinone form.

D. Regulation through Feedback Amplification and Inhibition

Once the formation of the clot (thrombus) begins, clot formation is accelerated in an almost explosive manner by a number of processes collectively termed feedback amplification.

1. THE ROLE OF THROMBIN IN REGULATION

Thrombin has both a prothrombotic regulatory role (feedback amplification) and an antithrombotic regulatory role (feedback inhibition). The prothrombotic action is initiated when thrombin stimulates its own formation by activating Factors V, VIII, and XI, thereby accelerating the rate of clot formation (see Fig. 45.3). Thrombin also promotes clot formation by activating platelet aggregation, stimulating the release of Factor VIII from vWF, and cleaving Factor XIII to Factor XIIIa.

Antithrombotic effects of thrombin result from its binding to an endothelial cell receptor called thrombomodulin (Fig. 45.6). Thrombomodulin abolishes the clotting function of thrombin and allows thrombin to activate protein C, which has anti-coagulant effects.

2. PROTEINS S AND C

Protein C and its cofactor protein S serve to suppress the activity of the coagulation cascade. After activation, protein C forms a complex with protein S. Protein S anchors the activated protein C complex (APC) to the clot through Ca^{++}/γ -carboxyglutamate binding to platelet phospholipids. The APC destroys the active blood coagulation cofactors Factor VIIIa and Factor Va by proteolytic cleavage, decreasing the production of thrombin. The APC also stimulates endothelial cells to increase secretion of the prostaglandin PGI_2 , which reduces platelet aggregation.

3. SERPINS

Many proteases of the blood coagulation enzyme system are serine proteases. Because uncontrolled proteolytic activity would be destructive, modulating

mechanisms control and limit intravascular proteolysis. The serpins (**serine protease inhibitors**) are a group of naturally occurring inhibitory proteins present in the plasma at high concentration (approximately 10% of the plasma proteins are serpins). Eight major inhibitors have been found that share a common mechanism of action and inhibit proteases involved in coagulation and clot dissolution (fibrinolysis). Each inhibitor possesses a reactive site that appears to be an ideal substrate for a specific serine protease and thus acts as a trap for that protease. The bound serine protease attacks a peptide bond located at a critical amino acid residue within the serpin and forms a tight enzyme–inhibitor complex.

The activity of thrombin is controlled by the serpin antithrombin III (ATIII). Regulation of blood coagulation at the level of thrombin is critical because this enzyme affects the pathways of both coagulation and fibrinolysis (see section F). One molecule of ATIII irreversibly inactivates one molecule of thrombin through reaction of an arginine residue in ATIII with the active-site serine residue of thrombin. ATIII–thrombin complex formation is markedly enhanced in the presence of heparin. Heparin is a glycosaminoglycan (see Chapter 49) found in the secretory granules of mast cells and in the loose connective tissue around small vascular beds. Heparin binds to lysyl residues on ATIII and dramatically accelerates its rate of binding to thrombin. This is because of an allosteric alteration in ATIII such that the position of the critical arginine residue of ATIII is more readily available for interaction with thrombin. The formation of the ATIII–thrombin complex releases the heparin molecule so that it can be reused, and therefore, the function of heparin is catalytic. Thrombin that is attached to a surface, for example, to thrombomodulin on the endothelial cell membrane, is no longer participating in clot formation and is not readily attacked by ATIII or the ATIII–heparin complex. The ATIII–heparin complex also can inactivate the serine protease Factors XIIIa, XIa, IXa and Xa, but has no effect on Factor VIIa or activated protein C.

E. Thromboresistance of Vascular Endothelium

Endothelial cells of blood vessels provide a selectively permeable barrier between the circulating blood and the tissues. The normal endothelial cell lining neither activates coagulation nor supports platelet adhesion; thus, it is called a nonthrombogenic surface. The thromboresistance of normal vascular endothelium is contributed to by several properties. Endothelial cells are highly negatively charged, a feature that may repel the negatively charged platelets. Endothelial cells synthesize prostaglandin I_2 (PGI_2) and nitric oxide, vasodilators and powerful inhibitors of platelet aggregation. PGI_2 synthesis is stimulated by thrombin, epinephrine, and local vascular injury. Endothelial cells also synthesize two cofactors that each inhibit the action of thrombin, thrombomodulin and heparan sulfate. Heparan sulfate is a glycosaminoglycan similar to heparin that potentiates antithrombin III, but not as efficiently. The inactivation of thrombin is accelerated by heparan sulfate present on the endothelial cell surface. Thus, the intact endothelium has the capability of modifying thrombin action and inhibiting platelet aggregation.

F. Fibrinolysis

After successful formation of a hemostatic plug, further propagation of the clot must be prevented. This is accomplished in part by switching off blood coagulation and in part by turning on fibrinolysis. Fibrinolysis involves the degradation of fibrin in a clot by plasmin, a serine protease that is formed from its zymogen, plasminogen. Plasminogen is a circulating serum protein that has a high affinity for fibrin, promoting the incorporation of plasminogen in the developing clot. The activity of plasminogen is mediated by proteins known as plasminogen activators. The conversion of plasminogen to plasmin by plasminogen activators can occur both in the liquid phase of the blood and at the clot surface; however, the latter process is by

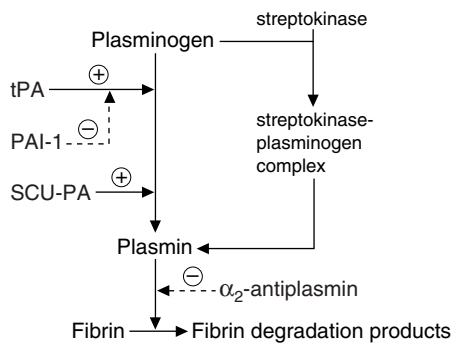


Fig. 45.7. Regulation of plasmin activation. Plasminogen can be activated by either t-PA or scu-PA (+). PAI-1 blocks t-PA action (-). Streptokinase binding to plasminogen allows autocatalysis to form plasmin. Circulating α_2 -antiplasmin blocks (-) the activity of any soluble plasmin that may be in the blood.



Both streptokinase and t-PA have been approved for the treatment of myocardial infarction. Both reduce mortality. Although there are more side effects associated with the use of streptokinase, it is substantially cheaper than t-PA.

far more efficient. Activated protein C (APC), in addition to turning off the blood coagulation cascade, also stimulates the release of plasminogen activator from tissues (t-PA, tissue plasminogen activator) and simultaneously inactivates an inhibitor of plasminogen activator, PAI-1.

Plasminogen activator release can lead to plasmin formation in the circulation. However, the circulating plasmin is rapidly inactivated by binding with α_2 -antiplasmin, a circulating protease inhibitor. Clot-bound plasmin is not readily inactivated by α -antiplasmin. Thus, plasminogen binding to fibrin facilitates its activation to plasmin, protects it from blood serpins, and localizes it on the fibrin substrate for subsequent efficient proteolytic attack. This mechanism allows for dissolution of fibrin in pathologic thrombi or oversized hemostatic plugs, and at the same time prevents degradation of fibrinogen in the circulating blood.

Two endogenous plasminogen activators are most important; both are synthesized in a variety of cells. Tissue plasminogen activator (t-PA) is chiefly produced by the vascular endothelial cells, has a high binding affinity for fibrin, and plays a major role in fibrinolysis. Single-chain urokinase (scu-PA), is synthesized in most cells and tissues and has a moderate affinity for fibrin. Streptokinase, the bacterial exogenous plasminogen activator from β -hemolytic streptococci, is not an enzyme but an allosteric modifier of human plasminogen that allows an autocatalytic reaction such that plasmin is formed. In vivo, physical stress, hypoxia, and large numbers of low-molecular-weight organic compounds promote increased synthesis and release of t-PA and scu-PA from tissues into the blood. The balance between release of plasminogen activators, the availability of fibrin, and inhibitors of the activators and plasmin determines regulation of the fibrinolytic response, as indicated in Figure 45.7.

G. Regulation of Fibrinolysis

Anti-activators regulate interaction of plasminogen in blood with plasminogen activators in a dynamic equilibrium. Even if minute amounts of plasmin are generated (e.g., after release of vascular plasminogen activator after stress), the enzyme is probably inactivated by antiplasmin. On activation of the blood coagulation system, a fibrin clot is formed, which not only strongly binds t-PA and plasminogen from blood but also accelerates the rate of plasmin activation. The clot-bound plasmin is protected from inhibitors while attached to fibrin. The enzyme is inactivated by α_2 -antiplasmin and α_2 -macroglobulin after proteolytic dissolution of fibrin and its liberation into the liquid phase of blood. Thus, the fibrin network catalyzes both initiation and regulation of fibrinolysis.

CLINICAL COMMENTS



Sloe Klotter has hemophilia A, the most frequently encountered serious disorder of blood coagulation in humans, occurring in 1 in every 10,000 males. The disease is transmitted with an X-linked pattern of inheritance.

The most common manifestations of hemophilia A are those caused by bleeding into soft tissues (hematomas) such as muscle or into body spaces such as the peritoneal cavity or the lumen of the gastrointestinal tract. When bleeding occurs repeatedly into joints (hemarthrosis), the joint may eventually become deformed and immobile.

In the past, bleeding episodes have been managed primarily by administration of Factor VIII, sometimes referred to as antihemophilia cofactor. Unfortunately, the concentration of Factor VIII in plasma is quite low (0.3 nM compared with 8,800 nM for fibrinogen), requiring that it be prepared from multiple human donors. Before donor screening and virus inactivation procedures during preparation

essentially eliminated transmission with blood transfusions, more than 50% of hemophiliac patients treated with Factor VIII during the 1980s in Western Europe or North America became infected with HIV. Recombinant Factor VIII is now available for clinical use.



Another X-linked bleeding disorder is hemophilia B, which is caused by mutations in the gene for Factor IX. Lack of Factor IX activity leads to an inability to convert prothrombin to thrombin, and impaired clotting.

BIOCHEMICAL COMMENTS



A number of drugs have been developed that inhibit the blood coagulation cascade. Such drugs are useful in cases in which patients develop spontaneous thrombi, which, if left untreated, would result in a fatal pulmonary embolism. There are three major classes of such drugs; the heparins, vitamin K antagonists, and specific inhibitors of thrombin.

Heparin will bind to and activate ATIII, which leads to thrombin inactivation. ATIII also blocks the activity of Factors VIIIa, IXa, Xa, and XIa. Heparin can be administered in either of two forms: unfractionated, or high-molecular-weight (HMW) heparin, and fractionated, or low-molecular-weight (LMW) heparin. HMW heparin is a heterogeneous mixture of glycosaminoglycans, with an average chain length of 45 monosaccharides with an average molecular weight of 15 kDa (the range is 3–30 kDa). LMW heparins are fragments of HMW heparin, containing fewer than 18 monosaccharides with an average molecular weight of 4 to 5 kDa.

HMW heparin will bind to plasma proteins and cell surfaces in addition to its prime target, ATIII. Because different individuals synthesize different levels of plasma proteins, the use of this form of heparin as an anticoagulant requires constant monitoring of the patient to ensure that the correct dosage has been given such that spontaneous thrombi do not develop, but not so much that spontaneous bleeding occurs. LMW heparin has fewer nonspecific interactions than HMW heparin, and its effects are easier to predict on patients, so that constant monitoring is not required.

A major complication of heparin therapy is heparin-induced thrombocytopenia (HIT, excessive blood clotting with a reduction in the number of circulating platelets). This unexpected result of heparin treatment is caused by heparin binding to a platelet protein, platelet factor 4 (PF4), which induces a conformational change in PF4 such that the immune system believes the complex is foreign. Thus, antibodies are developed against the heparin–PF4 complex. When the antibodies bind to the platelets, the platelets become activated, and thrombi develop. Treatment consists of removing the heparin and using a different form of anti-thrombotic agent.

The classic vitamin K antagonist is warfarin. Warfarin acts by blocking the vitamin K reductase enzymes required to regenerate active vitamin K (see Fig. 45.5). This results in reduced γ -carboxylation of Factors II, VII, IX, and X. In the absence of γ -carboxylation, the factors cannot bind calcium nor form the complexes necessary for the coagulation cascade to be initiated. However, warfarin also blocks the activity of proteins S and C, so both blood clotting and the regulation of clotting are impaired by warfarin administration.

Both heparin and warfarin therapy suffer from their lack of specificity, so drugs specific for single steps in the blood coagulation pathway have been sought and identified. Analysis of heparin potentiation of Factor Xa binding to ATIII showed that a unique pentasaccharide sequence was required. An appropriate pentasaccharide, named fondaparinux, was developed that would specifically enhance ATIII interactions with Factor Xa (Fig. 45.8). Fondaparinux stimulates the binding of ATIII to Factor Xa by 300-fold and is specific for Factor Xa inhibition. Fondaparinux does not affect thrombin or platelet activity, and it is not an activating agent of platelets. Because fondaparinux does not bind to PF4, HIT is not a complication with this therapy.

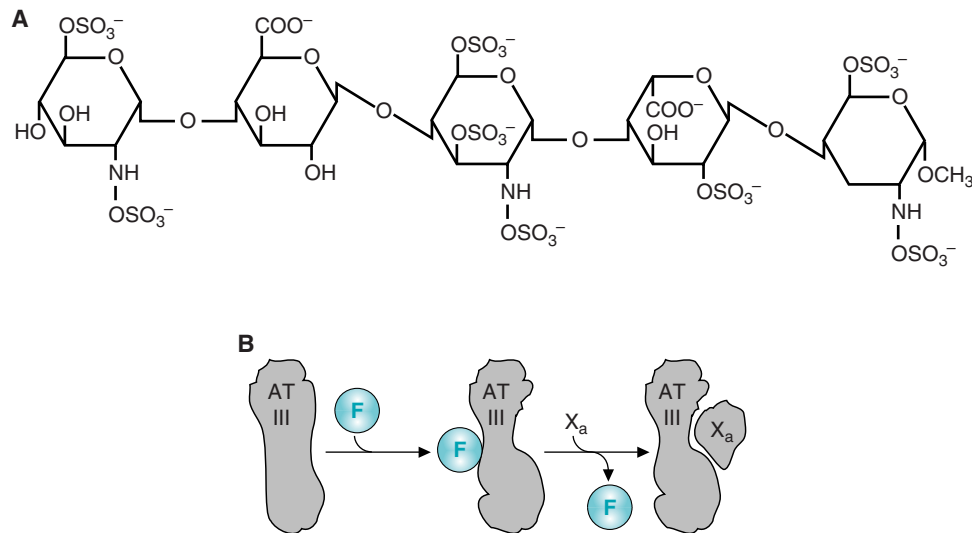


Fig. 45.8. A. Structure of fondaparinux. B. Mechanism of fondaparinux action. The drug (shown in blue) binds to ATIII, which induces a conformational change such that Factor Xa can now bind to ATIII. Once Factor Xa binds, and is inactivated, the drug is released and can activate another molecule of ATIII.

Direct thrombin inhibitors are based on the hirudin molecules, which were initially discovered in leeches and other blood-sucking organisms. These organisms would not be able to feed if the blood clotted at the site of the puncture wound, so the organisms secrete thrombin inhibitors to prevent clotting from occurring. Hirudin treatment itself is dangerous in that formation of the hirudin–thrombin complex is irreversible, and use of the drug requires constant monitoring of the patient. Thus, to overcome this problem, rational drug design based on the hirudin structure was used, and a synthetic 20–amino acid peptide known as bivalirudin was synthesized. This agent has a high binding affinity and specificity for thrombin although its effects on thrombin are transient (not irreversible), making this a safer agent for long-term use.

Suggested References

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REVIEW QUESTIONS—CHAPTER 45

- The edema observed in patients with non-calorie protein malnutrition is due to which of the following?
 - Loss of muscle mass
 - Ingestion of excess carbohydrates
 - Increased fluid uptake
 - Reduced protein synthesis in the liver
 - Increased ketone body production

2. A recent surgery patient receiving warfarin therapy was found to be bleeding internally. The clotting process is impaired in this patient primarily because of which of the following?
 - (A) Inability of the liver to synthesize clotting factors
 - (B) A specific inhibition of Factor XIII activation
 - (C) An inability to form clotting factor complexes on membranes
 - (D) A reduction of plasma calcium levels
 - (E) An enhancement of protein C activity
3. An inactivating mutation in which of the following proenzymes would be expected to lead to thrombosis, uncontrolled blood clotting?
 - (A) Factor XIII
 - (B) Prothrombin
 - (C) Protein C
 - (D) Factor VIII
 - (E) Tissue factor
4. Classical hemophilia A results in an inability to directly activate which of the following factors?
 - (A) Factor II
 - (B) Factor IX
 - (C) Factor X
 - (D) Protein S
 - (E) Protein C
5. Hemophilia B results in an inability to directly activate which of the following factors?
 - (A) Factor II
 - (B) Factor IX
 - (C) Factor X
 - (D) Protein S
 - (E) Protein C

46 Liver Metabolism

*The liver is strategically interposed between the general circulation and the digestive tract. It receives 20 to 25% of the volume of blood leaving the heart each minute (the cardiac output) through the portal vein (which delivers absorbed nutrients and other substances from the gastrointestinal tract to the liver) and through the hepatic artery (which delivers blood from the general circulation back to the liver). Potentially toxic agents absorbed from the gut or delivered to the liver by the hepatic artery must pass through this **metabolically active organ** before they can reach the other organs of the body. The liver's relatively large size (approximately 3% of total body weight) allows extended residence time within the liver for nutrients to be properly metabolized as well as for potentially harmful substances to be detoxified and prepared for excretion into the urine or feces. Among other functions, therefore, the liver, along with the kidney and gut, is an **excretory organ**, equipped with a broad spectrum of detoxifying mechanisms. It has the capacity, for example, to carry out **metabolic conversion pathways** as well as **secretory systems** that allow the excretion of potentially toxic compounds. Concurrently, the liver contains highly specific and selective **transport mechanisms** for essential nutrients that are required not only to sustain its own energy but to provide physiologically important substrates for the systemic needs of the organism. In addition to the myriad of transport processes within the sinusoidal and canalicular plasma membrane sheets (see below), intracellular hepatocytic transport systems exist in organelles such as **endosomes**, **mitochondria**, **lysosomes**, as well as the **nucleus**. The sequential transport steps carried out by these organelles include (1) **uptake**, (2) **intracellular binding and sequestration**, (3) **metabolism**, (4) **sinusoidal secretion**, and (5) **biliary excretion**. The rate of hepatobiliary transport is determined, in part, by the rate of activity of each of these steps. The overall transport rate is also determined by such factors as hepatic blood flow, plasma protein binding, and the rate of canalicular reabsorption. The various aspects of the major metabolic processes performed by the liver have been discussed in greater detail elsewhere in this text. These sources are referred to as the broad spectrum of the liver's contributions to overall health and disease are described.*



THE WAITING ROOM



Jean Ann Tonich's family difficulties continued, and, in spite of a period of sobriety lasting 6 months, she eventually started drinking increasing amounts of gin again in an effort to deal with her many anxieties. Her appetite for food declined slowly as well. She gradually withdrew from much of the

social support system that her doctors and friends had attempted to build during her efforts for rehabilitation. Upper mid-abdominal pain became almost constant, and she noted an increasing girth and distention of her abdomen. Early one morning, she was awakened in excruciating pain in her upper abdomen. She vomited dark-brown “coffee ground” material followed by copious amounts of bright red blood. She called a friend, who rushed her to the hospital emergency room.



Amy Biasis, a 23-year-old missionary, was brought to the hospital emergency room complaining of the abrupt onset of fever, chills, and severe pain in the right upper quadrant of her abdomen. The pain was constant in nature and radiated to her right shoulder top. She vomited undigested food twice in the hour before arriving at the emergency room. This did not relieve her pain.

Her medical history indicated that, while serving as a missionary in western Belize, Central America, 2 months earlier, she had a 3-day illness that included fever, chills, and mild but persistent diarrhea. A friend of Amy's there, a medical missionary, had given her an unidentified medication for 7 days. Amy's diarrhea slowly resolved, and she felt well again until her current abdominal symptoms began.

On physical examination, she appeared toxic and had a temperature of 101°F. She was sweating profusely. Her inferior anterior liver margin was palpable three fingerbreadths below the right rib cage, suggestive of an enlarged liver. The liver edge was rounded and tender. Gentle first percussion of the lower posterior right rib cage caused severe pain. Routine laboratory studies were ordered, and a computed tomogram (CT) of the upper abdomen was scheduled to be done immediately.

I. LIVER ANATOMY

The human liver consists of two lobes, each containing multiple lobules and sinusoids. The liver receives 75% of its blood supply from the portal vein, which carries blood returning to the heart from the small intestine, stomach, pancreas and spleen. The remaining 25% of the liver's blood supply is arterial, carried to the liver by the hepatic artery.

Blood from both the portal vein and hepatic artery empty into a common conduit, mixing their contents as they enter the liver sinusoids (Fig. 46.1). The sinusoids are expandable vascular channels that run through the hepatic lobules. They are lined with endothelial cells that have been described as “leaky” because, as blood flows through the sinusoids, the contents of the plasma have relatively free access to the hepatocytes, which are located on the other side of the endothelial cells.

The liver is also an exocrine organ, secreting bile into the biliary drainage system. The hepatocytes secrete bile into the bile canniculus, whose contents flow parallel to that in the sinusoids but in the opposite direction. The canniculi empty into the bile ducts. The lumina of the bile ducts then fuse, forming the common bile duct. The common duct then releases bile into the duodenum. Some of the liver's effluent is stored in the gallbladder and discharged into the duodenum postprandially to aid in digestion.

The entire liver surface is covered by a capsule of connective tissue that branches and extends throughout the liver. This capsule provides support for the blood vessels, lymphatic vessels, and bile ducts that permeate the liver. In addition, this connective tissue sheet subdivides the liver lobes into the smaller lobules.

II. LIVER CELL TYPES

The primary cell type of the liver is the hepatocyte. Hepatocytes, also known as the hepatic parenchymal cells, form the liver lobules. Eighty percent of the liver volume is composed of hepatocytes, but only 60% of the total number of cells in the liver

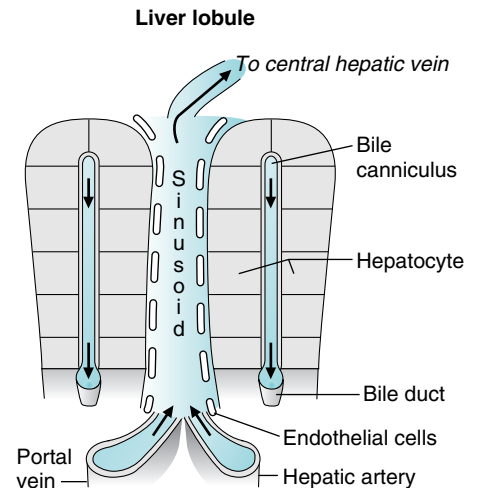


Fig. 46.1. Schematic view of liver anatomy.

are hepatocytes. The other 40% of the cells are the nonparenchymal cells, which constitute the lining cells of the walls of the sinusoids. The lining cells comprise the endothelial cells, Kupffer cells, and hepatic stellate cells. In addition, intrahepatic lymphocytes, which include pit cells (liver-specific natural killer cells) are also present in the sinusoidal lining.

A. Hepatocytes

The hepatocyte is the cell that carries out the many functions of the liver. Almost all pathways of metabolism are represented in the hepatocyte and these pathways are controlled through the actions of hormones that bind to receptors located on the plasma membrane of their cells. Although normally quiescent cells with low turnover and a long life span, hepatocytes can be stimulated to grow if damage occurs to other cells in the liver. The liver mass has a relatively constant relationship to the total body mass of adult individuals. Deviation from the normal or optimal ratio (caused, for example, by a partial hepatectomy or significant hepatic cell death or injury) is rapidly corrected by hepatic growth caused by a proportional increase in hepatocyte replication.



The reports of **Amy Biasis'** initial laboratory studies showed an elevation in her serum hepatic transaminases, her serum alkaline phosphatases, as well as her serum total bilirubin level.

B. Endothelial Cells

The sinusoidal endothelial cells constitute the lining cells of the sinusoid. Unlike endothelial cells in other body tissues, these cells contain fenestrations with a mean diameter of 100 nm. They do not, therefore, form a tight basement membrane barrier between themselves and the hepatocytes. In this way, they allow for free diffusion of small molecules to the hepatocytes but not of particles the size of chylomicrons (chylomicron remnants, however, which are smaller than chylomicrons, do have free passage to the hepatocyte). The endothelial cells are capable of endocytosing many ligands and also may secrete cytokines when appropriately stimulated. Because of their positioning, lack of tight junctions, and absence of a tight basement membrane, the liver endothelial cells do not present a significant barrier against the movement of the contents of the sinusoids into hepatocytes. Their fenestrations or pores further promote the free passage of blood components through this membrane into the liver parenchymal cells.

C. Kupffer Cells

These cells are located within the sinusoidal lining. They contain almost one quarter of all the lysosomes of the liver. The Kupffer cells are tissue macrophages with both endocytotic and phagocytic capacity. They phagocytose many substances such as denatured albumin, bacteria, and immune complexes. They protect the liver from gut-derived particulate materials and bacterial products. On stimulation by immunomodulators, these cells secrete potent mediators of the inflammatory response and play a role in liver immune defense through the release of cytokines that lead to the inactivation of substances considered foreign to the organism. The Kupffer cells also remove damaged erythrocytes from the circulation.

D. Hepatic Stellate Cells

The stellate cells are also called perisinusoidal or Ito cells. There are approximately 5 to 20 of these cells per 100 hepatocytes. The stellate cells are lipid-filled cells (the primary storage site for vitamin A). They also control the turnover of hepatic connective tissue and extracellular matrix and regulate the contractility of the sinusoids. When cirrhosis of the liver is present, the stellate cells are stimulated by various signals to increase their synthesis of extracellular matrix material. This, in turn, diffusely infiltrates the liver, eventually interfering with the function of the hepatocytes.

E. Pit Cells

The hepatic pit cells, also known as liver-associated lymphocytes, are natural killer cells, which are a defense mechanism against the invasion of the liver by potentially toxic agents, such as tumor cells or viruses.

III. MAJOR FUNCTIONS OF THE LIVER

A. The Liver Is a Central Receiving and Recycling Center for the Body

The liver can carry out a multitude of biochemical reactions. This is necessary because of its role in constantly monitoring, recycling, modifying, and distributing all of the various compounds absorbed from the digestive tract and delivered to the liver. If any portion of an ingested compound is potentially useful to that organism, the liver will retrieve this portion and convert it to a substrate that can be used by hepatic and nonhepatic cells. At the same time, the liver removes many of the toxic compounds that are ingested or produced in the body and targets them for excretion in the urine or in the bile.

As mentioned previously, the liver receives nutrient-rich blood from the enteric circulation through the portal vein; thus, all of the compounds that enter the blood from the digestive tract pass through the liver on their way to other tissues. The enterohepatic circulation allows the liver first access to nutrients to fulfill specific functions (such as the synthesis of blood coagulation proteins, heme, purines, and pyrimidines) and first access to ingested toxic compounds (such as ethanol) and to such potentially harmful metabolic products (such as NH_4^+ produced from bacterial metabolism in the gut).

In addition to the blood supply from the portal vein, the liver receives oxygen-rich blood through the hepatic artery; this arterial blood mixes with the blood from the portal vein in the sinusoids. This unusual mixing process gives the liver access to various metabolites produced in the periphery and secreted into the peripheral circulation, such as glucose, individual amino acids, certain proteins, iron–transferrin complexes, and waste metabolites as well as potential toxins produced during substrate metabolism. As mentioned, fenestrations in the endothelial cells, combined with gaps between the cells, the lack of a basement membrane between the endothelial cells and the hepatocytes, and low portal blood pressure (which results in slow blood flow) contribute to the efficient exchange of compounds between sinusoidal blood and the hepatocyte and clearance of unwanted compounds from the blood. Thus, large molecules targeted for processing, such as serum proteins and chylomicron remnants, can be removed by hepatocytes, degraded, and their components recycled. Similarly, newly synthesized molecules, such as very-low-density lipoprotein (VLDL) and serum proteins, can be easily secreted into the blood. In addition, the liver can convert all of the amino acids found in proteins into glucose, fatty acids, or ketone bodies. The secretion of VLDL by the liver not only delivers excess calories to adipose tissue for storage of fatty acids in triacylglycerol, but it also delivers phospholipids and cholesterol to tissues that are in need of these compounds for synthesis of cell walls as well as other functions. The secretion of glycoproteins by the liver is accomplished through the liver's gluconeogenic capacity as well as its access to a variety of dietary sugars to form the oligosaccharide chains, as well as its access to dietary amino acids with which it synthesizes proteins. Thus, the liver has the capacity to carry out a large number of biosynthetic reactions. It has the biochemical wherewithal to synthesize a myriad of compounds from a broad spectrum of precursors. At the same time, the liver metabolizes compounds into biochemically useful products. Alternatively, it has the ability to degrade and excrete those compounds presented to it that cannot be further used by the body.



The CT scan of **Amy Biasis'** upper abdomen showed an elevated right hemidiaphragm as well as several cystic masses in her liver, the largest of which was located in the superior portion of the right lobe. Her clinical history as well as her history of possible exposure to various parasites while working in a part of Belize, Central America, that is known to practice substandard sanitation, prompted her physicians to order a titer of serum antibodies against the parasite *Entamoeba histolytica* in addition to measuring serum antibodies against other invasive parasites.



A knowledge of functional characteristics of liver cells has been used to design diagnostic agents that can be used to determine the normalcy of specific biochemical pathways of the hepatocytes. These “tailor-made” pharmaceuticals can be designed to be taken up by one or more of the available transport mechanisms available to the liver. For example, receptor-related endocytic processes can be used as targets to probe specific receptor-mediated transport functions of the liver cells. The asialoglycoprotein receptor, also known as the hepatic-binding protein, has been used in this diagnostic approach. The substrate $^{99}\text{Tc}^m$ -galactosyl-neoglycoalbumin (NGA) was developed as a specific ligand for selective uptake via this specific hepatic receptor. The timing and extent of the assimilation of this probe into the hepatocytes, as determined by imaging the liver at precise intervals after the administration of this isotope, yields an estimate of hepatic blood flow as well as the transport capacity of this specific hepatic transporter protein.



Antibody titers against *Entamoeba histolytica* by use of an enzyme immunoassay were strongly reactive in **Amy Biasis’** blood. A diagnosis of **amoebiasis** was made. Her physicians started nitroimidazole amoebicides intravenously in a dose of 500 mg every 6 hours for 10 days. By the third day of treatment, Amy began to feel noticeably better. Her physicians told her that they expected a full clinical response in 95% of patients with amoebic liver abscesses treated in this way, although her multiple hepatic abscesses adversely affected her prognosis to a degree.

Table 46.1. Examples of Enzymes Used in Biotransformation of Xenobiotic Compounds

Acetyltransferase
Amidase-esterase
Dehydrogenase
Flavin-containing mono-oxygenase
Glutathione-S-transferase
Methyl transferase
Mixed-function oxidase
Reductase
Sulfotransferase
UDP-glucosyltransferase
UDP-glucuronosyltransferase

Each of the liver cells described contains specialized transport and uptake mechanisms for enzymes, infectious agents, drugs, and other xenobiotics that specifically target these substances to certain liver cell types. These are accomplished by linking these agents covalently by way of biodegradable bonds to their specific carrier. The latter then determines the particular fate of the drug by using specific cell recognition, uptake, transport, and biodegradation pathways.

B. Inactivation and Detoxification of Xenobiotic Compounds and Metabolites

Xenobiotics are compounds that have no nutrient value (cannot be used by the body for energy requirements) and are potentially toxic. They are present as natural components of foods or they may be introduced into foods as additives or through processing. Pharmacologic and recreational drugs are also xenobiotic compounds. The liver is the principal site in the body for the degradation of these compounds. Because many of these substances are lipophilic, they are oxidized, hydroxylated, or hydrolyzed by enzymes in phase I reactions. Phase I reactions introduce or expose hydroxyl groups or other reactive sites that can be used for conjugation reactions (the phase II reactions). The conjugation reactions add a negatively charged group such as glycine or sulfate to the molecule. Many xenobiotic compounds will be transformed through several different pathways. A general scheme of inactivation is shown in Figure 46.2.

The conjugation and inactivation pathways are similar to those used by the liver to inactivate many of its own metabolic waste products. These pathways are intimately related to the biosynthetic cascades that exist in the liver. The liver can synthesize the precursors that are required for conjugation and inactivation reactions from other compounds. For example, sulfation is used by the liver to clear steroid hormones from the circulation. The sulfate used for this purpose can be obtained from the degradation of cysteine or methionine.

The liver, kidney, and intestine are the major sites in the body for biotransformation of xenobiotic compounds. Many xenobiotic compounds contain aromatic rings (such as benzopyrene in tobacco smoke) or heterocyclic ring structures (such as the nitrogen-containing rings of nicotine or pyridoxine) that we are unable to degrade or recycle into useful components. These structures are hydrophobic, causing the molecules to be retained in adipose tissue unless they are sequestered by the liver, kidney, or intestine for biotransformation reactions. Sometimes, however, the phase I and II reactions backfire, and harmless hydrophobic molecules are converted to toxins or potent chemical carcinogens.

1. CYTOCHROME P450 AND XENOBIOTIC METABOLISM

The toxification/detoxification of xenobiotics is accomplished through the activity of a group of enzymes with a broad spectrum of biologic activity. Some examples of enzymes involved in xenobiotic transformation are described in Table 46.1. Of the wide variety of enzymes that are involved in xenobiotic metabolism, only the cytochrome P450-dependent monooxygenase system is discussed here. The cytochrome P450-dependent monooxygenase enzymes are determinants in oxidative, peroxidative, and reductive degradation of exogenous (chemicals, carcinogens, and pollutants, etc.) and endogenous (steroids, prostaglandins,

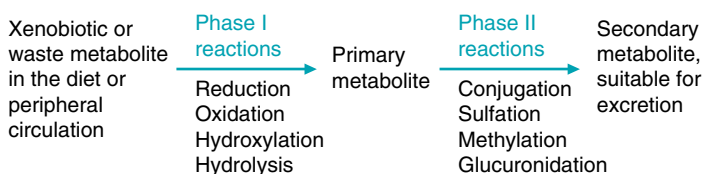


Fig. 46.2. General scheme of xenobiotic detoxification.

retinoids, etc.) substances. The key enzymatic constituents of this system are the flavo protein NADPH-cytochrome P450 oxidoreductase and cytochrome P450 (Fig. 46.3). The latter is the terminal electron acceptor and substrate-binding site of the microsomal mixed-function oxidase complex, a very versatile catalytic system. The system got its name in 1962, when Omura and Sato found a pigment with unique spectral characteristics derived from liver microsome of rabbits. When reduced and complexed with carbon monoxide, it exhibited a spectral absorbance maximum at 450 nm.

The major role of the cytochrome P450 enzymes (see Chapter 25) is to oxidize substrates and introduce oxygen to the structure. Similar reactions can be carried out by other flavin monooxygenases that do not contain cytochrome P450.

The cytochrome P450 enzyme family contains at least 100 to 150 different isozymes with at least 40% sequence homology. These isozymes have different but overlapping specificities. The human enzymes are generally divided into six major subfamilies, and each of these is further subdivided. For example, in the naming of the principal enzyme involved in the oxidation of ethanol to acetaldehyde, CYP2E1, the CYP denotes the cytochrome P450 family, the 2 denotes the subfamily, the E denotes ethanol, and the 1 denotes the specific isozyme.

The CYP3A4 isoform accounts for 60% of CYP450 enzymes in the liver and 70% of cytochrome enzymes in gut wall enterocytes. It metabolizes the greatest number of drugs in humans. Specific drugs are substrates for CYP3A4. The concomitant ingestion of two CYP3A4 substrates could potentially induce competition for the binding site, which, in turn, could alter the blood levels of these two agents. The drug with the highest affinity for the enzyme would be preferentially metabolized, whereas the metabolism (and degradation) of the other drug would be reduced. The latter drug's concentration in the blood would then rise.

Moreover, many substances or drugs impair or inhibit the activity of the CYP3A4 enzyme, thereby impairing the body's ability to metabolize a drug. The lipid-lowering agents known as the statins (HMGCoA reductase inhibitors) require CYP3A4 for degradation. Appropriate drug treatment and dosing takes into account the normal degradative pathway of the drug. However, grapefruit juice is a potent inhibitor of CYP3A4-mediated drug metabolism. Evidence suggests that if a statin is regularly taken with grapefruit juice, its level in the blood may increase as much as 15-fold. This marked increase in plasma concentration could increase the muscle and liver toxicity of the statin in question, because side effects of the statins appear to be dose-related.

The cytochrome P450 isozymes all have certain features in common:

1. They all contain cytochrome P450, oxidize the substrate, and reduce oxygen.
2. They all have a flavin-containing reductase subunit that uses NADPH, and not NADH, as a substrate.
3. They are all found in the smooth endoplasmic reticulum and are referred to as microsomal enzymes (for example, CYP2E1 is also referred to as the microsomal ethanol oxidizing system, MEOS).
4. They are all bound to the lipid portion of the membrane, probably to phosphatidylcholine.
5. They are all inducible by the presence of their own best substrate and somewhat less inducible by the substrates for other P450 isozymes.
6. They all generate a reactive free radical compound as an intermediate in the reaction.

2. EXAMPLES OF CYTOCHROME P450 DETOXIFICATION REACTIONS

i. Vinyl Chloride

The detoxification of vinyl chloride provides an example of effective detoxification by a P450 isozyme (ethanol detoxification was previously discussed in Chapter 25). Vinyl chloride is used in the synthesis of plastics and can cause angiosarcoma in the liver of exposed workers. It is activated in a phase I reaction to a reactive epoxide by

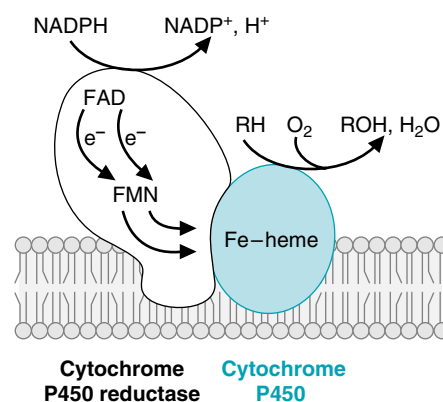


Fig. 46.3. General structure of the P450 enzymes. O₂ binds to the P450 Fe-heme in the active site and is activated to a reactive form by accepting electrons. The electrons are donated by the cytochrome P450 reductase, which contains an FAD plus an FMN or Fe-S center to facilitate the transfer of single electrons from NADPH to O₂. The P450 enzymes involved in steroidogenesis have a somewhat different structure. For CYP2E1, RH is ethanol (CH₃CH₂OH), and ROH is acetaldehyde (CH₃COH).

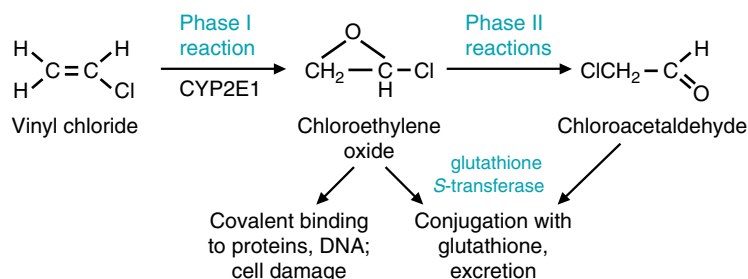


Fig. 46.4. Detoxification of vinyl chloride.

a hepatic P450 isozyme (CYP2E1), which can react with guanine in DNA or other cellular molecules. However, it also can be converted to chloroacetate, conjugated with reduced glutathione, and excreted in a series of phase II reactions (Fig. 46.4).

ii. Aflatoxin B1

Aflatoxin B1 is an example of a compound that is made more toxic by a cytochrome P450 reaction (CYP2A1). Current research suggests that ingested aflatoxin B1 in contaminated food (it is produced by a fungus [*Aspergillus flavus*] that grows on peanuts that may have been stored in damp conditions) is directly involved in hepatocarcinogenesis in humans by introducing a G > T mutation into the *p53* gene. Aflatoxin is metabolically activated to its 8,9 epoxide by two different isozymes of cytochrome P450. The epoxide modifies DNA by forming covalent adducts with guanine residues. In addition, the epoxide can combine with lysine residues within proteins and thus is also a hepatotoxin.

iii. Acetaminophen

Acetaminophen (Tylenol) is an example of a xenobiotic that is metabolized by the liver for safe excretion; however, it can be toxic if ingested in high doses. The pathways for acetaminophen metabolism are shown in Fig. 46.5 As shown in the

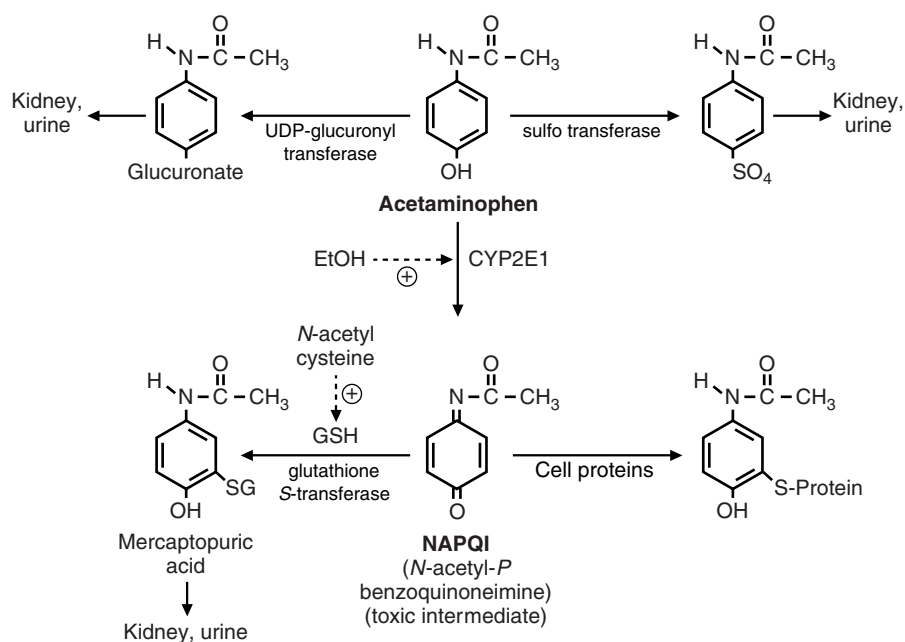


Fig. 46.5. Pathways of acetaminophen detoxification. N-acetyl cysteine stimulates the production of glutathione, thereby reducing the levels of NAPQI, which can damage cellular proteins. Ethanol upregulates CYP2E1 activity (the MEOS).

figure, acetaminophen can be glucuronylated or sulfated for safe excretion by the kidney. However, a cytochrome P450 enzyme produces the toxic intermediate N-acetyl-p-benzoquinoneimine (NAPQI), which can be excreted safely in the urine after conjugation with glutathione.

NAPQI is a dangerous and unstable metabolite that can damage cellular proteins and lead to death of the hepatocyte. Under normal conditions, when acetaminophen is taken in the correct therapeutic amounts, less than 10% of the drug forms NAPQI, an amount that can be readily handled by the glutathione detoxifying system (phase II reactions). However, when taken at doses that are potentially toxic, the sulfo-transferase and glucuronyl transferase systems are overwhelmed, and more acetaminophen is metabolized through the NAPQI route. When this occurs, the levels of glutathione in the hepatocyte are insufficient to detoxify NAPQI, and hepatocyte death can result.

The enzyme that produces NAPQI, CYP2E1, is induced by alcohol (see Chapter 25, MEOS). Thus, individuals who chronically abuse alcohol have an increased sensitivity to acetaminophen toxicity, because a higher percentage of acetaminophen metabolism is directed toward NAPQI, as compared with an individual with low levels of CYP2E1. Therefore, even recommended therapeutic doses of acetaminophen can be toxic to these individuals.

An effective treatment for acetaminophen poisoning involves the use of N-acetyl cysteine. This compound supplies cysteine as a precursor for increased glutathione production, which, in turn, enhances the phase II reactions, which reduces the levels of the toxic intermediate.

C. Regulation of Blood Glucose Levels

One of the primary functions of the liver is to maintain blood glucose concentrations within the normal range. The manner in which the liver accomplishes this has been the subject of previous chapters (26, 31, and 36). In brief, the pancreas monitors blood glucose levels and secretes insulin when blood glucose levels rise and glucagon when such levels decrease. These hormones initiate regulatory cascades that affect liver glycogenolysis, glycogen synthesis, glycolysis, and gluconeogenesis. In addition, sustained physiologic increases in growth hormone, cortisol, and catecholamine secretion help to sustain normal blood glucose levels during fasting.

When blood glucose levels drop, glycolysis and glycogen synthesis are inhibited, and gluconeogenesis and glycogenolysis are activated. Concurrently, fatty acid oxidation is activated to provide energy for glucose synthesis. During an overnight fast, blood glucose levels are primarily maintained by glycogenolysis and, if gluconeogenesis is required, the energy (6 ATP are required to produce one molecule of glucose from two molecules of pyruvate) is obtained by fatty acid oxidation. On insulin release, the opposing pathways are activated such that excess fuels can be stored either as glycogen or fatty acids. The pathways are regulated by the activation or inhibition of two key kinases, the cyclic adenosine monophosphate (cAMP)-dependent protein kinase, and the AMP-activated protein kinase (see Fig. 36.11 for a review of these pathways). Recall that the liver can export glucose because it is one of only two tissues that express glucose-6-phosphatase.

D. Synthesis and Export of Cholesterol and Triacylglycerol

When food supplies are plentiful, hormonal activation leads to fatty acid, triacylglycerol, and cholesterol synthesis. A high dietary intake and intestinal absorption of cholesterol will compensatorily reduce the rate of hepatic cholesterol synthesis, in which case the liver acts as a recycling depot for sending excess dietary cholesterol to the peripheral tissue when needed as well as accepting cholesterol from these tissues when required. The pathways of cholesterol metabolism were discussed in Chapter 34.



Numerous other factors, beside insulin and glucagon, can affect liver glucose metabolism, as has been described in Chapter 43.

E. Ammonia and the Urea Cycle

The liver is the primary organ for synthesizing urea and, as such, is the central depot for the disposition of ammonia in the body. Ammonia groups travel to the liver on glutamine and alanine, and the liver converts these ammonia nitrogens to urea for excretion in the urine. The reactions of the urea cycle were discussed in Chapter 38.

Table 46.2 lists some of the important nitrogen-containing compounds that are primarily synthesized or metabolized by the liver.

F. Ketone Body Formation

The liver is the only organ that can produce ketone bodies, yet it is one of the few that cannot use these molecules for energy production. Ketone bodies are produced when the rate of glucose synthesis is limited (i.e., substrates for gluconeogenesis are limited), and fatty acid oxidation is occurring rapidly. Ketone bodies can cross the blood-brain barrier and become a major fuel for the nervous system under conditions of starvation. Ketone body synthesis and metabolism have been described in Chapter 23.

G. Nucleotide Biosynthesis

The liver can synthesize and salvage all ribonucleotides and deoxyribonucleotides for other cells to use. Certain cells have lost the capacity to produce nucleotides de

Table 46.2. Nitrogen-Containing Products Produced by the Liver

Product	Precursors	Tissues	Function
Creatine	Arginine, glycine, and S-adenosyl methionine (SAM)	Liver	Forms creatine phosphate in muscle for energy storage. Excreted as creatinine.
Glutathione	Glutamate, cysteine, glycine	All tissues but highest use in the liver	Protection against free radical injury by reduction of hydrogen peroxide and lipid peroxides. In liver and kidney forms mercapturic acids.
Purines	Glycine, glutamine, aspartate, carbon dioxide, tetrahydrofolate, PRPP	Liver, small amounts in brain and cells of the immune system	Adenine and guanine nucleosides and nucleotides. DNA, RNA, and coenzymes, and energy-transferring nucleotides.
Pyrimidines	Aspartate, glutamine, carbon dioxide	Liver, small amounts in brain and cells of the immune system	Uracil, thymine and cytosine
Sialic acid (NANA), other amino sugars	Glutamine	Most cells	In the liver, synthesis of oligosaccharide chains on secreted proteins. Most cells, glycoproteins, proteoglycans, and glycolipids.
Sulfated compounds	Cysteine	Liver and kidney produce sulfate	Many cells use sulfate in blood for formation of PAPS, which transfers sulfate to proteoglycans, drugs, and xenobiotics
Taurine	Cysteine	Liver	Conjugated bile salts
Glycocholic acid, and glycocheno-Deoxycholic acid	Glycine, bile salts	Liver	Conjugated bile salts are excreted into the bile and assist in the absorption of lipids and fat-soluble vitamins through the formation of micelles
Sphingosine	Serine and palmitoyl CoA	Liver, brain, and other tissues	Precursor of sphingolipids found in myelin and other membranes
Heme	Glycine and succinyl CoA	Liver, bone marrow	Heme from liver is incorporated into cytochromes. Heme from bone marrow is incorporated into hemoglobin.
Glycine conjugates of xenobiotic compounds	Glycine, medium-size hydrophobic carboxylic acids	Liver, kidney	Inactivation and targeting toward urinary excretion
Niacin	Tryptophan, glutamine	Liver	NAD, NADP coenzymes for oxidation reactions
One-carbon methyl donors for tetrahydrofolate and SAM	Glycine, serine, histidine, methionine	Most cells, but highest in liver	Choline, phosphatidylcholine, purine and pyrimidine synthesis, inactivation of waste metabolites and xenobiotics through methylation.

novo but can use the salvage pathways to convert free bases to nucleotides. The liver can secrete free bases into the circulation for these cells to use for this purpose. Nucleotide synthesis and degradation are discussed in Chapter 41.

H. Synthesis of Blood Proteins

The liver is the primary site of the synthesis of circulating proteins such as albumin and the clotting factors. When liver protein synthesis is compromised, the protein levels in the blood are reduced. Hypoproteinemia may lead to edema because of a decrease in the protein-mediated osmotic pressure in the blood. This, in turn, causes plasma water to leave the circulation and enter (and expand) the interstitial space, causing edema.

Most circulating plasma proteins are synthesized by the liver. Therefore, the hepatocyte has a well-developed endoplasmic reticulum, Golgi system, and cellular cytoskeleton, all of which function in the synthesis, processing, and secretion of proteins. The most abundant plasma protein produced by the liver is albumin, which represents 55 to 60% of the total plasma protein pool. Albumin serves as a carrier for a large number of hydrophobic compounds, such as fatty acids, steroids, hydrophobic amino acids, vitamins, and pharmacologic agents. It is also an important osmotic regulator in the maintenance of normal plasma osmotic pressure. The other proteins synthesized by the liver are, for the most part, glycoproteins. They function in hemostasis, transport, protease inhibition, and ligand binding, as well as secretagogues for hormone release. The acute phase proteins that are part of the immune response and the body's response to many forms of "injury" are also synthesized in the liver. Table 46.3 lists some of the proteins and their functions.

I. The Synthesis of Glycoproteins and Proteoglycans

The liver, because it is the site of synthesis of most of the blood proteins (including the glycoproteins), has a high requirement for the sugars that go into the oligosaccharide portion of glycoproteins (The synthesis of glycoproteins is discussed in Chapter 30.). These include mannose, fructose, galactose, and amino sugars.

One of the intriguing aspects of the hepatic biosynthetic pathways that use carbohydrate in the synthesis of these compounds is that the liver is not dependent on either dietary glucose or hepatic glucose to generate the precursor intermediates for these pathways. This is because the liver can generate carbohydrates from dietary amino acids (which enter gluconeogenesis generally as pyruvate or an intermediate of the TCA cycle), lactate (generated from anaerobic glycolysis in other tissues), and glycerol (generated by the release of free fatty acids from the adipocyte). Of course, if dietary carbohydrate is available, the liver can use that source as well.

Most of the sugars secreted by the liver are O-linked, that is, the carbohydrate is attached to the protein at its anomeric carbon through a glucosidic link to the $-OH$ of a serine or a threonine residue. This is in contrast to the N-linked arrangement in which there is an N-glycosyl link to the amide nitrogen of an asparagine residue (Fig. 46.6). A particularly important O-linked sugar is N-acetylneuraminic acid (NANA or sialic acid), a nine-carbon sugar that is synthesized from fructose-6-phosphate and phosphoenolpyruvate (see Fig. 30.8). As circulating proteins age,



Cirrhosis of the liver results in portal hypertension, which because of increasing back pressure into the esophageal veins promotes the development of dilated thin-walled esophageal veins (varices). At the same time, synthesis of blood coagulation proteins by the liver and required vitamin K-dependent reactions are greatly diminished (resulting in a prolonged prothrombin time; which, in turn, increases clotting time). When the esophageal varices rupture, massive bleeding into the thoracic or abdominal cavity as well as the stomach may occur. Much of the protein content of the blood entering the gastrointestinal tract is metabolized by intestinal bacteria, releasing ammonium ion, which enters the portal vein. Because hepatocellular function has been compromised, the urea cycle capacity is inadequate, and the ammonium ion enters the peripheral circulation, thereby contributing to hepatic encephalopathy (brain toxicity due to elevated ammonia levels).

Table 46.3. A Partial List of Proteins Synthesized in the Liver

Type of Protein	Examples
Blood coagulation	Blood coagulation factors: fibrinogen, prothrombin, Factors V, VII, IX and X. Also α -2 macroglobulin.
Metal-binding proteins	Transferrin (iron), ceruloplasmin (copper), haptoglobin (heme), hemopexin (heme)
Lipid transport	Apoprotein B-100, apoprotein A-1
Protease inhibitor	α 1-Antitrypsin

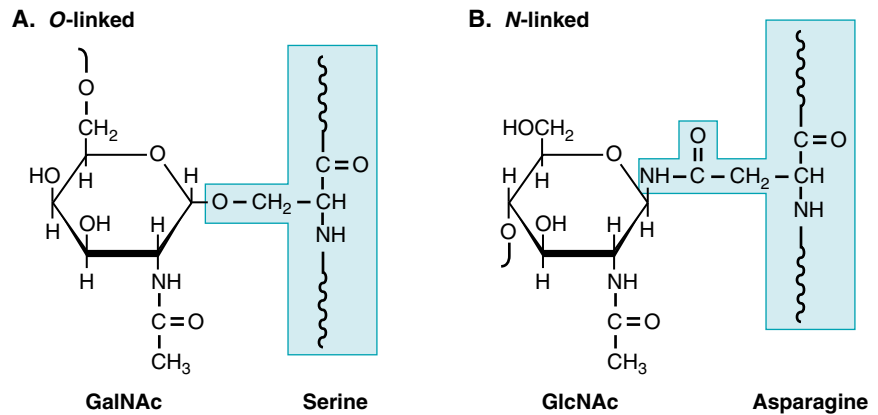


Fig. 46.6. The general configuration of O-linked and N-linked glycoproteins.

NANA (sialic acid) residues are lost from the serum proteins. This change signals their removal from the circulation and their eventual degradation. An asialoglycoprotein receptor on the liver cell surface binds such proteins, and the receptor—ligand complex is endocytosed and transported to the lysosomes. The amino acids from the degraded protein are then recycled within the liver.

J. The Pentose Phosphate Pathway

The major functions of the pentose phosphate pathway (see Chapter 29) are the generation of NADPH and five-carbon sugars. All cell types, including the red blood cell, can carry out this pathway because they need to generate NADPH so that the activity of glutathione reductase, the enzyme that catalyzes the conversion of oxidized glutathione (GSSG) back to reduced glutathione (GSH) can be maintained. Without the activity of this enzyme, the protection against free radical injury is lost. All cells also need this pathway for the generation of ribose, especially those cells that are dividing rapidly or have high rates of protein synthesis.

The liver has a much greater demand for NADPH than do most other organs. It uses NADPH for the biosynthesis of fatty acids and cholesterol, which the liver must make to produce phospholipids, and for the synthesis of VLDL and bile salts. It also uses NADPH for other biosynthetic reactions, such as that of proline synthesis. NADPH is also used by mixed-function oxidases such as cytochrome P450 that are involved in the metabolism of xenobiotics and of a variety of pharmaceuticals. Because the liver participates in so many reactions capable of generating free radicals, the liver uses more glutathione and NADPH to maintain glutathione reductase and catalase activity than any other tissue. Consequently, the concentration of glucose-6-phosphate dehydrogenase (the rate-limiting and regulated enzyme in the pentose phosphate pathway) is high in the liver, and the rate of flux through this pathway may be as high as 30% of the rate of flux through glycolysis.

IV. FUELS FOR THE LIVER

The reactions used to modify and inactivate dietary toxins and waste metabolites are energy requiring, as are the reactions used by anabolic (biosynthetic) pathways such as gluconeogenesis and fatty acid synthesis. Thus, the liver has a high energy requirement and consumes approximately 20% of the total oxygen used by the body. The principle forms in which energy is supplied to these reactions is the high-energy phosphate bonds of adenosine triphosphate (ATP), uridine triphosphate (UTP), and guanosine triphosphate (GTP), reduced NADPH, and acyl-CoA thioesters. The energy for the formation of these compounds is obtained directly

Table 46.4. Major Fates of Carbohydrates in the Liver

- Storage as Glycogen
- Glycolysis to pyruvate
 - Followed by oxidation to carbon dioxide in the TCA cycle
 - Precursors for the synthesis of glycerol-3-phosphate (the backbone of triacylglycerols and other glycolipids), sialic acid, and serine
 - Entry into the TCA cycle and exit as citrate, followed by conversion to acetyl CoA, malonyl CoA, and entry into fatty acid synthesis and secretion as VLDL
 - Synthesis of phospholipids and other lipids from triacylglycerols
- Conversion to mannose, sialic acid, and other sugars necessary for the synthesis of oligosaccharides for glycoproteins, including those secreted into blood
- Synthesis of acid sugars for proteoglycan synthesis and formation of glucuronides
- Oxidation in the pentose phosphate pathway for the formation of NADPH (necessary for biosynthetic reactions such as fatty acid synthesis, glutathione reduction, and other NADPH-utilizing detoxification reactions)

from oxidative metabolism, the TCA cycle, or the electron transport chain and oxidative phosphorylation. After a mixed meal containing carbohydrate, the major fuels used by the liver are glucose, galactose, and fructose. If ethanol is consumed, the liver is the major site of ethanol oxidation, yielding principally acetate and then acetyl CoA. During an overnight fast, fatty acids become the major fuel for the liver. They are oxidized to carbon dioxide or ketone bodies. The liver also can use all of the amino acids as fuels, converting many of them to glucose. The urea cycle disposes of the ammonia that is generated from amino acid oxidation.

A. Carbohydrate Metabolism in the Liver

After a carbohydrate-containing meal, glucose, galactose, and fructose enter the portal circulation and flow to the liver. This organ serves as the major site in the body for the utilization of dietary galactose and fructose. It metabolizes these compounds by converting them to glucose and intermediates of glycolysis. Their fate is essentially the same as that of glucose (Table 46.4).

B. Glucose as a Fuel

The entry of glucose into the liver is dependent on a high concentration of glucose in the portal vein after a high-carbohydrate meal. Because the K_m for both the glucose transporter (GLUT2) and glucokinase is so high (approximately 10 mM), glucose will enter the liver principally after its concentration rises to 10 to 40 mM in the portal blood and not at the lower 5-mM concentration in the hepatic artery. The increase in insulin secretion that follows a high-carbohydrate meal will promote the conversion of glucose to glycogen. In addition, the rate of glycolysis will be increased (PFK-2 is active; thus, PFK-1 is activated by fructose-2, 6 bisphosphate) such that acetyl CoA can be produced for fatty acid synthesis (acetyl CoA carboxylase will be activated by citrate; see Chapter 33). Thus, after a high-carbohydrate meal, the liver uses glucose as its major fuel, while activating the pathways for glycogen and fatty acid synthesis.

The rate of glucose utilization by the liver is determined, in part, by the level of activity of glucokinase. Glucokinase activity is regulated by a glucokinase regulatory protein (RP, Fig. 46.7), which is located in the nucleus. In the absence of glucose, glucokinase is partially sequestered within the nucleus, bound to RP, in an inactive form. High concentrations of fructose 6-phosphate promote the interaction of glucokinase with RP, whereas high levels of either glucose or fructose 1-phosphate block glucokinase from binding to RP and promote the dissociation of the complex. Thus, as glucose levels rise in the cytoplasm and nucleus (because of increased blood glucose levels after a meal, for example), there is a significant enhancement of glucose phosphorylation as glucokinase is released from the nucleus, travels to the cytoplasm, and phosphorylates glucose.



Why would you expect fructose 1-phosphate levels to promote the dissociation of glucokinase from regulatory protein (RP)?



The role of glucokinase RP is very complex. Mice that have been genetically engineered to no longer express the RP (a knockout mouse) display reduced levels of total glucokinase activity in the liver. This is attributable to the finding that RP is important in the post-transcriptional processing of the mRNA for glucokinase. In the absence of RP, less glucokinase is produced. These mice, therefore, have no glucokinase in the nucleus, a reduced cytoplasmic glucokinase content, and inefficient glucose phosphorylation in the liver when glucose levels rise.



Fructose-1-phosphate is produced from fructose metabolism. The major dietary source of fructose, the ingestion of which would lead to increased fructose 1-phosphate levels, is sucrose. Sucrose is a disaccharide of glucose and fructose. Thus, an elevation of fructose 1-phosphate usually indicates an elevation of glucose levels as well.

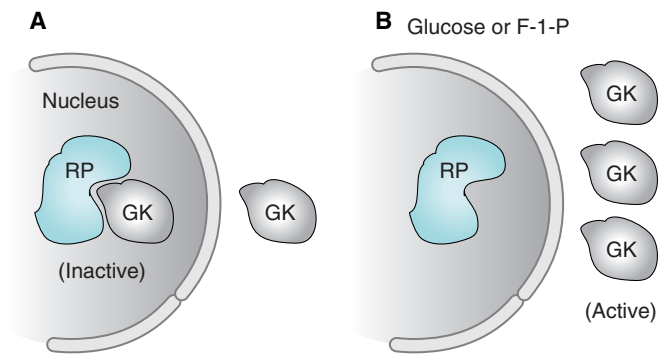


Fig. 46.7. Regulation of glucokinase by regulatory protein (RP). RP is localized to the nucleus, and in the absence of glucose or presence of fructose 6-phosphate, most glucokinase is translocated to the nucleus and binds RP. This leads to the formation of the inactive form of glucokinase. When glucose or fructose-1-phosphate levels rise, glucokinase is released from RP. It then translocates to the cytoplasm and actively converts glucose to glucose 6-phosphate.

The major regulatory step for liver glycolysis is the PFK-1 step. Even under fasting conditions, the ATP concentration in the liver (approximately 2.5 mM) is sufficiently high to inhibit PFK-1 activity. Thus, liver glycolysis is basically controlled by modulating the levels of fructose 2,6-bisphosphate, the product of the PFK-2 reaction. As fructose 2,6-bisphosphate levels increase (which would occur in the presence of insulin) the rate of glycolysis increases; when glucagon levels increase and protein kinase A is activated such that PFK-2 is phosphorylated and inactive, glycolysis will slow down, and gluconeogenesis will be enhanced (see Chapters 22 and 31).

C. Lipid Metabolism

Long-chain fatty acids are a major fuel for the liver during periods of fasting, when they are released from adipose tissue triacylglycerols and travel to the liver as fatty acids bound to albumin.

Within the liver, they bind to fatty acid-binding proteins and are then activated on the outer mitochondrial membrane, the peroxisomal membrane, and the smooth endoplasmic reticulum by fatty acyl CoA synthetases. The fatty acyl group is transferred from CoA to carnitine for transport through the inner mitochondrial membrane, where it is reconverted back into fatty acyl CoA and oxidized to acetyl CoA in the β -oxidation spiral (see Chapter 23).

The enzymes in the pathways of fatty acid activation and β -oxidation (the synthetases, the carnitine acyltransferases, and the dehydrogenases of β -oxidation) are somewhat specific for the length of the fatty acid carbon chain. The chain length specificity is divided into enzymes for long-chain fatty acids (C20 to approximately C12), medium-chain (approximately C12 to C4), and short-chain (C4–C2). The major lipids oxidized in the liver as fuels are the long-chain fatty acids (palmitic, stearic, and oleic acids), because these are the lipids that are synthesized in the liver, are the major lipids ingested from meat or dairy sources, and are the major form of fatty acids present in adipose tissue triacylglycerols. The liver, as well as many other tissues, uses fatty acids as fuels when the concentration of the fatty acid-albumin complex is increased in the blood.

1. MEDIUM-CHAIN LENGTH FATTY ACID OXIDATION

The liver and certain cells in the kidney are the major sites for the oxidation of medium-chain-length fatty acids. These fatty acids usually enter the diet of infants

in maternal milk as medium-chain-length triacylglycerols (MCT). In the intestine, the MCT are hydrolyzed by gastric lipase, bile salt-dependent lipases, and pancreatic lipase more readily than long-chain triacylglycerols. Within the enterocytes, they are neither reconverted to triacylglycerols nor incorporated into chylomicrons. Instead, they are directly released into the portal circulation (fatty acids of approximately 8-carbon chain lengths or less are water-soluble). In the liver, they diffuse through the inner mitochondrial membrane and are activated to acyl CoA derivatives by medium-chain-length fatty acid activating enzyme (MMFAE), a family of similar isozymes present only in liver and kidney. The medium-chain fatty acyl-CoA is then oxidized by the normal route, beginning with medium-chain-length acyl CoA dehydrogenase (MCAD; see Chapter 23).

2. PEROXISOMAL OXIDATION OF VERY-LONG-CHAIN FATTY ACIDS

Peroxisomes are present in greater number in the liver than in other tissues. Liver peroxisomes contain the enzymes for the oxidation of very-long-chain fatty acids such as C24:0 and phytanic acid, for the cleavage of the cholesterol side chain necessary for the synthesis of bile salts, for a step in the biosynthesis of ether lipids, and for several steps in arachidonic acid metabolism. Peroxisomes also contain catalase and are capable of detoxifying hydrogen peroxide.

Very-long-chain fatty acids of C20 to C26 or greater are activated to CoA derivatives by very-long-chain acyl CoA synthetase present in the peroxisomal membrane. The very-long-chain acyl CoA derivatives are then oxidized in liver peroxisomes to the 8-carbon octanoyl CoA level. In contrast to mitochondrial β -oxidation, the first enzyme in peroxisomal β -oxidation introduces a double bond and generates hydrogen peroxide instead of FAD(2H). The remainder of the cycle, however, remains the same, releasing NADH and acetyl CoA. Peroxisomal catalase inactivates the hydrogen peroxide, and the acetyl CoA can be used in biosynthetic pathways such as those of cholesterol and dolichol synthesis.

The octanoyl CoA that is the end-product of peroxisomal oxidation leaves the peroxisomes and the octanoyl group is transferred through the inner mitochondrial membrane by medium-chain-length acylcarnitine transferase. In the mitochondria, it enters the regular β -oxidation pathway, beginning with medium-chain-length acyl CoA dehydrogenase (MCAD).

3. PEROXISOME PROLIFERATOR ACTIVATED RECEPTORS

The peroxisome proliferator activated receptors (PPAR) play an important role in liver metabolism. These receptors obtained their name from the finding that certain agonists were able to induce the proliferation of peroxisomes in liver. These agonists included hypolipidemic agents, nonsteroidal anti-inflammatory agents, and environmental toxins. The receptors that bind these agents, the PPAR, are members of a nuclear receptor family and, when activated, stimulate new gene transcription. Within the liver, the major form of PPAR directly regulates the activity of genes that are involved in fatty acid uptake and β - and ω -oxidation of fatty acids.

There are three major PPAR isoforms, α , δ/β , and γ . The major form found in the liver is the α form. Fatty acids are an endogenous ligand for PPAR α , such that when the level of fatty acids in the circulation is increased (with a concurrent increase in the fatty acid content of hepatocytes), there is increased gene transcription for those proteins involved in regulating fatty acid metabolism (Table 46.5). Genetically altered mice have been generated that lack PPAR α . These “knockout” mice exhibit no abnormal phenotype when fed a normal diet. When fasted, however, or when fed a high-fat diet, these mice develop severe fatty infiltration of the liver. The inability to increase the rate of fatty acid oxidation in this organ leads to excessive fatty acid buildup in the hepatocytes. It also leads to an insufficient energy supply with which to make glucose (leading to hypoglycemia) as well as an inability to



Medium-chain triglycerides (MCT) are important components of nutritional supplements used in patients with digestive disorders. They therefore can be employed as an easily absorbed source of calories in patients who have a gastrointestinal (GI) disorder that may result in malabsorption of nutrients. These diseases include pancreatic insufficiency, intraluminal bile salt deficiency due to cholestatic liver disease, biliary obstruction, ileal disease or resection, and disease causing obstruction of intestinal lymphatics. Remember, however, that MCT do not contain polyunsaturated fatty acids that can be used for synthesis of eicosanoids (see Chapter 35).



Zellweger's (cerebrohepato renal) syndrome occurs in individuals with a rare inherited absence of peroxisomes in all tissues. Patients accumulate C26-C38 polyenoic acids in brain tissue owing to defective peroxisomal oxidation of the very-long-chain fatty acids synthesized in the brain for myelin formation. In liver, bile acid and ether lipid synthesis are affected, as is the oxidation of very-long-chain fatty acids.

Table 46.5. Genes Regulated by Activation of PPAR- α

Fatty acid transport proteins
The mitochondrial and peroxisomal enzymes of fatty acid oxidation
Carnitine palmitoyl transferase I
HMG-CoA synthase
Apoprotein CIII (suppression)



The fibrates (e.g., clofibrate) are a class of drugs that bind to PPARs to elicit changes in lipid metabolism. They are typically prescribed for individuals with elevated triglyceride levels because they increase the rate of triglyceride oxidation. This, in turn, leads to a reduction in serum triacylglycerol levels. Fibrates, through PPAR- α stimulation, also suppress apolipoprotein CIII synthesis and stimulate LPL activity. Apo CIII normally inhibits LPL activity, so by reducing CIII synthesis overall, LPL activity is increased. Apo CIII also blocks apolipoprotein E on IDL particles, causing the IDL particles to accumulate because they cannot be taken up by the apo E receptor in the liver. The suppression of apo CIII levels allows more IDL to be endocytosed, thereby also reducing circulating triacylglycerol levels.



Reye's syndrome is characterized clinically by vomiting with signs of progressive central nervous system damage. In addition, there are signs of hepatic injury and hypoglycemia. There is mitochondrial dysfunction with decreased activity of hepatic mitochondrial enzymes. Hepatic coma may occur as serum ammonia levels rise. It is epidemiologically associated with the consumption of aspirin by children during a viral illness, but it may occur in the absence of exposure to salicylates. The incidence in the United States has decreased dramatically since the 1980s, when parents were made aware of the dangers of giving aspirin to children to reduce fever. Reye's syndrome is not necessarily confined to children. In patients who die of this disease, the liver at autopsy shows swollen and disrupted mitochondria and extensive accumulation of lipid droplets with fatty vacuolization of cells in both the liver and the renal tubules.

produce ketone bodies. In normal fasted mice or mice fed a high-fat diet, fatty acids would eventually stimulate their own oxidation, via peroxisome proliferation and by induction of other enzymes needed for their oxidation. The knockout mice cannot make these compensations.

4. XENOBIOTICS METABOLIZED AS FATTY ACIDS

The liver uses the pathways of fatty acid metabolism to detoxify very hydrophobic and lipid-soluble xenobiotics that, like fatty acids, either have carboxylic acid groups or can be metabolized to compounds that contain carboxylic acids. Benzoate and salicylate are examples of xenobiotics that are metabolized in this way. Benzoate is naturally present in plant foods and is added to foods such as sodas as a preservative. Its structure is similar to salicylic acid (which is derived from the degradation of aspirin). Salicylic acid and benzoate are similar in size to medium-chain-length fatty acids and are activated to an acyl CoA derivative by MMFAE (Fig. 46.8). The acyl group is then conjugated with glycine, which targets the compound for urinary excretion. The glycine derivatives of salicylate and benzoate are called salicylurate and hippurate, respectively. Salicylurate is the major urinary metabolite of aspirin in humans. Benzoate has been administered to treat hyperammonemia associated with congenital defects, because urinary hippurate excretion tends to lower the free ammonia pool. Aspirin cannot be used for this purpose because it is toxic in the large doses required.

5. THE METABOLISM OF LIPIDS IN LIVER DISEASE

Chronic parenchymal liver disease is associated with relatively predictable changes in plasma lipids and lipoproteins. Some of these changes are related to a reduction in the activity of lecithin cholesterol acyltransferase (LCAT). This plasma enzyme is synthesized and glycosylated in the liver; then enters the blood, where it catalyzes the transfer of a fatty acid from the 2-position of lecithin to the 3β -OH group of free cholesterol to produce cholesterol ester and lysolecithin. As expected, in severe parenchymal liver disease, in which LCAT activity is decreased, plasma levels of cholesterol ester are reduced and free cholesterol levels normal or increased.

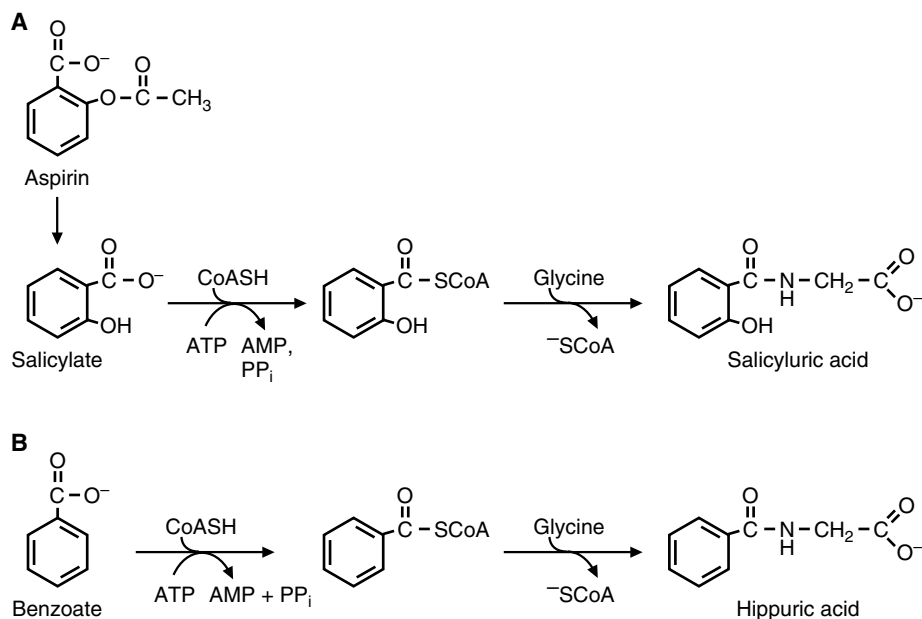


Fig. 46.8. Benzoate and salicylate metabolism.

Plasma triacylglycerols are normally cleared by peripheral lipases (lipoprotein lipase or LPL and hepatic triglyceride lipase or HTGL). Because the activities of both LPL and HTGL are reduced in patients with hepatocellular disease, a relatively high level of plasma triacylglycerols may be found in both acute and chronic hepatitis, in patients with cirrhosis of the liver, and in patients with other diffuse hepatocellular disorders.

With low LCAT activity and the elevated triacylglycerol level described, low-density lipoprotein (LDL) particles have an abnormal composition. They are relatively triacylglycerol rich and cholesterol ester poor.

High-density lipoprotein (HDL) metabolism may be abnormal in chronic liver disease as well. For example, because the conversion of HDL₃ (less antiatherosclerotic) to HDL₂ (more antiatherosclerotic) is catalyzed by LCAT, the reduced activity of LCAT in patients with cirrhosis leads to a decrease in the HDL₂:HDL₃ ratio. Conversely, the conversion of HDL₂ to HDL₃ requires hepatic lipases. If the activity of this lipase is reduced, one would expect an elevation in the HDL₂:HDL₃ ratio. Because the HDL₂:HDL₃ ratio is usually elevated in cirrhosis, the lipase deficiency appears to be the more dominant of the two mechanisms. These changes may result in an overall increase in serum total HDL levels. How this affects the efficiency of the reverse cholesterol transport mechanism and the predisposition to atherosclerosis is not fully understood.

With regard to triacylglycerol levels in patients with severe parenchymal liver disease, the hepatic production of the triacylglycerol-rich, very-low-density lipoprotein (VLDL) particle is impaired. Yet the total level of plasma triacylglycerols remains relatively normal because the LDL particle in such patients is triacylglycerol-rich, for reasons that have not been fully elucidated.

Non-esterified fatty acid (NEFA) levels are elevated in patients with cirrhosis. This change might be expected because basal hepatic glucose output is low in these patients. As a result, more NEFA are presumably required (via increased lipolysis) to meet the fasting energy requirements of peripheral tissues.

D. Amino Acid Metabolism in the Liver

The liver is the principle site of amino acid metabolism in humans. It essentially balances the free amino acid pool in the blood through the metabolism of amino acids supplied by the diet after a protein-containing meal and through metabolism of amino acids supplied principally by skeletal muscles during an overnight fast. In an adult who is no longer growing linearly, the total protein content of the body on a daily basis is approximately constant, such that the net degradation of amino acids (either to other compounds or used for energy) is approximately equal to the amount consumed. The key points concerning hepatic amino acid metabolism are the following:

1. The liver contains all the pathways for catabolism of all of the amino acids and can oxidize most of the carbon skeletons to carbon dioxide. A small proportion of the carbon skeletons are converted to ketone bodies. The liver also contains the pathways for converting amino acid carbon skeletons to glucose (gluconeogenesis) that can be released into the blood.
2. Because the liver is the principle site of amino acid catabolism, it also contains the urea cycle, the pathway that converts toxic ammonium ion to nontoxic urea. The urea is then excreted in the urine.
3. After a mixed or high-protein meal, the gut uses dietary aspartate, glutamate, and glutamine as a fuel (during fasting the gut uses glutamine from the blood as a major fuel). Thus, the ingested acidic amino acids do not enter the general circulation. The nitrogen from gut metabolism of these amino acids is passed to the liver as citrulline or ammonium ion via the portal vein.
4. The branched-chain amino acids (valine, leucine, and isoleucine) can be used by most cell types as a fuel, including cells of the gut and skeletal muscle. After a

high-protein meal, most of the branched-chain amino acids are not oxidized by the liver (because of very low activity of the branched-chain amino acid transaminase) and instead enter the peripheral circulation to be used as a fuel by other tissues or for protein synthesis (these amino acids are essential amino acids). The liver does, however, take up whatever amino acids it needs to carry out its own protein synthesis.

5. Most tissues transfer the amino acid nitrogen to the liver to dispose of as urea. They, therefore, produce either alanine (from the pyruvate–glucose–alanine cycle, in skeletal muscle, kidney, and intestinal mucosa) or glutamine (skeletal muscle, lungs, neural tissues) or serine (kidney), which are released into the blood and taken up by the liver.
6. The liver uses amino acids for the synthesis of proteins that it requires as well as for the synthesis of proteins to be used elsewhere. For example, the liver uses the carbon skeletons and nitrogens of amino acids for the synthesis of nitrogen-containing compounds such as heme, purines, and pyrimidines. The amino acid precursors for these compounds are all nonessential, because they can be synthesized in the liver.

E. Amino Acid Metabolism in Liver Disease

The concentration of amino acids in the blood of patients with liver disease is often elevated. This change is, in part, attributable to a significantly increased rate of protein turnover (general catabolic effect seen in severely ill patients) as well as to impaired amino acid uptake by the diseased liver. It is unlikely that the increased levels are due to degradation of liver protein and the subsequent release of amino acids from the failing hepatocyte into the blood. This is true because the total protein content of the liver is only approximately 300 g. To account for the elevated amino acid levels in the blood, the entire protein content of the liver would have to be degraded within 6 to 8 hours to account for the increased protein turnover rates found. Because 18 to 20 times more protein is present in skeletal muscle (greater mass), the muscle is probably the major source of the elevated plasma levels of amino acids seen in catabolic states such as cirrhosis of the liver.

In cirrhotic patients, such as **Jean Ann Tonich**, the fasting blood α -amino nitrogen level is elevated as a result of reduced clearance. Urea synthesis is reduced as well.

The plasma profile of amino acids in cirrhosis characteristically shows an elevation in aromatic amino acids, phenylalanine and tyrosine, and in free tryptophan and methionine. The latter changes may be caused by impaired hepatic utilization of these amino acids as well as to portosystemic shunting. Although the mechanism is not known, a reduction in fasting plasma levels of the branched-chain amino acids (BCAA) is also seen in cirrhotic patients. These findings, however, must be interpreted with caution because most of the free amino acid pool in humans is found in the intracellular space. Therefore, changes seen in their plasma concentrations do not necessarily reflect their general metabolic fate. Yet the elevation in aromatic amino acids and the suppression of the level of BCAAs in the blood of cirrhotics have been implicated in the pathogenesis of hepatic encephalopathy.

V. DISEASES OF THE LIVER

Diseases of the liver can be clinically and biochemically devastating, because no other organ can compensate for the loss of the multitude of functions that the liver normally performs. Alcohol-induced liver disease has been discussed in Chapter 25. A number of diseases can lead to hepatic fibrosis (see Biochemical Comments)



Unlike **Amy Biasis**, whose hepatic amoebic disorder was more localized (abscesses), **Jean Anne Tonich** had a diffuse hepatic disease, known as alcohol-induced cirrhosis (historically referred to as “Laennec’s cirrhosis”). The latter is characterized by diffuse fine scarring, a fairly uniform loss of hepatic cells, and the formation of small regenerative nodules (sometimes referred to as “micronodular cirrhosis”). With continued alcohol intake, fibroblasts and activated stellate cells deposit collagen at the site of persistent injury. This leads to the formation of weblike septa of connective tissue in periportal and pericentral zones. These eventually connect portal triads and central veins. With further exposure to alcohol, the liver shrinks and then becomes nodular and firm as “end-stage” cirrhosis develops. Unless successfully weaned from alcohol, these patients eventually die of liver failure. **Amy Biasis**, however, can probably look forward to enjoying normal liver function after successful amoebicidal therapy without evidence of residual hepatic scarring.

and cirrhosis. When this occurs to a great enough extent, liver function becomes inadequate for life. Signs and symptoms of liver disease include elevated levels of the enzymes alanine aminotransferase (ALT) and aspartate aminotransferase (AST) in the plasma (due to hepatocyte injury or death with a consequent release of these enzymes into the blood), jaundice (an accumulation of bilirubin in the blood caused by inefficient bilirubin glucuronidation by the liver; see Chapter 45), increased clotting times (the liver has difficulty producing clotting factors for secretion), edema (reduced albumin synthesis by the liver leads to a reduction in osmotic pressure in the blood), and hepatic encephalopathy (reduced urea cycle activity leading to excessive levels of ammonia and other toxic compounds in the central nervous system).



Alanine aminotransferase and aspartate aminotransferase are the modern versions of older names.

ALT was formally known as serum glutamic pyruvate transaminase (SPGT), and AST was formally known as serum glutamic oxaloacetic transaminase (SGOT).

CLINICAL COMMENTS



Patients with cirrhosis of the liver who have no known genetic propensity to glucose intolerance, such as **Jean Ann Tonich**, tend to have higher blood glucose levels than do normal subjects in both fasting and fed states. The mechanisms that may increase glucose levels in the fasting state include a reduction in the metabolic clearance rate of glucose by 25 to 40% compared with normal subjects. This reduction in glucose clearance results, in part, from increased oxidation of fatty acids and ketone bodies and the consequent decrease in glucose oxidation by peripheral tissues in cirrhosis patients. This is suggested by the discovery that plasma non-esterified fatty acid (NEFA) levels are high in many patients with hepatocellular dysfunction, in part because of decreased hepatic clearance of NEFA and in part because of increased adipose tissue lipolysis. Another possible explanation for the reduction in whole body glucose utilization in cirrhotic patients relates to the finding that ketone body production is increased in some patients with cirrhosis. This could lead to enhanced utilization of ketone bodies for fuel by the central nervous system in such patients, thereby reducing the need for glucose oxidation by the highly metabolically active brain.

After glucose ingestion (fed state), many patients with liver disease have abnormally elevated blood glucose levels (“hepatogenous diabetes”). Using World Health Organization (WHO) criteria, 60 to 80% of cirrhotic patients have varying degrees of glucose intolerance, and overt diabetes mellitus occurs 2 to 4 times as often in cirrhotics than it does in subjects without liver disease. The proposed mechanisms include a degree of insulin resistance in peripheral tissues; however, as the cirrhotic process progresses, they develop a marked impairment of insulin secretion as well. Although the mechanisms are not well understood, this decrease in insulin secretion leads to increased hepatic glucose output (leading to fasting hyperglycemia) and reduced suppression of hepatic glucose output after meals, leading to postprandial hyperglycemia as well. If the patient has an underlying genetic predisposition to diabetes mellitus, the superimposition of the mechanisms outlined above will lead to an earlier and more significant breakdown in glucose tolerance in these specific patients.

BIOCHEMICAL COMMENTS



Extensive and progressive fibrosis of the hepatic parenchyma leads to cirrhosis of the liver, a process that has many causes. The development of fibrosis requires the activities of hepatic stellate cells, cytokines, proteases, and protease inhibitors.

A major change that occurs when fibrosis is initiated is that the normally “sparse” or “leaky” basement membrane between the endothelial cells and the hepatocyte is replaced with a high-density membrane containing fibrillar collagen. This occurs because of both an increased synthesis of a different type of collagen than is normally produced and a reduction in the turnover rate of existing extracellular matrix components.

The supportive tissues of the normal liver contain an extracellular matrix that, among other proteins, includes type IV collagen (which does not form fibers), glycoproteins, and proteoglycans. After a sustained insult to the liver, a threefold to eightfold increase occurs in extracellular matrix components, some of which contain fibril-producing collagen (types I and III), glycoproteins, and proteoglycans. The accumulation of these fibril-producing compounds leads to a loss of endothelial cell fenestrations and, therefore, a loss of the normal sieve-like function of the basement membranes. These changes interfere with normal transmembrane metabolic exchanges between the blood and hepatocytes.

The hepatic stellate cell is the source of the increased and abnormal collagen production. These cells are activated by growth factors whose secretion is induced by injury to the hepatocytes or endothelial cells. Growth factors involved in cellular activation include TGF- β 1 (which is derived from the endothelial cells, Kupffer cells, and platelets) and platelet-derived growth factor (PDGF) and epidermal growth factor (EGF) from platelets. The release of PDGF stimulates stellate cell proliferation and, in the process, increases their synthesis and release of extracellular matrix materials and remodeling enzymes. These enzymes include matrix metalloproteinases (MMPs) and tissue inhibitors of MMPs, as well as converting (activating) enzymes. This cascade leads to the degradation of the normal extracellular matrix and replacement with a much denser and more rigid type of matrix material. These changes are, in part, the result of an increase in the activity of tissue inhibitors of MMP's for the new collagen relative to the original collagen in the extracellular matrix.

One consequence of the increasing stiffness of the hepatic vascular channels through which hepatic blood must flow is a greater resistance to the free flow of blood through the liver as a whole. Resistance to intrahepatic blood flow is also increased by a loss of vascular endothelial cell fenestrations, loss of free space between the endothelial cells and the hepatocytes (space of Disse), and even loss of vascular channels per se. This increased vascular resistance leads to an elevation in intrasinusoidal fluid pressure. When this intrahepatic (portal) hypertension reaches a critical threshold, the shunting of portal blood away from the liver (portosystemic shunting) further contributes to hepatic dysfunction. If the portal hypertension cannot be reduced, portal blood will continue to bypass the liver and return to the heart through the normally low-pressure esophageal veins. When this increasing intraesophageal venous pressure becomes severe enough, the walls of these veins thin dramatically and expand to form varices, which may suddenly burst, causing life-threatening esophageal variceal hemorrhage. This is a potentially fatal complication of cirrhosis of the liver.

Suggested Reading

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REVIEW QUESTIONS—CHAPTER 46

1. Drinking grapefruit juice while taking statins can lead to potentially devastating side effects. This is due to a component of grapefruit juice doing which of the following?
 - (A) Interfering with hepatic uptake of statins
 - (B) Accelerating the conversion of the statin to a more toxic form
 - (C) Inhibiting the inactivation of statins
 - (D) Upregulating the HMG CoA reductase
 - (E) Downregulating the HMG CoA reductase
2. Which one of the following characteristics of cytochrome P450 enzymes is correct?
 - (A) They are all found in the Golgi apparatus and are referred to as microsomal enzymes.
 - (B) They all contain a flavin-containing reductase unit that uses NADH and not NADPH as a source of electrons.
 - (C) They are all inducible by oxygen, which binds to the iron of the cytochrome.
 - (D) They all oxidize the substrate on which they act.
 - (E) They all generate a free radical compound as final products of the reaction.
3. Fairly predictable changes occur in the various metabolic pathways of lipid metabolism in patients with moderately advanced hepatocellular disease. Which one of the following changes would you expect to see under these conditions?
 - (A) The activity of plasma lecithin cholesterol acyltransferase (LCAT) is increased.
 - (B) Serum cholesterol esters are increased.
 - (C) Hepatic triglyceride lipase (HTGL) activity is increased.
 - (D) Serum triacylglycerol levels are increased.
 - (E) Serum nonesterified fatty acid levels are decreased.
4. After a 2-week alcoholic binge, **Jean Ann Tonich** ingested some Tylenol to help her with a severe headache. She took three times the suggested dose because of the severity of the pain. However, within 24 hours Jean Ann became very lethargic, vomited frequently, and developed severe abdominal pain. The symptoms Jean Ann is experiencing are attributable to a reaction to the Tylenol due to which of the following?
 - (A) The hypoglycemia experienced by the patient
 - (B) Ethanol-induced inhibition of Tylenol metabolism
 - (C) The hyperglycemia experienced by the patient
 - (D) Ethanol-induced acceleration of Tylenol metabolism
 - (E) Acetaminophen inhibition of VLDL secretion by the liver
5. An individual displays impaired glucose tolerance; blood glucose levels remain elevated after a meal for a longer time than is normal, although they do eventually go down to fasting levels. The patient has a normal release of insulin from the pancreas in response to elevated blood glucose levels. Fibroblasts obtained from the patient display normal levels of insulin binding to its receptor, and normal activation of the intrinsic tyrosine kinase activity associated with the insulin receptor. Analysis of glucose 6-phosphate formation within the fibroblasts, however, indicated a much slower rate of formation than in fibroblasts obtained from a normal control. A possible mutation that could lead to these results is which of the following?
 - (A) A decrease in the K_m of glucokinase
 - (B) An increase in the V_{max} of glucokinase
 - (C) A nonfunctional glucokinase regulatory protein
 - (D) An increase in hexokinase activity
 - (E) A decrease in hexokinase activity

47 Metabolism of Muscle at Rest and During Exercise

*There are three types of muscle cells: **smooth**, **skeletal**, and **cardiac**. In all types of muscle, contraction occurs via an actin/myosin sliding filament system, which is regulated by oscillations in **intracellular calcium levels**.*

*Muscle cells use stored glycogen and circulating glucose, fatty acids, and amino acids as energy sources. Muscle glycolysis is regulated differently from the liver, with the key difference being the regulation of **phosphofructokinase-2 (PFK-2)**. Muscle PFK-2 is not inhibited by phosphorylation; cardiac PFK-2 is actually activated by an insulin-stimulated protein kinase. Thus, under conditions in which liver PFK-2 is inactive, and glycolysis is running slowly, muscle glycolysis is either unaffected, or even stimulated, depending on the isoform of PFK-2 being expressed.*

*Although muscle cells do not synthesize fatty acids, they do contain an isozyme of **acetyl CoA carboxylase (ACC-2)** to regulate the **rate of fatty acid oxidation**. ACC-2 produces malonyl CoA, which inhibits carnitine palmitoyl transferase I, thereby blocking fatty acid entry into the mitochondria. Muscle also contains **malonyl CoA decarboxylase**, which catalyzes the conversion of malonyl CoA to acetyl CoA and carbon dioxide. Thus, both the synthesis and degradation of malonyl CoA is carefully regulated in muscle cells to balance glucose and fatty acid oxidation. Both allosteric and covalent means of regulation are employed. Citrate activates ACC-2, and phosphorylation of ACC-2 by the adenosine monophosphate (AMP)-activated protein kinase inhibits ACC-2 activity. Phosphorylation of malonyl CoA decarboxylase by the AMP-activated protein kinase activates the enzyme, further enhancing fatty acid oxidation when energy levels are low.*

*Muscles use **creatine phosphate** to store high-energy bonds. Creatine is derived from arginine and glycine in the kidney, and the guanidinoacetate formed is methylated (using S-adenosyl methionine) in the liver to form creatine. The enzyme **creatine phosphokinase (CPK)** then catalyzes the reversible transfer of a high-energy phosphate from adenosine triphosphate (ATP) to creatine, forming creatine-phosphate and adenosine diphosphate (ADP). Creatine phosphate is unstable and spontaneously cyclizes to form **creatinine**, which is excreted in the urine. The spontaneous production of creatinine occurs at a constant rate and is proportional to body muscle mass. Thus, the amount of creatinine excreted each day (the creatinine clearance rate) is constant and can be used as an indicator of the normalcy of the excretory function of the kidneys.*

*Skeletal muscle cells can be subdivided into **type I** and **type II fibers**. Type I fibers are **slow-twitch** fibers that use primarily **oxidative metabolism** for energy, whereas the type II fibers (**fast-twitch**) use **glycolysis** as their primary energy-generating pathway.*

***Glucose transport** into muscle cells can be stimulated during exercise because of the activity of the **AMP-activated protein kinase**. Fatty acid uptake into exercising muscle is dependent on the levels of circulating fatty acids, which are increased by **epinephrine** release.*



THE WAITING ROOM



Rana Felya, a 9-year-old girl, complained of a severe pain in her throat and difficulty in swallowing. She had chills, sweats, headache, and a fever of 102.4° F. When her symptoms persisted for several days, her mother took her to the pediatrician, who found diffuse erythema (redness) in her posterior pharynx (throat) with yellow exudates (patches) on her tonsils. Large, tender lymph nodes were present under her jaw on both sides of her neck. A throat culture was taken, and therapy with penicillin was begun.

Although the sore throat and fever improved, 8 days after the onset of the original infection, Rana's eyes and legs became swollen and her urine suddenly turned the color of "Coca-Cola." Her blood pressure was elevated. Protein and red blood cells were found in her urine. Her serum creatinine level was elevated at 1.8 mg/dL (reference range = 0.3–0.7 for a child). Because the throat culture grew out group A β -hemolytic streptococci, the doctor ordered a Streptozyme test. This test was positive for antibodies to streptolysin O and several other streptococcal antigens. As a result, a diagnosis of acute poststreptococcal glomerulonephritis was made. Supportive therapy, including bed rest and treatment for hypertension, was initiated.

I. MUSCLE CELL TYPES

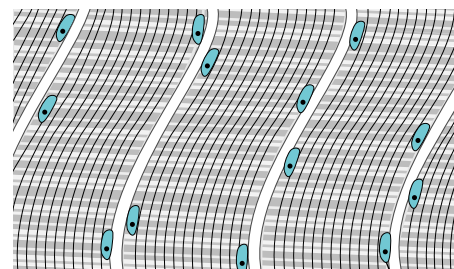
Muscle consists of three different types: skeletal, smooth, and cardiac (Fig. 47.1). The metabolism of each is similar, but the functions of the muscle are quite different.

A. Skeletal Muscle

Skeletal muscles are those muscles that are attached to bone and facilitate the movement of the skeleton. Skeletal muscles are found in pairs, which are responsible for opposing, coordinated directions of motion on the skeleton. The muscles appear striated under the microscope, and are controlled voluntarily (you think about moving a specific muscle group, and then it happens).

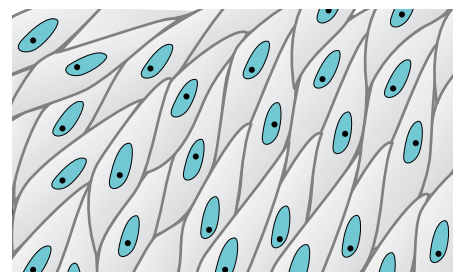
Skeletal muscle cells are long, cylindrical fibers that run the length of the muscle. The fibers are multinucleated because of cell fusion during embryogenesis. The cell membrane surrounding the fibers is called the sarcolemma, and the sarcoplasm is the intracellular milieu, which contains the proteins, organelles, and contractile apparatus of the cell. The sarcoplasmic reticulum is analogous to the endoplasmic reticulum in other cell types and is an internal membrane system that runs throughout the length of the muscle fiber. Another membrane structure, the transverse tubules (T-tubules), are thousands of invaginations of the sarcolemma that tunnel from the surface toward the center of the muscle fiber to make contact with the terminal cisterns of the sarcoplasmic reticulum. Because the T tubules are open to the outside of the muscle fiber and filled with extracellular fluid, the muscle action potential that propagates along the surface of the muscle fiber's sarcolemma travels into the T tubules and to the sarcoplasmic reticulum.

The striations in skeletal muscle are attributable to the presence and organization of myofibrils in the cells. Myofibrils are thread-like structures consisting of thin and thick filaments. The contractile proteins actin and myosin are contained within the filaments; myosin within the thick filaments, actin within the thin filaments. The sliding of these filaments relative to each other, using myosin-catalyzed ATP hydrolysis as an energy source, allows for the contraction and relaxation of the muscle (see Fig. 19.4).



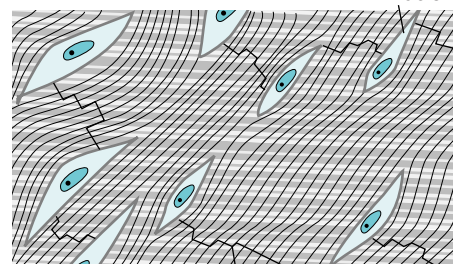
A. Skeletal muscle

Nuclei



B. Smooth muscle

Nuclei



C. Cardiac muscle

Intercalated disks

Fig. 47.1. Structures of the three different muscle types. Smooth Cardiac. Adapted from Basic Histology, Text and Atlas, Junqueira and Carneiro, 10th Ed. Lange, McGraw-Hill, 2002.



Duchenne's muscular dystrophy is caused by the absence of the protein dystrophin, which is a structural protein located in the sarcolemma. Dystrophin is required to maintain the integrity of the sarcolemma, and when absent leads to a loss of muscle function, caused by breakdown of the sarcolemma. The gene is X-linked, and mutations that lead to Duchenne's muscular dystrophy generally result from large deletions of the gene, such that dystrophin is absent from the membrane. Becker's muscular dystrophy, a milder form of disease, is caused by point mutations in the dystrophin gene. In Becker's muscular dystrophy, dystrophin is present in the sarcolemma, but in a mutated form.

Muscle fibers can be classified as either fast-twitch or slow-twitch. The slow-twitch fibers, or type I fibers (also called slow-oxidative), contain large amounts of mitochondria and myoglobin (giving them a red color), utilize respiration and oxidative phosphorylation for energy, and are relatively resistant to fatigue. Compared with fast-twitch fibers, their glycogen content is low. The slow-twitch fibers develop force slowly but maintain contractions longer than fast-twitch muscle.

The fast-twitch fibers, or type II, can be subdivided as type IIa or IIb. Type IIb fibers (also called fast-glycolytic) have few mitochondria and low levels of myoglobin (hence, they appear white). They are rich in glycogen and use glycogenolysis and glycolysis as their primary energy source. These muscles are prone to fatigue, because a continued reliance on glycolysis to produce ATP leads to an increase in lactic acid levels, resulting in a drop in the intracellular pH. As the pH drops, the ability of the muscle to produce ATP also diminishes. However, fast-twitch muscle can develop greater forces than slow-twitch muscle, such that contractions occur more rapidly. Type IIa fibers (also called fast-oxidative glycolytic) have properties of both type I and IIb fibers and thus display functional characteristics of both fiber types. The properties of types I, IIa, and IIb fibers are summarized in Table 47.1.

Muscles are a mixture of the different fiber types, but depending on the function a muscle could have a preponderance of one fiber type over another. Type I fibers are found in postural muscles such as the psoas in the back musculature or the soleus in the leg. The percentage of type I to type II will vary with the muscle. The triceps, which functions phasically, has 32.6% type I, whereas the soleus, which functions tonically, has 87.7% type I. Type II fibers are more prevalent in the large muscles of the limbs that are responsible for sudden, powerful movements. Extraocular muscles would also have more of these fibers than type I.

B. Smooth Muscle Cells

Smooth muscle cells are found in the digestive system, blood vessels, bladder, airways, and uterus. The cells have a spindle shape with a central nucleus (see Fig. 47.1B). The designation of smooth refers to the fact that these cells, which contain a single nucleus, display no striations under the microscope. The contraction of smooth muscle is controlled involuntarily (the cells contract and relax without any conscious attempt to have them do so; examples of smooth muscle activity include moving food

Table 47.1. Properties of Muscle Fiber Types

Type I Fibers	Type II Fibers	
	Type IIa	Type IIb
• Slow-twitch (slow speed of contraction) • Slow-oxidative (low glycogen content) • High myoglobin content (appear red) • Small fiber diameter • Increased concentration of capillaries surrounding muscle (greater oxygen delivery) • High capacity for aerobic metabolism • High resistance to fatigue • Used for prolonged, aerobic exercise	• Intermediate-twitch (fast speed of contraction) • Fast-oxidative glycolytic fibers (intermediate glycogen levels) • Intermediate fiber diameter • High myoglobin content (appear red) • Increased oxidative capacity on training • Intermediate resistance to fatigue	• Fast-twitch (fast speed of contraction) • Fast-glycolytic (high glycogen content) • Low myoglobin content (appear white) • Low mitochondrial content • Limited aerobic metabolism • Large fiber diameter • More sensitive to fatigue as compared with other fiber types • Least efficient use of energy, primarily glycolytic • Used for sprinting and resistance tasks

along the digestive tract, altering the diameter of the blood vessels, and expelling urine from the bladder). In contrast to skeletal muscle, these cells have the ability to maintain tension for extended periods, and do so efficiently, with a low use of energy.

C. Cardiac Muscle Cells

The cardiac cells are similar to skeletal muscle in that they are striated (contain fibers), but like smooth muscle cells they are regulated involuntarily (we do not have to think about making our heart beat). The cells are quadrangular in shape (see Fig. 47.1C) and form a network with multiple other cells through tight membrane junctions and gap junctions. The multicellular contacts allow the cells to act as a common unit and to contract and relax synchronously. Cardiac muscle cells are designed for endurance and consistency. They depend on aerobic metabolism for their energy needs because they contain many mitochondria and very little glycogen. These cells thus generate only a small amount of their energy from glycolysis using glucose derived from glycogen.

II. NEURONAL SIGNALS TO MUSCLE

For an extensive review of how muscle contracts or a detailed view of the signaling to allow muscle contraction, consult a medical physiology book. Only a brief overview is presented here.

The nerve–muscle cell junction is called the neuromuscular junction (Fig. 47.2). When appropriately stimulated, the nerve cell releases acetylcholine at the junction, which binds to acetylcholine receptors on the muscle membrane. This binding stimulates the opening of sodium channels on the sarcolemma. The massive influx of sodium ions results in the generation of an action potential in the sarcolemma at the edges of the motor end plate of the neuromuscular junction. The action potential sweeps across the surface of the muscle fiber and down the transverse tubules to the sarcoplasmic reticulum, where it initiates the release of calcium from its lumen, via the ryanodine receptor (Fig. 47.3). The calcium ion binds to troponin, resulting in a conformational change in the troponin–tropomyosin complexes such that they move away from the myosin-binding sites on the actin. When the binding site becomes available, the myosin head attaches to the myosin-binding site on the actin. The binding is followed by a conformational change (pivoting) in the myosin head, which shortens the sarcomere. After the pivoting, ATP binds the myosin head, which detaches from the actin and is available to bind another myosin-binding site on the



A reduced flow of oxygen-rich blood to the heart muscle may lead to a myocardial infarction (heart attack). The amount of ATP that can be generated by glycolysis alone is not sufficient to meet the energy requirements of the contracting heart.



The ryanodine receptors are calcium release channels found in the endoplasmic reticulum and sarcoplasmic reticulum of muscle cells. One type of receptor can be activated by a depolarization signal (depolarization-induced calcium release). Another receptor type is activated by calcium ions (calcium-induced calcium release). The proteins received their name because they bind ryanodine, a toxin obtained from the stem and roots of the plant *Ryania speciosa*. Ryanodine inhibits sarcoplasmic reticulum calcium release, and acts as a paralytic agent. It was first used commercially in insecticides.

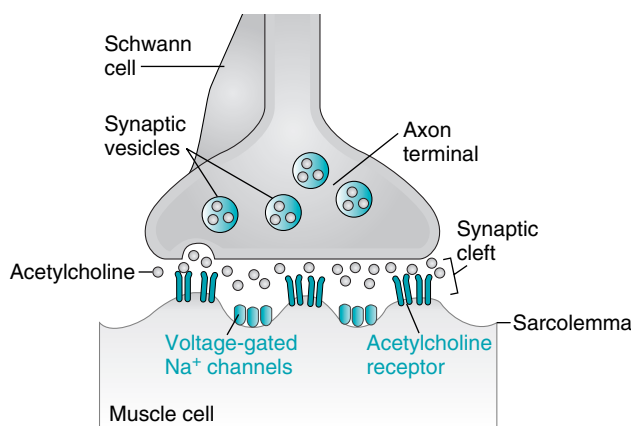


Fig. 47.2. The neuromuscular junction. When appropriately stimulated, the synaptic vesicles, containing acetylcholine, fuse with the axonal membrane and release acetylcholine into the synaptic cleft. The acetylcholine binds to its receptors on the muscle cells, which will initiate signaling for muscle contraction.

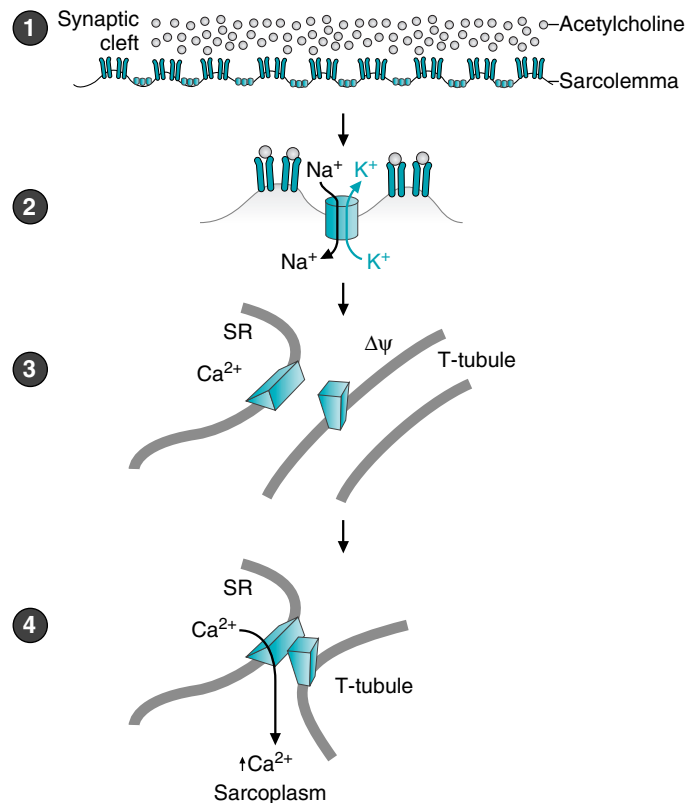


Fig. 47.3. Events leading to sarcoplasmic reticulum calcium release in skeletal muscle. 1. Acetylcholine, released at the synaptic cleft, binds to acetylcholine receptors on the sarcolemma, leading to a change of conformation of the receptors such that they now act as an ion pore. This allows sodium to enter the cell and potassium to leave. 2. The membrane polarization that results from these ion movements is transmitted throughout the muscle fiber by the T-tubule system. 3. A receptor in the T-tubules (the dihydropyridine receptor, DHPR) is activated by membrane polarization (a voltage-gated activation) such that activated DHPR physically binds to and activates the ryanodine receptor in the sarcoplasmic reticulum (depolarization-induced calcium release). 4. The activation of the ryanodine receptor, which is a calcium channel, leads to calcium release from the SR into the sarcoplasm. In cardiac muscle, activation of DHPR leads to calcium release from the T-tubules, and this small calcium release is responsible for the activation of the cardiac ryanodine receptor (calcium-induced calcium release) to release large amounts of calcium into the sarcoplasm.



Acetylcholine levels in the neuromuscular junction are rapidly reduced by the enzyme acetylcholinesterase. A number of nerve gas poisons act to inhibit acetylcholinesterase (such as sarin and VX), such that muscles are continuously stimulated to contract. This leads to blurred vision, bronchoconstriction, seizures, respiratory arrest, and death. The poisons are covalent modifiers of acetylcholinesterase; therefore, recovery from exposure to such poisons requires the synthesis of new enzyme. A new generation of acetylcholinesterase inhibitors, which act reversibly (i.e., they do not form covalent bonds with the enzyme), are now being used to treat dementia, in particular dementia as brought about by Alzheimer's disease.

actin. As long as calcium ion and ATP remain available, the myosin heads will repeat this cycle of attachment, pivoting, and detachment (Fig. 47.4). This movement requires ATP, and when ATP levels are low (such as occurs during ischemia), the ability of the muscle to relax or contract is compromised. As the calcium release channel closes, the calcium is pumped back into the sarcoplasmic reticulum against its concentration gradient by use of the energy-requiring protein SERCA (sarcoplasmic reticulum Ca^{2+} ATPase), and contraction stops. This basic process occurs in all muscle cell types, with some slight variations between cell types.

III. GLYCOLYSIS AND FATTY ACID METABOLISM IN MUSCLE CELLS

The pathways of glycolysis and fatty acid oxidation in muscle are the same as has been previously described (see Chapters 22 and 23). The difference between muscles and other tissues is how these pathways are regulated.

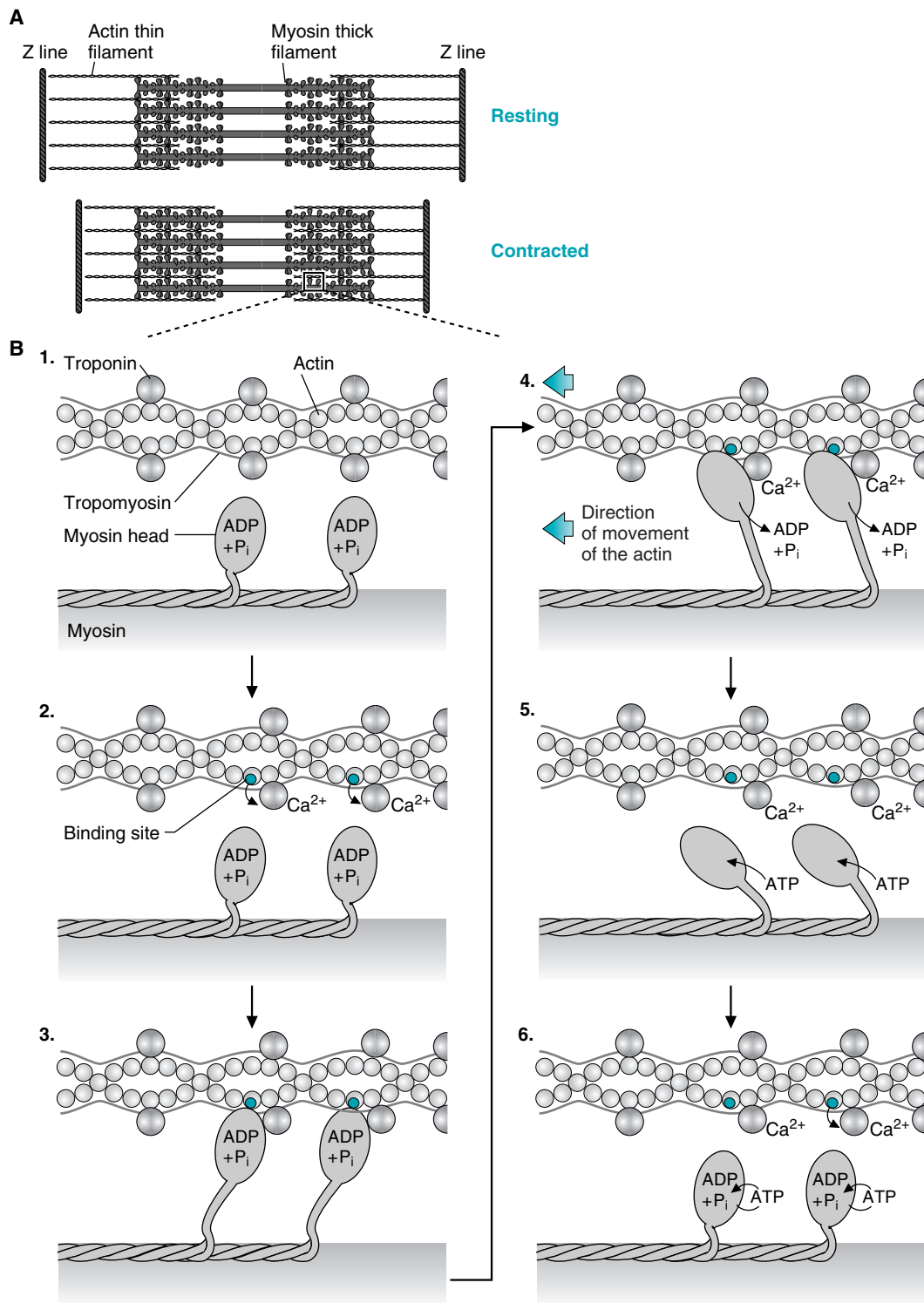


Fig. 47.4. An overview of muscle contraction. **A.** Muscle contraction. During muscle contraction, the myosin head binds to the actin thin filament. A pivoting of the myosin head toward the center of the sarcomere pulls the Z-lines closer together, with subsequent shortening of the sarcomere. **B.** A closer look at myosin–actin interactions. 1. A resting sarcomere. The troponin–tropomyosin complex is blocking the myosin-binding sites on the actin. The myosin head is already energized to power a contraction. 2. Exposure of the active site. After calcium binding to troponin, a conformational change in the troponin molecule pulls the troponin away from the binding site. 3. Cross-bridge attachment. Once the binding sites on the actin are exposed, the myosin head binds to it. 4. Myosin head pivoting. After cross-bridge attachment, the energy stored in the myosin head is released, and the myosin head pivots toward the center of the sarcomere (power stroke). Now the ADP and phosphate bound to the myosin head are released. 5. Detachment of the cross-bridge. Now a molecule of ATP binds to the myosin head with simultaneous detachment of the myosin head from the binding site on the actin molecule. 6. Reactivation of the myosin head. The ATPase activity of the myosin head hydrolyzes the ATP into ADP and phosphate. The energy released from the hydrolysis of this high-energy bond is used to re-energize the myosin head, and the entire cycle can be repeated as long as calcium is present and there are sufficient ATP reserves.



ACC-1 is the isozyme of acetyl-coA carboxylase, which is used to synthesize malonyl-CoA for fatty acid synthesis. Muscle cells do not synthesize fatty acids; however, they do carefully regulate the oxidation of fatty acids through the synthesis and destruction of malonyl-CoA. ACC-1 is a cytosolic protein, whereas ACC-2 is mitochondrial, closely linked to CPT-I in the outer mitochondrial membrane. Mice that have been bred to lack ACC-2 have a 50% reduction of fat stores as compared with “normal” mice. This was shown to be attributable to a 30% increase in skeletal muscle fatty acid oxidation because of the dysregulation of CPT-I, because malonyl CoA could not be produced to regulate the rate at which fatty acid oxidation occurred.



Under conditions in which ketone bodies are produced, fatty acid levels in the plasma are also elevated. Because the heart preferentially burns fatty acids as a fuel rather than the ketone bodies produced by the liver, the ketone bodies are spared for use by the nervous system.

Phosphofructokinase 2 (PFK-2) is negatively regulated by phosphorylation in the liver (the enzyme that catalyzes the phosphorylation is the cyclic adenosine monophosphate [cAMP]–dependent protein kinase). However, in skeletal muscle, PFK-2 is not regulated by phosphorylation. This is because the skeletal muscle isozyme of PFK-2 lacks the regulatory serine residue, which is phosphorylated in the liver. However, the cardiac isozyme of PFK-2 is phosphorylated and activated by a kinase cascade initiated by insulin. This allows the heart to activate glycolysis and to use blood glucose when blood glucose levels are elevated.

Fatty acid uptake by muscle requires the participation of fatty acid-binding proteins and the usual enzymes of fatty acid oxidation. Fatty acyl-CoA uptake into the mitochondria is controlled by malonyl-CoA, which is produced by an isozyme of acetyl-coA carboxylase (ACC-2; the ACC-1 isozyme is found in liver and adipose tissue and is used for fatty acid biosynthesis). ACC-2 is inhibited by phosphorylation by the AMP-activated protein kinase (AMP-PK) such that when energy levels are low the levels of malonyl CoA will drop, allowing fatty acid oxidation by the mitochondria. In addition, muscle cells also contain the enzyme malonyl CoA decarboxylase, which is activated by phosphorylation by the AMP-PK. Malonyl CoA decarboxylase converts malonyl CoA to acetyl CoA, thereby relieving the inhibition of carnitine palmitoyl transferase I (CPT-I) and stimulating fatty acid oxidation (Fig. 47.5). Muscle cells do not synthesize fatty acids; the presence of acetyl CoA carboxylase in muscle is exclusively for regulatory purposes.

IV. FUEL UTILIZATION IN CARDIAC MUSCLE

A. Normal Conditions

The heart primarily uses fatty acids (60–80%), lactate, and glucose (20–40%) as its energy sources. Ninety-eight percent of cardiac ATP is generated by oxidative means; 2% is derived from glycolysis. The lactate used by the heart is taken up by a monocarboxylate transporter in the cell membrane that is also used for the transport of ketone bodies. However, ketone bodies are not a preferred fuel for the heart, because the heart prefers to use fatty acids.

Lactate is generated by red blood cells and working skeletal muscle. When the lactate is used by the heart, it is oxidized to carbon dioxide and water, following the

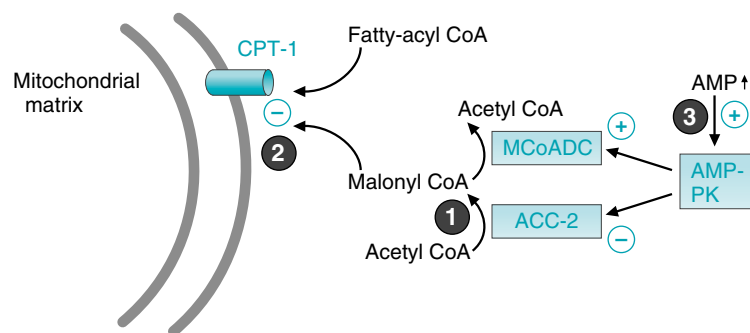


Fig. 47.5. Regulation of fatty acyl CoA entry into muscle mitochondria. 1. Acetyl CoA carboxylase-2 (ACC-2) converts acetyl CoA to malonyl CoA, which inhibits carnitine palmitoyl transferase I (CPT-I), thereby blocking fatty acyl CoA entry into the mitochondria. 2. However, as energy levels drop, AMP levels rise because of the activity of the adenylate kinase reaction. 3. The increase in AMP levels activates the AMP-activated protein kinase (AMP-PK), which phosphorylates and inactivates ACC-2, and also phosphorylates and activates malonyl CoA decarboxylase (MCoADC). The decarboxylase converts malonyl CoA to acetyl CoA, thereby relieving the inhibition of CPT-1, and allowing fatty acyl CoA entry into the mitochondria. This allows the muscle to generate ATP via the oxidation of fatty acids.

pathway lactate to pyruvate, pyruvate to acetyl-coA, acetyl CoA oxidation in the TCA cycle, and ATP synthesis through oxidative phosphorylation. An alternative fate for lactate is its utilization in the reactions of the Cori cycle in the liver.

Glucose transport into the cardiocyte occurs via both GLUT1 and GLUT4 transporters, although approximately 90% of the transporters are GLUT4. Insulin stimulates an increase in the number of GLUT4 transporters in the cardiac cell membrane, as does myocardial ischemia. This ischemia-induced increase in GLUT4 transporter number is additive to the effect of insulin on the translocation of GLUT4 transporters to the plasma membrane.

Fatty acid uptake into cardiac muscle is similar to that for other muscle cell types and requires fatty acid-binding proteins and carnitine palmitoyl transferase I for transfer into the mitochondria. Fatty acid oxidation in cardiac muscle cells is regulated by altering the activities of ACC-2 and malonyl CoA decarboxylase.

B. Ischemic Conditions

When blood flow to the heart is interrupted, the heart switches to anaerobic metabolism. The rate of glycolysis increases, but the accumulation of protons (via lactate formation) is detrimental to the heart. Ischemia also increases the levels of free fatty acids in the blood and, surprisingly, when oxygen is reintroduced to the heart, the high rate of fatty acid oxidation in the heart is detrimental to the recovery of the damaged heart cells. Fatty acid oxidation occurs so rapidly that NADH accumulates in the mitochondria, leading to a reduced rate of NADH shuttle activity, an increased cytoplasmic NADH level, and lactate formation, which generates more protons. In addition, fatty acid oxidation increases the levels of mitochondrial acetyl CoA, which inhibits pyruvate dehydrogenase, leading to cytoplasmic pyruvate accumulation and lactate production. As lactate production increases and the intracellular pH of the heart drops, it is more difficult to maintain ion gradients across the sarcolemma. ATP hydrolysis is required to repair these gradients, which are essential for heart function. However, the use of ATP for gradient repair reduces the amount of ATP available for the heart to use in contraction, which, in turn, compromises the ability of the heart to recover from the ischemic event.



A new class of drugs, known as partial fatty acid oxidation (pFOX) inhibitors, are being developed to reduce the extensive fatty acid oxidation in heart after an ischemic episode (the prototype drug is ranolazine). The reduction in fatty acid oxidation induced by the drug will allow glucose oxidation to occur and reduce lactate buildup in the damaged heart muscle. Other possible targets of such drugs, which have yet to be developed, include ACC-2, malonyl CoA decarboxylase, and carnitine palmitoyl transferase I.

V. FUEL UTILIZATION IN SKELETAL MUSCLE

Skeletal muscles use many fuels to generate ATP. The most abundant immediate source of ATP is creatine phosphate. ATP also can be generated from glycogen stores either anaerobically (generating lactate) or aerobically, in which case pyruvate is converted to acetyl CoA for oxidation via the TCA cycle. All human skeletal muscles have some mitochondria and thus are capable of fatty acid and ketone body oxidation. Skeletal muscles are also capable of completely oxidizing the carbon skeletons of alanine, aspartate, glutamate, valine, leucine, and isoleucine, but not other amino acids. Each of these fuel oxidation pathways plays a somewhat unique role in skeletal muscle metabolism.

A. ATP and Creatine Phosphate

ATP is not a good choice as a molecule to store in quantity for energy reserves. Many reactions are allosterically activated or inhibited by ATP levels, especially those that generate energy. Muscle cells solve this problem by storing high-energy phosphate bonds in the form of creatine phosphate. When energy is required, creatine phosphate will donate a phosphate to ADP, to regenerate ATP for muscle contraction (Fig. 47.6).

Creatine synthesis begins in the kidney and is completed in the liver. In the kidney, glycine combines with arginine to form guanidinoacetate. In this reaction, the guanidinium group of arginine (the group that also forms urea), is transferred to

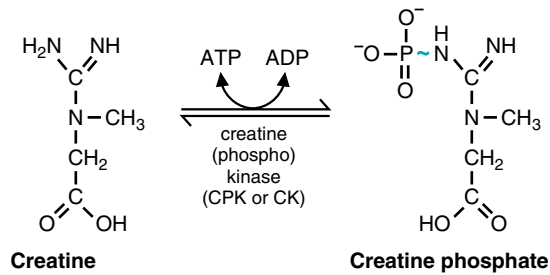


Fig. 47.6. The creatine phosphokinase reaction. The high-energy bond is the unusual nitrogen–phosphate bond, as indicated by the blue squiggle.

glycine, and the remainder of the arginine molecule is released as ornithine. Guanidinoacetate then travels to the liver, where it is methylated by S-adenosyl methionine to form creatine (Fig 47.7).

The creatine formed is released from the liver and travels through the bloodstream to other tissues, particularly skeletal muscle, heart, and brain, where it reacts with ATP to form the high-energy compound creatine phosphate (see Fig. 47.6). This reaction, catalyzed by creatine phosphokinase (CK, also abbreviated as CPK), is reversible. Therefore, cells can use creatine phosphate to regenerate ATP.

Creatine phosphate serves as a small reservoir of high-energy phosphate that can readily regenerate ATP from ADP. As a result, it plays a particularly important role in muscle during exercise. It also carries high-energy phosphate from

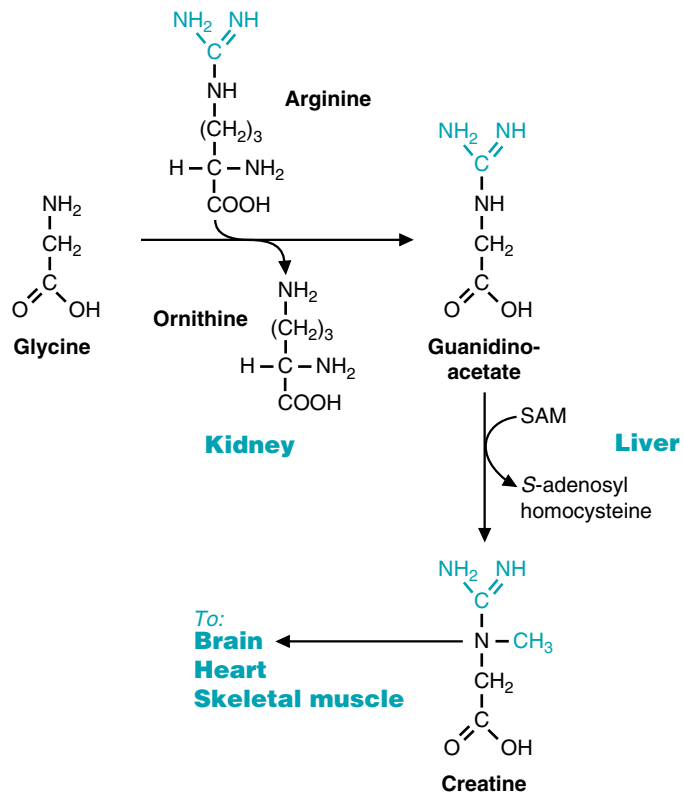


Fig. 47.7. The synthesis of creatine from arginine, glycine, and S-adenosyl methionine. Synthesis originates in the kidney and is completed in the liver.

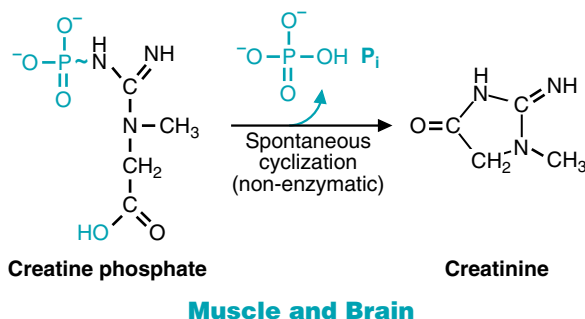


Fig. 47.8. The spontaneous production of creatinine from creatine phosphate.

mitochondria, where ATP is synthesized, to myosin filaments, where ATP is used for muscle contraction.

Creatine phosphate is an unstable compound. It spontaneously cyclizes, forming creatinine (Fig. 47.8). Creatinine cannot be further metabolized and is excreted in the urine. The amount of creatinine excreted each day is constant and depends on body muscle mass. Therefore, it can be used as a gauge for determining the amounts of other compounds excreted in the urine as well as an indicator of renal excretory function. The daily volume of urine is determined by such factors as the volume of blood reaching the renal glomeruli and the amount of renal tubular fluid reabsorbed from the tubular urine back into the interstitial space of the kidneys over time. At any given moment, the concentration of a compound in a single urine specimen does not give a good indication of the total amount that is being excreted on a daily basis. However, if the concentration of the compound is divided by the concentration of creatinine, the result provides a better indication of the true excretion rate.

B. Fuel Utilization at Rest

Muscle fuel utilization at rest is dependent on the serum levels of glucose, amino acids, and fatty acids. If blood glucose and amino acids are elevated, glucose will be converted to glycogen, and branched-chain amino acid metabolism will be high. Fatty acids will be used for acetyl CoA production and will satisfy the energy needs of the muscle under these conditions.

There is a balance between glucose oxidation and fatty acid oxidation, which is regulated by citrate. When the muscle cell has adequate energy, citrate leaves the mitochondria and activates ACC-2, which produces malonyl CoA. The malonyl CoA inhibits carnitine palmitoyl transferase-1, thereby reducing fatty acid oxidation by the muscle. Malonyl CoA decarboxylase is also inactive, because the AMP-PK is not active in the fed state. Thus, the muscle regulates its oxidation of glucose and fatty acids in part through monitoring of cytoplasmic citrate levels.

C. Fuel Use during Starvation

As blood glucose levels drop, insulin levels drop. This reduces the levels of GLUT4 transporters in the muscle membrane, and glucose use by muscle drops significantly. This conserves glucose for use by the nervous system and red blood cells. In cardiac muscle, PFK-2 is phosphorylated and activated by insulin. The lack of insulin results in a reduced use of glucose by these cells as well. Pyruvate dehydrogenase is inhibited by the high levels of acetyl CoA and NADH being produced by fatty acid oxidation.

Fatty acids become the muscle's preferred fuel under starvation conditions. The AMP-PK is active because of lower than normal ATP levels, ACC-2 is inhibited, and malonyl CoA decarboxylase is activated, thereby retaining full activity of CPT-1.



Each kidney normally contains approximately one million glomerular units. Each unit is supplied by arterial blood via the renal arteries and acts as a "filter." Metabolites such as creatinine leave the blood by passing through pores or channels in the glomerular capillaries and enter the fluid within the proximal kidney tubule for eventual excretion in the urine. When functionally intact, these glomerular tissues are impermeable to all but the smallest of proteins. When acutely inflamed, however, this barrier function is lost, and albumin and other proteins may appear in the urine.

The marked inflammatory changes in the glomerular capillaries that accompany post-streptococcal glomerulonephritis significantly reduce the flow of blood to the filtering surfaces of these vessels. As a result, creatinine, urea, and other circulating metabolites that are filtered into the urine at a normal rate (the glomerular filtration rate or GFR) in the absence of kidney disease now fail to reach the filters, and, therefore, they accumulate in the plasma.

These changes explain **Rena Felya's** laboratory profile during her acute inflammatory glomerular disease.



Muscle and brain cells contain large amounts of creatine phosphokinase (CK), and damage to these cells causes the enzyme to leak into the blood. Serum CK is measured to diagnose and evaluate patients who have had strokes and heart attacks. The presence of 5% or more of the CK in the blood as the muscle isoform is indicative of a heart attack (see Chapters 8 and 9).



Recall that in prolonged starvation muscle proteolysis is induced to provide substrates for gluconeogenesis by the liver. This does not, however, alter the use of fatty acids by the muscle for its own energy needs under these conditions.

The lack of glucose reduces the glycolytic rate, and glycogen synthesis does not occur because of the inactivation of glycogen synthase by epinephrine-stimulated phosphorylation.

D. Fuel Utilization during Exercise

The rate of ATP utilization in skeletal muscle during exercise can be as much as 100 times greater than that in resting skeletal muscles; thus, the pathways of fuel oxidation must be rapidly activated during exercise to respond to the much greater demand for ATP. ATP and creatine phosphate would be rapidly used up if they were not continuously regenerated. The synthesis of ATP occurs from glycolysis (either aerobic or anaerobic) and oxidative phosphorylation (which requires a constant supply of oxygen).

Anaerobic glycolysis is especially important as a source of ATP in three conditions. The first is during the initial period of exercise before exercise-stimulated increase in blood flow and substrate and oxygen delivery begin, allowing aerobic processes to occur. The second condition in which anaerobic glycolysis is important is exercise by muscle containing predominately fast-twitch glycolytic muscle fibers, because these fibers have low oxidative capacity and generate most of their ATP through glycolysis. The third condition is during strenuous activity, when the ATP demand exceeds the oxidative capacity of the tissue, and the increased ATP demand is met by anaerobic glycolysis.

1. ANAEROBIC GLYCOLYSIS AT THE ONSET OF EXERCISE

During rest, most of the ATP required in all types of muscle fibers is obtained from aerobic metabolism. However, as soon as exercise begins, the demand for ATP increases. The amount of ATP present in skeletal muscle could sustain exercise for only 1.2 seconds if not regenerated, and the amount of phosphocreatine could sustain exercise for only 9 seconds if it were not regenerated. It takes longer than 1 minute for the blood supply to exercising muscle to increase significantly due to vasodilation, and therefore oxidative metabolism of blood-borne glucose and fatty acids cannot increase rapidly at the onset of exercise. Thus, for the first few minutes of exercise, the conversion of glycogen to lactate provides a considerable portion of the ATP requirement.

2. ANAEROBIC GLYCOLYSIS IN THE TYPE IIB FAST-TWITCH GLYCOLYTIC FIBER

Although the human has no muscles that consist entirely of this fiber type, many animals do. Examples are white abdominal muscles of fish and the pectoral muscle of game birds (turkey white meat). These muscles contract rapidly and vigorously (the fast twitch refers to the time to peak tension), but only for short periods. Thus, they are used for activities such as flight in birds and sprinting and weight-lifting in humans.

In such muscles, the glycolytic capacity is high because the enzymes of glycolysis are present in large amounts (thus, the overall $V_{\max}$ [maximum velocity] is large). The levels of hexokinase, however, are low, so very little circulating glucose is used. The low levels of hexokinase in fast-twitch glycolytic fibers prevent the muscle from drawing on blood glucose to meet this high demand for ATP, thus avoiding hypoglycemia. Glucose 6-phosphate, formed from glycogenolysis, further inhibits hexokinase. The tissues rely on endogenous fuel stores (glycogen and creatine phosphate) to generate ATP, following the pathway of glycogen breakdown to glucose 1-phosphate, the conversion of glucose 1-phosphate to glucose 6-phosphate, and the metabolism of glucose 6-phosphate to lactate. Thus, anaerobic glycolysis is the main source of ATP during exercise of these muscle fibers.

3. ANAEROBIC GLYCOLYSIS FROM GLYCOGEN

Glycogenolysis and glycolysis during exercise are activated together because both PFK-1 (the rate-limiting enzyme of glycolysis) and glycogen phosphorylase b (the inhibited form of glycogen phosphorylase) are allosterically activated by AMP.

AMP is an ideal activator because its concentration is normally kept low by the adenylate kinase (also called myokinase in muscle) equilibrium [$2 \text{ ADP} \leftrightarrow \text{AMP} + \text{ATP}$]. Thus, whenever ATP levels decrease slightly, the AMP concentration increases manyfold (Fig. 47.9)

Starting from a molecule of glucose 1-phosphate derived from glycogenolysis, three ATP molecules are produced in anaerobic glycolysis, as compared with 32 to 34 moles of ATP in aerobic glycolysis. To compensate for the low ATP yield of anaerobic glycolysis, fast-twitch glycolytic fibers have a much higher content of glycolytic enzymes, and the rate of glucose 6-phosphate utilization is more than 12 times as fast as slow-twitch fibers.

Muscle fatigue during exercise generally results from a lowering of the pH of the tissue to approximately 6.4. Both aerobic and anaerobic metabolism lowers the pH. Both the lowering of pH and lactate production can cause pain.

Metabolic fatigue also can occur once muscle glycogen is depleted. Muscle glycogen stores are used up in less than 2 minutes of anaerobic exercise. If you do pushups, you can prove this to yourself. The muscle used in pushups, a high-strength exercise, is principally fast-twitch glycolytic fibers. Time yourself from the start of your pushups. No matter how well you have trained, you probably cannot do pushups for as long as 2 minutes. Furthermore, you will feel the pain as the muscle pH drops as lactate production continues.

The regulation of muscle glycogen metabolism is complex. Recall that glycogen degradation in muscle is not sensitive to glucagon (muscles lack glucagon receptors), so there is little change in muscle glycogen stores during overnight fasting or long-term fasting, if the individual remains at rest. Glycogen synthase is inhibited during exercise but can be activated in resting muscle by the release of insulin after a high-carbohydrate meal. Unlike the liver form of glycogen phosphorylase, the muscle isozyme contains an allosteric site for AMP binding. When AMP binds to muscle glycogen phosphorylase b, the enzyme is activated even though it is not phosphorylated. Thus, as muscle begins to work and the myosin-ATPase hydrolyzes existing ATP stores to ADP, AMP will begin to accumulate (due to the myokinase reaction), and glycogen degradation will be enhanced. The activation of muscle glycogen phosphorylase b is further enhanced by the release of Ca^{2+} from the sarcoplasmic reticulum, which occurs when muscles are stimulated to contract. The increase in sarcoplasmic Ca^{2+} also leads to the allosteric activation of glycogen phosphorylase kinase (through binding to the calmodulin subunit of the enzyme), which phosphorylates muscle glycogen phosphorylase b, fully activating it. And, finally, during intense exercise, epinephrine release stimulates the activation of adenylate cyclase in muscle cells, thereby activating the

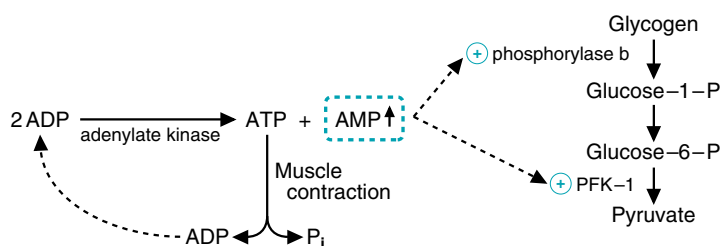


Fig. 47.9. Activation of muscle glycogenolysis and glycolysis by AMP. As muscle contracts, ATP is converted to ADP and P_i . In the adenylate kinase reaction, two ADP react to form ATP and AMP. The ATP is used for contraction. As AMP accumulates, it activates glycogenolysis and glycolysis.

cAMP-dependent protein kinase (see Fig. 28.10). Protein kinase A phosphorylates and fully activates glycogen phosphorylase kinase such that continued activation of muscle glycogen phosphorylase can occur. The hormonal signal is slower than the initial activation events triggered by AMP and calcium (Fig. 47.10).

4. ANAEROBIC GLYCOLYSIS DURING HIGH-INTENSITY EXERCISE

Once exercise begins, the electron transport chain, the TCA cycle, and fatty acid oxidation are activated by the increase of ADP and the decrease of ATP. Pyruvate dehydrogenase remains in the active, nonphosphorylated state as long as NADH can be reoxidized in the electron transport chain and acetyl CoA can enter the TCA cycle. However, even though mitochondrial metabolism is working at its maximum capacity, additional ATP may be needed for very strenuous, high-intensity exercise. When this occurs, ATP is not being produced rapidly enough to meet the muscle's needs, and AMP begins to accumulate. Increased AMP levels activate PFK-1 and glycogenolysis, thereby providing additional ATP from anaerobic glycolysis (the additional pyruvate

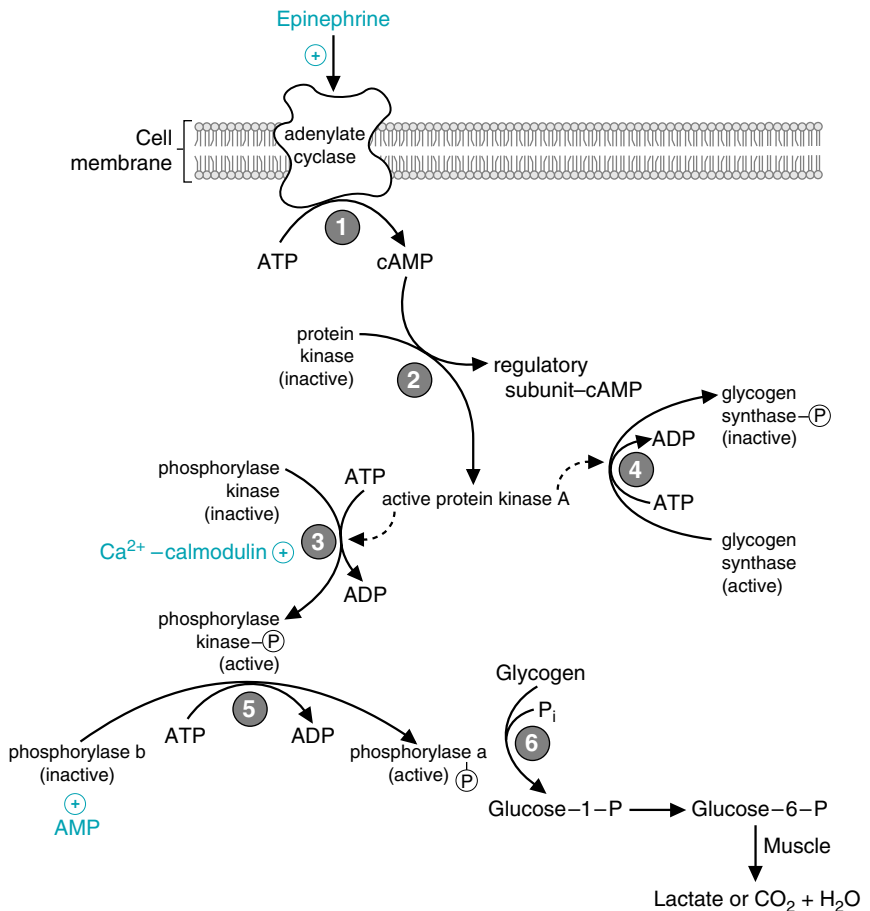


Fig. 47.10. Stimulation of glycogenolysis in muscle by epinephrine. 1. Epinephrine binding to its receptor leads to the activation of adenylate cyclase, which increases cAMP levels. 2. cAMP binds to the regulatory subunits of protein kinase A, thereby activating the catalytic subunits. 3. Active protein kinase A phosphorylates and activates phosphorylase kinase. Phosphorylase kinase also can be activated partially by the Ca^{2+} -calmodulin complex as Ca^{2+} levels increase as muscles contract. 4. Protein kinase A phosphorylates and inactivates glycogen synthase. 5. Active phosphorylase kinase converts glycogen phosphorylase b to glycogen phosphorylase a. 6. Glycogen degradation forms glucose 1-phosphate, which is converted to glucose 6-phosphate, which enters the glycolytic pathway for energy production.

produced does not enter the mitochondria but rather is converted to lactate such that glycolysis can continue). Thus, under these conditions, most of the pyruvate formed by glycolysis enters the TCA cycle whereas the remainder is reduced to lactate to regenerate NAD^+ for continued use in glycolysis.

5. FATE OF LACTATE RELEASED DURING EXERCISE

The lactate that is released from skeletal muscles during exercise can be used by resting skeletal muscles or by the heart, a muscle with a large amount of mitochondria and very high oxidative capacity. In such muscles, the NADH/NAD^+ ratio will be lower than in exercising skeletal muscle, and the lactate dehydrogenase reaction will proceed in the direction of pyruvate formation. The pyruvate that is generated is then converted to acetyl CoA and oxidized in the TCA cycle, producing energy by oxidative phosphorylation.

The second potential fate of lactate is that it will return to the liver through the Cori cycle, where it will be converted to glucose (see Fig. 22.10).

VI. MILD AND MODERATE-INTENSITY LONG-TERM EXERCISE

A. Lactate Release Decreases with Duration of Exercise

Mild to moderate-intensity exercise can be performed for longer periods than can high-intensity exercise. This is because of the aerobic oxidation of glucose and fatty acids, which generates more energy per fuel molecule than anaerobic metabolism, and which also produces acid at a slower rate than anaerobic metabolism. Thus, during mild and moderate-intensity exercise, the release of lactate diminishes as the aerobic metabolism of glucose and fatty acids becomes predominant.

B. Blood Glucose as a Fuel

At any given time during fasting, the blood contains only approximately 5 g glucose, enough to support a person running at a moderate pace for a few minutes. Therefore, the blood glucose supply must be constantly replenished. The liver performs this function by processes similar to those used during fasting. The liver produces glucose by breaking down its own glycogen stores and by gluconeogenesis. The major source of carbon for gluconeogenesis during exercise is, of course, lactate, produced by the exercising muscle, but amino acids and glycerol are also used (Fig. 47.11). Epinephrine released during exercise stimulates liver glycogenolysis and gluconeogenesis by causing cAMP levels to increase.

During long periods of exercise, blood glucose levels are maintained by the liver through hepatic glycogenolysis and gluconeogenesis. The amount of glucose that the liver must export is greatest at higher work loads, in which case the muscle is using a greater proportion of the glucose for anaerobic metabolism. With increasing duration of exercise, an increasing proportion of blood glucose is supplied by gluconeogenesis. However, for up to 40 minutes of mild exercise, glycogenolysis is mainly responsible for the glucose output of the liver. However, after 40 to 240 minutes of exercise, the total glucose output of the liver decreases. This is caused by the increased utilization of fatty acids, which are being released from adipose tissue triacylglycerols (stimulated by epinephrine release). Glucose uptake by the muscle is stimulated by the increase in AMP levels and the activation of the AMP-activated protein kinase, which stimulates the translocation of GLUT4 transporters to the muscle membrane.

The hormonal changes that direct the increased hepatic glycogenolysis, hepatic gluconeogenesis, and adipose tissue include a decrease in insulin and an increase in glucagon, epinephrine, and norepinephrine. Plasma levels of growth hormone, cortisol, and thyroid-stimulating hormone (TSH) also increase and may make a contribution to



If **Otto Shape** runs at a pace at which his muscles require approximately 500 Calories per hour, how long could he run on the amount of glucose that is present in circulating blood? Assume the blood volume is 5 liters.



Remember from Chapter 1 that a food Calorie is equivalent to 1 kcal of energy. One gram of glucose can give rise to 4 kcal of energy, so at a rate of consumption of 500 Calories per hour we have the following:

$$\begin{aligned} (500 \text{ Calories/hr}) \times (1 \text{ gram glucose}/4 \\ \text{Calories energy}) \times (1 \text{ hour}/60 \text{ minutes}) = \\ 2 \text{ grams of glucose/min.} \end{aligned}$$

Thus, Otto must use 2 grams of glucose per minute to run at his current pace. In the fasting state, blood glucose levels are approximately 90 mg/dL, or 900 mg/L. Because blood volume is estimated at 5 liters, Otto has 4.5 grams glucose available. If not replenished, that amount of glucose would only support 2.5 minutes of running, at 2 grams glucose per minute.

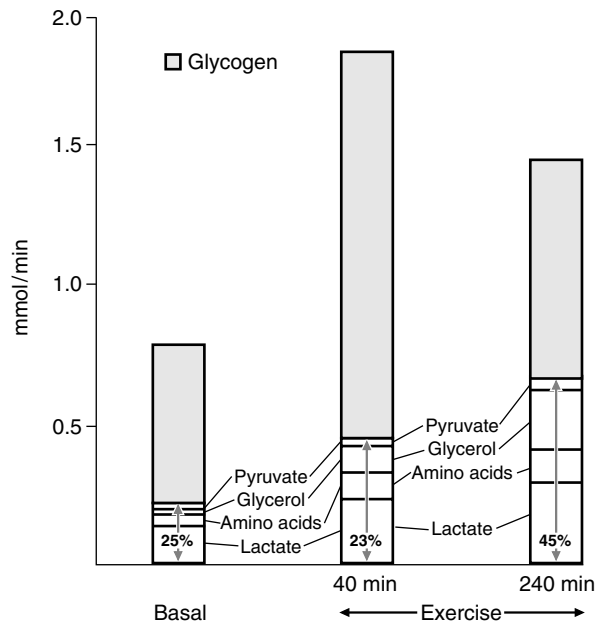


Fig. 47.11. Production of blood glucose by the liver from various precursors during rest and during prolonged exercise. The shaded area represents the contribution of liver glycogen to blood glucose, and the open area represents the contribution of gluconeogenesis. From Wahren J, et al. In: Howald H, Poortmans JR, eds. *Metabolic Adaptation to Prolonged Physical Exercise*. Cambridge, MA: Birkhauser, 1973:148.

fuel mobilization as well (see Chapter 43). The activation of hepatic glycogenolysis occurs through glucagon and epinephrine release. Hepatic gluconeogenesis is activated by the increased supply of precursors (lactate, glycerol, amino acids, and pyruvate), the induction of gluconeogenic enzymes by glucagon and cortisol (this only occurs in prolonged exercise), and the increased supply of fatty acids to provide the ATP and NADH needed for gluconeogenesis and the regulation of gluconeogenic enzymes.

C. Free Fatty Acids as a Source of ATP

The longer the duration of the exercise, the greater the reliance of the muscle on free fatty acids for the generation of ATP (Fig. 47.12). Because ATP generation from

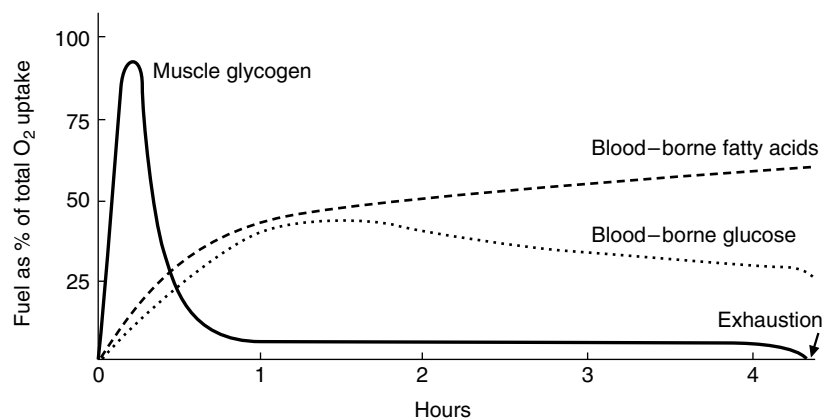


Fig. 47.12. Fuels used during exercise. The pattern of fuel utilization changes with the duration of the exercise. From Felig P, et al. *Endocrinology & Metabolism*. New York: McGraw-Hill, 1981:796.

free fatty acids depends on mitochondria and oxidative phosphorylation, long-distance running uses muscles that are principally slow-twitch oxidative fibers, such as the gastrocnemius. It is also important to realize that resting skeletal muscle uses free fatty acids as a principle fuel. At almost anytime except the postprandial state (right after eating), free fatty acids are the preferred fuel for skeletal muscle.

The preferential utilization of fatty acids over glucose as a fuel in skeletal muscle depends on the following factors:

1. The availability of free fatty acids in the blood, which depends on their release from adipose tissue triacylglycerols by hormone-sensitive lipase. During prolonged exercise, the small decrease of insulin, and increases of glucagon, epinephrine and norepinephrine, cortisol, and possibly growth hormone all activate adipocyte tissue lipolysis.
2. Inhibition of glycolysis by products of fatty acid oxidation. Pyruvate dehydrogenase activity is inhibited by acetyl CoA, NADH, and ATP, all of which are elevated as fatty acid oxidation proceeds. As AMP levels drop, and ATP levels rise, PFK-1 activity is decreased (see Chapter 22).
3. Glucose transport may be reduced during long-term exercise. Glucose transport into skeletal muscles via the GLUT 4 transporter is greatly activated by either insulin or exercise. During long-term exercise, the effect of falling insulin levels or increased fatty acid levels may counteract the stimulation of glucose transport by the exercise itself.
4. Ketone body oxidation also increases during exercise. Their utilization as a fuel is dependent on their rate of production by the liver. Ketone bodies are, however, never a major fuel for skeletal muscle (muscles prefer free fatty acids).
5. Acetyl-CoA carboxylase (isozyme ACC-2) must be inactivated for the muscle to use fatty acids. This occurs as the AMP-PK is activated and phosphorylates ACC-2, rendering it inactive.

D. Branched-Chain Amino Acids

Branched-chain amino acid oxidation has been estimated to supply a maximum of 20% of the ATP supply of resting muscle. Oxidation of branched-chain amino acids in muscle serves two functions. The first is the generation of ATP, and the second is the synthesis of glutamine, which effluxes from the muscle. The highest rates of branched-chain amino acid oxidation occur under conditions of acidosis, in which there is a higher demand for glutamine to transfer ammonia to the kidney and to buffer the urine as ammonium ion during proton excretion. Recall that glutamine synthesis occurs from the carbon skeletons of branched-chain amino acid oxidation (valine and isoleucine) after the initial five steps of the oxidative pathway.

E. The Purine Nucleotide Cycle

Exercise increases the activity of the purine nucleotide cycle, which converts aspartate to fumarate plus ammonia (see Fig. 41.13). The ammonia is used to buffer the proton production and lactate production from glycolysis, and the fumarate is recycled and can form glutamine.

F. Acetate

Acetate is an excellent fuel for skeletal muscle. It is treated by the muscle as a very-short-chain fatty acid. It is activated to acetyl CoA in the cytosol and then transferred into the mitochondria via acetylcarnitine transferase, an isozyme of carnitine palmitoyl transferase. Sources of acetate include the diet (vinegar is acetic acid) and acetate produced in the liver from alcohol metabolism. Certain commercial power bars for athletes contain acetate.

VII. METABOLIC EFFECTS OF TRAINING ON MUSCLE METABOLISM

The effect of training depends, to some extent, on the type of training. In general, training increases the muscle glycogen stores and increases the number and size of mitochondria. The fibers thus increase their capacity for generation of ATP from oxidative metabolism and their ability to use fatty acids as a fuel. The winners in marathon races seem to use muscle glycogen more efficiently than others.

Training to improve strength, power, and endurance of muscle performance is called resistance training. Its goal is to increase the size of the muscle fibers (hypertrophy of the muscle). Muscle fibers can develop a maximal force of 3 to 4 kg/cm² of muscle area. Thus, if one can increase their muscle size from 80 to 120 cm², the maximal resistance that could be lifted would increase from 240 to 360 kg. Hypertrophy occurs by increased protein synthesis in the muscle and a reduction in existing protein turnover.

CLINICAL COMMENTS



Poststreptococcal glomerulonephritis (PSGN) may follow pharyngeal or cutaneous infection with one of a limited number of “nephritogenic” strains of group A β -hemolytic streptococci. The pathogenesis of PSGN involves a host immune (antibody) response to one or more of the enzymes secreted by the bacterial cells. The antigen–antibody complexes are deposited on the tissues of glomerular units, causing a local acute inflammatory response. Hypertension may occur as a consequence of sodium and water retention caused by an inability of the inflamed glomerular units to filter sodium and water into the urine. Proteinuria is usually mild if the immune response is self-limited.

Overall, one of the most useful clinical indicators of glomerular filtration rate in both health and disease is the serum creatinine concentration. The endogenous production of creatinine, which averages approximately 15 mg/kg of body weight per day, is correlated with muscle mass and, therefore, tends to be constant for a given individual if renal function is normal. Any rise in serum creatinine in patients such as **Rena Felya**, therefore, can be assumed to result from decreased excretion of this metabolite into the urine. The extent of the rise in the blood is directly related to the severity of the pathologic process involving the glomerular units within the kidneys.

BIOCHEMICAL COMMENTS



The SERCA pump is a transmembrane protein of 110 kDa present in several different isoforms throughout the body. Three genes encode SERCA proteins, designated SERCA1, SERCA2, and SERCA3. The SERCA1 gene produces two alternatively spliced transcripts, SERCA1a and SERCA1b. SERCA1b is expressed in the fetal and neonatal fast-twitch skeletal muscles, and is replaced by SERCA1a in adult fast-twitch muscle. The SERCA2 gene also undergoes alternative splicing, producing the SERCA2a and SERCA2b isoforms. The SERCA2b isoform is expressed in all cell types and is associated with inositol trisphosphate (IP₃)-regulated calcium stores. SERCA2a is the primary isoform expressed in cardiac tissue. SERCA3 produces at least five different alternatively spliced isoforms, which are specifically expressed in different tissues.

SERCA2a plays an important role in cardiac contraction and relaxation. Contraction is initiated by the release of calcium from intracellular stores, whereas relaxation

occurs as the calcium is re-sequestered in the sarcoplasmic reticulum, in part mediated by the SERCA2a protein. The SERCA2a pump is regulated, in part, by its association with the protein phospholamban (PLN). PLN is a pentameric molecule consisting of five identical subunits of molecular weight 22,000 Daltons. PLN associates with SERCA2a in the sarcoplasmic reticulum and reduces its pumping activity. Because new contractions cannot occur until cytosolic calcium has been re-sequestered into the sarcoplasmic reticulum, a reduction in SERCA2a activity increases the relaxation time. However, when called on, the heart can increase its rate of contractions by inhibiting the activity of phospholamban. This is accomplished by phosphorylation of phospholamban by protein kinase A (PKA). Epinephrine release stimulates the heart to beat faster. This occurs through epinephrine binding to its receptor, activating a G protein, which leads to adenylyl cyclase activation, elevation of cAMP levels, and activation of protein kinase A. PKA phosphorylates PLN, thereby reducing its association with SERCA2a and relieving the inhibition of pumping activity. This results in reduced relaxation times and more frequent contractions.

Mutations in PLN lead to cardiomyopathies, primarily an autosomal dominant form of dilated cardiomyopathy. This mutation leads to an arginine in place of cysteine at position 9 in PLN, which forms an inactive complex with PKA and blocks PKA phosphorylation of PLN. Individuals with this form of PLN develop cardiomyopathy in their teens. In this condition, the cardiac muscle does not pump well (because of the constant inhibition of SERCA2a), and becomes enlarged (dilated). Because of the poor pumping action of the heart, fluid can build up in the lungs. The pulmonary congestion results in a sense of breathlessness (left heart failure). Eventually, progressive left heart failure leads to fluid accumulation in other tissues and organs of the body, such as the legs and ankles (right heart failure).

Suggested References

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REVIEW QUESTIONS—CHAPTER 47

1. The process of stretching before exercise has which of the following biochemical benefits?
 - (A) Stimulates the release of epinephrine
 - (B) Activates glycolysis in the liver
 - (C) Increases blood flow to the muscles
 - (D) Activates glycolysis in the muscles
 - (E) Stimulates glycogenolysis in the liver
2. The major metabolic fuel for participating in a prolonged aerobic exercise event is which of the following?
 - (A) Liver glycogen
 - (B) Muscle glycogen
 - (C) Brain glycogen
 - (D) Adipose triacylglycerol
 - (E) Red blood cell-produced lactate

3. A 24-hour urine collection showed that an individual's excretion of creatinine was much lower than normal. Decreased excretion of creatinine could be caused by which of the following?
- (A) A decreased dietary intake of creatine
 - (B) A higher than normal muscle mass resulting from weight lifting
 - (C) A genetic defect in the enzyme that converts creatine phosphate to creatinine
 - (D) Kidney failure
 - (E) A vegetarian diet
4. In the biosynthetic pathways for the synthesis of heme, creatine, and guanine, which one of the following amino acids directly provides carbon atoms that appear in the final product?
- (A) Serine
 - (B) Aspartate
 - (C) Cysteine
 - (D) Glutamate
 - (E) Glycine
5. In skeletal muscle, increased hydrolysis of ATP during muscular contraction leads to which of the following?
- (A) A decrease in the rate of palmitate oxidation to acetyl CoA
 - (B) A decrease in the rate of NADH oxidation by the electron transport chain
 - (C) The activation of PFK-1
 - (D) An increase in the proton gradient across the inner mitochondrial membrane.
 - (E) The activation of glycogen synthase

48 Metabolism of the Nervous System

The nervous system consists of various cell types. The most abundant cell in the nervous system is the **glial cell**, which consists of **astrocytes** and **oligodendrocytes** in the **central nervous system**, and **Schwann cells** in the **peripheral nervous system**. These cells provide support for the neurons and synthesize the protective **myelin sheath** that surrounds the axons emanating from the neurons. **Microglial cells** in the nervous system act as immune cells, destroying and ingesting foreign organisms that enter the nervous system. The interface between the brain parenchyma and the cerebrospinal fluid compartment is formed by the **ependymal cells**, which line the cavities of the brain and spinal cord. These cells use their cilia to allow for the circulation of the **cerebrospinal fluid (CSF)**, which bathes the cells of the central nervous system.

The cells of the brain are separated from free contact with the rest of the body by the **blood-brain barrier**. The capillaries of the brain exhibit features, such as tight endothelial cell junctions, that restrict their permeability to metabolites in the blood. This protects the brain from compounds that might be toxic or otherwise interfere with nerve impulse transmission. It also affects the entry of precursors for brain metabolic pathways such as fuel metabolism and neurotransmitter synthesis.

Neurotransmitters can be divided structurally into two categories: **small nitrogen-containing neurotransmitters** and **neuropeptides**. The small nitrogen-containing neurotransmitters are generally synthesized in the **presynaptic terminal** from amino acids and intermediates of glycolysis and the TCA cycle. They are retained in storage vesicles until the neuron is depolarized. The **catecholamine** neurotransmitters (**dopamine**, **norepinephrine**, and **epinephrine**) are derived from tyrosine. **Serotonin** is synthesized from **tryptophan**. **Acetylcholine** is synthesized from **choline**, which can be supplied from the diet or is synthesized and stored as part of **phosphatidylcholine**. **Glutamate** and its neurotransmitter derivative, **γ -aminobutyric acid (GABA)**, are derived from **α -ketoglutarate** in the TCA cycle. **Glycine** is synthesized in the brain from **serine**. The synthesis of the neurotransmitters is regulated to correspond to the rate of depolarization of the individual neurons. A large number of cofactors are required for the synthesis of neurotransmitters, and deficiencies of **pyridoxal phosphate**, **thiamine-pyrophosphate**, and **vitamin B12** result in a variety of neurologic dysfunctions.

Brain metabolism has a high **requirement for glucose and oxygen**. **Deficiencies of either (hypoglycemia or hypoxia)** affect brain function because they influence adenosine triphosphate (ATP) production and the supply of precursors for neurotransmitter synthesis. **Ischemia** elicits a condition in which increased **calcium levels**, **swelling**, **glutamate excitotoxicity**, and **nitric oxide** generation affect brain function, and can lead to a stroke. The generation of **free radicals** and abnormalities in nitric oxide production are important players in the pathogenesis of a variety of **neurodegenerative** diseases.

Because of the restrictions posed by the blood-brain barrier to the entry of a variety of substances into the central nervous system, the brain generally synthesizes and

*degrades its own lipids. Essential fatty acids can enter the brain, but the more common fatty acids do not. The turnover of lipids at the synaptic membrane is very rapid, and the neuron must replace those lipids lost during exocytosis. The glial cells produce the **myelin** sheath, which is composed primarily of lipids. These lipids are of a different composition than those of the neuronal cells. Because there is considerable lipid synthesis and turnover in the brain, this organ is sensitive to **disorders of peroxisomal function** (Refsum's disease; interference in very-long-chain fatty acid oxidation and α -oxidation) and **lysosomal diseases** (mucopolysaccharidoses; inability to degrade complex lipids and glycolipids).*



THE WAITING ROOM



Katie Colamin, a 34-year-old dress designer, developed alarming palpitations of her heart while bending forward to pick up her cat. She also developed a pounding headache and sweated profusely. After 5 to 10 minutes, these symptoms subsided. One week later, her aerobic exercise instructor, a registered nurse, noted that Katie grew very pale and was tremulous during exercise. The instructor took Katie's blood pressure, which was 220 mm Hg systolic (normal, up to 120 at rest) and 132 mm Hg diastolic (normal, up to 80 at rest). Within 15 minutes, Katie recovered, and her blood pressure returned to normal. The instructor told Katie to see her physician the next day.

The doctor told Katie that her symptom complex coupled with severe hypertension strongly suggested the presence of a tumor in the medulla of one of her adrenal glands (a pheochromocytoma) that was episodically secreting large amounts of catecholamines, such as norepinephrine (noradrenaline) and epinephrine (adrenaline). Her blood pressure was normal until moderate pressure to the left of her umbilicus caused Katie to suddenly develop a typical attack, and her blood pressure rose rapidly. She was immediately scheduled for a magnetic resonance imaging (MRI) study of her adrenal glands. The MRI showed a $3.5 \times 2.8 \times 2.6$ cm mass in the left adrenal gland, with imaging characteristics typical of a pheochromocytoma.



Ivan Applebod's brother, **Evan Applebod**, was 6 feet tall and weighed 425 pounds. He had only been successful in losing weight once in his life, in 1977. Evan's weight was not usually a concern for him, but in 1997 he had become concerned when it became difficult for him to take walks or go fishing because of joint pain in his knees. He was also suffering from symptoms suggestive of a peripheral neuropathy, manifest primarily as tingling in his legs. He had failed in all previous dieting attempts and was desperate now to lose weight. The physician placed Evan on a new drug, Redux, which had just been approved for use as a weight loss agent, and a slightly restricted low-fat, low-calorie diet. In 4 months, Evan's weight dropped from 425 pounds to 335 pounds, his total cholesterol dropped from 250 to 185, and his serum triglycerides dropped from 375 to 130. However, Redux was withdrawn from the market by its manufacturer late in 1997 because of its toxicity. Evan was then placed on Prozac, a drug used primarily as an antidepressant and less commonly as an appetite suppressant.

I. CELL TYPES OF THE NERVOUS SYSTEM

The nervous system consists of neurons, the cells that transmit signals, and supporting cells, the neuroglia. The neuroglia consists of oligodendrocytes and astrocytes

(collectively known as glial cells), microglial cells, ependymal cells, and Schwann cells. The neuroglia are designed to support and sustain the neurons and do so by surrounding neurons and holding them in place, supplying nutrients and oxygen to the neurons, insulating neurons so their electrical signals are more rapidly propagated, and cleaning up any debris that enters the nervous system. The central nervous system (CNS) consists of the brain and spinal cord. This system integrates all signals emanating from the peripheral nervous system (PNS). The PNS is composed of all neurons lying outside of the CNS.

A. Neurons

Neurons consist of a cell body (soma) from which long (axons) and short (dendrites) extensions protrude. Dendrites receive information from the axons of other neurons, whereas the axons transmit information to other neurons. The axon–dendrite connection is known as a synapse (Fig. 48.1). Most neurons contain multiple dendrites, each of which can receive signals from multiple axons. This configuration allows a single neuron to integrate information from multiple sources. Although neurons also contain just one axon, most axons branch extensively and distribute information to multiple targets (divergence). The neurons transmit signals by changes in the electrical potential across their membrane. Signaling across a synapse requires the release of neurotransmitters that, when bound to their specific receptors, initiate an electrical signal in the receiving or target cell. Neurons are terminally differentiated cells and, as such, have little capability for division. Because of this, injured neurons have a limited capacity to repair themselves and frequently undergo apoptosis (programmed cell death) when damaged.

B. Neuroglial Cells

1. ASTROCYTES

The astrocytes are found in the CNS and are star-shaped cells that provide physical and nutritional support for neurons. During development of the CNS, the astrocytes guide neuronal migration to their final adult position and form a matrix that keeps neurons in place. These cells serve several functions, including the ability to phagocytose debris left behind by cells, to provide lactate (from glucose metabolism) as

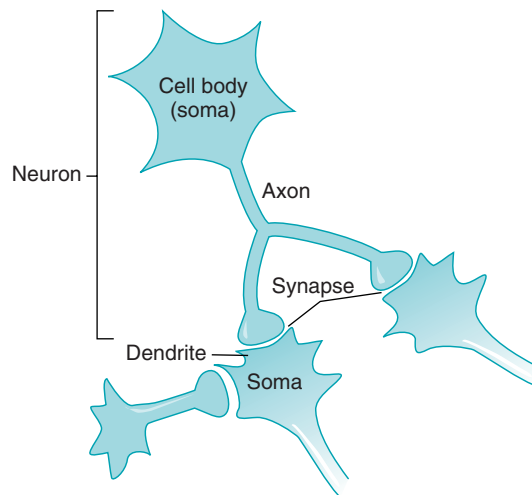


Fig. 48.1. A neuron consists of a cell body (soma) with short extensions (dendrites) and a long extension (axon). The axon–dendrite interface is the synapse. A soma may receive input from multiple axons.

a carbon source for the neurons, and to control the brain extracellular ionic environment. Astrocytes help to regulate the content of the extracellular fluid (ECF) by taking up, processing, and metabolizing nutrients and waste products.

2. OLIGODENDROCYTES

The oligodendrocyte provides the myelin sheath that surrounds the axon, acting as an “insulation” for many of the neurons in the CNS. The myelin sheath consists of a lipid–protein covering of the axons (see section V.B. for a description of the composition and synthesis of the myelin sheath). Oligodendrocytes can form myelin sheaths around multiple neurons in the CNS by sending out processes that bind to the axons on target neurons. The speed with which a neuron conducts its electrical signal (action potential) is directly proportional to the degree of myelination. Oligodendrocytes, along with the astrocytes, form a supporting matrix for the neurons. Oligodendrocytes have a limited capacity for mitosis, and if damaged, do not replicate. If this occurs, demyelination of the axons may occur, resulting in abnormalities in signal conduction along that axon (see Biochemical Comments).

3. SCHWANN CELLS



Peripheral axons can regenerate if Schwann cells are available to guide the growth of the axon. There is a synergistic interaction between the Schwann cells, secreted growth factors, and the axon that allows the axons to reconnect to its appropriate target.

Schwann cells are the supporting cells of the PNS. Like oligodendrocytes, Schwann cells form myelin sheaths around the axons, but unlike the oligodendrocytes, Schwann cells only myelinate one axon. Schwann cells also clean up cellular debris in the PNS.

4. MICROGLIAL CELLS

These are the smallest type of glial cells in the nervous system. They serve as immunologically responsive cells similar to the action of macrophages in the circulation. Microglial cells destroy invading microorganisms and phagocytose cellular debris.

5. EPENDYMAL CELLS



For many years, it had been believed that damaged neurons in the CNS could not regenerate, for it was thought that there were no pluripotent stem cells (cells that could differentiate into various cell types found in the CNS) in the CNS. However, recent data suggest that cells found within the ependymal layer can act as neural stem cells, which under appropriate stimulation can regenerate neurons. Such a finding opens up a large number of potential treatments for diseases that alter neuronal cell function.

The ependymal cells are ciliated cells that line the cavities (ventricles) of the CNS and the spinal cord. In some areas of the brain, the ependymal cells are functionally specialized to elaborate and secrete cerebrospinal fluid (CSF) into the ventricular system. The beating of the ependymal cilia allow for efficient circulation of the CSF throughout the CNS. The CSF acts as both a shock absorber protecting the CNS from mechanical trauma and a system for the removal of metabolic wastes. The CSF can be aspirated from the spinal canal and analyzed to determine whether disorders of CNS function, with their characteristic CSF changes, are present.

II. THE BLOOD-BRAIN BARRIER

A. Capillary Structure

In the capillary beds of most organs, a rapid passage of molecules occurs from the blood through the endothelial wall of the capillaries into the interstitial fluid. Thus, the composition of interstitial fluid resembles that of blood, and specific receptors or transporters in the plasma membrane of the cells being bathed by the interstitial fluid may directly interact with amino acids, hormones, or other compounds from the blood. In the brain, transcapillary movement of substrates in the peripheral circulation into the brain is highly restricted by the blood-brain barrier. This barrier limits the accessibility of blood-borne toxins and other potentially harmful compounds to the neurons of the CNS.

The blood-brain barrier begins with the endothelial cells that form the inner lining of the vessels supplying blood to the CNS (Fig. 48.2). Unlike the endothelial cells of other organs, these cells are joined by tight junctions that do not permit the movement of polar molecules from the blood into the interstitial fluid bathing the neurons. They also lack mechanisms for transendothelial transport that are present in other capillaries of the body. These mechanisms include fenestrations (“windows” or pores that span the endothelial lining and permit the rapid movement of molecules across membranes) or transpinocytosis (vesicular transport from one side of the endothelial cell to another).

The endothelial cells actively, as well as passively, serve to protect the brain. Because they contain a variety of drug-metabolizing enzyme systems similar to the drug-metabolizing enzymes found in the liver, the endothelial cells can metabolize neurotransmitters and toxic chemicals and, therefore, form an enzymatic barrier to entry of these potentially harmful substances into the brain. They actively pump hydrophobic molecules that diffuse into endothelial cells back into the blood (especially xenobiotics) with P-glycoproteins, which act as transmembranous, ATP-dependent efflux pumps. Although lipophilic substances, water, oxygen, and carbon dioxide can readily cross the blood-brain barrier by passive diffusion, other molecules depend on specific transport systems. Differential transporters on the luminal and abluminal endothelial membranes can transport compounds into, as well as out of, the brain.

Further protection against the free entry of blood-borne compounds into the CNS is provided by a continuous collagen-containing basement membrane that completely surrounds the capillaries. The basement membrane appears to be surrounded by the foot processes of astrocytes. Thus, compounds must pass through endothelial cell membranes, the enzymatic barrier in the endothelial cells, the basement membrane, and possibly additional cellular barriers formed by the astrocytes to reach the neurons in the brain.

B. Transport through the Blood-Brain Barrier

Many nonpolar substances, such as drugs and inert gases, probably diffuse through the endothelial cell membranes. A large number of other compounds are transported through the endothelial capillaries by facilitative transport, whereas others, such as nonessential fatty acids, cannot cross the blood-brain barrier. Essential fatty acids, however, are transported across the barrier.

1. FUELS

Glucose, which is the principle fuel of the brain, is transported through both endothelial membranes by facilitated diffusion via the GLUT-1 transporter (see Fig. 27.4). GLUT-3 transporters present on the neurons then allow the neurons to transport the glucose from the ECF. Glial cells express GLUT-1 transporters. Although the rate of glucose transport into the ECF normally exceeds the rate required for energy metabolism by the brain, glucose transport may become rate-limiting as blood glucose levels fall below the normal range. Thus, individuals begin to experience hypoglycemic symptoms at approximately 60 mg/dL, as the glucose levels are reduced to the K_m , or below the K_m values of the GLUT-1 transporters in the endothelial cells of the barrier.

Monocarboxylic acids, including L-lactate, acetate, pyruvate, and the ketone bodies acetoacetate and β -hydroxybutyrate, are transported by a separate stereospecific system that is slower than the transport system for glucose. During starvation, when the level of ketone bodies in the blood is elevated, this transporter is upregulated. Ketone bodies are important fuels for the brain of both the adult and the neonate during prolonged starvation (greater than 48 hours).

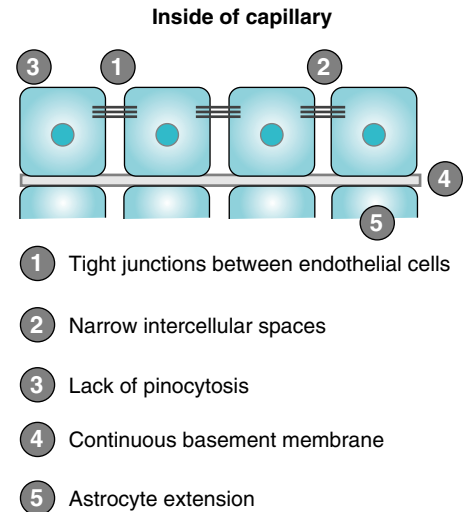


Fig. 48.2. The blood-brain barrier. Compounds in the blood cannot freely pass into the brain; they must traverse the endothelial cells, basement membrane, and astrocytes, by use of specific carriers to gain access to the brain. Very lipophilic molecules may pass through all of these membranes in the absence of a carrier.



A number of disorders of glucose transport across the blood-brain barrier are known. The most common of these is facilitated glucose transporter protein type 1 (GLUT1) deficiency syndrome. In this disorder, GLUT1 transporters are impaired, which results in a low glucose concentration in the CSF (a condition known as hypoglycorrachia). A diagnostic indication of this disorder is that in the presence of normal blood glucose levels the ratio of CSF glucose to blood glucose levels is less than 0.4. Clinical features are variable but include seizures, developmental delay, and a complex motor disorder. These symptoms are the result of inadequate glucose levels within the brain. The disorder can be treated by prescribing a ketogenic diet (high-fat, low-carbohydrate). This will force the patient to produce ketone bodies, which are easily transported into the CNS and can spare the brain's requirement for glucose as an energy source.



The finding that the large, neutral amino acids (LNAA) have a common carrier system across the blood-brain barrier suggests that if one amino acid is in excess, it can, by competitive inhibition, result in a lower transport of the other amino acids. This suggests that the mental retardation that results from untreated PKU and maple syrup urine disease (see Chapter 39) may be attributable to the high levels of either phenylalanine or branched-chain amino acids in the blood. These high levels overwhelm the LNAA carrier, such that excessive levels of the damaging amino acid enter the CNS. In support of this theory is the finding that treatment of PKU patients with large doses of LNAA lacking phenylalanine resulted in a decrease of phenylalanine levels in the CSF and brain, with an improvement in their cognitive functions as well.

2. AMINO ACIDS AND VITAMINS

Large neutral amino acids (LNAA, such as phenylalanine, leucine, tyrosine, isoleucine, valine, tryptophan, methionine, and histidine) rapidly enter the CSF via a single amino acid transporter. (L-[leucine preferring]-system amino acid transporter). Many of these compounds are essential in the diet and must be imported for protein synthesis or as precursors of neurotransmitters. Because a single transporter is involved, these amino acids compete with each other for transport into the brain.

The entry of small neutral amino acids, such as alanine, glycine, proline, and γ -aminobutyrate (GABA), is markedly restricted because their influx could dramatically change the content of neurotransmitters (see section III). They are synthesized in the brain, and some are transported out of the CNS and into the blood via the A- (alanine-preferring)-system carrier. Vitamins have specific transporters through the blood-brain barrier as they do in most tissues.

3. RECEPTOR-MEDIATED TRANSCYTOSIS

Certain proteins, such as insulin, transferrin, and insulin-like growth factors, cross the blood-brain barrier by receptor-mediated transcytosis. Once the protein binds to its membrane receptor, the membrane containing the receptor-protein complex is endocytosed into the endothelial cell to form a vesicle. It is released on the other side of the endothelial cell. Absorptive-mediated transcytosis also can occur. It differs from receptor-mediated transcytosis in that the protein binds nonspecifically to the membrane and not to a distinct receptor.

III. SYNTHESIS OF SMALL NITROGEN-CONTAINING NEUROTRANSMITTERS

Molecules that serve as neurotransmitters fall into two basic structural categories: (1) small nitrogen-containing molecules, and (2) neuropeptides. The major small nitrogen-containing molecule neurotransmitters include glutamate, γ -aminobutyric acid (GABA), glycine, acetylcholine, dopamine, norepinephrine, serotonin, and histamine. Additional neurotransmitters that fall into this category include epinephrine, aspartate, and nitric oxide. In general, each neuron synthesizes only those neurotransmitters that it uses for transmission through a synapse or to another cell. The neuronal tracts are often identified by their neurotransmitter; for example, a dopaminergic tract synthesizes and releases the neurotransmitter dopamine.

Neuropeptides are usually small peptides, which are synthesized and processed in the CNS. Some of these peptides have targets within the CNS (such as endorphins, which bind to opioid receptors and block pain signals), whereas others are released into the circulation to bind to receptors on other organs (such as growth hormone and thyroid-stimulating hormone). Many neuropeptides are synthesized as a larger precursor, which is then cleaved to produce the active peptides. Until recently, the assumption was that a neuron only synthesized and released a single neurotransmitter. More recent evidence suggests that a neuron may contain (1) more than one small molecule neurotransmitter, (2) more than one neuropeptide neurotransmitter, or (3) both types of neurotransmitters. The differential release of the various neurotransmitters is the result of the neuron altering its frequency and pattern of firing.

A. General Features of Neurotransmitter Synthesis

A number of features are common to the synthesis, secretion, and metabolism of most small nitrogen-containing neurotransmitters (Table 48.1). Most of these neurotransmitters are synthesized from amino acids, intermediates of glycolysis and the TCA cycle, and O_2 in the cytoplasm of the presynaptic terminal. The rate of synthesis is

Table 48.1. Features Common to Neurotransmitters^a

- Synthesis from amino acid and common metabolic precursors usually occurs in the cytoplasm of the presynaptic nerve terminal. The synthetic enzymes are transported by fast axonal transport from the cell body, where they are synthesized, to the presynaptic terminal.
- The synthesis of the neurotransmitter is regulated to correspond to the rate of firing of the neuron, both acutely and through long-term enhancement of synaptic transmission.
- The neurotransmitter is actively taken up into storage vesicles in the presynaptic terminal.
- The neurotransmitter acts at a receptor on the postsynaptic membrane.
- The action of the neurotransmitter is terminated through reuptake into the presynaptic terminal, diffusion away from the synapse, or enzymatic inactivation. The enzymatic inactivation may occur in the postsynaptic terminal, the presynaptic terminal, or an adjacent astrocyte or microglial cell.
- The blood-brain barrier affects the supply of precursors for neurotransmitter synthesis.

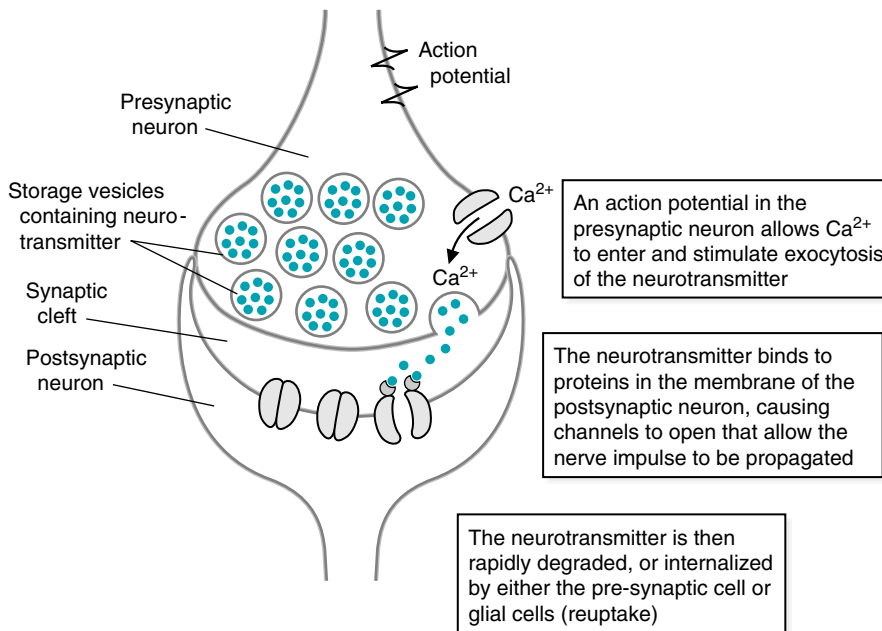
^aNot all neurotransmitters exhibit all of these features. Nitric oxide is an exception to most of these generalities. Some neurotransmitters (epinephrine, serotonin, and histamine) are also secreted by cells other than neurons. Their synthesis and secretion by non-neuronal cells follows other principles.

generally regulated to correspond to the rate of firing of the neuron. Once synthesized, the neurotransmitters are transported into storage vesicles by an ATP-requiring pump linked with the proton gradient. Release from the storage vesicle is triggered by the nerve impulse that depolarizes the postsynaptic membrane and causes an influx of Ca^{2+} ions through voltage-gated calcium channels. The influx of Ca^{2+} promotes fusion of the vesicle with the synaptic membrane and release of the neurotransmitter into the synaptic cleft. The transmission across the synapse is completed by binding of the neurotransmitter to a receptor on the postsynaptic membrane (Fig. 48.3).

The action of the neurotransmitter is terminated through reuptake into the presynaptic terminal, uptake into glial cells, diffusion away from the synapse, or enzymatic inactivation. The enzymatic inactivation may occur in the postsynaptic terminal, the presynaptic terminal or an adjacent astrocyte microglia cell, or in endothelial cells in the brain capillaries.



Drugs have been developed that block neurotransmitter uptake into storage vesicles. Reserpine, which blocks catecholamine uptake into vesicles, had been used as an antihypertensive and antiepileptic drug for many years, but it was noted that a small percentage of patients on the drug became depressed and even suicidal. Animals treated with reserpine showed signs of lethargy and poor appetite, similar to depression in humans. Thus, a link was forged between monoamine release and depression in humans.

**Fig. 48.3.** Action of neurotransmitters.

Not all neurotransmitters exhibit all of these features. Nitric oxide, because it is a gas, is an exception to most of these generalities. Some neurotransmitters are synthesized and secreted by both neurons and other cells (e.g., epinephrine, serotonin, histamine).

B. Dopamine, Norepinephrine, and Epinephrine

1. SYNTHESIS OF THE CATECHOLAMINE NEUROTRANSMITTERS

These three neurotransmitters are synthesized in a common pathway from the amino acid L-tyrosine. Tyrosine is supplied in the diet or is synthesized in the liver from the essential amino acid phenylalanine by phenylalanine hydroxylase (see Chapter 39). The pathway of catecholamine biosynthesis is shown in Figure 48.4.

The first and rate-limiting step in the synthesis of these neurotransmitters from tyrosine is the hydroxylation of the tyrosine ring by tyrosine hydroxylase, a tetrahydrobiopterin (BH₄)-requiring enzyme. The product formed is dihydroxyphenylalanine or DOPA. The phenyl ring with two adjacent OH groups is a catechol, and hence dopamine, norepinephrine, and epinephrine are called catecholamines.

The second step in catecholamine synthesis is the decarboxylation of DOPA to form dopamine. This reaction, like many decarboxylation reactions of amino acids, requires pyridoxal phosphate. Dopaminergic neurons (neurons using dopamine as a neurotransmitter) stop the synthesis at this point, because these neurons do not synthesize the enzymes required for the subsequent steps.

Neurons that secrete norepinephrine synthesize it from dopamine in a hydroxylation reaction catalyzed by dopamine β-hydroxylase (DBH). This enzyme is present only within the storage vesicles of these cells. Like tyrosine hydroxylase, it is a mixed-function oxidase that requires an electron donor. Ascorbic acid (vitamin C) serves as the electron donor and is oxidized in the reaction. Copper (Cu²⁺) is a bound cofactor required for the electron transfer.

Although the adrenal medulla is the major site of epinephrine synthesis, it is also synthesized in a few neurons that use epinephrine as a neurotransmitter. These neurons contain the above pathway for norepinephrine synthesis and in addition contain the enzyme that transfers a methyl group from SAM to norepinephrine to form epinephrine. Thus, epinephrine synthesis is dependent on the presence of adequate levels of B12 and folate (see Chapter 40).

2. STORAGE AND RELEASE OF CATECHOLAMINES

Ordinarily, only low concentrations of catecholamines are free in the cytosol, whereas high concentrations are found within the storage vesicles. Conversion of tyrosine to L-DOPA and that of L-DOPA to dopamine occurs in the cytosol. Dopamine is then taken up into the storage vesicles. In norepinephrine-containing neurons, the final β-hydroxylation reaction occurs within the vesicles.

The catecholamines are transported into vesicles by the protein VMAT2 (vesicle membrane transporter 2) (Fig. 48.5). The vesicle transporters contain 12 transmembrane domains and are homologous to a family of bacterial drug resistance transporters, including the P-glycoprotein. The mechanism that concentrates the catecholamines in the storage vesicles is an ATP-dependent process linked to a proton pump (secondary active transport). Protons are pumped into the vesicles by a vesicular-ATPase (v-ATPase). The protons then exchange for the positively charged catecholamine via the transporter VMAT2. The influx of the catecholamine is thus driven by the H⁺ gradient across the membrane. The intravesicular concentration of catecholamines is approximately 0.5 M, roughly 100 times the cytosolic concentration. In the vesicles, the catecholamines exist in a complex with ATP and acidic proteins known as chromogranins.



Chromogranins are required for the biogenesis of the secretory vesicle. When released from the vesicle, chromogranins can be proteolytically clipped to form bioactive peptides. Elevated levels of chromogranins in the circulation may be found in patients harboring neuroendocrine tumors, such as a pheochromocytoma.

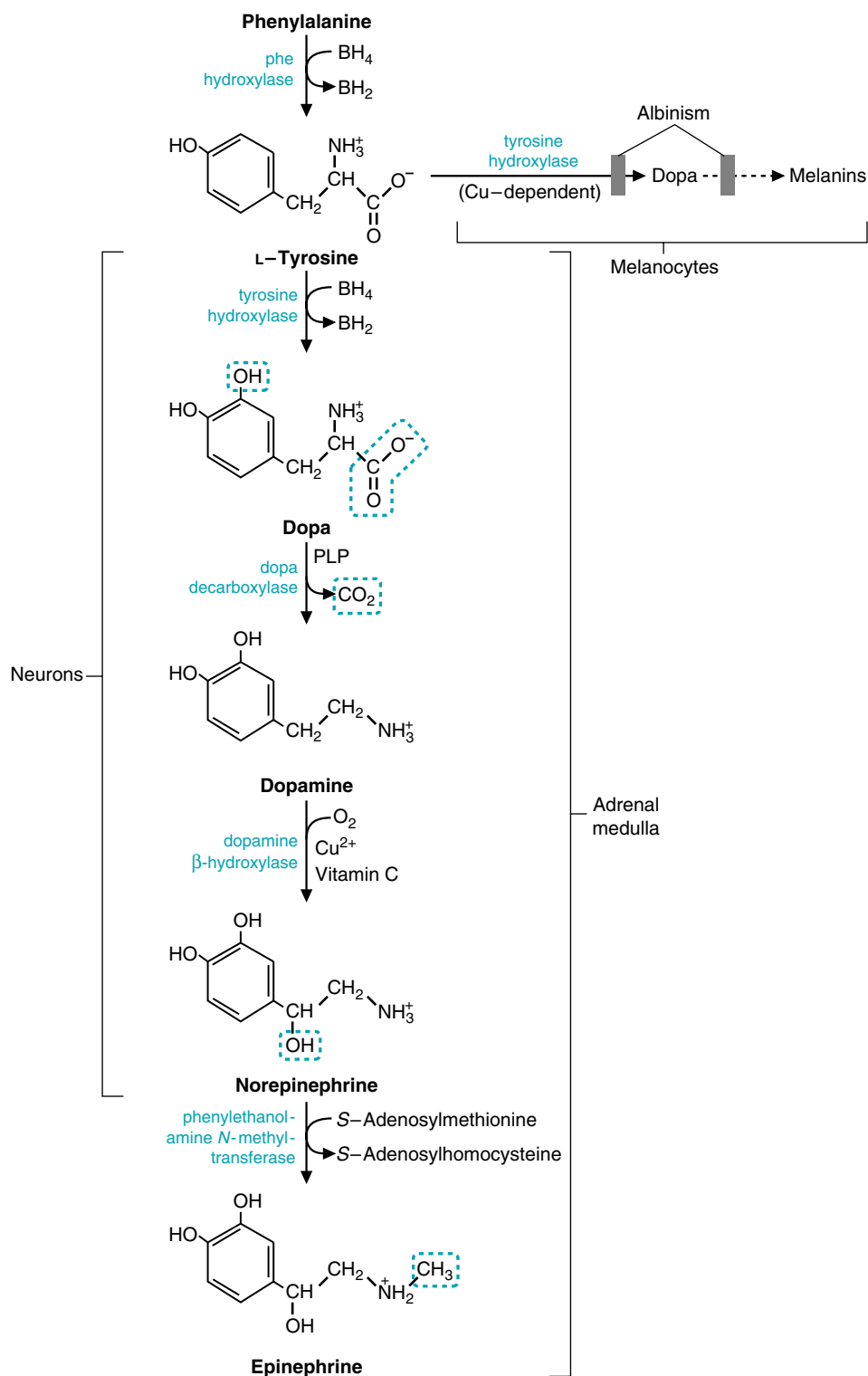


Fig. 48.4. The pathways of catecholamine and melanin biosynthesis. PLP = pyridoxal phosphate. BH_4 = tetrahydrobiopterin. The shaded boxes indicate the enzymes, which, when defective, lead to albinism.



In albinism, either the copper-dependent tyrosine hydroxylase of melanocytes (which is distinct from the tyrosine hydroxylase found in the adrenal medulla) or other enzymes that convert tyrosine to melanins may be defective. Individuals with albinism suffer from a lack of pigment in the skin, hair, and eyes, and they are sensitive to sunlight.

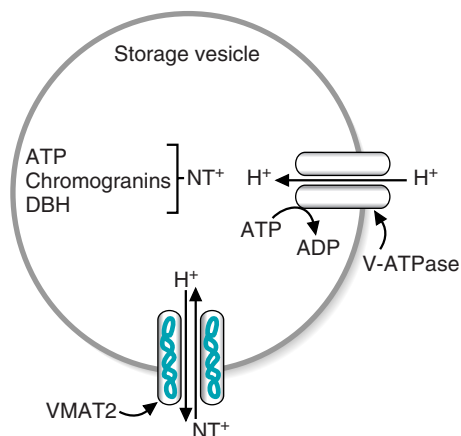
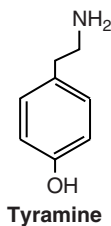


Fig. 48.5. Transport of catecholamines into storage vesicles. This is a secondary active transport based on the generation of a proton gradient across the vesicular membrane. NT⁺ = positively charged neurotransmitter (catecholamine); DBH = dopamine β -hydroxylase; VMAT2 = vesicle membrane transporter 2; V-ATPase = vesicular ATPase.



Tyramine is a degradation product of tyrosine that can lead to headaches, palpitations, nausea and vomiting, and elevated blood pressure if present in large quantities. Tyramine mimics norepinephrine and binds to norepinephrine receptors, stimulating them. Tyramine is inactivated by MAO-A, but if a person is taking an MAO inhibitor, foods containing tyramine should be avoided.



The vesicles play a dual role: they maintain a ready supply of catecholamines at the nerve terminal that is available for immediate release, and they mediate the process of release. When an action potential reaches the nerve terminal, Ca^{2+} channels open, allowing an influx of Ca^{2+} , which promotes the fusion of vesicles with the neuronal membrane. The vesicles then discharge their soluble contents, including the neurotransmitters, ATP, chromogranins, and DBH, into the extraneuronal space by the process of exocytosis. In some cases, the catecholamines affect other neurons. In other cases, they circulate in the blood and initiate responses in peripheral tissues.

3. INACTIVATION AND DEGRADATION OF CATECHOLAMINE NEUROTRANSMITTERS

The action of catecholamines is terminated through reuptake into the presynaptic terminal and diffusion away from the synapse. Degradative enzymes are present in the presynaptic terminal, and in adjacent cells, including glial cells and endothelial cells.

Two of the major reactions in the process of inactivation and degradation of catecholamines are catalyzed by monamine oxidase (MAO) and catechol-O-methyltransferase (COMT). MAO is present on the outer mitochondrial membrane of many cells and oxidizes the carbon containing the amino group to an aldehyde, thereby releasing ammonium ion (Fig. 48.6). In the presynaptic terminal, MAO inactivates catecholamines that are not protected in storage vesicles. (Thus, drugs that deplete storage vesicles indirectly increase catecholamine degradation.) There are two isoforms of MAO with different specificities of action: MAO-A preferentially deaminates norepinephrine and serotonin, whereas MAO-B acts on a wide spectrum of phenylethylamines (phenylethyl refers to a $-\text{CH}_2-$ group linked to a phenyl ring). MAO in the liver and other sites protects against the ingestion of dietary biogenic amines such as the tyramine found in many cheeses.

COMT is also found in many cells, including the erythrocyte. It works on a broad spectrum of extraneuronal catechols and those that have diffused away from the synapse. COMT transfers a methyl group from SAM to a hydroxyl group on the catecholamine or its degradation product (see Fig. 48.6). Because the inactivation reaction requires SAM, it is indirectly dependent on vitamins B12 and folate. The action of MAO and COMT can occur in almost any order, thereby resulting in a large number of degradation products and intermediates, many of which appear in the urine. Cerebrospinal homovanillylmandelic acid (HVA) is an indicator of dopamine degradation. Its concentration is decreased in the brain of patients with Parkinson's disease.



Katie Colamin's doctor ordered plasma catecholamine (epinephrine, norepinephrine, and dopamine) levels and also had Katie collect a 24-hour urine specimen for the determination of catecholamines and their degradation products. All of these tests showed unequivocal elevations of these compounds in Katie's blood and urine. Katie was placed on phenoxybenzamine, an α_1 - and α_2 -adrenergic receptor antagonist that blocks the pharmacologic effect of the elevated catecholamines on these receptors. After ruling out evidence to suggest metastatic disease to the liver or other organs (in case Katie's tumor was malignant), the doctor referred Katie to a surgeon with extensive experience in adrenal surgery.

The catecholamines exert their physiologic and pharmacologic effects by circulating in the bloodstream to target cells whose plasma membranes contain catecholamine receptors. This interaction initiates a biochemical cascade leading to responses that are specific for different types of cells. Patients such as **Katie Colamin** experience palpitations, excessive sweating, hypertensive headaches, and a host of other symptoms when a catecholamine-producing tumor of the adrenal medulla suddenly secretes supraphysiologic amounts of epinephrine or norepinephrine into the venous blood draining the neoplasm.

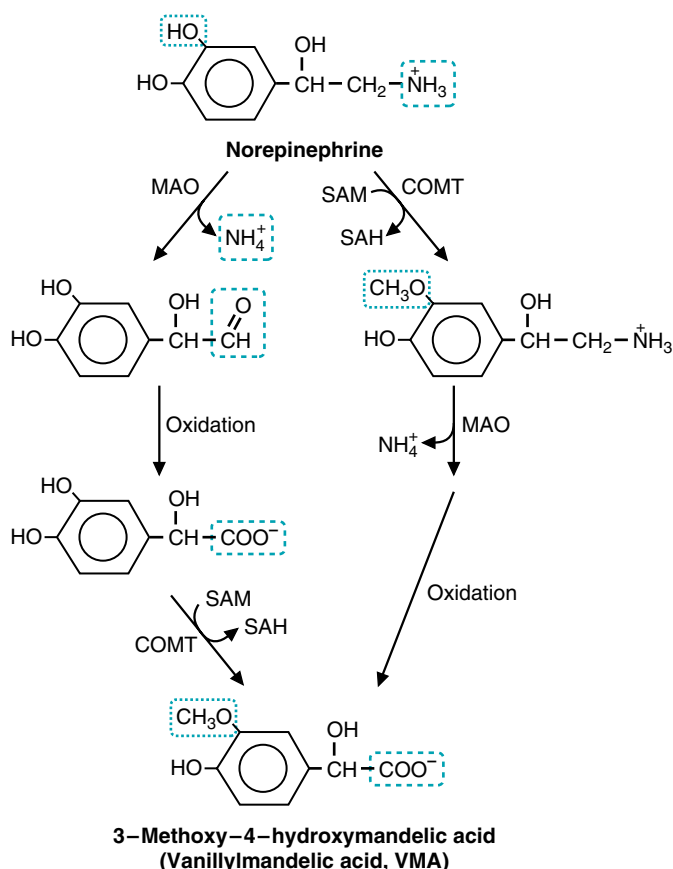


Fig. 48.6. Inactivation of catecholamines. Methylation and oxidation may occur in any order. Methylated and oxidized derivatives of norepinephrine and epinephrine are produced, and 3-methoxy-4-hydroxymandelic acid is the final product. These compounds are excreted in the urine. MAO = monoamine oxidase; COMT = catechol O-methyltransferase; SAM = S-adenosylmethionine; SAH = S-adenosylhomocysteine.

4. REGULATION OF TYROSINE HYDROXYLASE

Efficient regulatory mechanisms coordinate the synthesis of catecholamine neurotransmitters with the rate of firing. Tyrosine hydroxylase, the first committed step and rate-limiting enzyme in the pathway, is regulated by feedback inhibition that is coordinated with depolarization of the nerve terminal. Tyrosine hydroxylase is inhibited by free cytosolic catecholamines that compete at the binding site on the enzyme for the pterin cofactor (tetrahydrobiopterin, BH_4 ; see Chapter 39).

Depolarization of the nerve terminal activates tyrosine hydroxylase. Depolarization also activates a number of protein kinases (including protein kinase C, protein kinase A [the cAMP-dependent protein kinase] and CAM kinases [Ca^{2+} -calmodulin-dependent kinases]) that phosphorylate tyrosine hydroxylase. These activation steps result in an enzyme that binds BH_4 more tightly, making it less sensitive to end-product inhibition.

In addition to these short-term regulatory processes, a long-term process involves alterations in the amounts of tyrosine hydroxylase and dopamine β -hydroxylase present in nerve terminals. When sympathetic neuronal activity is increased for a prolonged period, the amounts of mRNA coding for tyrosine hydroxylase and dopamine β -hydroxylase are increased in the neuronal perikarya (the cell body of the neuron). The increased gene transcription may be the result of phosphorylation of CREB (cAMP response element binding protein; see Chapter 26) by



In addition to the catecholamines, serotonin is also inactivated by monoamine oxidase. The activity of a number of antipsychotic drugs are based on inhibiting MAO. The first generation of drugs (exemplified by iproniazid, which was originally developed as an anti-tuberculosis drug and was found to induce mood swings in patients) were irreversible inhibitors of both the A and B forms of MAO. Although leading to a reduction in the severity of depression (by maintaining higher levels of serotonin), these drugs suffered from the “cheese” effect. Cheese and other foods that are processed over long periods (such as red wine) contain tyramine. Usually tyramine is inactivated by MAO-A, but if an individual is taking an MAO inhibitor, tyramine levels will increase. Tyramine induces the release of norepinephrine from storage vesicles, which leads to potentially life-threatening hypertensive episodes. When it was realized that MAO existed in two forms, selective irreversible inhibitors were developed; examples include clorgyline for MAO-A, and deprenyl for MAO-B. Deprenyl has been used to treat Parkinson’s disease (which is caused by a lack of dopamine, which is also inactivated by MAO). Deprenyl, however, is not an antidepressant. Clorgyline is an antidepressant but suffers from the “cheese” effect. This led to the development of the third generation of MAO inhibitors, which are reversible inhibitors of the enzyme, as typified by moclobemide. Moclobemide is a specific, reversible inhibitor of MAO-A and is effective as an antidepressant. More importantly, because of the reversible nature of the drug, the “cheese” effect is not observed, because as tyramine levels increase, they displace the drug from MAO, and the tyramine is safely inactivated.

protein kinase A or by other protein kinases. CREB then binds to the CRE (cAMP response element) in the promoter region of the gene (similar to the mechanism for the induction of gluconeogenic enzymes in the liver). The newly synthesized enzyme molecules are then transported down the axon to the nerve terminals. The concentration of dopamine decarboxylase in the terminal does not appear to change in response to neuronal activity.

C. Metabolism of Serotonin

The pathway for the synthesis of serotonin from tryptophan is very similar to the pathway for the synthesis of norepinephrine from tyrosine (Fig. 48.7). The first enzyme of the pathway, tryptophan hydroxylase, uses an enzymic mechanism similar to that of tyrosine and phenylalanine hydroxylase and requires BH_4 to hydroxylate the ring structure of tryptophan. The second step of the pathway is a decarboxylation reaction

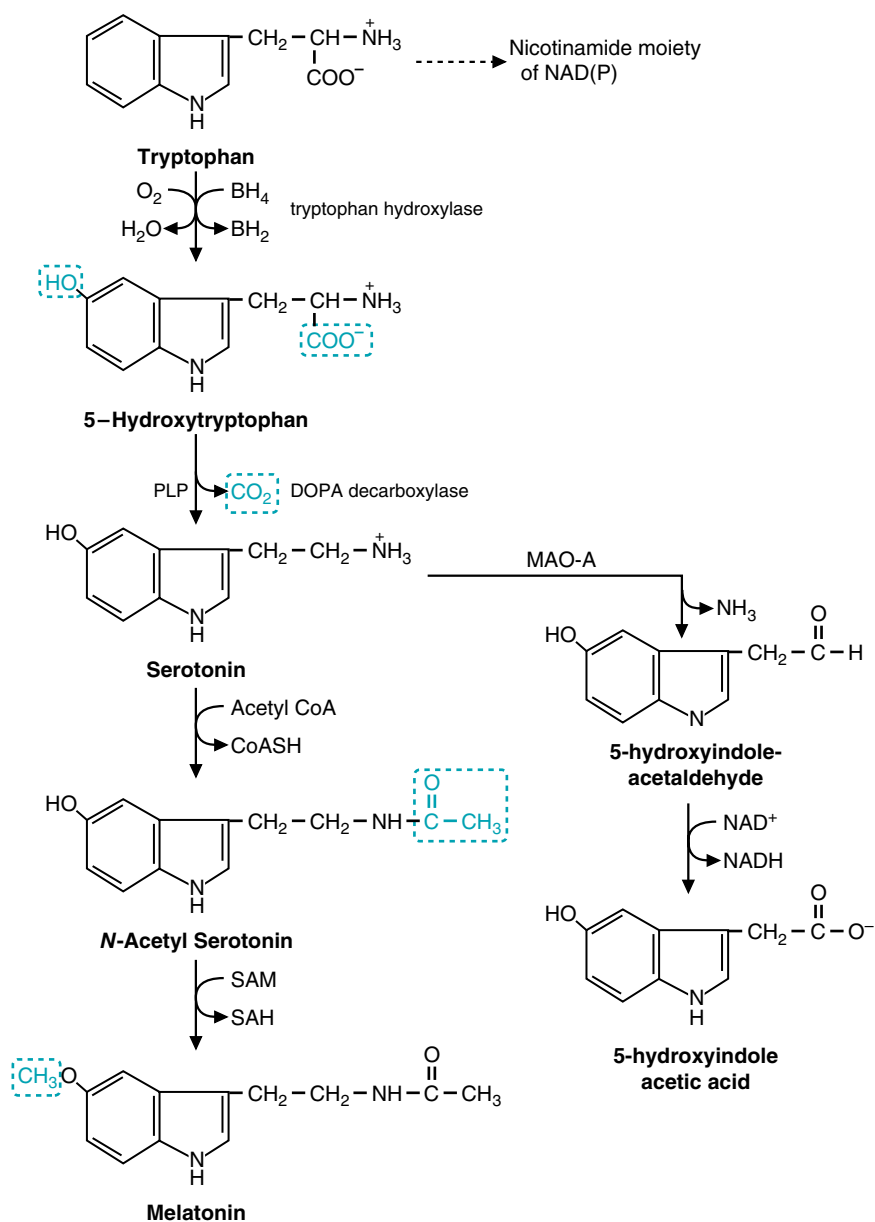


Fig. 48.7. Synthesis and inactivation of serotonin.

catalyzed by the same enzyme that decarboxylates DOPA. Serotonin, like the catecholamine neurotransmitters, can be inactivated by MAO.

The neurotransmitter melatonin is also synthesized from tryptophan (see Fig. 48.7). Melatonin is produced in the pineal gland in response to the light–dark cycle, its level in the blood rising in a dark environment. It is probably through melatonin that the pineal gland conveys information about light–dark cycles to the body, organizing seasonal and circadian rhythms. Melatonin also may be involved in regulating reproductive functions.

D. Metabolism of Histamine

Within the brain, histamine is produced both by mast cells and by certain neuronal fibers. Mast cells are a family of bone marrow–derived secretory cells that store and release high concentrations of histamine. They are prevalent in the thalamus, hypothalamus, dura mater, leptomeninges, and choroid plexus. Histaminergic neuronal cell bodies in the human are found in the tuberomammillary nucleus of the posterior basal hypothalamus. The fibers project into nearly all areas of the CNS, including the cerebral cortex, the brainstem, and spinal cord.

Histamine is synthesized from histidine in a single enzymatic step. The enzyme histidine decarboxylase requires pyridoxal phosphate, and its mechanism is very similar to that of DOPA decarboxylase (Fig. 48.8).

Like other neurotransmitters, newly synthesized neuronal histamine is stored within the nerve terminal vesicle. Depolarization of nerve terminals activates the exocytotic release of histamine by voltage-dependent as well as a calcium-dependent mechanism.

Once released from neurons, histamine is thought to activate both postsynaptic and presynaptic receptors. Unlike other neurotransmitters, histamine does not appear to be recycled into the presynaptic terminal to any great extent. However, astrocytes have a specific high-affinity uptake system for histamine and may be the major site of the inactivation and degradation of this monoamine.

The first step in the inactivation of histamine in the brain is methylation (see Fig. 48.8). The enzyme histamine methyltransferase transfers a methyl group from SAM to a ring nitrogen of histamine to form methylhistamine. The second step is oxidation by MAO-B, followed by an additional oxidation step. In peripheral tissues, histamine undergoes deamination by diamine oxidase, followed by oxidation to a carboxylic acid (see Fig. 48.8).

E. Acetylcholine

1. SYNTHESIS

The synthesis of acetylcholine from acetyl CoA and choline is catalyzed by the enzyme choline acetyltransferase (ChAT) (Fig. 48.9). This synthetic step occurs in the presynaptic terminal. The compound is stored in vesicles and later released through calcium-mediated exocytosis. Choline is taken up by the presynaptic terminal from the blood via a low-affinity transport system (high K_m) and from the synaptic cleft via a high-affinity transport mechanism (low K_m). It is also derived from the hydrolysis of phosphatidylcholine (and possibly sphingomyelin) in membrane lipids. Thus, membrane lipids may form a storage site for choline, and their hydrolysis, with the subsequent release of choline, is highly regulated.



It is believed that the vitamin B12 requirement for choline synthesis contributes to the neurologic symptoms of vitamin B12 deficiency. The methyl groups for choline synthesis are donated by SAM, which is converted to S-adenosylhomocysteine in the reaction. Recall that formation of SAM through recycling of homocysteine requires both tetrahydrofolate and vitamin B12 (unless extraordinary amounts of methionine are available to bypass the B12-dependent methionine synthase step).



Evan Applebod was placed on Redux, which increased the secretion of serotonin. Serotonin has been implicated in many processes, including mood control and appetite regulation. When serotonin levels are low, increased appetite, or depression, or both can occur. Thus, drugs that can increase serotonin levels may be able to control appetite and depression. Redux was a second-generation drug developed from fenfluramine, a known appetite suppressant. When first used, fenfluramine could not be resolved between two distinct optical isomers (d versus l). The l-isomer induced sleepiness, so to counteract this effect fenfluramine was often given with phentermine, which elevated epinephrine levels to counteract the drowsiness (the combination of drugs was known as fen/phen). Once the two isomers of fenfluramine could be resolved, Redux, dexfenfluramine, was developed. Because levels of serotonin have been linked to mood, many antidepressant drugs were developed that affect serotonin levels. The first of these were the MAO inhibitors; a second class are the tricyclics, and the third class are known as the SSRI (selective serotonin reuptake inhibitors). The SSRI block reuptake of serotonin from the synapse, leading to an elevated response to serotonin. Redux acted both as an SSRI but also increased the secretion of serotonin, leading to elevated levels of this compound in the synapse. None of the other drugs that affect serotonin levels have this effect.



Histamine elicits a number of effects on different tissues. Histamine is the major mediator of the allergic response and when released from mast cells (a type of white blood cell found in tissues) leads to vasodilation and an increase in the permeability of blood vessel walls. This leads to the allergic symptoms of a runny nose and watering eyes. When histamine is released in the lungs, the airways constrict in an attempt to reduce the intake of the allergic material. The ultimate result of this, however, is bronchospasm, which can lead to difficulty in breathing. In the brain, histamine is an excitatory neurotransmitter. Antihistamines block histamine from binding to its receptor. In the tissues, this will counteract histamine's effect on vasodilation and blood vessel wall permeability, but in the brain the effect is to cause drowsiness. The new generation of “non-drowsy” antihistamines have been modified such that they cannot pass through the blood-brain barrier. Thus, the effects on the peripheral tissues are retained with no effect on CNS histamine response.

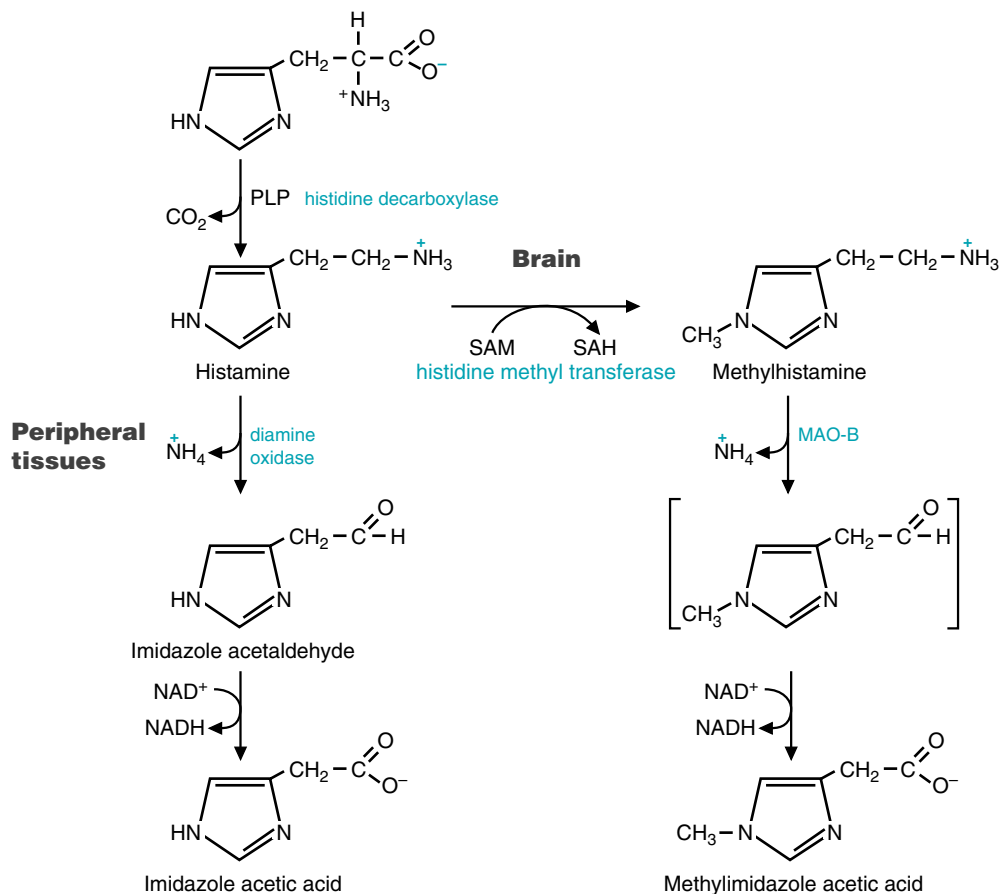


Fig. 48.8. Synthesis and inactivation of histamine; note the different pathways for brain and peripheral tissues. SAM = S-adenosylmethionine; SAH = S-adenosylhomocysteine.

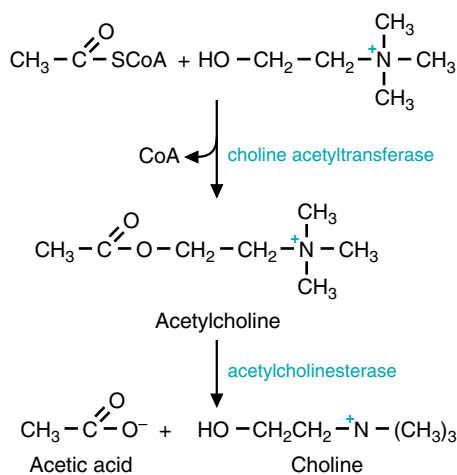


Fig. 48.9. Acetylcholine synthesis and degradation.

Choline is a common component of the diet but also can be synthesized in the human as part of the pathway for the synthesis of phospholipids (see Chapter 33). The only route for choline synthesis is via the sequential addition of three methyl groups from SAM to the ethanolamine portion of phosphatidylethanolamine to form phosphatidylcholine. Phosphatidylcholine is subsequently hydrolyzed to release choline or phosphocholine. Conversion of phosphatidylethanolamine to phosphatidylcholine occurs in many tissues, including liver and brain. This conversion is B6- and B12-dependent.

The acetyl group used for acetylcholine synthesis is derived principally from glucose oxidation to pyruvate and decarboxylation of pyruvate to form acetyl



The supply of choline in the brain can become rate-limiting for acetylcholine synthesis, and supplementation of the diet with lecithin (phosphatidylcholine) has been used to increase brain acetylcholine in patients suffering from tardive dyskinesia (often persistent involuntary movements of the facial muscles and tongue). The neonate has a very high demand for choline, for both brain lipid synthesis (phosphatidylcholine and sphingomyelin) and acetylcholine biosynthesis. High levels of phosphatidylcholine in the maternal milk and a high activity of a high-affinity transport system through the blood-brain barrier for choline in the neonate help to maintain brain choline concentrations. The fetus also has a high demand for choline, and there is a high-affinity transport system for choline across the placenta. The choline is derived from maternal stores, maternal diet, and synthesis primarily in the maternal liver. Because choline synthesis is dependent on folate and vitamin B12, the high fetal demand may contribute to the increased maternal requirement for both vitamins during pregnancy.

CoA via the pyruvate dehydrogenase reaction. This is because neuronal tissues have only a limited capacity to oxidize fatty acids to acetyl CoA so that glucose oxidation is the major source of acetyl groups. Pyruvate dehydrogenase is found only in mitochondria. The acetyl group is probably transported to the cytoplasm as part of citrate, which is then cleaved in the cytosol to form acetyl CoA and oxaloacetate.

2. INACTIVATION OF ACETYLCHOLINE

Acetylcholine is inactivated by acetylcholinesterase, which is a serine esterase that forms a covalent bond with the acetyl group. The enzyme is inhibited by a wide range of compounds (pharmacologic drugs and neurotoxins) that form a covalent bond with this reactive serine group. Neurotoxins such as Sarin (the gas used in Japanese subways by a terrorist group) and the nerve gas in the movie “The Rock” work through this mechanism. Acetylcholine is the major neurotransmitter at the neuromuscular junctions; inability to inactivate this molecule leads to constant activation of the nerve–muscle synapses, a condition that leads to varying degrees of paralysis.

F. Glutamate and GABA

1. SYNTHESIS OF GLUTAMATE

Glutamate functions as an excitatory neurotransmitter within the central nervous system, leading to the depolarization of neurons. Within nerve terminals, glutamate is generally synthesized *de novo* from glucose rather than taken up from the blood because its plasma concentration is low and it does not readily cross the blood-brain barrier.

Glutamate is primarily synthesized from the TCA cycle intermediate α -ketoglutarate (Fig. 48.10). This can occur via either of two routes. The first is via the enzyme glutamate dehydrogenase, which reduces α -ketoglutarate to glutamate, thereby incorporating free ammonia into the carbon backbone. The ammonia pool is provided by amino acid /neurotransmitter degradation or by diffusion of ammonia across the blood-brain barrier. The second route is through transamination reactions in which an amino group is transferred from other amino acids to α -ketoglutarate to form glutamate. Glutamate also can be synthesized from glutamine, using glutaminase. The glutamine is derived from glial cells as described in section F.2.

Like other neurotransmitters, glutamate is stored in vesicles, and its release is Ca^{2+} -dependent. It is removed from the synaptic cleft by high-affinity uptake systems present in nerve terminals and glial cells.

2. GABA

GABA (γ -aminobutyric acid) is the major inhibitory neurotransmitter in the central nervous system. Its functional significance is far-reaching, and altered GABA-ergic function plays a role in many neurologic and psychiatric disorders.

GABA is synthesized by the decarboxylation of glutamate (see Fig. 48.10) in a single step catalyzed by the enzyme glutamic acid decarboxylase (GAD). GABA is recycled in the central nervous system by a series of reactions called the GABA shunt, which conserves glutamate and GABA (see Fig. 48.10).

Much of the uptake of GABA occurs in glial cells. The GABA shunt in glial cells produces glutamate, which is converted to glutamine and transported out of the glial cells to neurons, where it is converted back to glutamate. Glutamine thus serves as a transporter of glutamate between cells in the CNS (see Chapter 42). Glial cells lack GAD and cannot synthesize GABA.



An inherited pyruvate dehydrogenase deficiency, a thiamine deficiency, or hypoxia deprives the brain of a source of acetyl CoA for acetylcholine synthesis, as well as a source of acetyl CoA for ATP generation from the TCA cycle.



The β -cells of the pancreas also express glutamate decarboxylase (GAD), and patients with type 1 diabetes express autoantigens against pancreatic GAD. Studies in a mouse model of type 1 diabetes have implicated GAD as a key autoantigen. If GAD expression is blocked in these mice by use of genetic tricks, diabetes does not develop. As usual, what occurs in the human condition is not as straightforward as it appears to be in animals, but there is a relationship between pancreatic GAD expression and the cause of type 1 diabetes in humans.



Tiagabine is a drug that inhibits the reuptake of GABA from the synapse, and it has been used to treat epilepsy as well as other convulsant disorders. Because GABA is an inhibitory neurotransmitter in the brain, its prolonged presence can block neurotransmission by other agents, thereby reducing the frequency of convulsions.

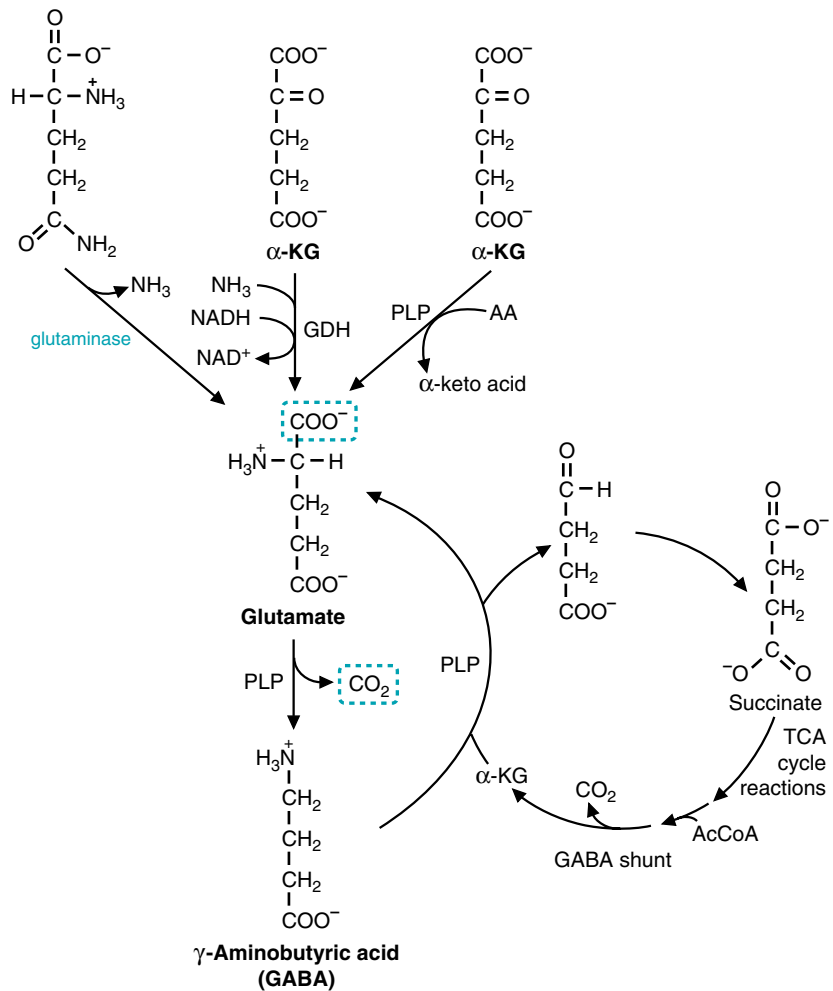


Fig. 48.10. Synthesis of glutamate and GABA and the GABA shunt. GDH = glutamate dehydrogenase; PLP = pyridoxal phosphate.

G. Other Amino Acid Neurotransmitters

1. ASPARTATE

Aspartate, like glutamate, is an excitatory neurotransmitter, but it functions in far fewer pathways. It is synthesized from the TCA cycle intermediate oxaloacetate via transamination reactions. Like glutamate synthesis, aspartate synthesis uses oxaloacetate that must be replaced through anaplerotic reactions. Aspartate cannot pass through the blood-brain barrier.

2. GLYCINE

Glycine is the major inhibitory neurotransmitter in the spinal cord. Most of the glycine in neurons is synthesized *de novo* within the nerve terminal from serine by the enzyme serine hydroxymethyltransferase, which requires folic acid. Serine, in turn, is synthesized from the intermediate 3-phosphoglycerate in the glycolytic pathway. The action of glycine is probably terminated via uptake by a high-affinity transporter.

3. CONVERSION OF ARGININE TO NITRIC OXIDE

Nitric oxide (NO) is a biologic messenger in a variety of physiologic responses, including vasodilation, neurotransmission, and the ability of the immune system to

kill tumor cells and parasites. NO is synthesized from arginine in a reaction catalyzed by NO synthase (see Fig. 24.10).

NO synthase exists as tissue-specific forms of two families of enzymes. The form present in macrophages is responsible for the overproduction of NO, leading to its cytotoxic actions on parasites and tumor cells. The enzyme present in nervous tissue, vascular endothelium, platelets, and other tissues is responsible for the physiologic responses to NO such as vasodilation and neural transmission. In target cells, NO activates a soluble guanylate cyclase, which results in increased cellular levels of cGMP (3', 5'-cyclic GMP)(Fig. 48.11). In smooth muscle cells, cGMP, like cAMP, activates one or more protein kinases, which are responsible for the relaxation of smooth muscle and the subsequent dilation of vessels. NO stimulates penile erection by acting as a neurotransmitter, stimulating smooth muscle relaxation that permits the corpus cavernosum to fill with blood. Nitric oxide can readily cross cell membranes because it is a gas. As a result, its effect may not necessarily be limited to the neuron that synthesizes it (Fig. 48.12). There is ample evidence that NO may function as a retrograde messenger that can influence neurotransmitter release from the presynaptic terminal after diffusing from the postsynaptic neuron (where it is synthesized). There is also evidence supporting retrograde messenger roles for both arachidonic acid and carbon monoxide in the CNS.

IV. METABOLIC ENCEPHALOPATHIES AND NEUROPATHIES

The brain has an absolute dependence on the blood for its supply of glucose and oxygen. It uses approximately 20% of the oxygen supply of the body. During the developmental period and during prolonged fasting, ketone bodies can be used as a fuel, but they cannot totally substitute for glucose. Glucose is converted to pyruvate in glycolysis, and the pyruvate is oxidized in the TCA cycle. Anaerobic glycolysis, with a yield of 2 ATPs/glucose, cannot sustain the ATP requirement of the brain, which can be provided only by the complete oxidation of glucose to CO_2 , which yields approximately 32 ATPs /glucose. However, during periods of mild hypoglycemia or mild hypoxia, decreased neurotransmitter synthesis contributes as much, if not more, to the development of symptoms as does an absolute deficiency of ATP for energy needs.

A. Hypoglycemic Encephalopathy

Hypoglycemia is sometimes encountered in medical conditions such as malignancies producing insulin, insulin-like growth factors, or chronic alcoholism. Early clinical signs in hypoglycemia reflect the appearance of physiologic protective mechanisms initiated by hypothalamic sensory nuclei such as sweating, palpitations, anxiety, and hunger. If unheeded, these symptoms give way to a more serious CNS disorder progressing through confusion and lethargy to seizures and eventually coma. Prolonged hypoglycemia can lead to irreversible brain damage.

During the progression of hypoglycemic encephalopathy, as blood glucose falls below 2.5 mM (45 mg/dL), the brain attempts to use internal substrates such as glutamate and TCA cycle intermediates as fuels. Because the pool size of these substrates is quite small, they are quickly depleted. If blood glucose levels continue to fall below 1 mM (18 mg/dL), ATP levels become depleted.

As the blood glucose drops from 2.5 to 2.0 mM (45 to 36 mg/dL, before EEG changes are observed), the symptoms appear to arise from decreased synthesis of neurotransmitters in particular regions of the brain rather than a global energy deficit. Figure 48.13 summarizes the relationship between the oxidation of glucose in glycolysis and the provision of precursors for the synthesis of neurotransmitters in different types of neurons.

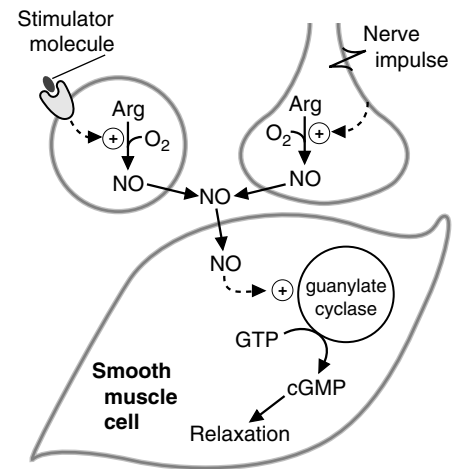


Fig. 48.11. Action of nitric oxide (NO) in vasodilation. The synthesis of NO occurs in response to a stimulator binding to a receptor on some cells or to a nerve impulse in neurons. NO enters smooth muscle cells, stimulating guanylate cyclase to produce cGMP, which causes smooth muscle cell relaxation. When the smooth muscle cells relax, blood vessels dilate.

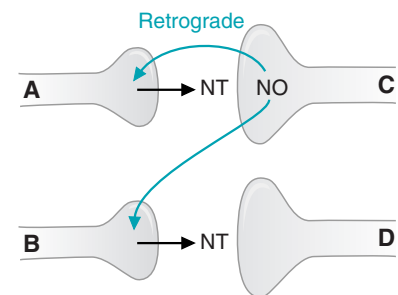


Fig. 48.12. NO as a retrograde messenger. Cell A releases a neurotransmitter, which stimulates cell C to produce NO. NO, being a gas, can diffuse back and regulate cell A's production and release of neurotransmitters. NO can also diffuse to cell B, and to stimulate cell B to produce a different neurotransmitter to elicit a response from cell D. NT = neurotransmitter.

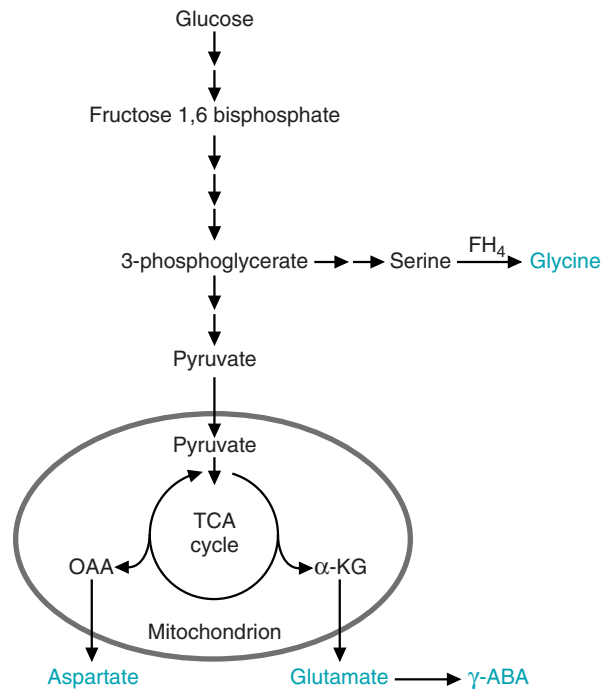


Fig. 48.13. Glucose metabolism leading to the synthesis of the neurotransmitters glycine, aspartate, glutamate, and GABA. As blood glucose levels drop and brain glucose levels diminish, synthesis of these neurotransmitters may be compromised.

As hypoglycemia progresses below 1 mM (18 mg/dL), and high-energy phosphate levels are depleted, the EEG becomes isoelectric, and neuronal cell death ensues. As is the case in some other metabolic encephalopathies, cell death is not global in distribution; rather, certain brain structures, in particular hippocampal and cortical structures, are selectively vulnerable to hypoglycemic insult. Pathophysiologic mechanisms responsible for neuronal cell death in hypoglycemia include the involvement of glutamate excitotoxicity. Glutamate excitotoxicity occurs when the cellular energy reserves are depleted. The failure of the energy-dependent reuptake pumps results in a buildup of glutamate in the synaptic cleft and overstimulation of the postsynaptic glutamate receptors. The prolonged glutamate receptor activation leads to prolonged opening of the receptor ion channel and the influx of lethal amounts of Ca^{+2} ion, which can activate cytotoxic intracellular pathways in the postsynaptic neuron.

B. Hypoxic Encephalopathy

Experimental studies with human volunteers show that cerebral energy metabolism remains normal when mild to moderate hypoxia (partial pressure of oxygen, or $\text{PaO}_2 = 25\text{--}40$ mm Hg) results in severe cognitive dysfunction. The diminished cognitive function is believed to result from impaired neurotransmitter synthesis. In mild hypoxia, cerebral blood flow increases to maintain oxygen delivery to the brain. In addition, anaerobic glycolysis is accelerated, resulting in maintenance of ATP levels. This occurs, however, at the expense of an increase of lactate production and a fall of pH. Acute hypoxia ($\text{PaO}_2 \leq 20$ mm Hg) generally results in a coma.

Hypoxia can result from insufficient oxygen reaching the blood (e.g., at high altitudes), severe anemia (e.g., iron deficiency), or a direct insult to the oxygen-utilizing capacity of the brain (e.g., cyanide poisoning). All forms of hypoxia result in diminished neurotransmitter synthesis. Inhibition of pyruvate dehydrogenase diminishes acetylcholine synthesis, which is acutely sensitive to hypoxia. Glutamate and

GABA synthesis, which depend on a functioning TCA cycle, are decreased as a result of elevated NADH levels, which inhibit TCA cycle enzymes. NADH levels are increased when oxygen is unavailable to accept electrons from the electron transport chain and NADH cannot be converted back into NAD^+ . Even the synthesis of catecholamine neurotransmitters may be decreased because the hydroxylase reactions require O_2 .

C. The Relationship Between Glutamate Synthesis and the Anaplerotic Pathways of Pyruvate Carboxylase and Methylmalonyl CoA Mutase

Synthesis of glutamate removes α -ketoglutarate from the TCA cycle, thereby decreasing the regeneration of oxaloacetate in the TCA cycle. Because oxaloacetate is necessary for the oxidation of acetyl CoA, oxaloacetate must be replaced by anaplerotic reactions. There are two major types of anaplerotic reactions: (1) pyruvate carboxylase and (2) the degradative pathway of the branched-chain amino acids, valine and isoleucine, which contribute succinyl CoA to the TCA cycle. This pathway uses B12 (but not folate) in the reaction catalyzed by methylmalonyl CoA mutase.

V. LIPID SYNTHESIS IN THE BRAIN AND PERIPHERAL NERVOUS SYSTEM

A number of features of lipid synthesis and degradation in the nervous system distinguish it from most other tissues. The first is that the portion of the neuronal cell membrane involved in synaptic transmission has a unique role and a unique composition. At the presynaptic terminal, the lipid composition is rapidly changing as storage vesicles containing the neurotransmitter fuse with the cell membrane and release their contents. Portions of the membrane are also lost as endocytotic vesicles. On the postsynaptic terminal, the membrane contains the receptors for the neurotransmitter as well as a high concentration of membrane signaling components, such as phosphatidylinositol. A second important feature of brain lipid metabolism is that the blood-brain barrier restricts the entry of nonessential fatty acids such as palmitate, which are released from adipose tissue or present in the diet. Conversely, essential fatty acids are taken up by the brain. Because of these considerations, the brain is constantly synthesizing those lipids (cholesterol, fatty acids, glycosphingolipids, and phospholipids), which it needs for various neurologic functions. Neuronal signaling also requires that non-neuronal glial cells synthesize myelin, a multilayered membrane that surrounds the axons of many neurons. Myelin is lipid rich and has a different lipid composition than the neuronal membranes. The white matter in the brain contains significantly more myelin than the gray matter; it is the presence of myelin sheaths that is responsible for the characteristic color differences that exist between the two types of brain tissue.

A. Brain Lipid Synthesis and Oxidation

Because the blood-brain barrier significantly inhibits the entry of certain fatty acids and lipids into the CNS, virtually all lipids found there must be synthesized within the CNS. The exceptions are the essential fatty acids (linoleic and linolenic acid), which do enter the brain, where they are elongated or further desaturated. The uptake of fatty acids into the CNS is insufficient to meet the energy demands of the CNS; hence the requirement for aerobic glucose metabolism. Thus, cholesterol, glycerol, and sphingolipids, glycosphingolipids, and cerebrosides are all synthesized using pathways previously discussed in this text. Of particular note is that very-long-chain fatty acids are synthesized in the brain, where they play a major role in myelin formation.

Oxidation and turnover of brain lipids occurs as described previously in the text (see Chapter 23). Peroxisomal fatty acid oxidation is important in the brain because the brain contains very-long-chain fatty acids and phytanic acid (from the diet), both of which are oxidized in the peroxisomes by α -oxidation. Thus, disorders that affect peroxisome biogenesis (such as Refsum's disease) severely affect brain cells because of the inability to metabolize both branched-chain and very-long-chain fatty acids. If there is a disorder in which the degradation of glycosphingolipids or mucopolysaccharides is reduced, lysosomes in brain cells will become engorged with partially digested glycolipids, leading to varying degrees of neurologic dysfunction.

B. Myelin Synthesis

A rapid rate of nerve conduction in the peripheral and central motor nerves depends on the formation of myelin, a multilayered lipid and protein structure that is formed from the plasma membrane of glial cells. In the peripheral nervous system, the Schwann cell is responsible for myelinating one portion of an axon of one nerve cell. The Schwann cell does this by wrapping itself around the axon multiple times such that a multilayered sheath of membrane surrounds the axon. In the central nervous system, the oligodendrocyte is responsible for myelination. Unlike the Schwann cell, oligodendrocytes can myelinate portions of numerous axons (up to 40), and do so by extending a thin process that wraps around the axon multiple times. Thus, CNS axons are only surrounded by the membranes of oligodendrocytes, whereas axons in the PNS are surrounded by the entire Schwann cell. A generalized view of myelination is depicted in Figure 48.14. To maintain the myelin structure, the oligodendrocyte synthesizes 4 times its own weight in lipids per day.

1. MYELIN LIPIDS

As the plasma membrane of the glial cell is converted into myelin, the lipid composition of the brain changes (Table 48.2). The lipid-to-protein ratio is greatly increased, as is the content of sphingolipids. The myelin is a tightly packed structure,

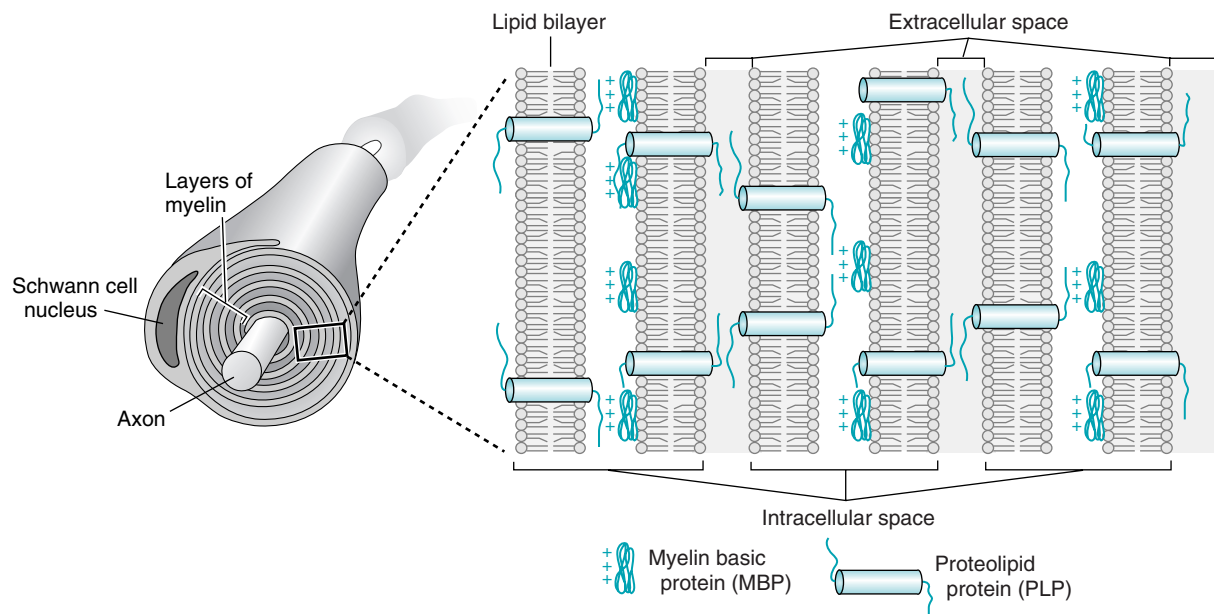


Fig. 48.14. A composite diagram indicating a Schwann cell that has wrapped around a portion of an axon, forming the myelin sheath. The expansion represents a portion of the myelin sheath. CNS myelin is shown, although it is similar to PNS myelin except that Po would take the place of PLP (proteolipid protein). Recall that there are multiple layers of membrane surrounding the axon; PLP protrudes into the extracellular space and aids in compaction of the membranes through hydrophobic interactions. MBPs help to stabilize the structure from within the membrane.

Table 48.2. Protein and Lipid Composition of CNS Myelin and Human Brain

Substance ^a	Myelin	White Matter	Gray Matter
Protein	30.0	39.0	55.3
Lipid	70.0	54.9	32.7
Cholesterol	27.7	27.5	22.0
Cerebroside	22.7	19.8	5.4
Sulfatide	3.8	5.4	1.7
Total galactolipid	27.5	26.4	7.3
Ethanolamine phosphatides	15.6	14.9	22.7
Phosphatidylcholine	11.2	12.8	26.7
Sphingomyelin	7.9	7.7	6.9
Phosphatidylserine	4.8	7.9	8.7
Phosphatidylinositol	0.6	0.9	2.7
Plasmalogens	12.3	11.2	8.8
Total phospholipids	43.1	45.9	69.5

^aProtein and lipid figures in percent dry weight; all others in percent total lipid weight

Data from W. Norton, in Siegel GJ, Albers RW, Agranoff BW, Katzman R, eds. *Basic Neurochemistry*, 3rd Ed. Boston: Little Brown & Co, 1981:77.

and there are significant hydrophobic interactions between the lipids and proteins to allow this to occur. Cerebrosides constitute approximately 16% of total myelin lipid and are almost completely absent from other cell-type membrane lipids. The predominant cerebroside, galactosylcerebroside, has a single sugar attached to the hydroxyl group of the sphingosine. In contrast, sphingomyelin, which one might guess is the predominant lipid of myelin, is present in roughly the same low concentration in all membranes. Galactocerebrosides pack more tightly together than phosphatidylcholine; the sugar, although polar, carries no positively charged amino group or negatively charged phosphate. The brain synthesizes very-long-chain fatty acids (greater than 20 carbons long); these long uncharged side chains develop strong hydrophobic associations, allowing a close packing of the myelin sheath. The high cholesterol content of the membrane also contributes to the tight packing, although the myelin proteins are also required to complete the tightness of the packing process.

2. MYELIN STRUCTURAL PROTEINS

The layers of myelin are held together by protein/lipid and protein/protein interactions, and any disruption can lead to demyelination of the membrane (see Biochemical Comments). Although numerous proteins are found in both the CNS and PNS, only the major proteins are discussed here. The major proteins in the CNS and PNS are different. In the CNS, two proteins constitute between 60 and 80% of the total proteins—proteolipid protein (PLP) and myelin basic proteins (MBP). The PLP is a very hydrophobic protein that forms large aggregates in aqueous solution and is relatively resistant to proteolysis. Its molecular weight, based on sequence analysis, is 30,000 Daltons. PLP is highly conserved in sequence amongst species. Its role is thought to be one of promoting the formation and stabilization of the multilayered myelin structure.

The MBPs are a family of proteins. Unlike PLP, MBPs are easily extracted from the membrane and are soluble in aqueous solution. The major MBP has no tertiary structure and has a molecular weight of 15,000 Daltons. MBP is located on the cytoplasmic face of myelin membranes. Antibodies directed against MBPs elicit experimental allergic encephalomyelitis (EAE), which has become a model system for understanding multiple sclerosis, a demyelinating disease. A model of how PLP and MBPs aid in stabilizing myelin is shown in Figure 48.14.

In the PNS, the major myelin protein is Po, a glycoprotein that accounts for greater than 50% of the PNS myelin protein content. The molecular weight of Po is 30,000, the same as PLP. Po is thought to play a similar structural role in maintaining

myelin structure, as does PLP in the CNS. Myelin basic proteins are also found in the PNS, with some similarities and differences to the MBPs found in the CNS. The major PNS-specific MBP has been designated P2.

CLINICAL COMMENTS



Catecholamines affect nearly every tissue and organ in the body. Their integrated release from nerve endings of the sympathetic (adrenergic) nervous system plays a critical role in the reflex responses we make to sudden changes in our internal and external environment. For example, under stress, catecholamines appropriately increase heart rate, blood pressure, myocardial (heart muscle) contractility, and conduction velocity of the heart.

Episodic, inappropriate secretion of catecholamines in pharmacologic amounts, such as occurs in patients with pheochromocytomas, like **Katie Colamin**, causes an often alarming array of symptoms and signs of a hyperadrenergic state.

Most of the signs and symptoms related to catecholamine excess can be masked by phenoxybenzamine, a long-acting α_1 - and α_2 -adrenergic receptor antagonist, combined with a β_1 - and β_2 -adrenergic receptor blocker such as propranolol. Pharmacologic therapy alone is reserved for patients with inoperable pheochromocytomas (e.g., patients with malignant tumors with metastases and patients with severe heart disease). Because of the sudden, unpredictable, and sometimes life-threatening discharges of large amounts of catecholamines from these tumors, definitive therapy involves surgical resection of the neoplasms(s) after appropriate preoperative preparation of the patient with the agents mentioned above. Katie's tumor was resected without intraoperative or postoperative complications. After surgery, she remained free of symptoms, and her blood pressure decreased to normal levels.



Evan Applebod, after stopping Redux, was placed on Prozac, an antidepressant that acts as a specific serotonin reuptake inhibitor (SSRI) but does not lead to increased synthesis or secretion of serotonin, as did dexfenfluramine in Redux. Thus, the mechanism of action of these two drugs is different, even if the end result (elevated levels of serotonin) is the same. Unfortunately, Prozac did not work as well for Mr. Applebod as did Redux, and he regained his 100 pounds within a year after switching medications. Redux was withdrawn from the market by its manufacturer because of reports of heart valve abnormalities in a small percentage of patients who had taken either phen/fen or Redux. Since then, the FDA has banned the use of Redux for weight loss because of the undesirable side effects. Other treatments, such as orlistat, a partial inhibitor of dietary fatty acid absorption from the gastrointestinal tract, are being tried in an effort to effect weight loss safely in patients like **Evan Applebod**.

BIOCHEMICAL COMMENTS



The importance of myelin in nerve transmission is underscored by the wide variety of demyelinating diseases, all of which lead to neurologic symptoms. The best known disease in this class is multiple sclerosis (MS). MS can be a progressive disease of the CNS in which demyelination of CNS neurons is the key anatomic and pathologic finding. The cause of MS has yet to be determined, although it is believed that an event occurs that triggers the formation of autoimmune antibodies directed against components of the nervous system. This

event could be a bacterial or viral infection that stimulates the immune system to fight off the invaders. Unfortunately, this stimulus may also trigger the autoimmune response that leads to the antibody-mediated demyelinating process. The unusual geographic distribution of MS is of interest. Patients are concentrated in northern and southern latitudes, yet its incidence is almost nil at the equator. Clinical presentation of MS varies widely. Most commonly it is a mild disease that has few or no obvious clinical manifestations. At the other end of the spectrum is a rapidly progressive and fatal disease. The most well-known presentation is the relapsing-remitting type. In this type, early in the course of the disease, the natural history is one of exacerbations, followed by remission. Eventually the CNS cannot repair the damage that has accumulated through the years, and remissions occur less and less frequently. Available treatments for MS target the relapsing-remitting type of disease.

The primary injury to the CNS in MS is the loss of myelin in the white matter, which interferes with nerve conduction along the demyelinated area (the insulator is lost). The CNS compensates by stimulating the oligodendrocyte to remyelinate the damaged axon, and when this occurs, remission is achieved. Often remyelination leads to a slowing in conduction velocity because of a reduced myelin thickness (speed is proportional to myelin thickness) or a shortening of the internodal distances (the action potential has to be propagated more times). Eventually, when it becomes too difficult to remyelinate large areas of the CNS, the neuron adapts by upregulating and redistributing along its membrane ion pumps, to allow nerve conductance along demyelinated axons. Eventually this adaptation also fails, and the disease progresses.

Treatment of MS is now based on blocking the action of the immune system. Because antibodies directed against cellular components appear to be responsible for the progression of the disease (regardless of how the autoantibodies were first generated), agents that interfere with immune responses have had various levels of success in keeping patients in remission for extended periods.

Other demyelinating diseases also exist, and their cause is much more straightforward. These are relatively rare disorders. In all of these diseases, there is no fully effective treatment for the patient. Inherited mutations in *Po* (the major PNS myelin protein) leads to a version of Charcot-Marie-Tooth polyneuropathy syndrome. The inheritance pattern for this disease is autosomal dominant, indicating that the expression of one mutated allele will lead to expression of the disease. Mutations in *PLP* (the major myelin protein in the CNS) lead to Pelizaeus-Merzbacher disease and X-linked spastic paraplegia type 2 disease. These diseases display a wide range of phenotypes, from a lack of motor development and early death (most severe) to mild gait disturbances. The phenotype displayed depends on the precise location of the mutation within the protein. An altered function of either *Po* or *PLP* leads to demyelination and its subsequent clinical manifestations.

Suggested Readings

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- Siegel G, Agranoff B, Albers RW, Molinoff P, eds. *Basic Neurochemistry*, 4th Ed. New York: Raven Press, 1989.
- Webster RA, ed. *Neurotransmitters, Drugs and Brain Function*. London: John Wiley and Sons Ltd, 2001.



REVIEW QUESTIONS—CHAPTER 48

1. A patient with a tumor of the adrenal medulla experienced palpitations, excessive sweating, and hypertensive headaches. His urine contained increased amounts of vanillylmandelic acid. His symptoms are probably caused by an overproduction of which of the following?
 - (A) Acetylcholine
 - (B) Norepinephrine and epinephrine
 - (C) DOPA and serotonin
 - (D) Histamine
 - (E) Melatonin
2. The two lipids found in highest concentration in myelin are which of the following?
 - (A) Cholesterol and cerebrosides such as galactosylceramide
 - (B) Cholesterol and phosphatidylcholine
 - (C) Galactosylceramide sulfatide and sphingomyelin
 - (D) Plasmalogens and sphingomyelin
 - (E) Triacylglycerols and lecithin
3. Myelin basic protein can best be described by which of the following?
 - (A) It is synthesized in Schwann cells, but not in oligodendrocytes.
 - (B) It is a transmembrane protein found only in peripheral myelin.
 - (C) It attaches the two extracellular leaflets together in central myelin.
 - (D) It contains basic amino acid residues that bind the negatively charged extracellular sides of the myelin membrane together.
 - (E) It contains lysine and arginine residues that bind the negatively charged intracellular sides of the myelin membrane together.
4. A patient presented with dysmorphia and cerebellar degeneration. Analysis of his blood indicated elevated levels of phytanic acid and very-long-chain fatty acids, but no elevation of palmitate. His symptoms are consistent with a defect in an enzyme involved in which of the following?
 - (A) α -Oxidation
 - (B) Mitochondrial β -oxidation
 - (C) Transport of enzymes into lysosomes
 - (D) Degradation of mucopolysaccharides
 - (E) Elongation of fatty acids
5. One of the presenting symptoms of vitamin B6 deficiency is dementia. This may result from an inability to synthesize serotonin, norepinephrine, histamine, and GABA from their respective amino acid precursors. This is because B6 is required for which type of reaction?
 - (A) Hydroxylation
 - (B) Transamination
 - (C) Deamination
 - (D) Decarboxylation
 - (E) Oxidation

49 The Extracellular Matrix and Connective Tissue

Many of the cells in tissues are embedded in an **extracellular matrix** that fills the spaces between cells and binds cells and tissue together. In so doing, the extracellular matrix aids in determining the shape of tissues as well as the nature of the partitioning between tissue types. In the skin, loose connective tissue beneath epithelial cell layers consists of an extracellular matrix in which fibroblasts, blood vessels, and other components are distributed (Fig 49.1). Other types of connective tissue, such as tendon and cartilage, consist largely of extracellular matrix, which is principally responsible for their structure and function. This matrix also forms the sheetlike **basal laminae**, or basement membranes, on which layers of epithelial cells rest, and which act as supportive tissue for muscle cells, adipose cells, and peripheral nerves.

Basic components of the extracellular matrix include fibrous structural proteins, such as **collagens**, **proteoglycans** containing long glycosaminoglycan chains attached to a protein backbone, and **adhesion proteins** linking components of the matrix to each other and to cells.

These **fibrous structural proteins** are composed of repeating elements that form a linear structure. **Collagens**, **elastin**, and **laminin** are the principal structural proteins of connective tissue.

Proteoglycans consist of a core protein covalently attached to many long, linear chains of **glycosaminoglycans**, which contain repeating disaccharide units. The repeating disaccharides usually contain a **hexosamine** and a **uronic acid**, and these sugars are frequently **sulfated**. Synthesis of the proteoglycans starts with the attachment of a sugar to a serine, threonine, or asparagine residue of the protein. Additional sugars, donated by **UDP-sugar precursors**, add sequentially to the nonreducing end of the molecule.

Proteoglycans, such as glycoproteins and glycolipids, are **synthesized in the endoplasmic reticulum (ER) and the Golgi complex**. The glycosaminoglycan chains of proteoglycans are **degraded by lysosomal enzymes** that cleave one sugar at a time from the nonreducing end of the chain. An inability to degrade proteoglycans leads to a set of diseases known as the **mucopolysaccharidoses**.

Adhesion proteins, such as **fibronectin** and **laminin**, are extracellular glycoproteins that contain separate distinct binding domains for proteoglycans, collagen, and fibrin. These domains allow these adhesion proteins to bind the various components of the extracellular matrix. They also contain specific binding domains for specific cell surface receptors known as the **integrins**. These integrins bind to **fibronectin** on the external surface, span the plasma membrane of cells, and adhere to proteins, which, in turn, bind to the intracellular **actin** filaments of the **cytoskeleton**. Integrins also provide a mechanism for signaling between cells via both internal signals as well as through signals generated via the extracellular matrix.

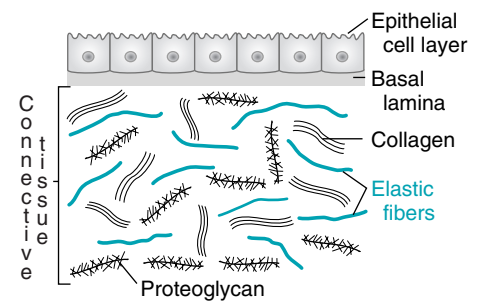


Fig. 49.1. An overview of connective tissue extracellular matrix. Supporting the epithelial cell layer is a basal lamina, beneath which are collagen, elastic fibers, and proteoglycans. The cell types present in connective tissue, such as fibroblasts and macrophages, have been removed from the diagram for clarity.

Cell movement within the extracellular matrix requires remodeling of the various components of the matrix. This is accomplished by a variety of **matrix metalloproteinases (MMPs)** and regulators of the MMPs, **tissue inhibitors of matrix metalloproteinases (TIMPs)**. Dysregulation of this delicate balance of the regulators of cell movement allows cancer cells to travel to other parts of the body (**metastasize**) as well as to spread locally to contiguous tissues.



THE WAITING ROOM



Sis Lupus (first introduced in Chapter 14) noted a moderate reduction in pain and swelling in the joints of her fingers when she was taking a 6-week course of high-dose prednisone, an anti-inflammatory steroid. As the dose of this drug was tapered to minimize its long-term side effects, however, the pain in the joints of her fingers returned, and, for the first time, her left knee became painful, swollen, and warm to the touch. Her rheumatologist described to her the underlying inflammatory tissue changes that her systemic lupus erythematosus (SLE) was causing in the joint tissues.



Ann Sulin complained of a declining appetite for food as well as severe weakness and fatigue. The reduction in her kidneys' ability to maintain normal daily total urinary net acid excretion contributed to her worsening metabolic acidosis. This plus her declining ability to excrete nitrogenous waste products, such as creatinine and urea, into her urine ("azotemia") are responsible for many of her symptoms. Her serum creatinine level was rising steadily. As it approached a level of 5 mg/dL, she developed a litany of complaints caused by the multisystem dysfunction associated with her worsening metabolic acidosis, retention of nitrogenous waste products, and so forth ("uremia"). Her physicians discussed with Ann the need to consider peritoneal dialysis or hemodialysis.

I. COMPOSITION OF THE EXTRACELLULAR MATRIX

A. Fibrous Proteins

1. COLLAGEN

Collagen, a family of fibrous proteins, is produced by a variety of cell types but principally by fibroblasts (cells found in interstitial connective tissue), muscle cells, and epithelial cells. Type I collagen [collagen(I)], the most abundant protein in mammals, is a fibrous protein that is the major component of connective tissue. It is found in the extracellular matrix (ECM) of loose connective tissue, bone, tendons, skin, blood vessels, and the cornea of the eye. Collagen(I) contains approximately 33% glycine and 21% proline and hydroxyproline. Hydroxyproline is an amino acid produced by posttranslational modification of peptidyl proline residues (see Chapter 7, section V.C., for an earlier introduction to collagen).

Procollagen(I), the precursor of collagen(I), is a triple helix composed of three polypeptide (pro- α) chains that are twisted around each other, forming a rope-like structure. Polymerization of collagen(I) molecules forms collagen fibrils, which provide great tensile strength to connective tissues (Fig. 49.2). The individual polypeptide chains each contain approximately 1,000 amino acid residues. The three polypeptide chains of the triple helix are linked by interchain hydrogen bonds.



Fig. 49.2. The triple helix of collagen.

Each turn of the triple helix contains three amino acid residues, such that every third amino acid is in close contact with the other two strands in the center of the structure. Only glycine, which lacks a side chain, can fit in this position, and indeed, every third amino acid residue of collagen is glycine. Thus, collagen is a polymer of (Gly-X-Y) repeats, where Y is frequently proline or hydroxyproline, and X is any other amino acid found in collagen.

Procollagen(I) is an example of a protein that undergoes extensive posttranslational modifications. Hydroxylation reactions produce hydroxyproline residues from proline residues and hydroxylysine from lysine residues. These reactions occur after the protein has been synthesized (Fig. 49.3) and require vitamin C (ascorbic acid) as a cofactor of the enzymes, for example, prolyl hydroxylases and lysyl hydroxylase. Hydroxyproline residues are involved in hydrogen bond formation that helps to stabilize the triple helix, whereas hydroxylysine residues are the sites of attachment of disaccharide moieties (galactose-glucose).

The side chains of lysine residues also may be oxidized to form the aldehyde, allysine. These aldehyde residues produce covalent cross-links between collagen molecules (Fig. 49.4). An allysine residue on one collagen molecule reacts with the amino group of a lysine residue on another molecule, forming a covalent Schiff base that is converted to more stable covalent cross-links. Aldol condensation also may occur between two allysine residues, which forms the structure lysinonorleucine.



The role of carbohydrates in collagen structure is still controversial. The hydroxyproline residues in collagen are required for stabilization of the triple helix by hydrogen bond formation. In the absence of vitamin C (scurvy), the melting temperature of collagen can drop from 42°C to 24°C, because of the loss of inter-strand hydrogen bond formation from the lack of hydroxyproline residues.

i. Types of Collagen

At least 19 different types of collagen have been characterized (Table 49.1). Although each type of collagen is found only in particular locations in the body, more than one type may be present in the ECM at a given location. The various types of collagen can be classified as fibril-forming (types I, II, III, V, and XI), network-forming (types IV, VIII and X), those that associate with fibril surfaces (types IX, XII, and XIV), those that are transmembrane proteins (types XIII and XVII), endostatin-forming (types XV and XVIII), and those that form periodic beaded filaments (type VI).

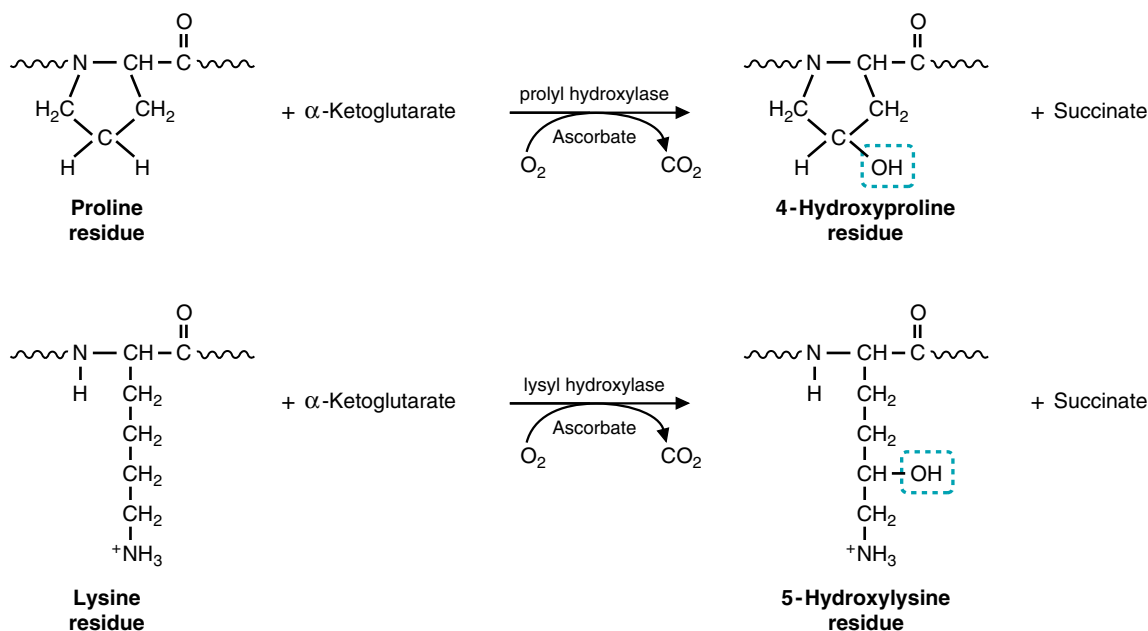


Fig. 49.3. Hydroxylation of proline and lysine residues in collagen. Proline and lysine residues within the collagen chains are hydroxylated by reactions that require vitamin C.

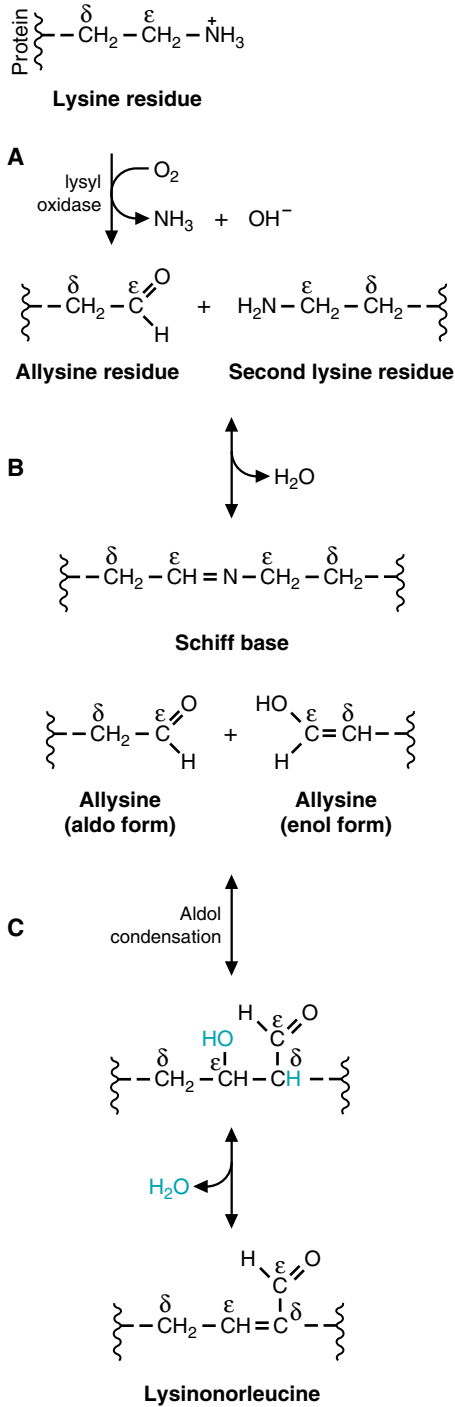


Fig. 49.4. Formation of cross-links in collagen. A. Lysine residues are oxidized to allysine (an aldehyde). Allysine may react with an unmodified lysine residue to form a Schiff base (B), or two allysine residues may undergo an aldol condensation (C).

Table 49.1. Types of Collagen

Collagen Type	Gene	Structural Details	Localization
I	Col1A1-Col1A2	Fibrils	Skin, tendon, bone, cornea
II	Col2A1	Fibrils	Cartilage, vitreous humour
III	Col3A1	Fibrils	Skin, muscle, associates with type I collagen
IV	Col4A1–Col4A6	Nonfibrillar, mesh collagen	All basal laminae (basement membranes)
V	Col5A1-Col5A3	Small fibers, N-terminal globular domains	Associates with type I collagen in most interstitial tissues
VI	Col6A1-Col6A3	Microfibrils, with both N and C-terminal globular domains	Associates with type I collagen in most interstitial tissues
VII	Col7A1	An anchoring collagen	Epithelial cells; dermal–epidermal junction
VIII	Col8A1-Col8A2	Nonfibrillar, mesh collagen	Cornea, some endothelial cells
IX	Col9A1-Col9A3	Fibril-associated collagens with interrupted triple helices (FACIT); N-terminal globular domain	Associates with type II collagen in cartilage and vitreous humour
X	Col10A1	Nonfibrillar, mesh collagen, with C-terminal globular domain	Growth plate, hypertrophic and mineralizing cartilage
XI	Col11A1-Col11A3	Small fibers	Cartilage, vitreous humor
XII	Col12A1	FACIT	Interacts with types I and II collagen in soft tissues
XIII	Col13A1	Transmembrane collagen	Cell surfaces, epithelial cells
XIV	Col14A1	FACIT	Soft tissue
XV	Col15A1	Endostatin-forming collagen	Endothelial cells
XVI	Col16A1	Other	Ubiquitous
XVII	Col17A1	Transmembrane collagen	Epidermal cell surface
XVIII	Col18A1	Endostatin-forming	Endothelial cells
XIX	Col19A1	Other	Ubiquitous

See the text for descriptions of the differences in types of collagen.

All collagens contain three polypeptide chains with at least one stretch of triple helix. The non-triple helical domains can be short (such as in the fibril-forming collagens) or can be rather large, such that the triple helix is actually a minor component of the overall structure (examples are collagen types XII and XIV). The FACIT (fibril-associated collagens with interrupted triple helices, collagen types IX, XII, and XIV) collagen types associate with fibrillar collagens, without themselves forming fibers. The endostatin-forming collagens are cleaved at their C-terminus to form endostatin, an inhibitor of angiogenesis. The network-forming collagens (type IV) form a mesh-like structure, because of large (approximately 230 amino acids) non-collagenous domains at the carboxy-terminal (Fig. 49.5). And finally, a number of collagen types are actually transmembrane proteins (XIII and XVII) found on epithelial or epidermal cell surfaces, which play a role in a number of cellular processes, including adhesion of components of the ECM to cells embedded within it.

Types I, II, and III collagens form fibrils that assemble into large insoluble fibers. The fibrils (see below) are strengthened through covalent cross-links between lysine residues on adjacent fibrils. The arrangement of the fibrils gives individual tissues their distinct characteristics. Tendons, which attach muscles to bones, contain collagen



 Endostatins block angiogenesis (new blood vessel formation) by inhibiting endothelial cell migration. Because endothelial cell migration and proliferation are required to form new blood vessels, inhibiting this action blocks angiogenesis. Tumor growth is dependent on a blood supply; inhibiting angiogenesis can reduce tumor cell proliferation.

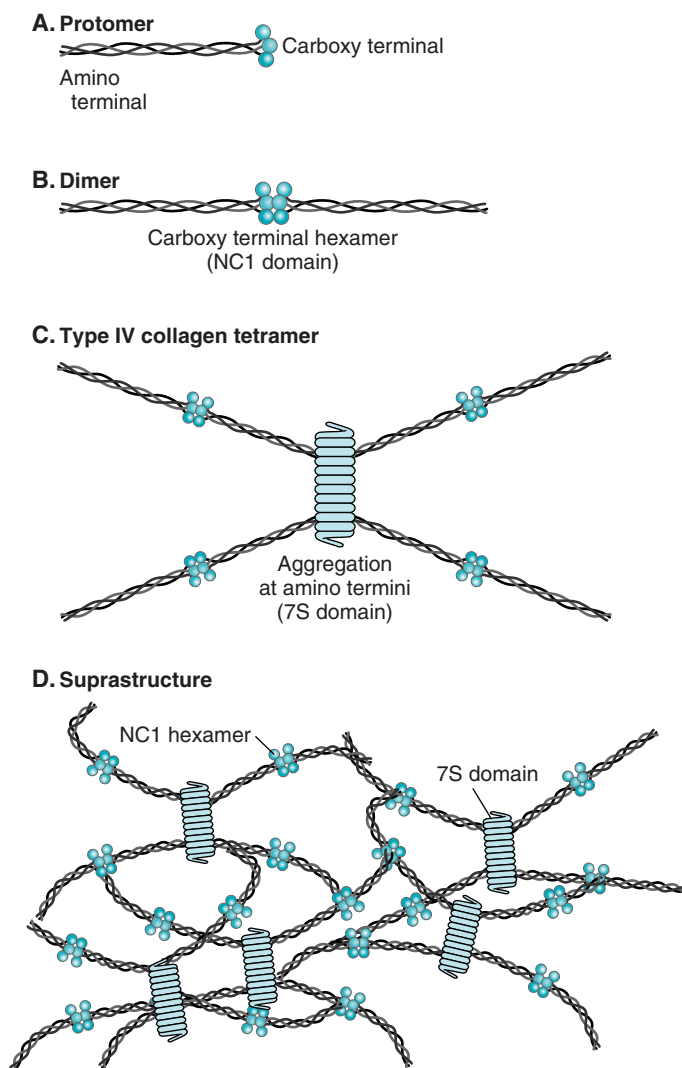


Fig. 49.5. Type IV collagen contains a globular carboxy-terminal domain (A), which forms tropocollagen dimers (hexamers of collagen, B). Four dimers associate at the amino-terminal domains to form a 7S domain (C), and the tetramers form a lattice (D), which provides structural support to the basal lamina.

fibrils aligned parallel to the long axis of the tendon, thus giving the tendon tremendous tensile strength.

The types of collagen that do not form fibrils perform a series of distinct roles. Fibril-associated collagens bind to the surface of collagen fibrils and link them to other matrix-forming components. The transmembrane collagens form anchoring fibrils that link components of the extracellular matrix to underlying connective tissue. The network-forming collagens (type IV) form a flexible collagen that is part of the basement membrane and basal lamina that surround many cells.

ii. Synthesis and Secretion of Collagen

Collagen is synthesized within the endoplasmic reticulum as a precursor known as procollagen. The presequence acts as the signal sequence for the protein and is cleaved, forming procollagen within the endoplasmic reticulum. From there it is transported to the Golgi apparatus (Table 49.2). Three procollagen molecules associate through formation of intrastrand disulfide bonds at the carboxy-terminus; once



One type of osteogenesis imperfecta (OI) is caused by a mutation in a gene that codes for collagen.

The phenotype of affected individuals varies greatly, depending on the location and type of mutation. See the Biochemical Comments for more information concerning this type of OI.

Table 49.2. Steps Involved in Collagen Biosynthesis

Location	Process
Rough endoplasmic reticulum	Synthesis of procollagen; insertion of the procollagen molecule into the lumen of the ER
Lumen of the ER	Hydroxylation of proline and lysine residues; glycosylation of selected hydroxylysine residues
Lumen of ER and Golgi apparatus	Self-assembly of the tropocollagen molecule, initiated by disulfide bond formation in the carboxy-terminal extensions; triple helix formation
Secretory vesicle	Procollagen prepared for secretion from cell
Extracellular	Cleavage of the propeptides, removing the amino- and carboxy-terminal extensions, and self-assembly of the collagen molecules into fibrils, and then fibers

these disulfides are formed, the three molecules can align properly to initiate formation of the triple helix. The triple helix forms from the carboxy-end toward the amino-end, forming tropocollagen. The tropocollagen contains a triple helical segment between two globular ends, the amino- and carboxy-terminal extensions. The tropocollagen is secreted from the cell, the extensions are removed using extracellular proteases, and the mature collagen takes its place within the ECM. The individual fibrils of collagen line up in a highly ordered fashion to form the collagen fiber.

2. ELASTIN

Elastin is the major protein found in elastic fibers, which are located in the ECM of connective tissue of smooth muscle cells, endothelial and microvascular cells, chondrocytes, and fibroblasts. Elastic fibers allow tissues to expand and contract; this is of particular importance to blood vessels, which must deform and reform repeatedly in response to the changes in intravascular pressure that occur with the contraction of the left ventricle of the heart. It is also important for the lungs, which stretch each time a breath is inhaled and return to their original shape with each exhalation. In addition to elastin, the elastic fibers contain microfibrils, which are composed of a number of acidic glycoproteins, the major ones being fibrillin-1 and fibrillin-2.

i. Tropoelastin

Elastin has a highly cross-linked, insoluble, amorphous structure. Its precursor, tropoelastin, is a molecule of high solubility, which is synthesized on the rough endoplasmic reticulum (RER) for eventual secretion. Tropoelastin contains two types of alternating domains. The first domain consists of a hydrophilic sequence rich in lysine and alanine residues. The second domain consists of a hydrophobic sequence rich in valine, proline, and glycine, which frequently occur in repeats of VPGVG or VGGVG. The protein contains approximately 16 regions of each domain, alternating throughout the protein (Fig. 49.6).

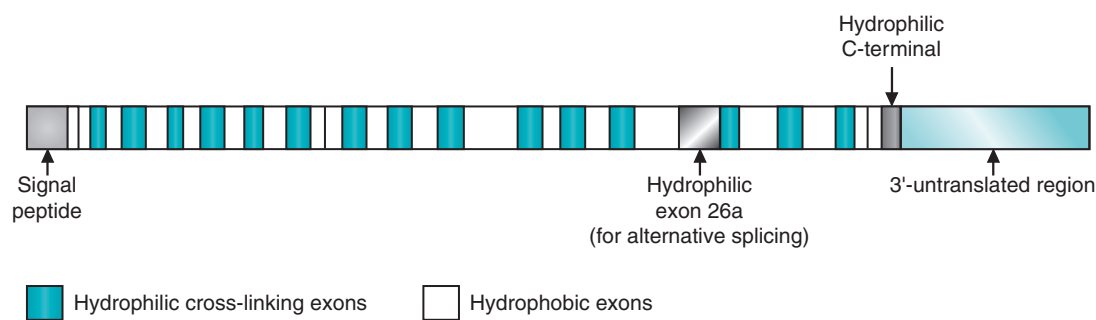


Fig. 49.6. The cDNA structure of elastin, indicating the repeating cross-linking and hydrophobic domains.

On secretion from the cell, the tropoelastin is aligned with the microfibrils, and lysyl oxidase initiates the reactions that cross-link elastin molecules, using lysine residues within the hydrophilic alternating domains in the proteins. This cross-linking reaction is the same as that which occurs in collagen. In this reaction, 2, 3, or 4 lysine residues are cross-linked to form a stable structure. The net result of the cross-linking is the generation of a fibrous mesh that encircles the cells.

ii. Elastic Properties of Elastin

Elastic fibers have the ability to stretch and then to reform without requiring an obvious energy source to do so. The mechanism by which this stretching and relaxing actively occurs is still controversial but does relate to the basic principles of protein folding described in Chapter 7. When the elastic fibers are stretched (such as when a breath is taken in and the lung fills up with air), the amorphous elastin structure is stretched. This stretching exposes the repeating hydrophobic regions of the molecule to the aqueous environment. This, in turn, leads to a decrease in the entropy of water, because the water molecules need to rearrange to form cages about each hydrophobic domain. When this stretching force within the lung is removed (e.g., when the subject exhales), the elastin takes on its original structure because of the increase in entropy that occurs because the water no longer needs to form cages about hydrophobic domains. Thus, the hydrophobic effect is the primary force that allows this stretched structure to reform. Elastin is inherently stable, with a half-life of up to 70 years.

3. LAMININ

After type IV collagen, laminin is the most abundant protein in basal laminae. Laminin provides additional structural support for the tissues through its ability to bind to type IV collagen, to other molecules present in the ECM, and to cell surface-associated proteins (the integrins, see section D).

i. Laminin Structure

Laminin is a heterotrimeric protein shaped, for the most part, like a cross (Fig. 49.7). The trimer is composed of α , β , and γ subunits. There are five possible α proteins (designated $\alpha 1$ – $\alpha 5$), three different versions of the β subunit ($\beta 1$ – $\beta 3$), and three different γ forms ($\gamma 1$ – $\gamma 3$). Thus, there is a potential for the formation of as many as 45 different combinations of these three subunits. However, only 12 have been discovered (designated laminins 1–12). Laminin 1, composed of $\alpha 1\beta 1\gamma 1$, is typical of this class of proteins. The major feature of the laminin structure is a coiled α -helix, which joins the three subunits together and forms a rigid rod. All three chains have extensions at the amino-terminal end. Only the α chain has a significant carboxy-terminal extension past the rod-like structure. It is the laminin extensions that allow laminin to bind to other components within the ECM and to provide stability for the structure. Components of the ECM that are bound by laminin include collagen, sulfated lipids, and proteoglycans.

ii. Laminin Biosynthesis

Like other secreted proteins, laminin is synthesized with a leader sequence targeting the three chains to the endoplasmic reticulum. Chain association occurs within the Golgi apparatus before secretion from the cell. After laminin is secreted by the cell, the amino terminal extensions promote self-association, as well as the binding to other ECM components. Disulfide linkages are formed to stabilize the trimer, but there is much less posttranslational processing of laminin than there is of collagen and elastin.



Supravalvular aortic stenosis (SVAS) results from an insufficiency of elastin in the vessel wall, leading to a narrowing of the large elastic arteries. Current theory suggests that the levels of elastin in the vessel walls may regulate the number of smooth muscle cell rings that encircle the vessel. If the levels of elastin are low, smooth muscle hypertrophy results, leading to a narrowing and stenosis of the artery.

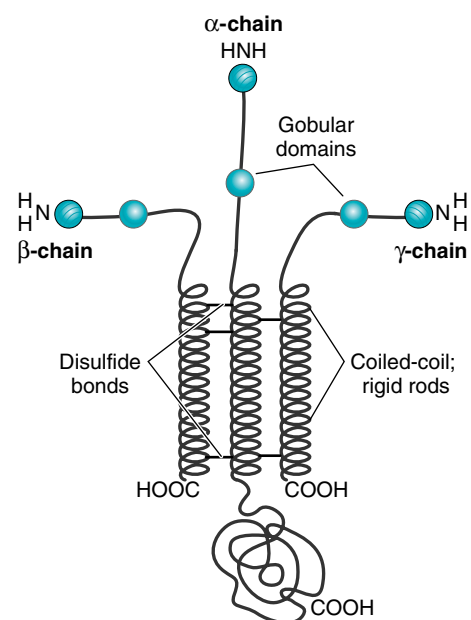


Fig. 49.7. The structure of laminin.



Defects in the structures of laminin 5 or laminin 6 (proteins that contribute to the cohesion of the dermis and epidermis) lead to the disorder referred to as junctional epidermolysis bullosa (JEB). In this disorder, there can be severe spontaneous blistering of the skin and mucous membranes. A severe form of the disease, JEB gravis, is often fatal early in life. Death occurs as a result of epithelial blistering of the respiratory, digestive, and genitourinary systems.

Congenital muscular dystrophy (CMD) results from a defect in laminin 2, which is a component of the bridge that links the muscle cell cytoskeleton to the extracellular matrix. Lack of this bridge triggers muscle cell apoptosis, which results in weakened muscles.



The ECM is not simply a glue that holds cells together; it also serves to keep cells from moving to other locations and to prevent large molecules and other particles, such as microorganisms, from reaching contiguous and distant cells. This confining property of the matrix is medically important. For example, infections spread, in part, because the infectious agent alters the “containing” capacity of the ECM. Cancer cells that metastasize (migrate to other tissues) can do so only by altering the integrity of the matrix. Diseases such as rheumatoid arthritis (an autoimmune destruction of articular and periarticular tissues) and osteoarthritis (degenerative joint disease often associated with aging) involve damage to the functional capacity of the matrix. Alterations in the structural characteristics of the matrix of the renal glomerulus may allow proteins to be excreted into the urine, an indication of inexorable decline in renal function. Genetic defects may cause components of the matrix to be structurally and functionally abnormal, resulting in connective tissue disorders such as the Ehlers-Danlos syndrome (caused by a number of mutations that affect specific collagen genes) and Marfan’s syndrome (a defect in the protein, fibrillin, in which over 330 different mutations, many of which give rise to different phenotypes, have been identified). Deficiencies of lysosomal enzymes involved in normal degradation of molecules of the matrix result in diseases such as the mucopolysaccharidoses.



The principal components of the matrix of cartilage are collagen and proteoglycans, both of which are produced and degraded by the chondrocytes that are embedded in this matrix. An autoimmune attack on articular proteins alters the balance between cartilage degradation and formation. The resulting loss of cartilage organization accompanied by an inflammatory response is responsible for the symptoms experienced by **Sis Lupus**.

The collagen component forms a network of fine fibrils that give shape to the cartilage. The proteoglycans embedded in the cartilage are responsible for its compressibility and its deformability.

B. Proteoglycans

The fibrous structural proteins of the ECM are embedded in gels formed from proteoglycans. Proteoglycans consist of polysaccharides called glycosaminoglycans (GAG) linked to a core protein. The GAGs are composed of repeating units of disaccharides. One sugar of the disaccharide is either N-acetylglucosamine or N-acetylgalactosamine, and the second is usually acidic (either glucuronic acid or iduronic acid). These sugars are modified by the addition of sulfate groups to the parent sugar. A proteoglycan may contain more than 100 GAG chains and consist of up to 95% oligosaccharide by weight.

The negatively charged carboxylate and sulfate groups on the proteoglycan bind positively charged ions and form hydrogen bonds with trapped water molecules, thereby creating a hydrated gel. The gel provides a flexible mechanical support to the ECM. The gel also acts as a filter that allows the diffusion of ions (e.g., Ca^{2+}), H_2O , and other small molecules, but slows diffusion of proteins and movement of cells. Hyaluronan is the only GAG that occurs as a single long polysaccharide chain and is the only GAG that is not sulfated.

1. STRUCTURE AND FUNCTION OF THE PROTEOGLYCANS

Proteoglycans are found in interstitial connective tissues, for example, the synovial fluid of joints, the vitreous humor of the eye, arterial walls, bone, cartilage, and cornea. They are major components of the ECM in these tissues. The proteoglycans interact with a variety of proteins in the matrix, such as collagen and elastin, fibronectin (which is involved in cell adhesion and migration), and laminin.

Proteoglycans are proteins that contain many chains of GAGs (formerly called mucopolysaccharides). Glycosaminoglycans are long, unbranched polysaccharides composed of repeating disaccharide units (Fig. 49.8). The repeating disaccharides usually contain an iduronic or uronic acid and a hexosamine and are frequently sulfated. Consequently, they carry a negative charge, are hydrated, and act as lubricants. After synthesis, proteoglycans are secreted from cells; thus, they function extracellularly. Because the long, negatively charged glycosaminoglycan chains repel each other, the proteoglycans occupy a very large space and act as “molecular sieves,” determining which substances enter or leave cells (Table 49.3). Their properties also give resilience and a degree of flexibility to substances such as cartilage, permitting compression and reexpansion of the molecule to occur.

At least seven types of glycosaminoglycans exist, which differ in the monosaccharides present in their repeating disaccharide units—chondroitin sulfate, dermatan sulfate, heparin, heparin sulfate, hyaluronic acid, and keratan sulfates I and II. Except for hyaluronic acid, the glycosaminoglycans are linked to proteins, usually attached covalently to serine or threonine residues (Fig. 49.9). Keratan sulfate I is attached to asparagine.

2. SYNTHESIS OF THE PROTEOGLYCANS

The protein component of the proteoglycans is synthesized on the ER. It enters the lumen of this organelle, where the initial glycosylations occur. UDP-sugars serve as



The long polysaccharide side chains of the proteoglycans in cartilage contain many anionic groups. This high concentration of negative charges attracts cations that create a high osmotic pressure within cartilage, drawing water into this specialized connective tissue and placing the collagen network under tension. At equilibrium, the resulting tension balances the swelling pressure caused by the proteoglycans. The complementary roles of this macromolecular organization give cartilage its resilience. Cartilage can thus withstand the compressive load of weight bearing and then reexpand to its previous dimensions when that load is relieved.

Table 49.3. Some Specific Functions of the Glycosaminoglycans and Proteoglycans

Glycosaminoglycan	Function
Hyaluronic acid	Cell migration in: Embryogenesis Morphogenesis Wound healing
Chondroitin sulfate proteoglycans	Formation of bone, cartilage, cornea
Keratan sulfate proteoglycans	Transparency of cornea
Dermatan sulfate proteoglycans	Transparency of cornea Binds LDL to plasma walls
Heparin	Anticoagulant (binds antithrombin III) Causes release of lipoprotein lipase from capillary walls
Heparan sulfate (syndecan)	Component of skin fibroblasts and aortic wall; commonly found on cell surfaces

the precursors that add sugar units, one at a time, first to the protein and then to the nonreducing end of the growing carbohydrate chain (Fig. 49.10). Glycosylation occurs initially in the lumen of the ER and subsequently in the Golgi complex. Glycosyltransferases, the enzymes that add sugars to the chain, are specific for the sugar being added, the type of linkage that is formed, and the sugars already present in the chain. Once the initial sugars are attached to the protein, the alternating action of two glycosyltransferases adds the sugars of the repeating disaccharide to the growing glycosaminoglycan chain. Sulfation occurs after addition of the sugar. 3'-Phosphoadenosine 5'-phosphosulfate (PAPS), also called active sulfate, provides the sulfate groups (see Fig. 33.34). An epimerase converts glucuronic acid residues to iduronic acid residues.

After synthesis, the proteoglycan is secreted from the cell. Its structure resembles a bottle brush, with many glycosaminoglycan chains extending from the core protein (Fig. 49.11). The proteoglycans may form large aggregates, noncovalently attached by a “link” protein to hyaluronic acid (Fig. 49.12). The proteoglycans interact with the adhesion protein, fibronectin, which is attached to the cell membrane protein integrin. Cross-linked fibers of collagen also associate with this complex, forming the ECM (Fig. 49.13).



The functional properties of a normal joint depend, in part, on the presence of a soft, well-lubricated, deformable, and compressible layer of cartilaginous tissue covering the ends of the long bones that constitute the joint.

In **Sis Lupus**’ case, the pathologic process that characterizes SLE disrupted the structural and functional integrity of her articular (joint) cartilage.

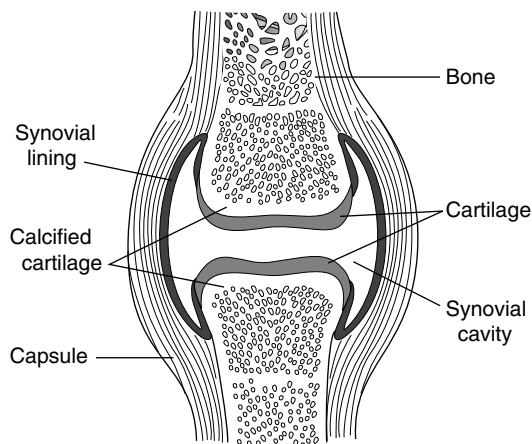
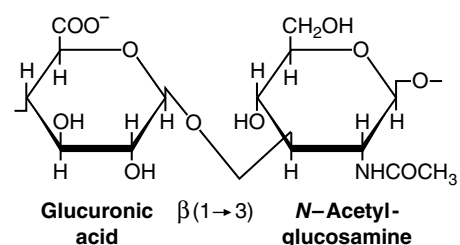
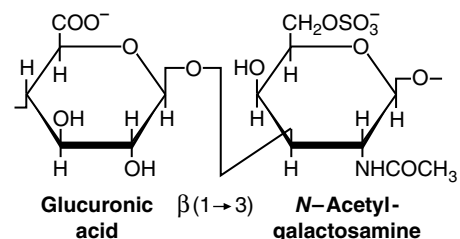
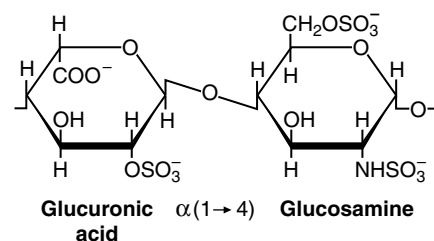
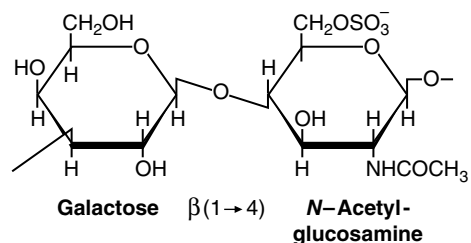
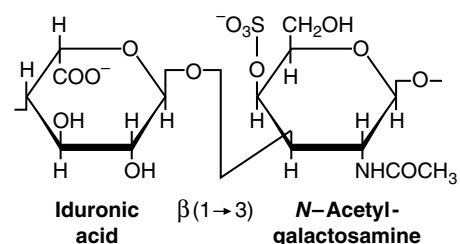
**Hyaluronate****Chondroitin 6-sulfate****Heparin****Keratan sulfate****Dermatan sulfate**

Fig. 49.8. Repeating disaccharides of some glycosaminoglycans. These repeating disaccharides usually contain an *N*-acetylated sugar and a uronic acid, which usually is glucuronic acid or iduronic acid. Sulfate groups are often present but are not included in the sugar names in this figure.

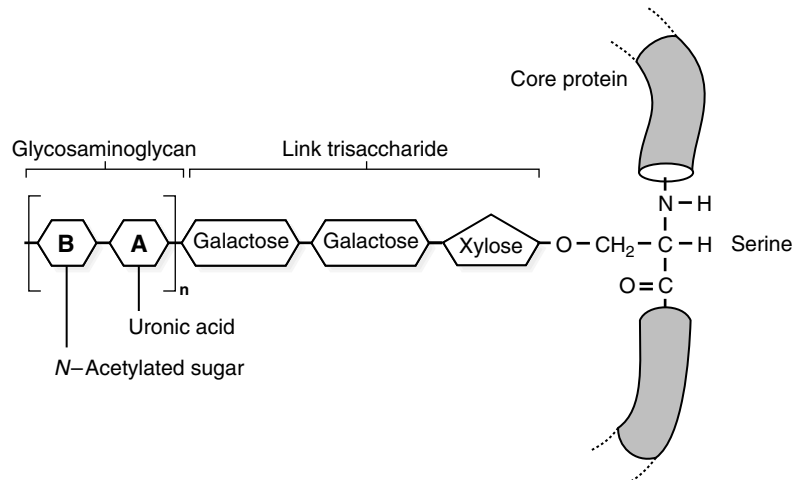


Fig. 49.9. Attachment of glycosaminoglycans to proteins. The sugars are linked to a serine or threonine residue of the protein. A and B represent the sugars of the repeating disaccharide.

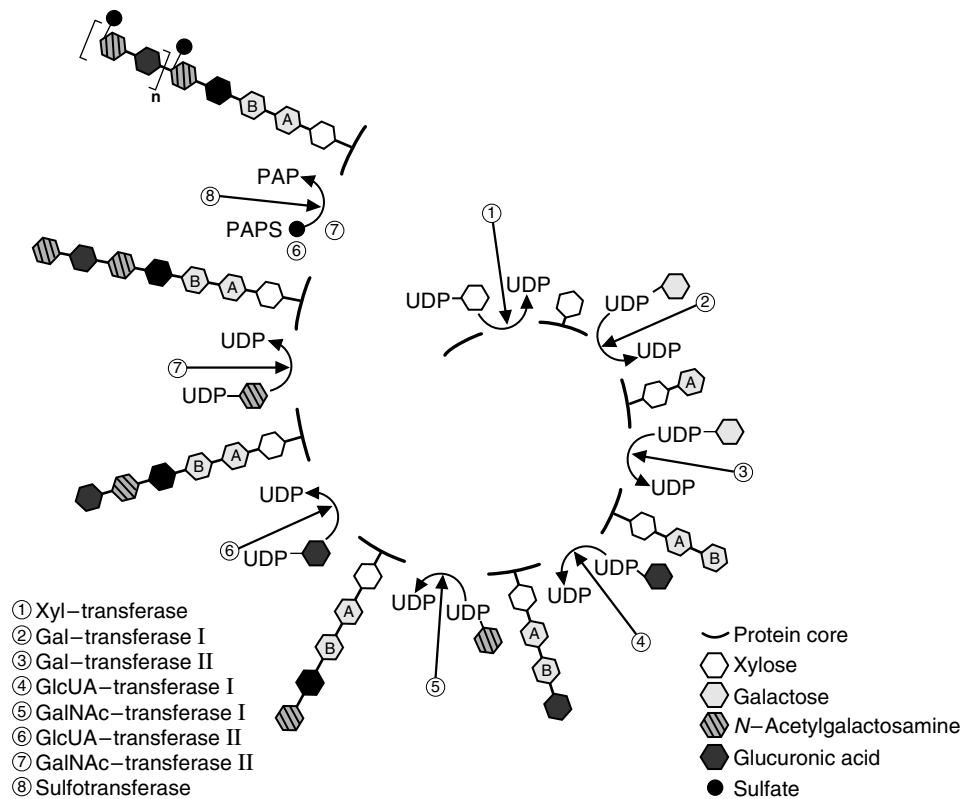


Fig. 49.10. Synthesis of chondroitin sulfate. Sugars are added to the protein one at a time, with UDP-sugars serving as the precursors. Initially a xylose residue is added to a serine in the protein. Then two galactose residues are added, followed by a glucuronic acid (GlcUA) and an N-acetylglucosamine (GalNAc). Subsequent additions occur by the alternating action of two enzymes that produce the repeating disaccharide units. One enzyme (6) adds GlcUA residues, and the other (7) adds GalNAc. As the chain grows, sulfate groups are added by phosphoadenosine phosphosulfate (PAPS). Modified from Roden L. In: Fishman WH, ed. *Metabolic Conjugation and Metabolic Hydrolysis*, vol II. Orlando, FL: Academic Press, 1970:401.

Table 49.4. Defective Enzymes in the Mucopolysaccharidoses

Disease	Enzyme Deficiency	Accumulated Products
Hunter	Iduronate sulfatase	Heparan sulfate, Dermatan sulfate
Hurler + Scheie	α -L-Iduronidase	Heparan sulfate, Dermatan sulfate
Maroteaux-Lamy	<i>N</i> -Acetylgalactosamine sulfatase	Dermatan sulfate
Mucopolidosis VII	β -Glucuronidase	Heparan sulfate, Dermatan sulfate
Sanfilippo A	Heparan sulfamidase	Heparan sulfate
Sanfilippo B	<i>N</i> -Acetylglucosaminidase	Heparan sulfate
Sanfilippo D	<i>N</i> -Acetylglucosamine 6-sulfatase	Heparin sulfate

These disorders share many clinical features, although there are significant variations between disorders, and even within a single disorder, based on the amount of residual activity remaining. In most cases, multiple organ systems are affected (with bone and cartilage being a primary target). For some disorders, there is significant neuronal involvement, leading to mental retardation.

3. DEGRADATION OF PROTEOGLYCANS

Lysosomal enzymes degrade proteoglycans, glycoproteins, and glycolipids, which are brought into the cell by the process of endocytosis. Lysosomes fuse with the endocytic vesicles, and lysosomal proteases digest the protein component. The carbohydrate component is degraded by lysosomal glycosidases.

Lysosomes contain both endoglycosidases and exoglycosidases. The endoglycosidases cleave the chains into shorter oligosaccharides. Then exoglycosidases, specific for each type of linkage, remove the sugar residues, one at a time, from the nonreducing ends.

Deficiencies of lysosomal glycosidases cause partially degraded carbohydrates from proteoglycans, glycoproteins, and glycolipids to accumulate within membrane-enclosed vesicles inside cells. These “residual bodies” can cause marked enlargement of the organ with impairment of its function.

In the clinical disorder known as the mucopolysaccharidoses (caused by accumulation of partially degraded glycosaminoglycans), deformities of the skeleton may occur (Table 49.4). Mental retardation often accompanies these skeletal changes.

II. INTEGRINS

Integrins are the major cellular receptors for ECM proteins and provide a link between the internal cytoskeleton of cells (primarily the actin microfilament system) and extracellular proteins, such as fibronectin, collagen, and laminin. Integrins

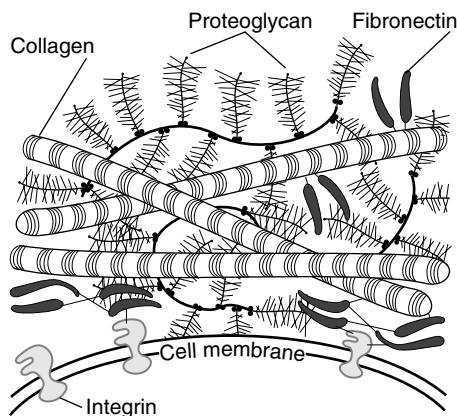


Fig. 49.13. Interactions between the cell membrane and the components of the extracellular matrix.

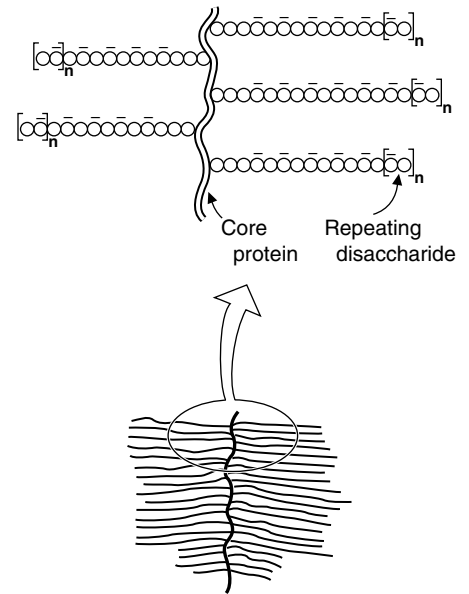


Fig. 49.11. “Bottle-brush” structure of a proteoglycan, with a magnified segment.

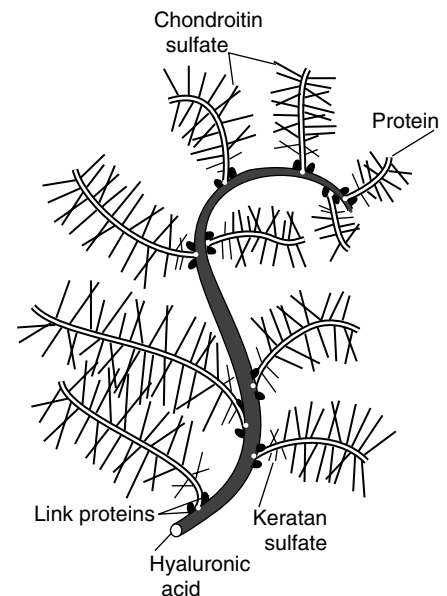


Fig. 49.12. Proteoglycan aggregate.

consist of an α and a β subunit. There are 18 distinct α and eight distinct β gene products. Twenty-four unique α/β dimers have been discovered. Mice have been genetically engineered to be unable to express many of the integrin genes (one gene at a time), and the phenotypes of these knockout mice vary from embryonic lethality (the $\alpha 5$ gene is an example) to virtually no observable defects (as exemplified by the $\alpha 1$ gene). In addition to anchoring the cell's cytoskeleton to the ECM, thereby providing a stable environment in which the cell can reside, the integrins are also involved in a wide variety of cell signaling options.

Certain integrins, such as those associated with white blood cells, are normally inactive because the white cell must circulate freely in the bloodstream. However, if an infection occurs, cells located in the area of the infection release cytokines, which activate the integrins on the white blood cells, allowing them to bind to vascular endothelial cells (leukocyte adhesion) at the site of infection. Leukocyte adhesion deficiency (LAD) is a genetic disorder that results from mutations in the $\beta 2$ integrin such that leukocytes cannot be recruited to the sites of infection. Conversely, drugs are now being developed to block either the $\beta 2$ or $\alpha 4$ integrins (on lymphocytes) to treat inflammatory and autoimmune disorders by interfering with the normal white cell response to cytokines.

Integrins can be activated by “inside-out” mechanisms, whereby intracellular signaling events activate the molecule, or “outside-in” mechanisms, in which a binding event with the extracellular portion of the molecule initiates intracellular signaling events. For those integrins that bind cells to ECM components, activation of specific integrins can result in migration of the affected cell through the ECM. This mechanism is operative during growth, during cellular differentiation, and in the process of metastasis of malignant cells to neighboring tissues.



Fibronectin was first discovered as a large, external transformation-sensitive protein (LETS), which was lost when fibroblasts were transformed into tumor cells. Many tumor cells secrete less than normal amounts of adhesion protein material, which allows for more movement within the extracellular milieu. This, in turn, increases the potential for the tumor cells to leave their original location and take root at another location within the body (metastasis).



Because MMPs degrade extracellular matrix (ECM) components, their expression is important to allow cell migration and tissue remodeling during growth and differentiation. In addition, many growth factors bind to ECM components and, as a bound component, do not exhibit their normal growth-promoting activity. Destruction of the ECM by the MMPs releases these growth factors, thereby allowing them to bind to cell surface receptors to initiate growth of tissues. Thus, coordinated expression of the MMPs is required for appropriate cell movement and growth. Cancer cells that metastasize require extensive ECM remodeling and usually use MMP activity to spread throughout the body.

III. ADHESION PROTEINS

Adhesion proteins are found in the ECM and link integrins to ECM components. Adhesion proteins, of which fibronectin is a prime example, are large multidomain proteins that allow binding to many different components simultaneously. In addition to integrin binding sites, fibronectin contains binding sites for collagen and glycosaminoglycans. As the integrin molecule is bound to intracellular cytoskeletal proteins, the adhesion proteins provide a bridge between the actin cytoskeleton of the cell and the cells' position within the ECM. Loss of adhesion protein capability can lead to either physiologic or abnormal cell movement. Alternative splicing of fibronectin allows many different forms of this adhesion protein to be expressed, including a soluble form (versus cell-associated forms), which is found in the plasma. The metabolic significance of these products remains to be determined.

IV. MATRIX METALLOPROTEINASES

The ECM contains a series of proteases known as the matrix metalloproteinases, or MMPs. These are zinc-containing proteases that use the zinc to appropriately position water to participate in the proteolytic reaction. At least 23 different types of human MMPs exist, and they cleave all proteins found in the ECM, including collagen and laminin.

A propeptide is present in newly synthesized MMPs that contains a critical cysteine residue. The cysteine residue in the propeptide binds to the zinc atom at the active site of the protease and prevents the propeptide from exhibiting proteolytic activity. Removal of the propeptide is required to activate the MMPs. Once activated, certain MMPs can activate other forms of MMP.

Regulation of MMP activity is quite complex. These regulatory processes include transcriptional regulation, proteolytic activation, inhibition by the circulating protein $\alpha 2$ -macroglobulin, and regulation by a class of inhibitors known as

tissue inhibitors of metalloproteinases, or TIMPs. It is important that the synthesis of TIMPs and MMPs be coordinately regulated, because dissociation of their expression can facilitate various clinical disorders, such as certain forms of cancer and atherosclerosis.

CLINICAL COMMENTS



Articular cartilage is a living tissue with a turnover time determined by a balance between the rate of its synthesis and that of its degradation (Fig. 49.14). The chondrocytes that are embedded in the matrix of intra-articular cartilage participate in both its synthesis and its enzymatic degradation. The latter occurs as a result of cleavage of proteoglycan aggregates by enzymes produced and secreted by the chondrocytes.

In SLE, the condition that affects **Sis Lupus**, this delicate balance is disrupted in favor of enzymatic degradation, leading to dissolution of articular cartilage and, with it, the loss of its critical cushioning functions. The underlying mechanisms responsible for this process in SLE include the production of antibodies directed against specific cellular proteins in cartilage as well as in other intra-articular tissues. The cellular proteins thus serve as the “antigens” to which these antibodies react. In this sense, SLE is an “autoimmune” disease because antibodies are produced by the host that attack “self” proteins. This process excites the local release of cytokines such as interleukin-1 (IL-1), which increases the proteolytic activity of the chondrocytes, causing further loss of articular proteins such as the proteoglycans. The associated inflammatory cascade is responsible for **Sis Lupus**’ joint pain.

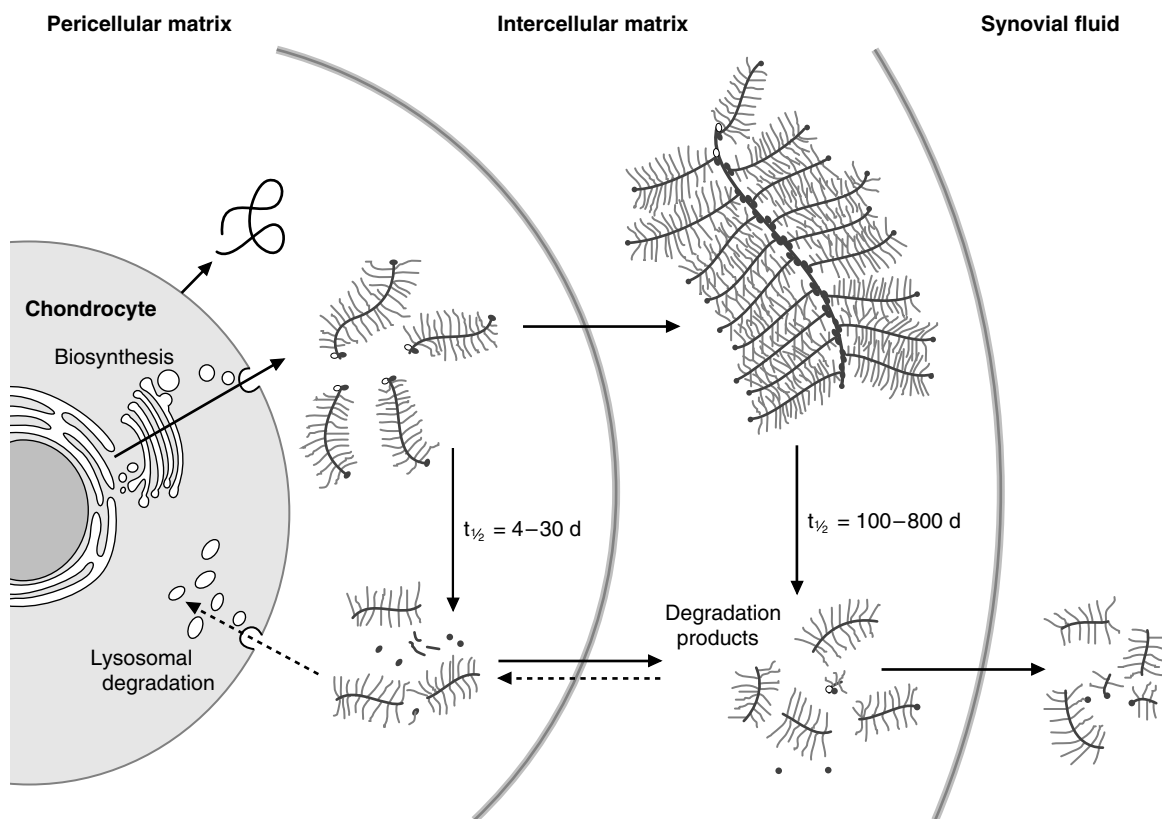


Fig. 49.14. Synthesis and degradation of proteoglycans by chondrocytes. From Cohen RD, et al. *The Metabolic Basis of Acquired Disease*, vol 2. London: Bailliere Tindall, 1990:1859.



The microvascular complications of both type 1 and type 2 diabetes mellitus involve the small vessels of the retina (diabetic retinopathy), the renal glomerular capillaries (diabetic nephropathy), and the vessels supplying blood to the peripheral nerves (autonomic neuropathy). The lack of adequate control of **Ann Sulin's** diabetic state over many years caused a progressive loss of the filtering function of the approximately one-and-one-half million glomerular capillary–mesangial units that are present in her kidneys.

Chronic hyperglycemia is postulated to be a major metabolic initiator or inducer of diabetic microvascular disease, including those renal glomerular changes that often lead to end-stage renal disease (“glucose toxicity”).

For a comprehensive review of the four postulated molecular mechanisms by which chronic hyperglycemia causes these vascular derangements, the reader is referred to an excellent review by Sheetz and King (see suggested references).

Regardless of which of the postulated mechanisms (increased flux through the aldose reductase or polyol pathway [see Chapter 30], the generation of advanced glycosylation end products [AGEs], the generation of reactive oxygen intermediates [see Chapter 24], or excessive activation of protein kinase C [see Chapter 18]) will eventually be shown to be the predominant causative mechanism, each can lead to the production of critical intracellular and extracellular signaling molecules (e.g., cytokines). These, in turn, can cause pathologic changes within the glomerular filtration apparatus that reduce renal function. These changes include: (1) increased synthesis of collagen, type IV, fibronectin, and some of the proteoglycans, causing the glomerular basement membrane (GBM; Fig. 49.15) to become diffusely thickened throughout the glomerular capillary network. This membrane thickening alters certain specific filtration properties of the GBM, preventing some of the metabolites that normally enter the urine from the glomerular capillary blood (via the fenestrated capillary endothelium) from doing so (a decline in glomerular filtration rate or GFR). As a result, these potentially toxic substances accumulate in the blood and contribute to the overall clinical presentation of advancing uremia. In spite of the

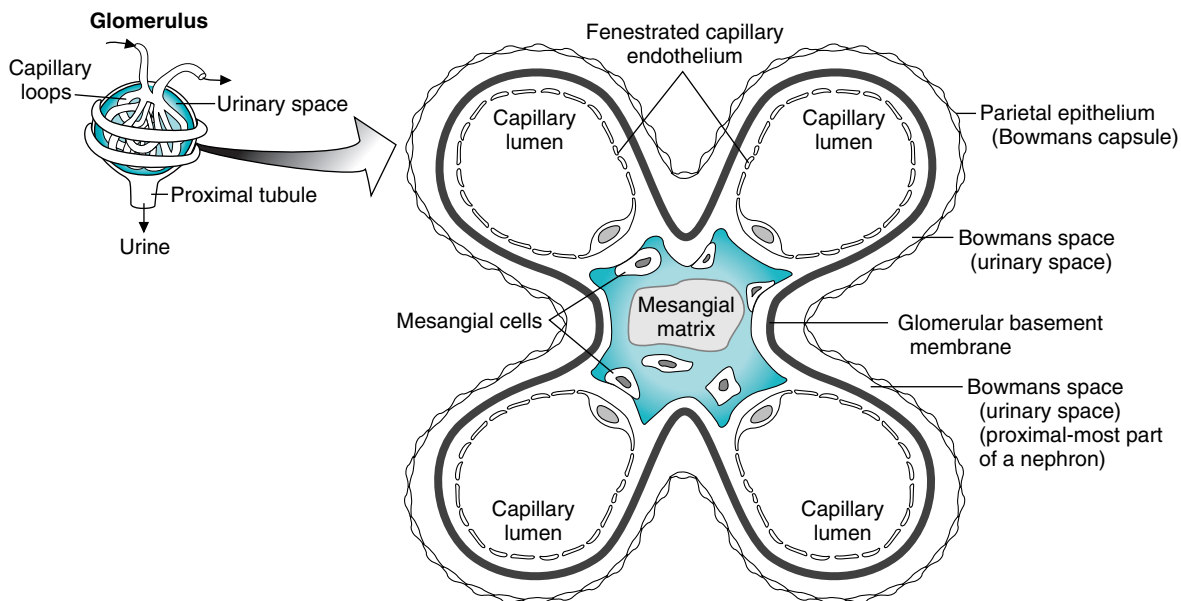


Fig. 49.15. A cross-section of a normal renal glomerulus showing four capillary tufts delivering blood to the glomerulus for filtration across the fenestrated capillary endothelium then through the glomerular basement membrane into the Bowman's space to form urine. The urine then enters the proximal tubule of the nephron. This filtration removes potentially toxic metabolic end products from the blood. The mesangium, by contracting and expanding, controls the efficiency of these filtering and excretory functions by regulating the hydraulic filtration pressures within the glomerulus. An intact basement membrane must be present to maintain the integrity of the filtering process.

thickening of the GBM, this membrane becomes “leaky” for some macromolecules (e.g., albumin) that normally do not enter the urine from the glomerular capillaries (microalbuminuria). Suggested mechanisms for this increased permeability or leakiness include reduced synthesis of the specific proteoglycan, heparan sulphate, as well as increased basement membrane production of vascular endothelium growth factor (VEGF), a known angiogenic and permeability factor; and expansion of the extracellular matrix in the mesangium. The mesangium consists of specialized tissue containing collagen, proteoglycans, and other macromolecules that surround the glomerular capillaries and that, through its gel-like and sieving properties, determine, in part, the glomerular capillary hydraulic filtration pressure as well as the functional status of the capillary endothelium–mesangial glomerular basement membrane filtration apparatus (see Fig. 49.15). As the mesangial tissue expands, the efficiency of glomerular filtration diminishes proportionately. The cause of these mesangial changes is, in part, the consequence of increased expression of certain growth factors, especially transforming growth factor β (TGF- β) and connective tissue growth factor (CTGF). Current therapeutic approaches in patients with early diabetic nephropathy include the use of antibodies that neutralize TGF- β .

BIOCHEMICAL COMMENTS



Osteogenesis imperfecta (OI) is a heterogenous group of diseases that have in common a defect in collagen production. This defect can be either of two types: The first type is associated with a reduction in the synthesis of normal collagen (due to a gene deletion or splice-site mutation). The second type is associated with the synthesis of a mutated form of collagen. Most of the mutations have a dominant-negative effect, leading to an autosomal dominant mode of transmission.

In the second type of OI, many of the known mutations involve substitutions of another amino acid for glycine. This results in an unstable collagen molecule, because glycine is the only amino acid that can fit between the other two chains within the triple helix of collagen. If the mutation is near the carboxy-terminal of the molecule, the phenotype of the disease is usually more severe than if the mutation is near the amino-terminal end (recall that triple helix formation proceeds from the carboxy- to the amino-terminal end of the molecule). Of interest are mutations that replace glycine with either serine or cysteine. Such mutations are more stable than expected, because of the hydrogen-bonding capabilities of serine and the ability of cysteine to form disulfide bonds. Both would aid in preventing the strands of the triple helix from unwinding.

Children with OI can be treated with a class of compounds known as bisphosphonates, which consist of two phosphates linked by a carbon or nitrogen bridge (thus, they are analogs of pyrophosphate, in which the two phosphates are linked by oxygen). Normal bone remodeling is the result of a coordinated “coupling” between osteoclast activity (cells that resorb bone) and osteoblast activity (cells that form bone). In OI, bone resorption outpaces bone formation because osteoclast activity is enhanced (perhaps because of the reduced levels of normal collagen present to act as nucleating sites for bone formation). This leads to a net loss of bone mass and fragility of the skeleton. Bisphosphonates inhibit osteoclast action with the potential to increase bone mass and its tensile strength.

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REVIEW QUESTIONS—CHAPTER 49

1. Individuals who develop scurvy suffer from sore and bleeding gums and loss of teeth. This is due, in part, to the synthesis of a defective collagen molecule. The step that is affected in collagen biosynthesis attributable to scurvy is which of the following?
 - (A) The formation of disulfide bonds, which initiates tropocollagen formation
 - (B) The formation of lysyl cross-links between collagen molecules
 - (C) Secretion of tropocollagen into the extracellular matrix
 - (D) The formation of collagen fibrils
 - (E) The hydroxylation of proline residues, which stabilizes the collagen structure
2. The underlying mechanism that allows elastin to exhibit elastic properties (expand and contract) is which of the following?
 - (A) Proteolysis during expansion, and resynthesis during contraction
 - (B) Breaking of disulfide bonds during expansion, reformation of these bonds during contraction
 - (C) A decrease in entropy during expansion, and an increase in entropy during contraction
 - (D) The breaking of salt bridges during expansion, and reformation of the salt bridges during contraction
 - (E) Hydroxylation of elastin during expansion, and decarboxylation of elastin during contraction
3. The underlying mechanism by which glycosaminoglycans allow for the formation of a gel-like substance in the extracellular matrix in which of the following?
 - (A) Charge attraction between glycosaminoglycan chains
 - (B) Charge repulsion between glycosaminoglycan chains
 - (C) Hydrogen bonding between glycosaminoglycan chains
 - (D) Covalent cross-linking between glycosaminoglycan chains
 - (E) Hydroxylation of adjacent glycosaminoglycan chains
4. The movement of tumor cells from their site of origin to other locations within the body requires the activity of which of the following proteins?
 - (A) Collagen
 - (B) Laminin
 - (C) Proteoglycans
 - (D) Elastin
 - (E) Matrix metalloproteinases
5. Fibronectin is frequently absent in malignant fibroblast cells. One of the major functions of fibronectin is which of the following?
 - (A) To inhibit the action of matrix metalloproteinases
 - (B) To coordinate collagen deposition within the extracellular matrix
 - (C) To fix the position of cells within the extracellular matrix
 - (D) To regulate glycosaminoglycan production
 - (E) To extend glycosaminoglycan chains using nucleotide sugars